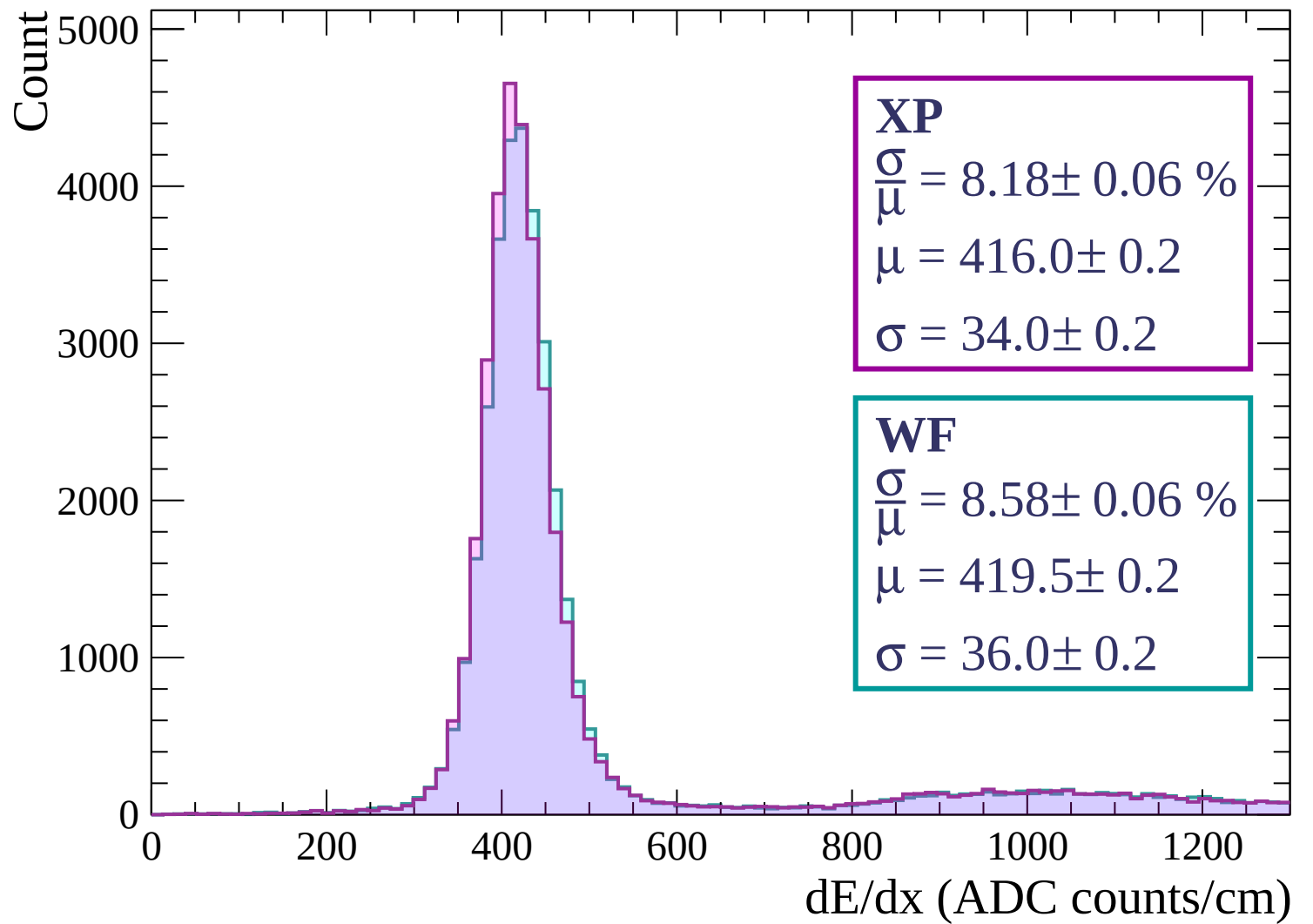
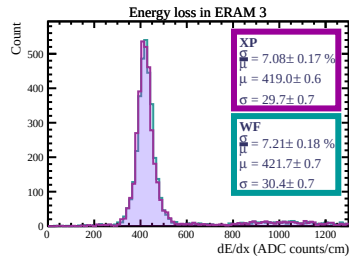
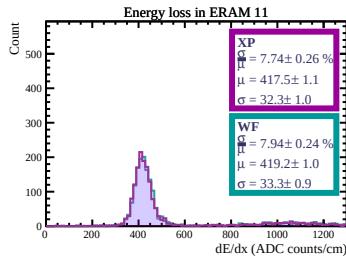
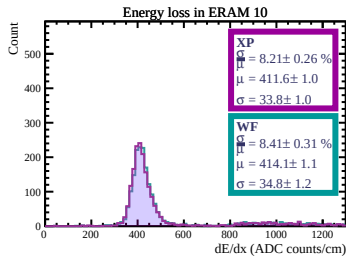
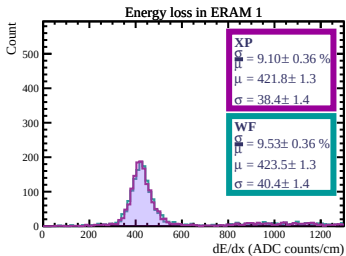
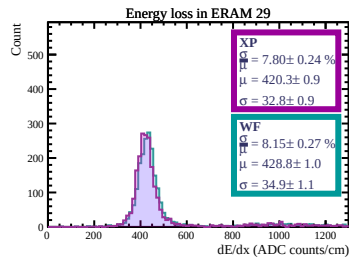
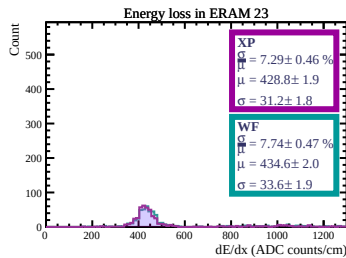
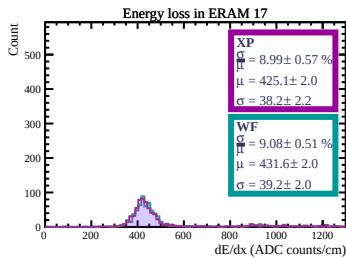
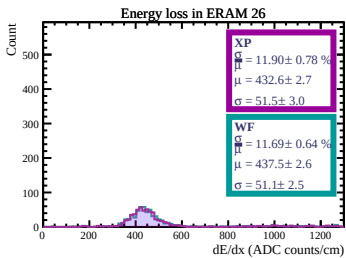
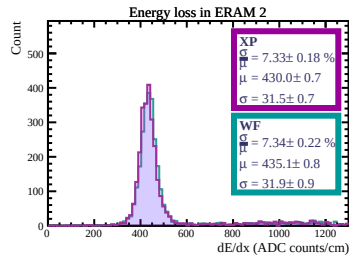
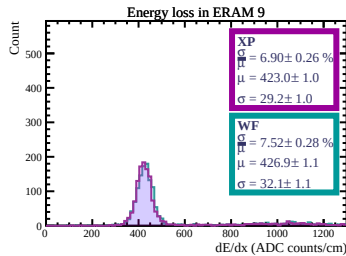
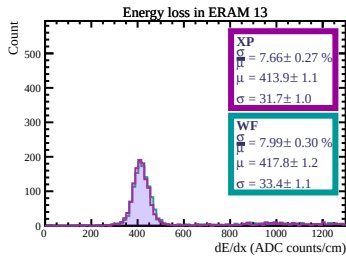
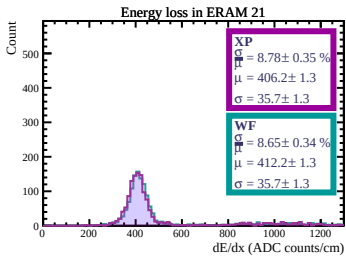
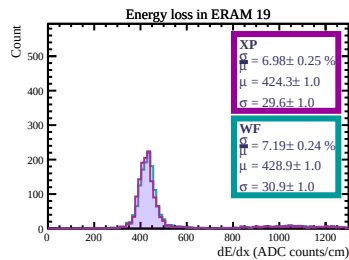
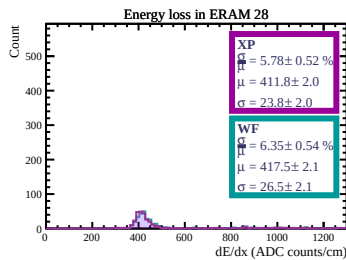
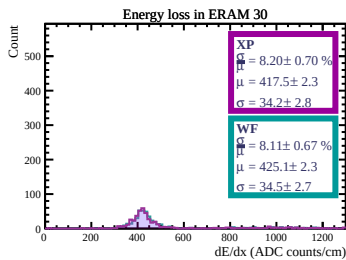
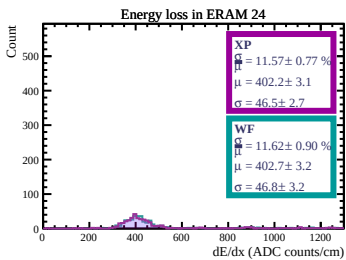
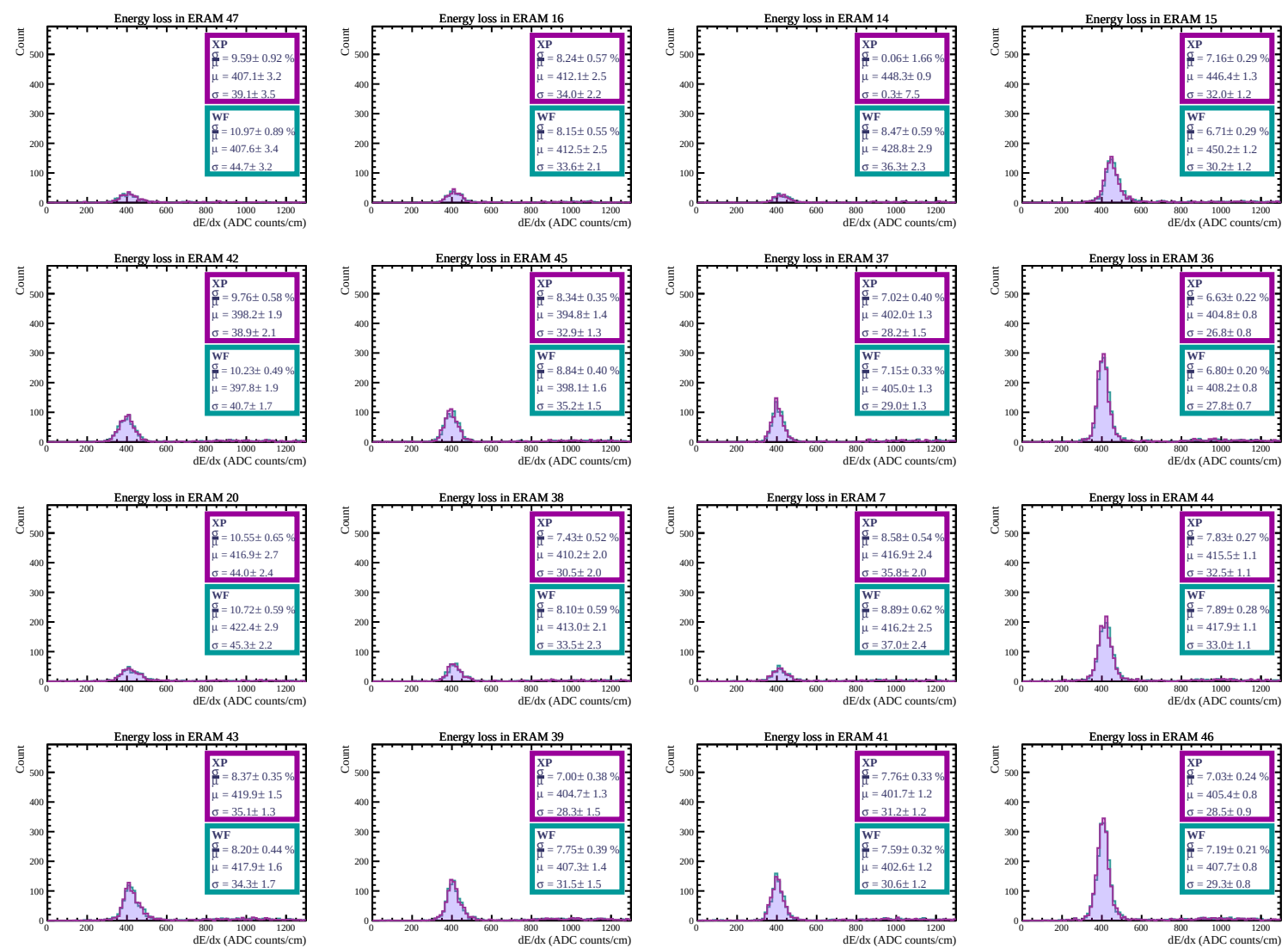
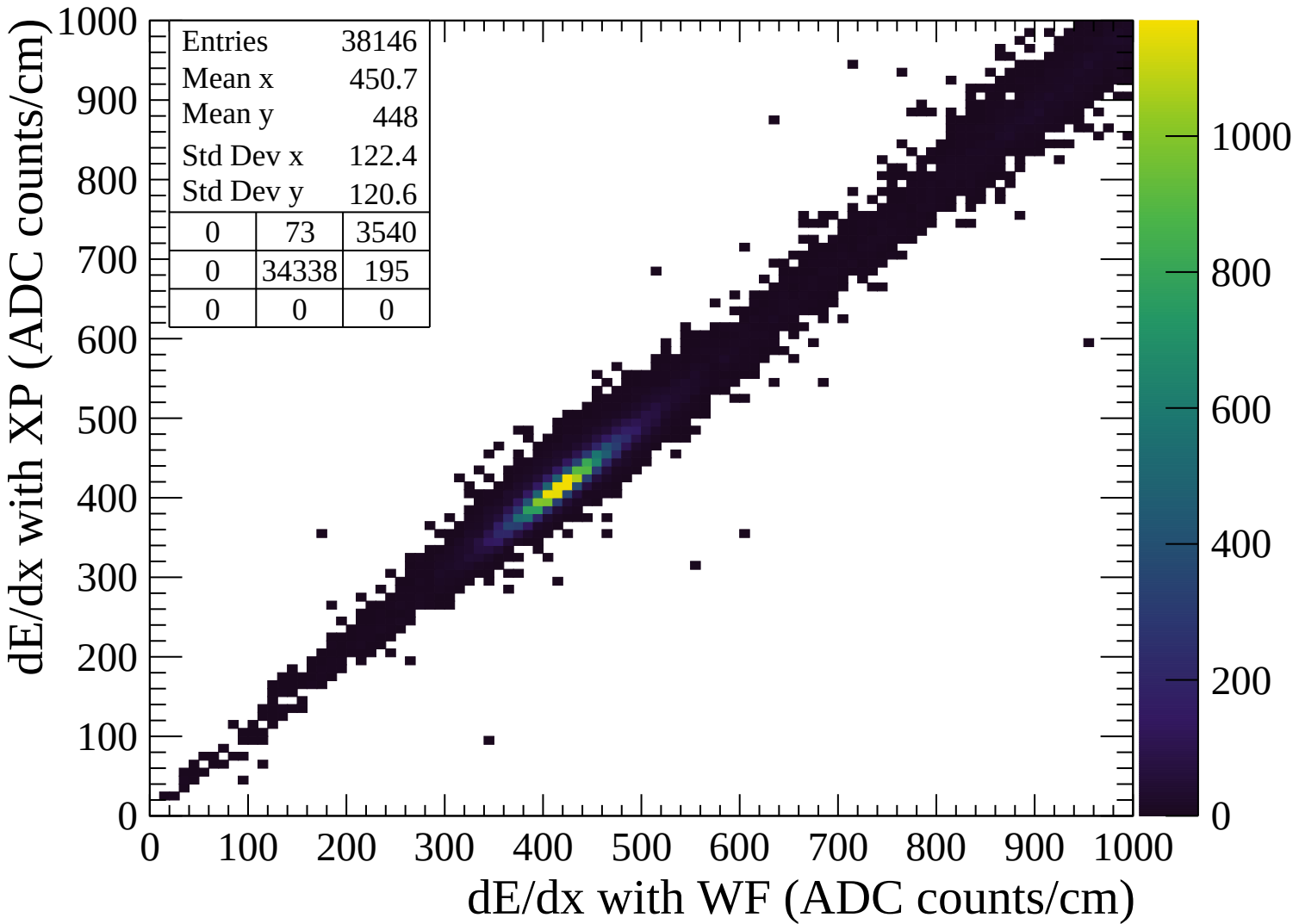
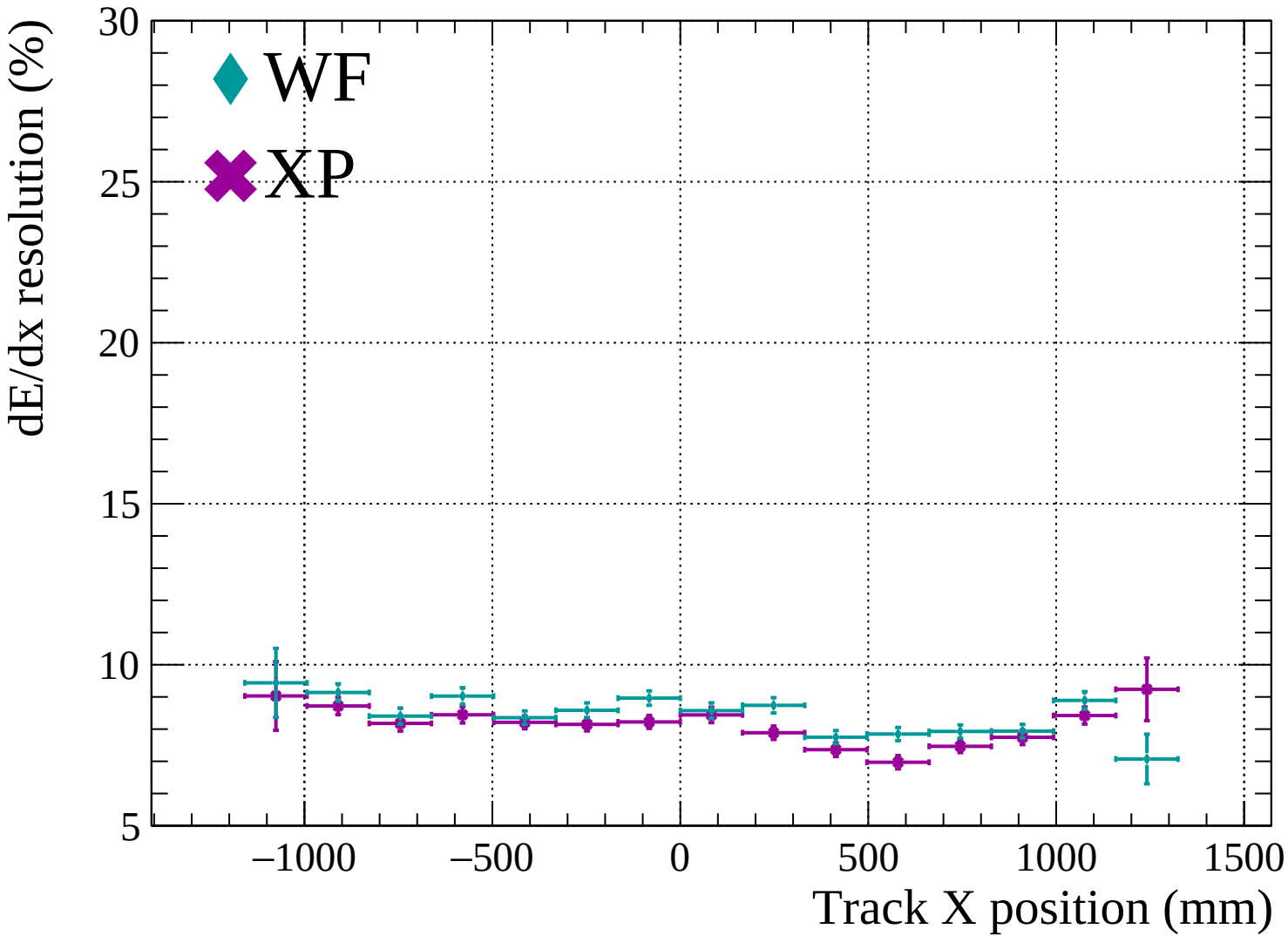


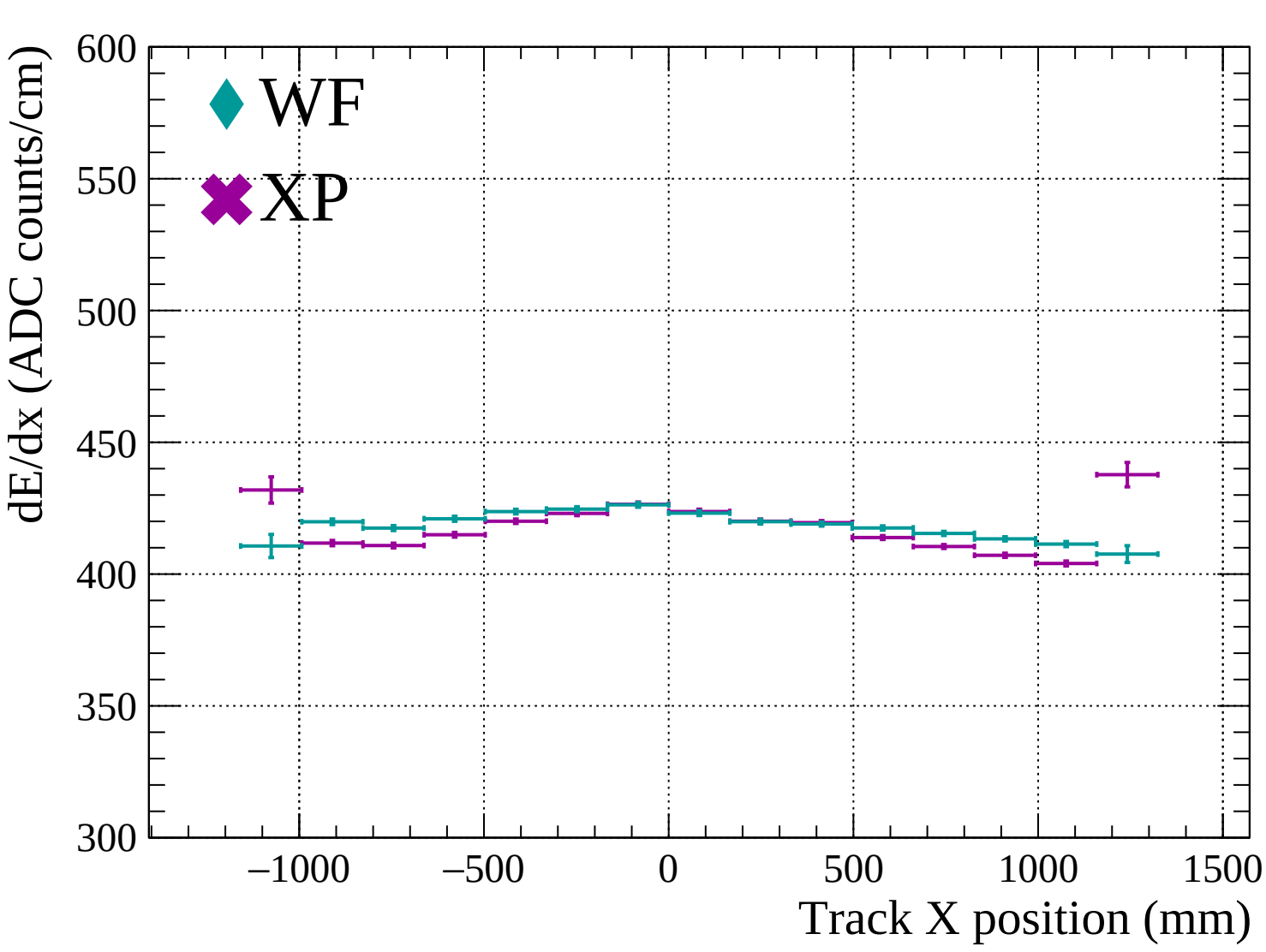


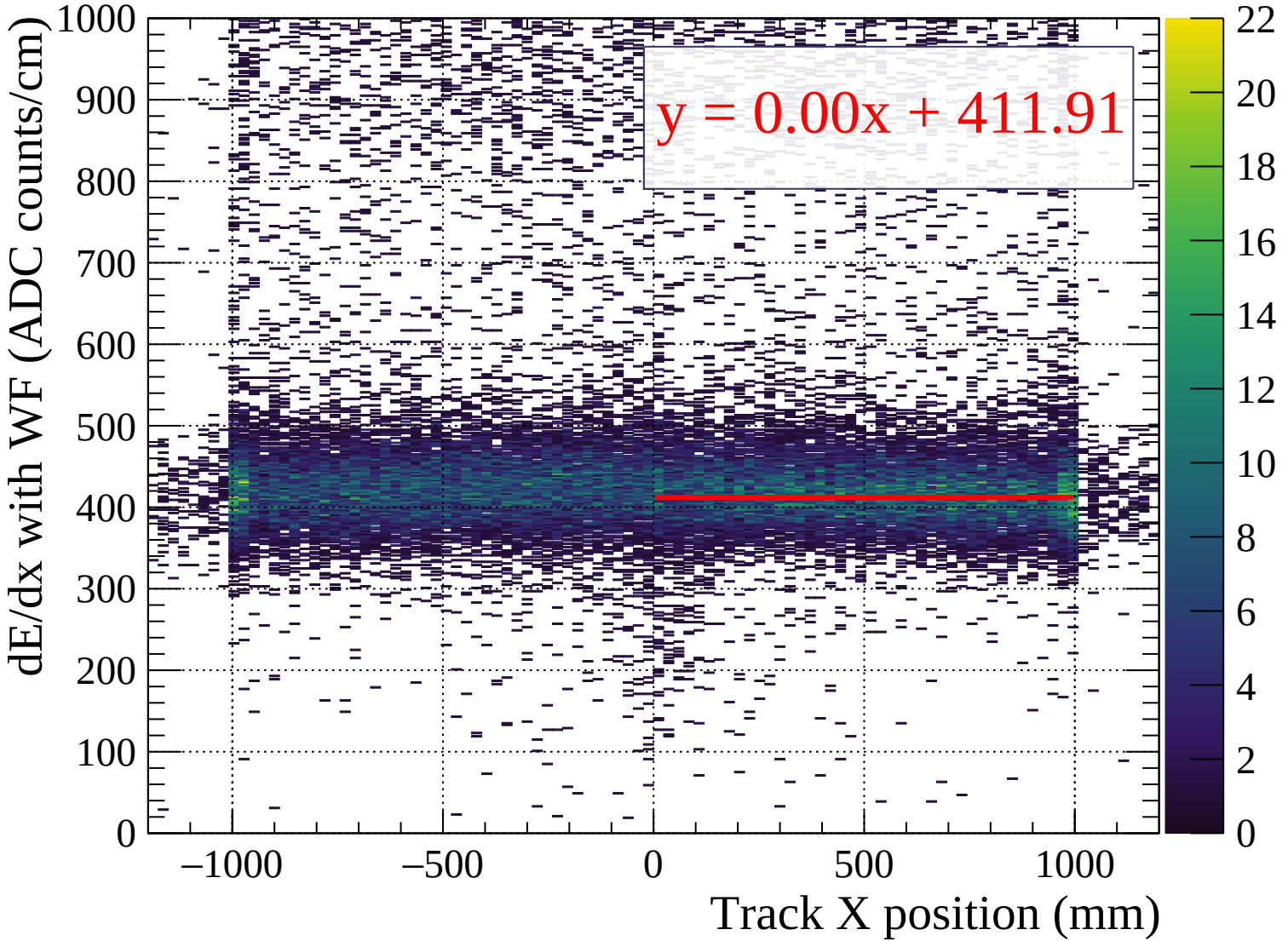


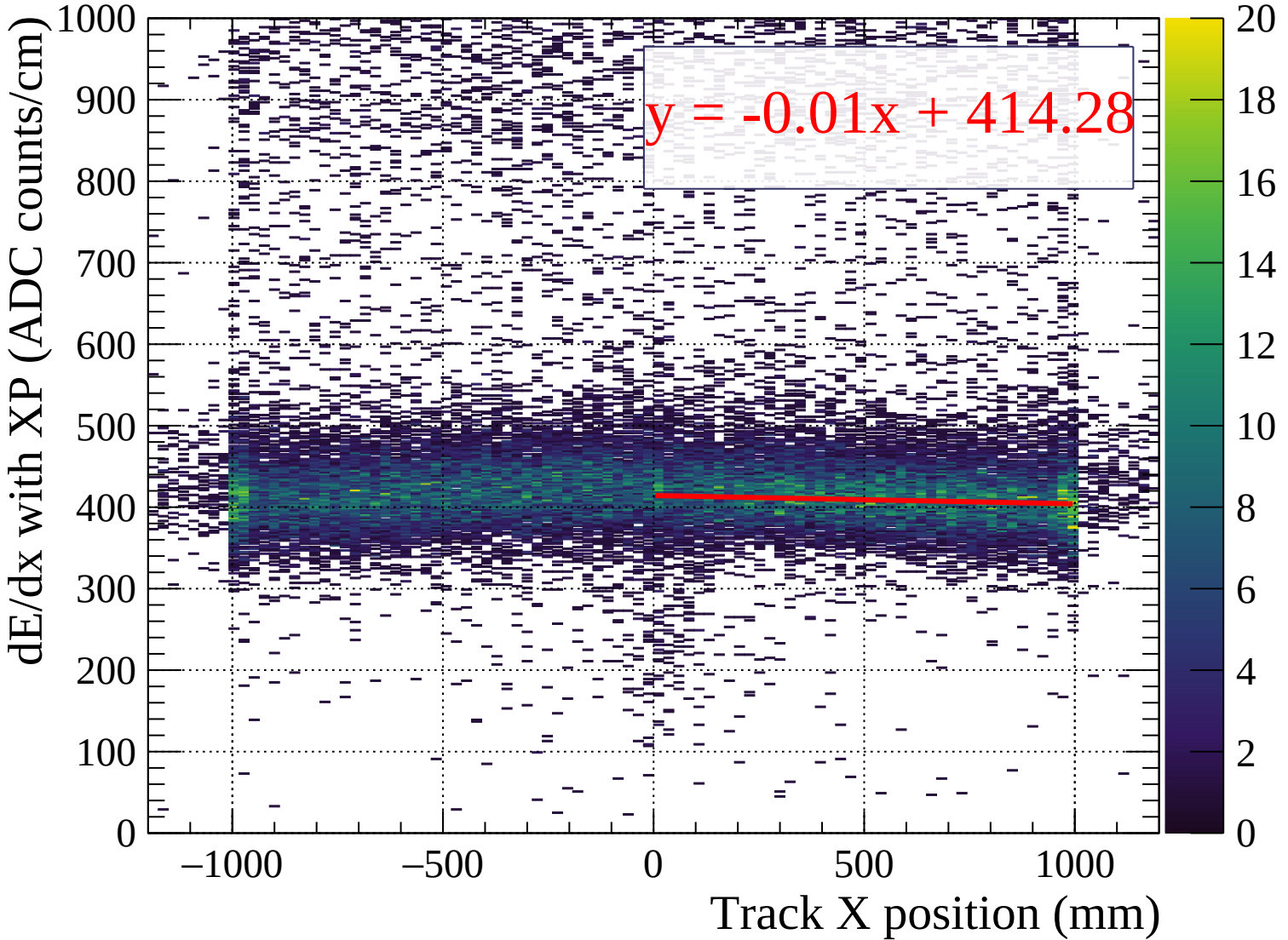


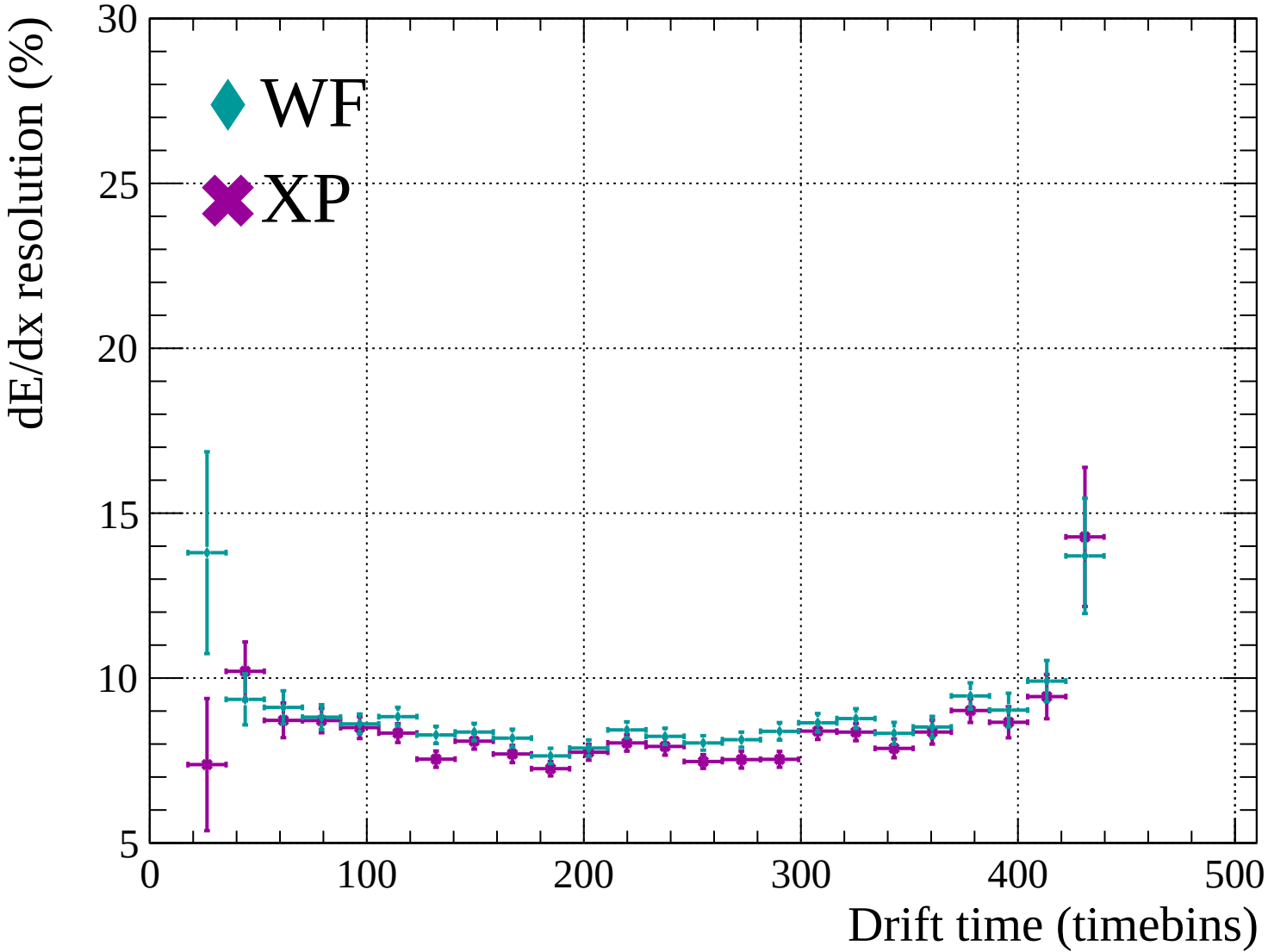


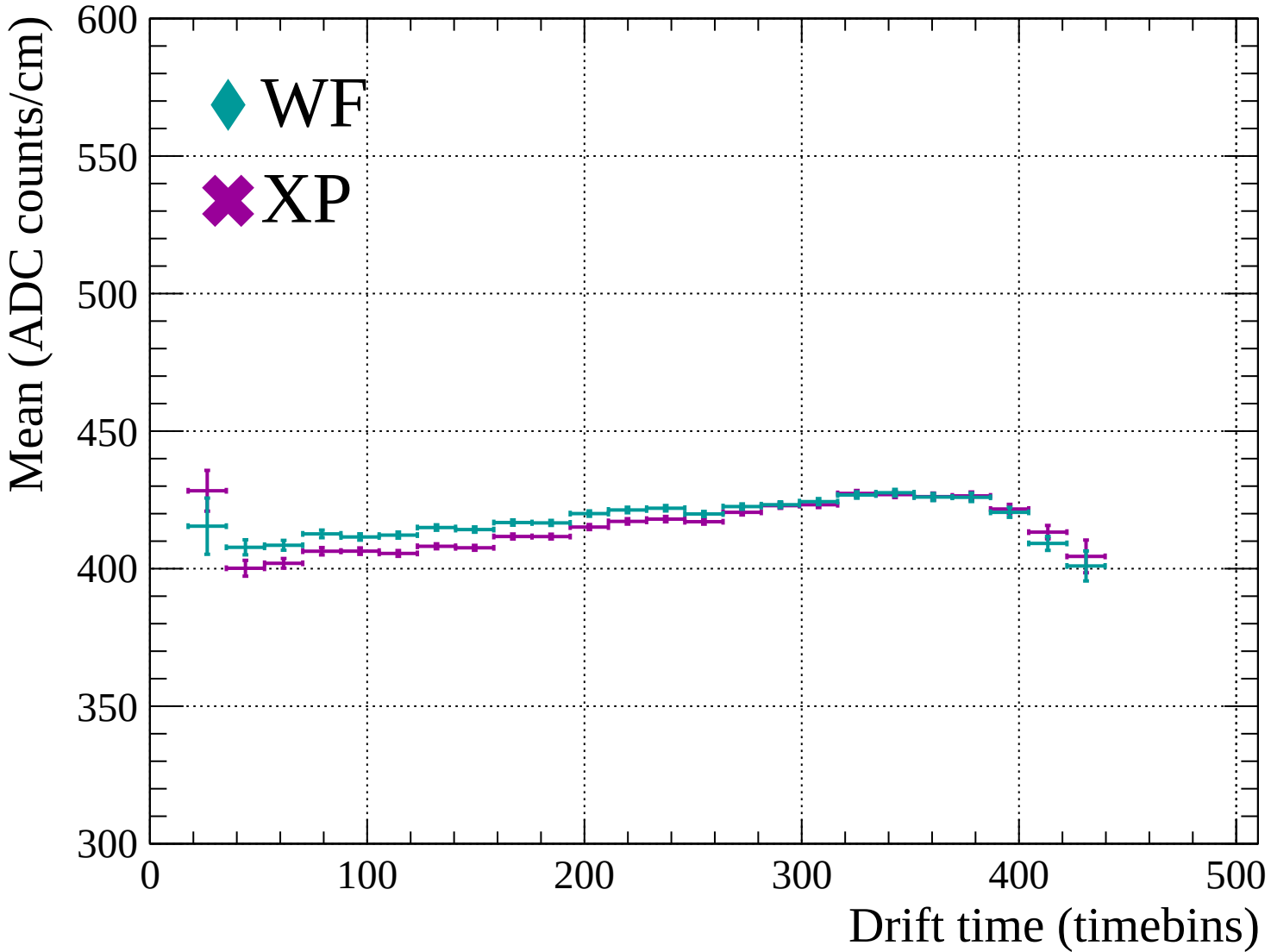


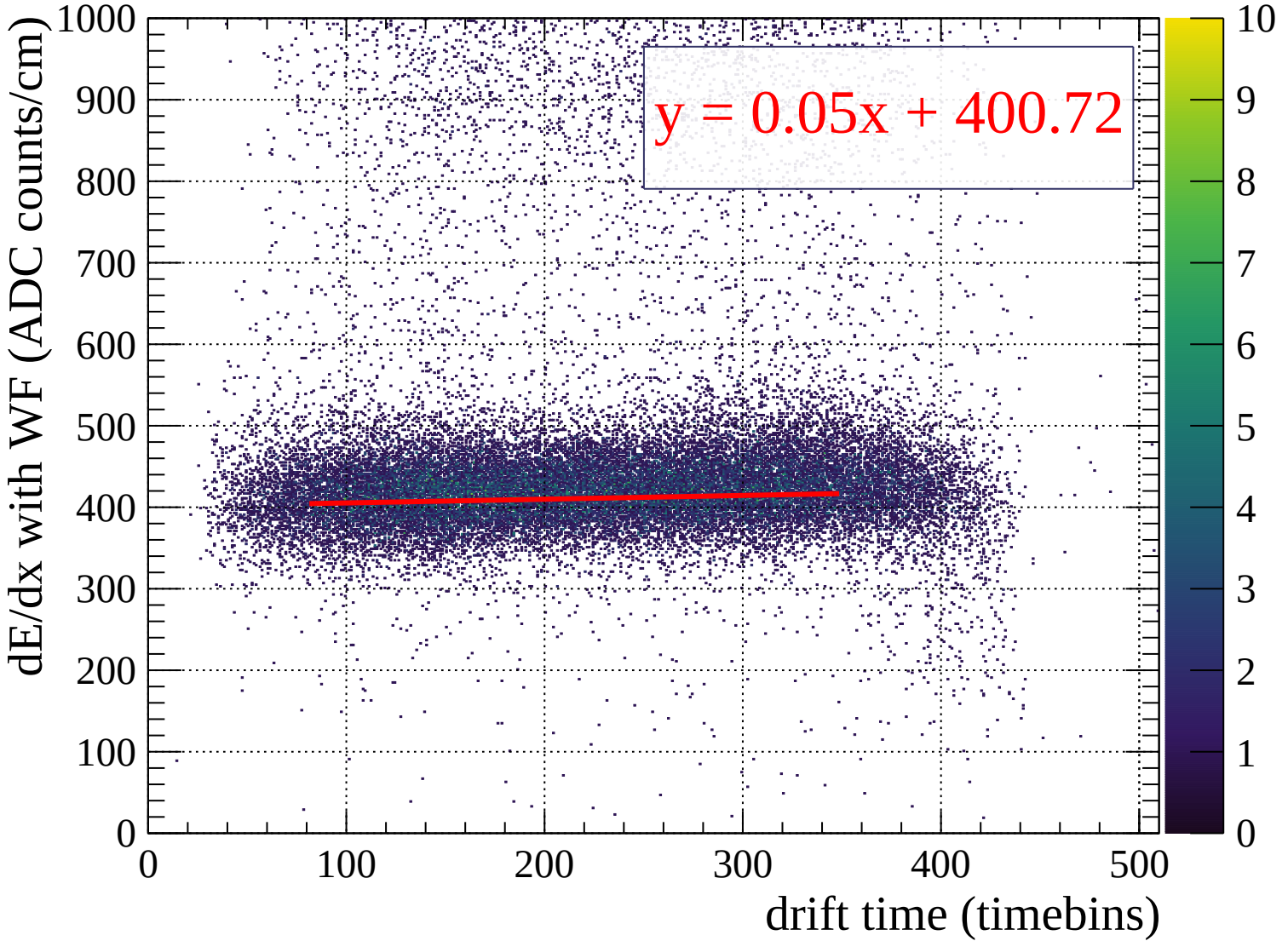


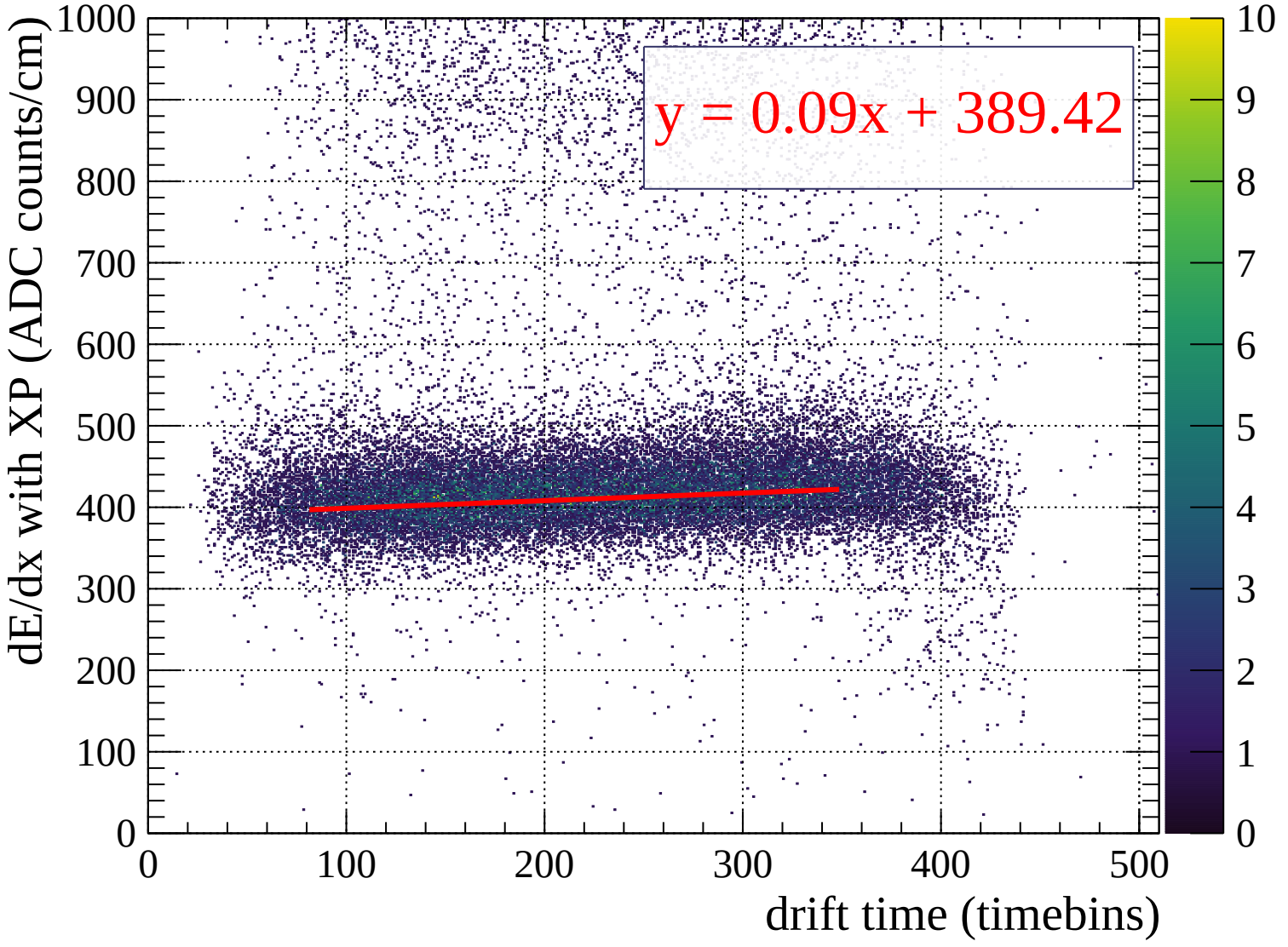


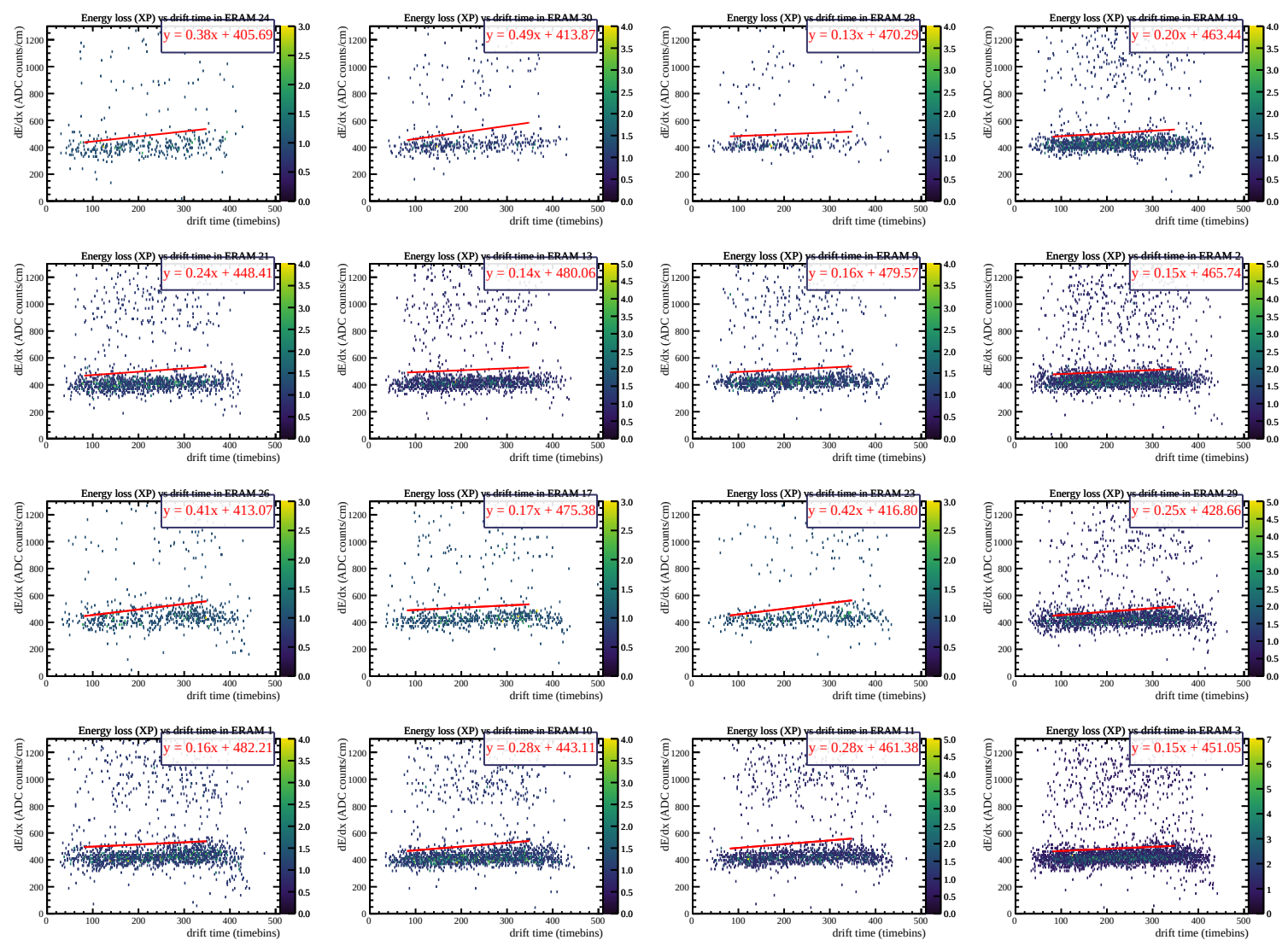


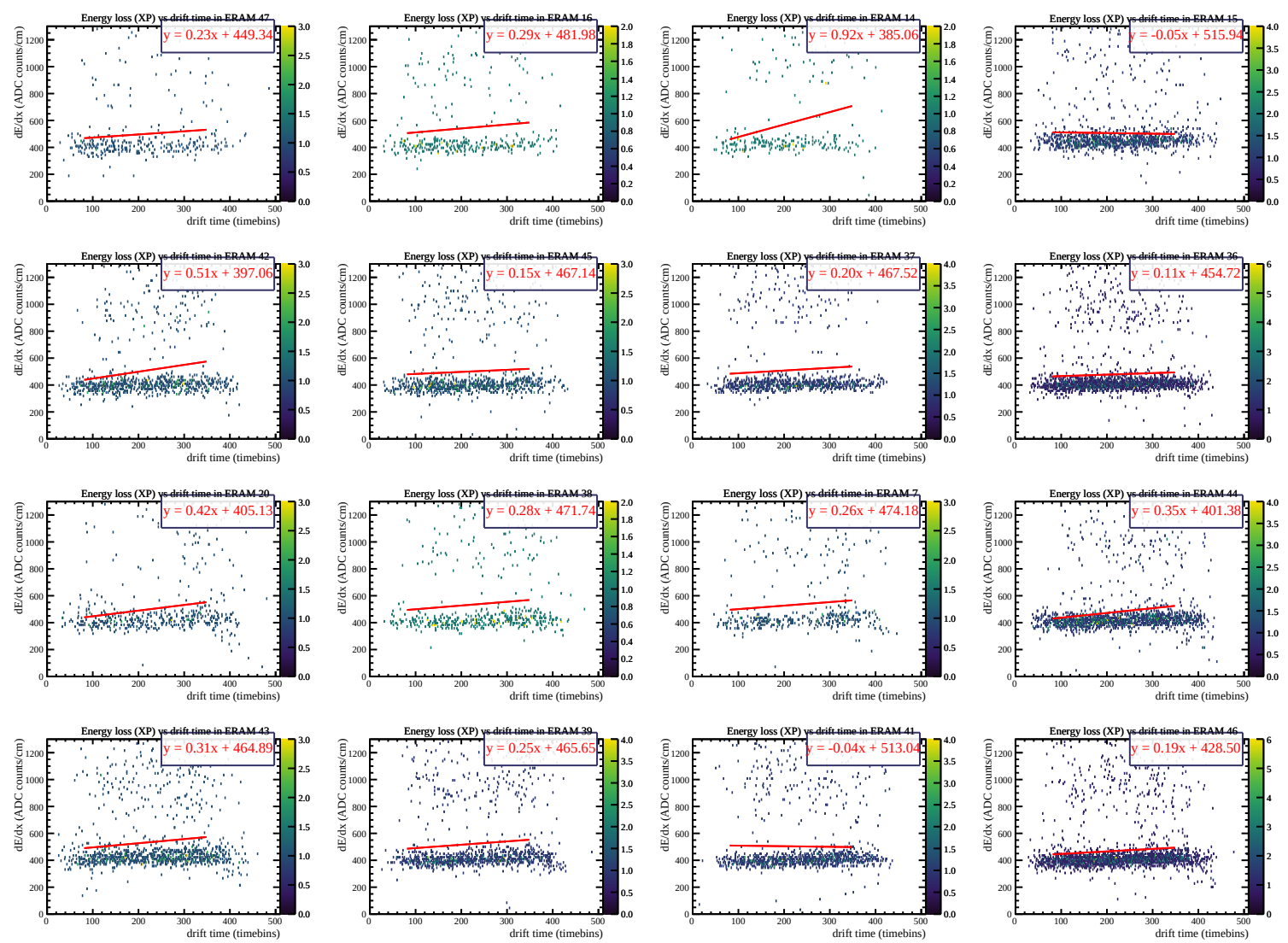


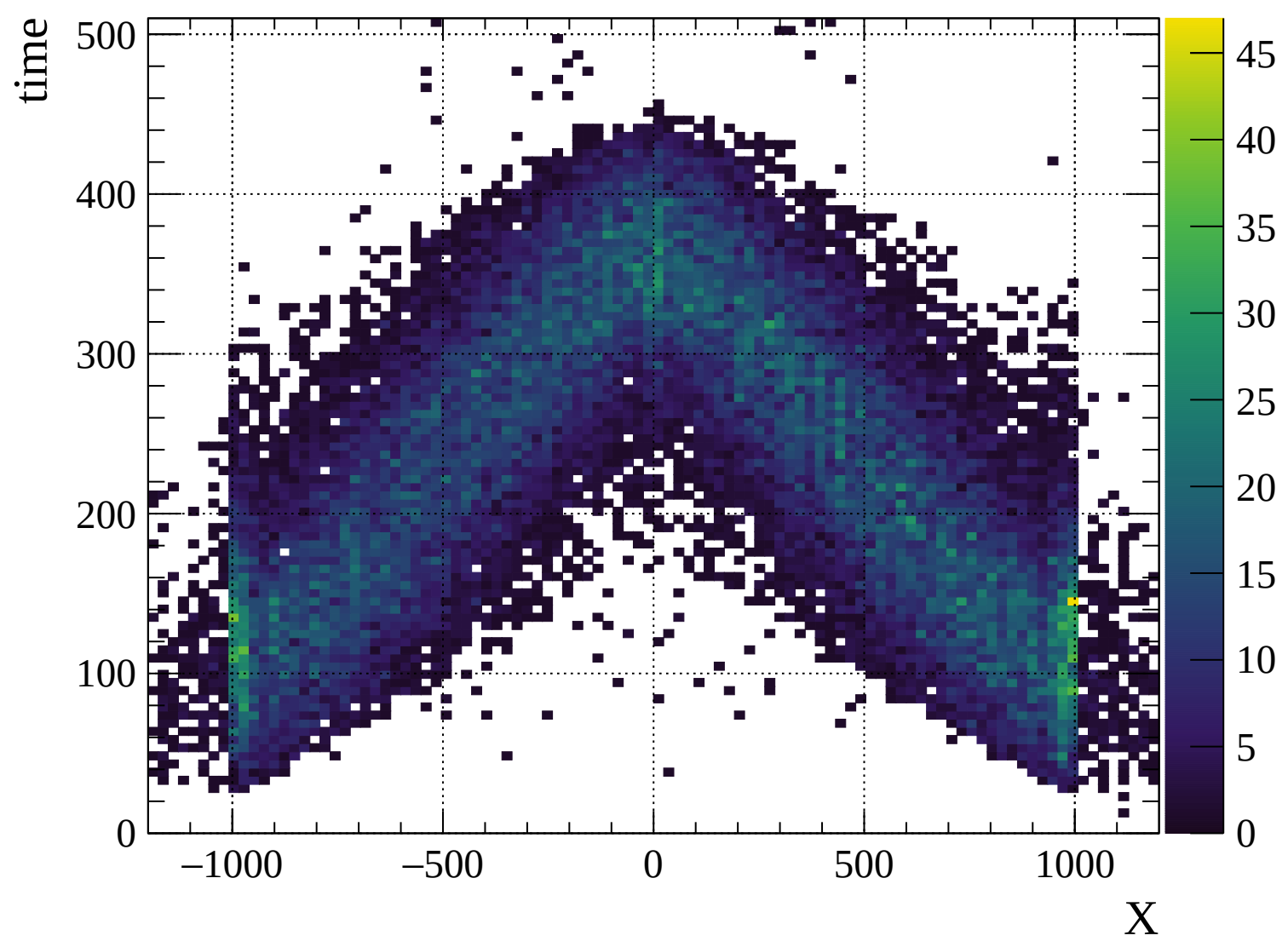


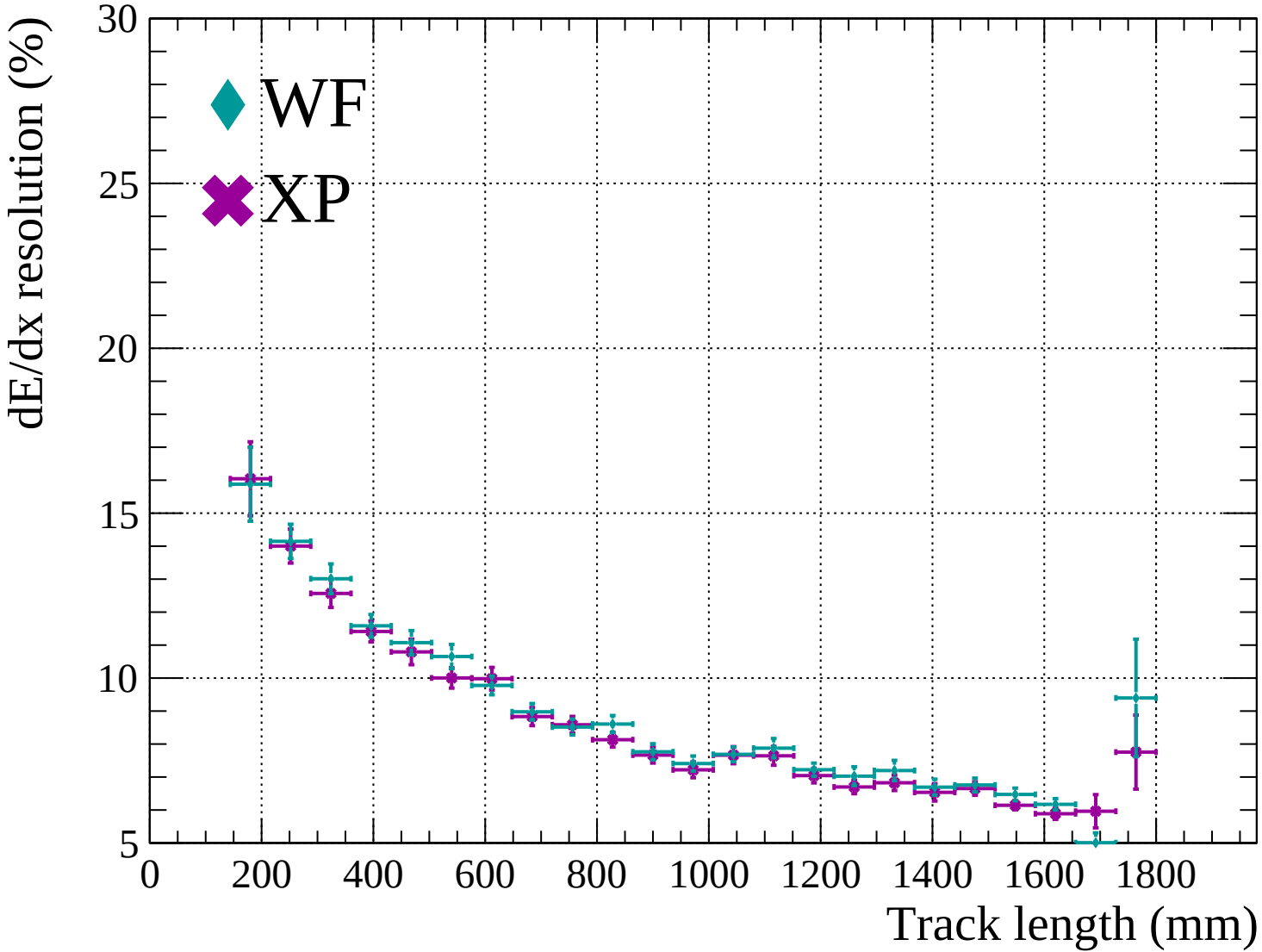


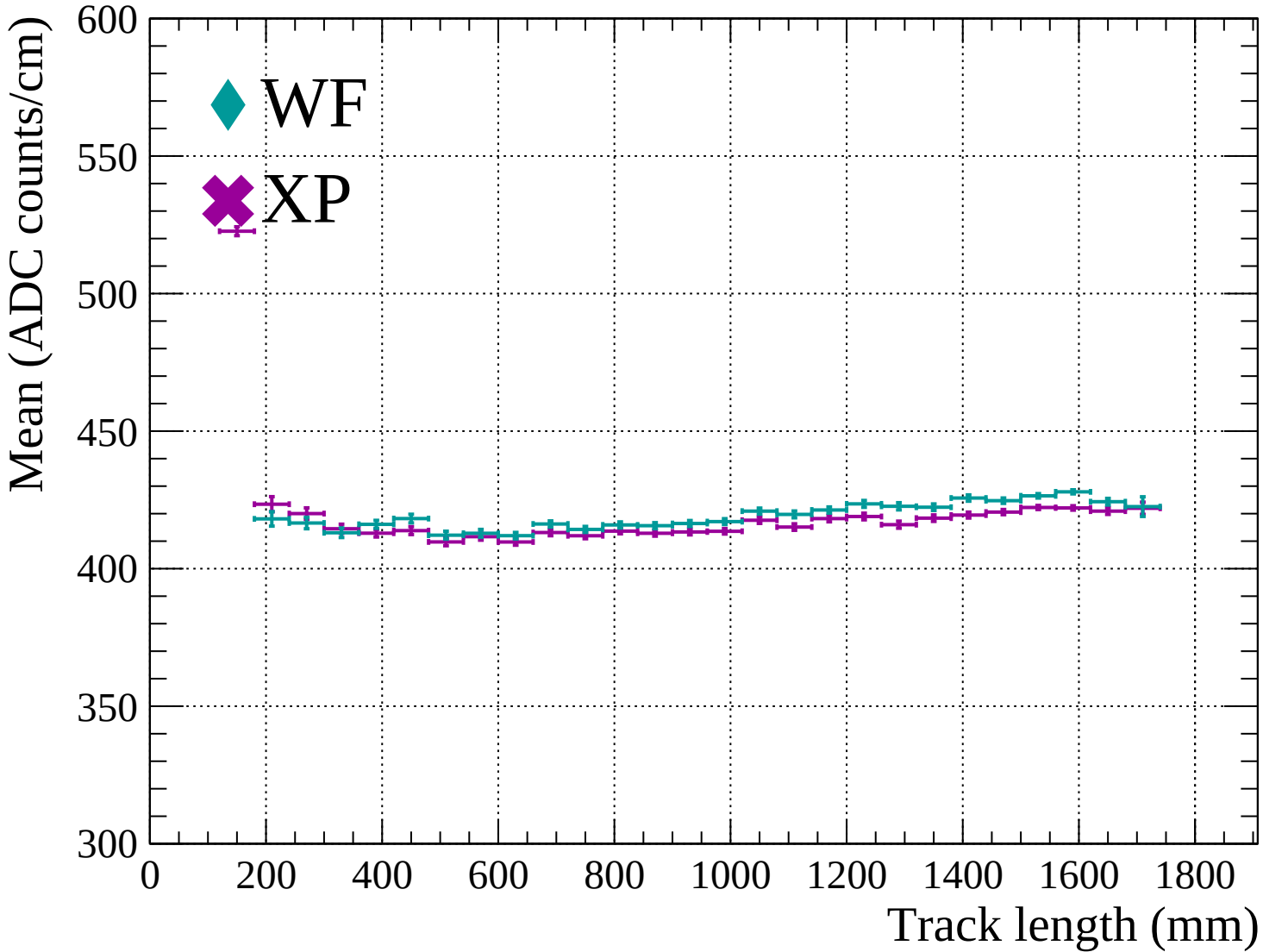


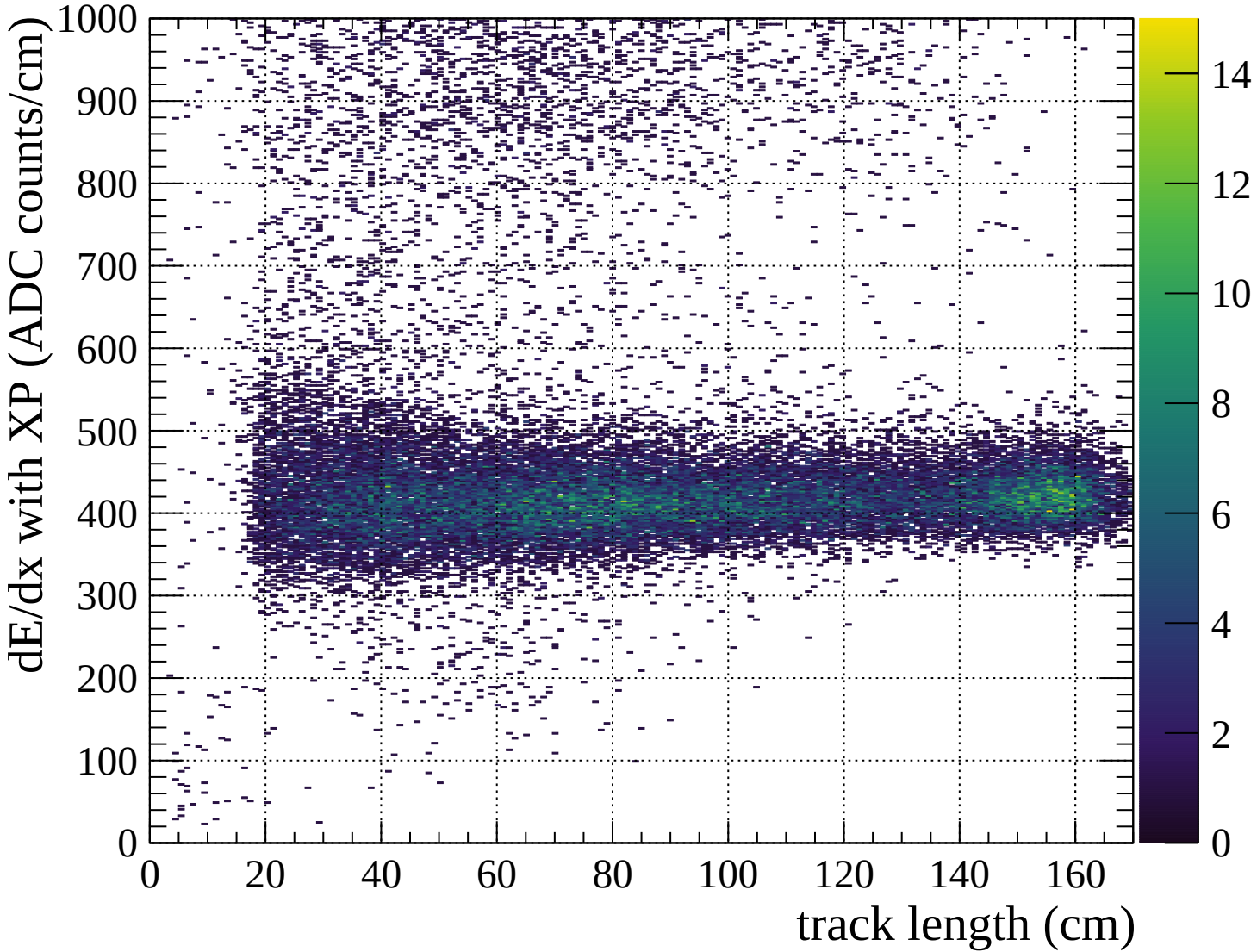


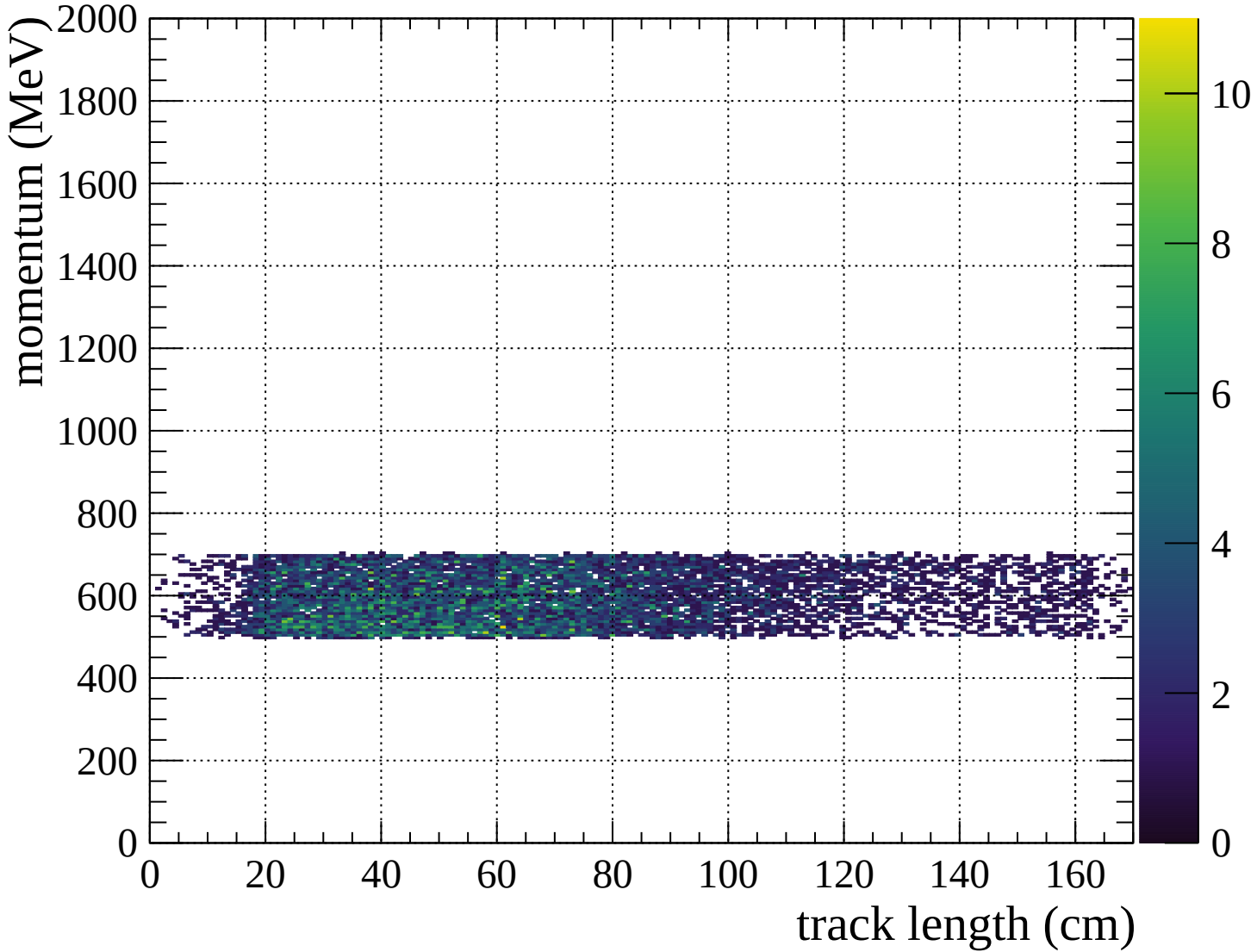


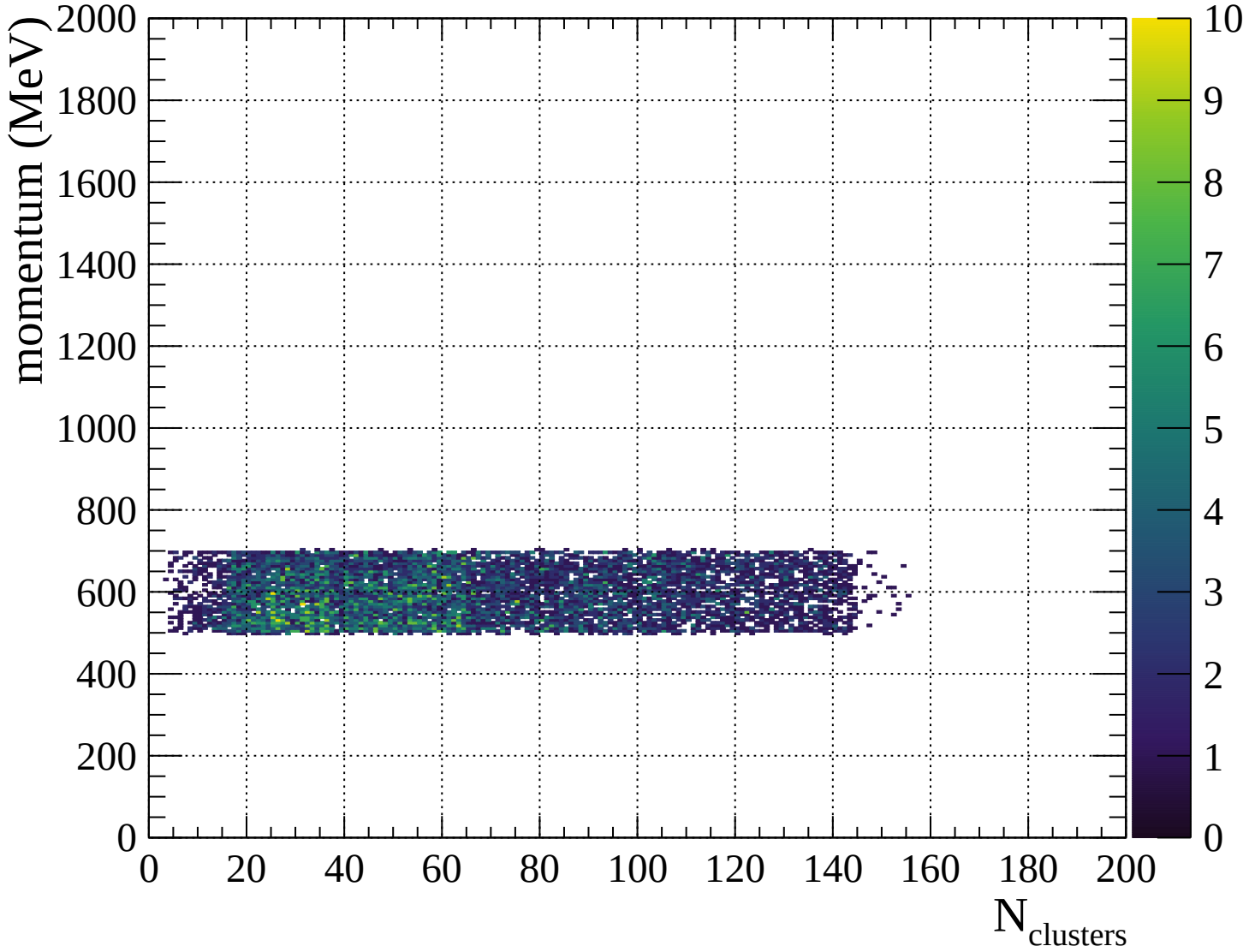


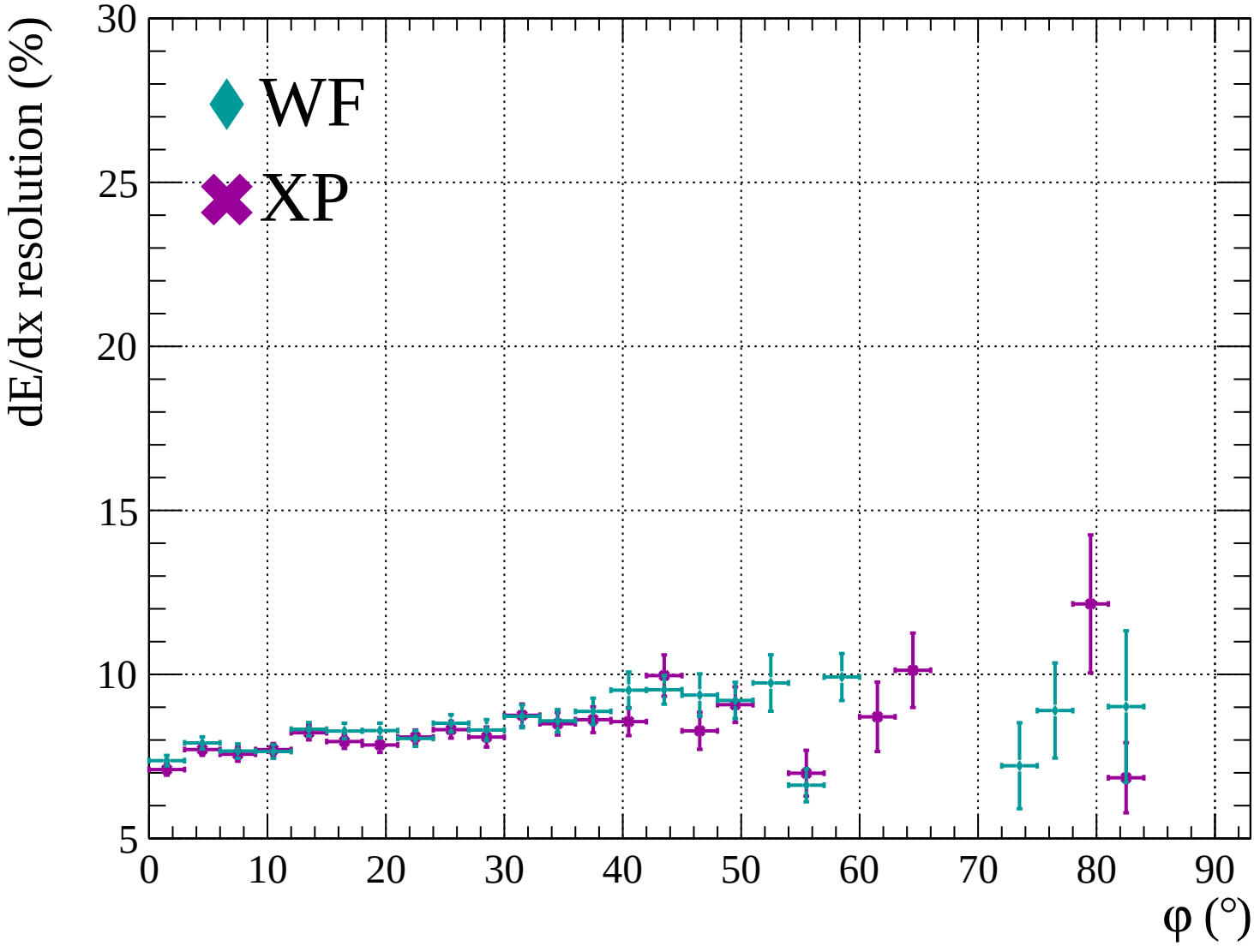


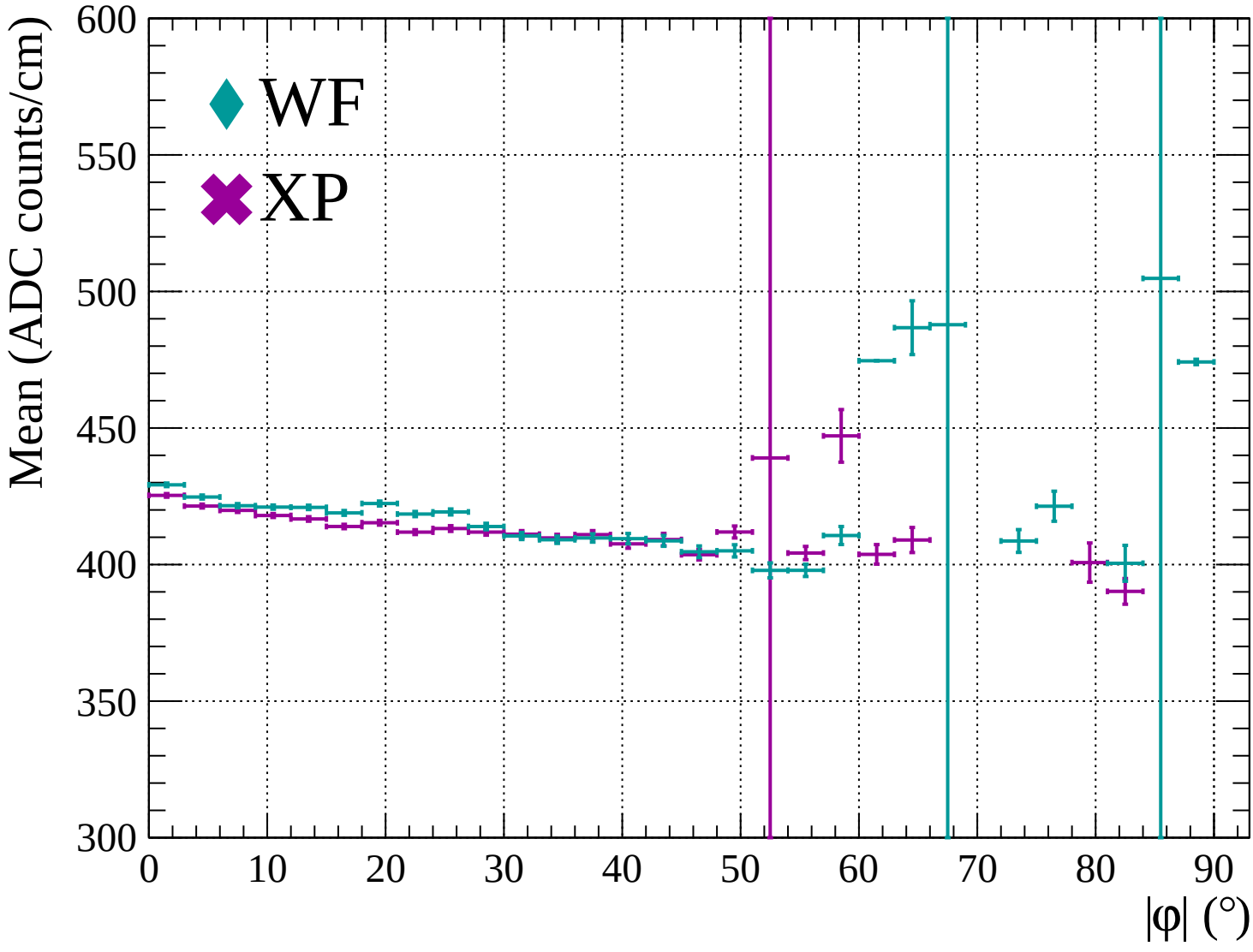


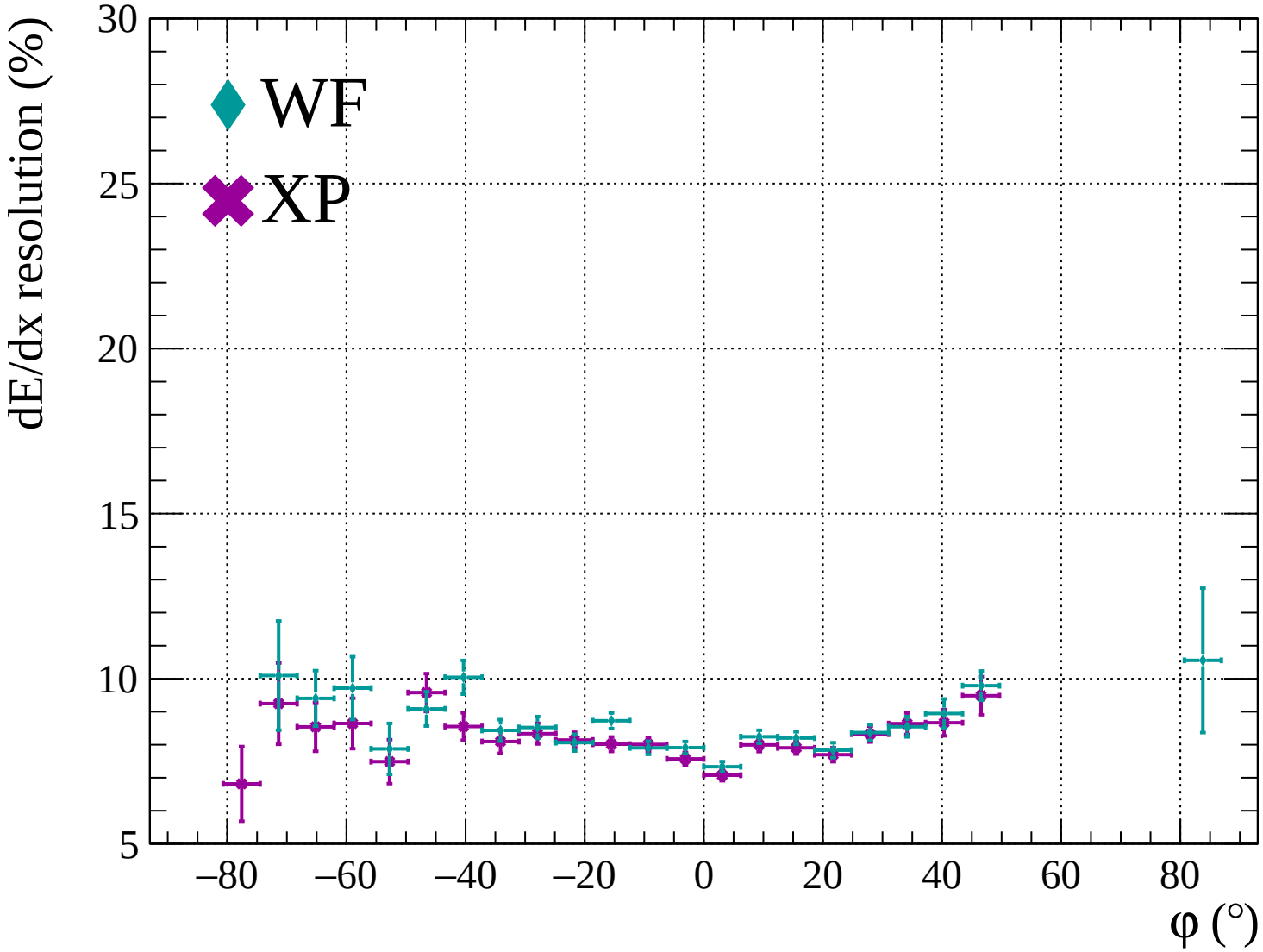


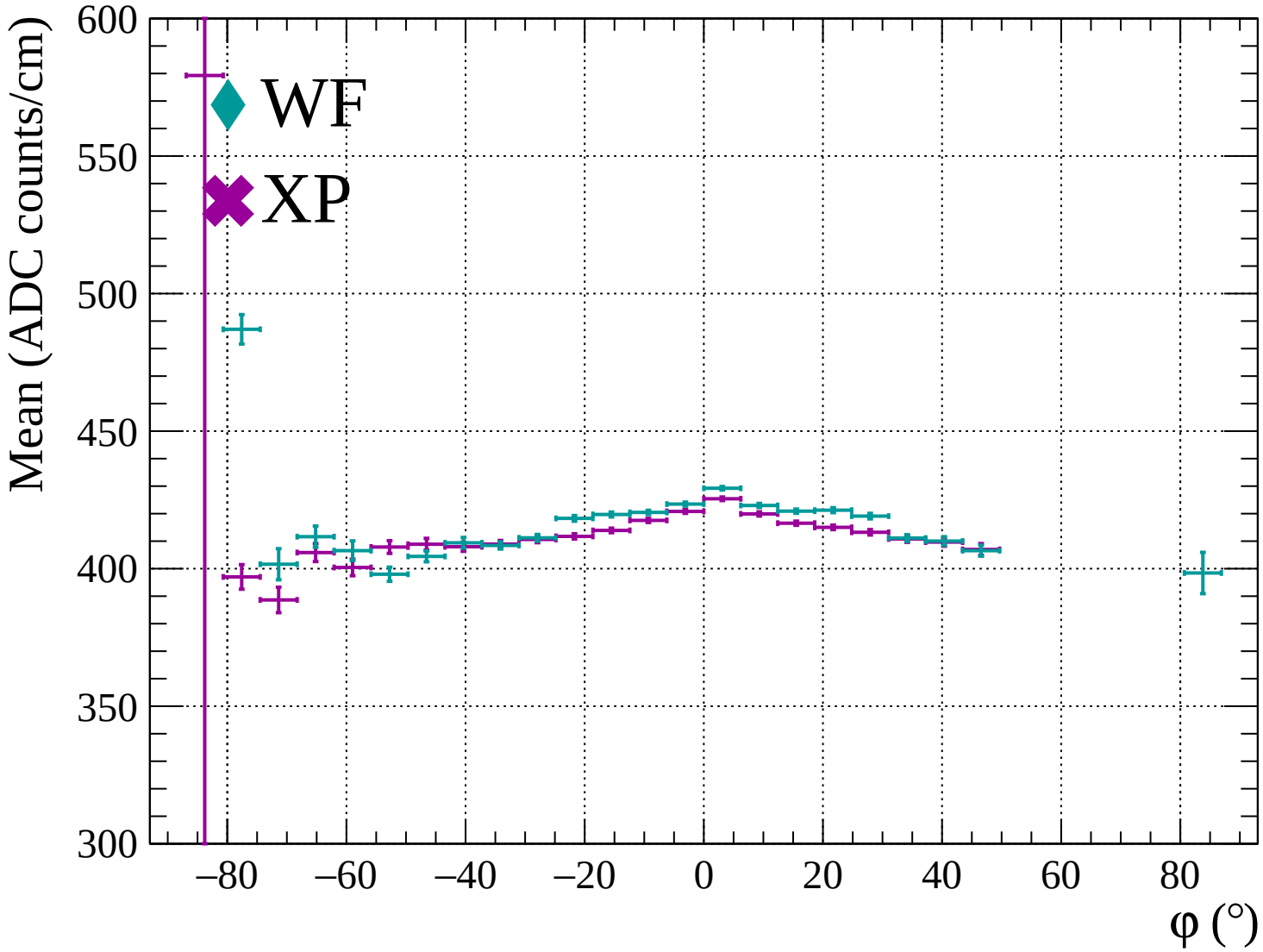


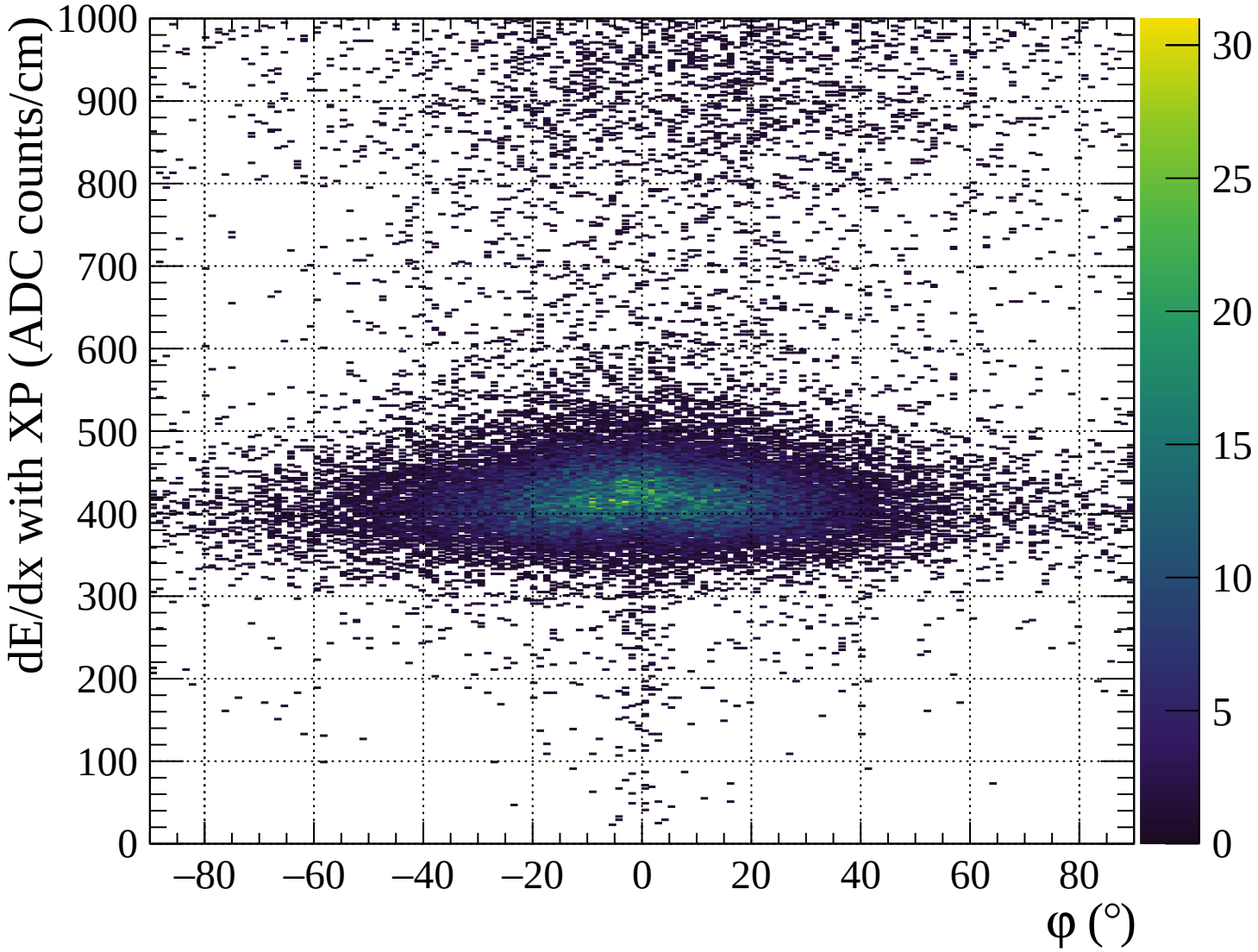


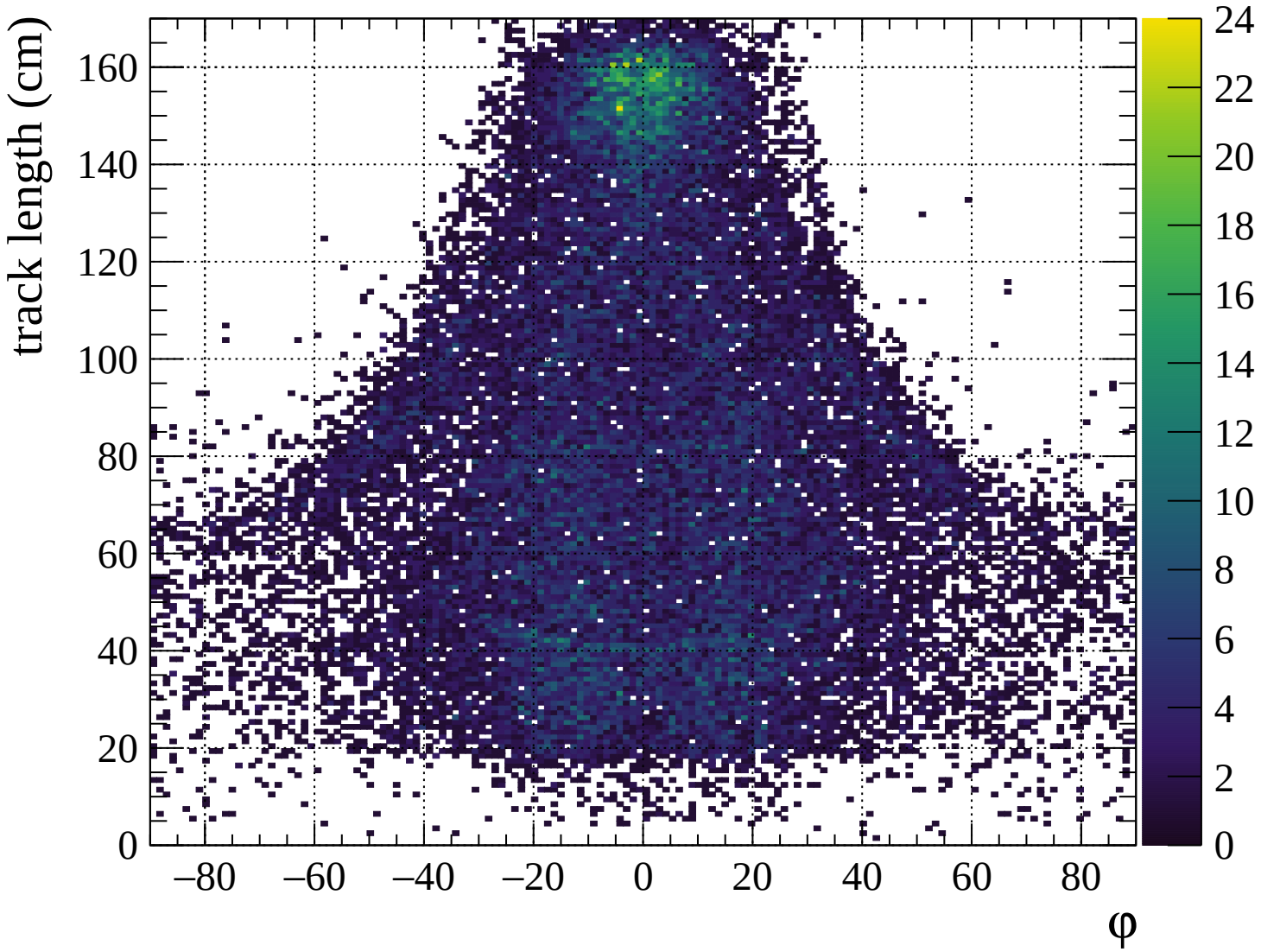


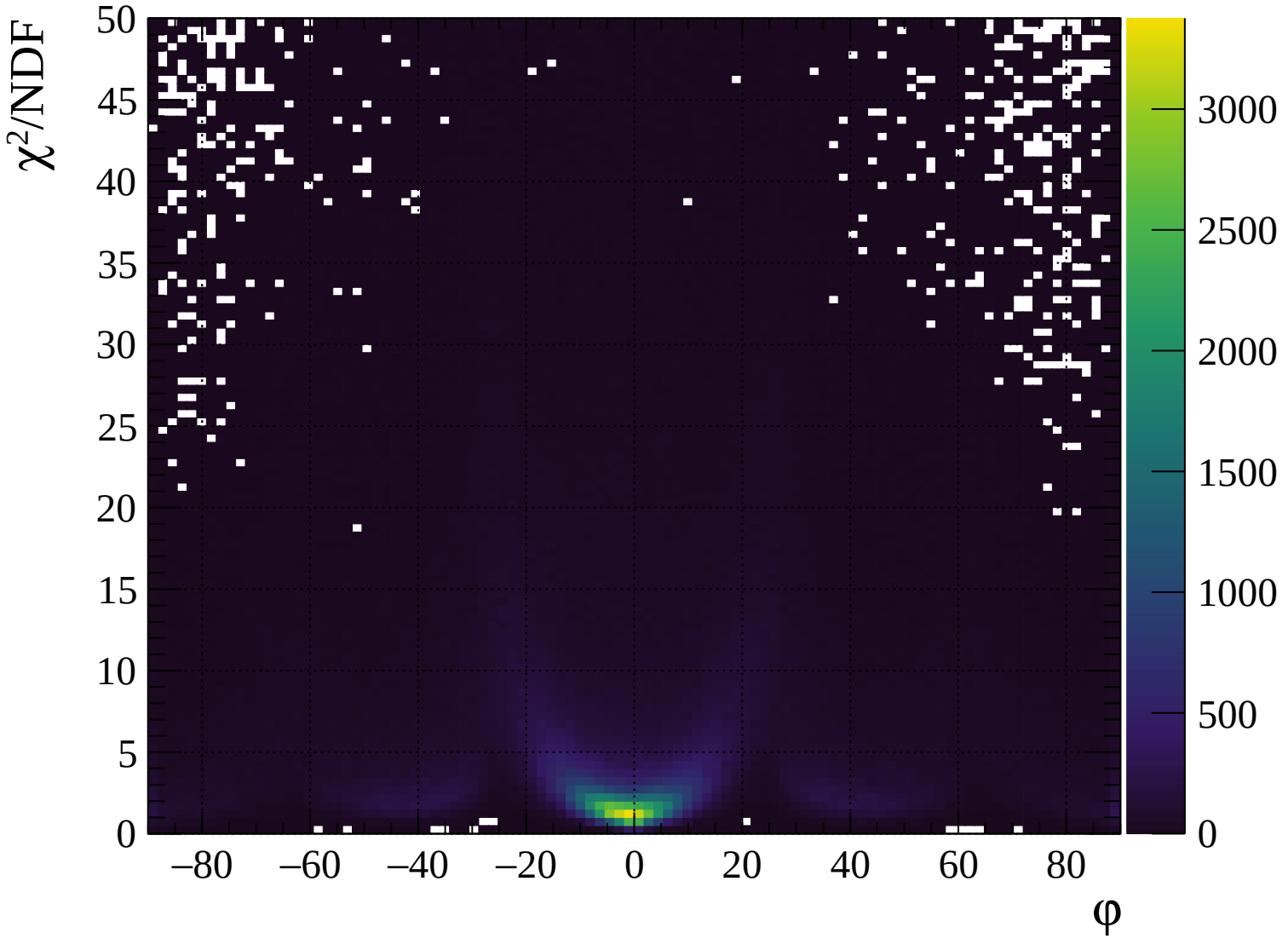


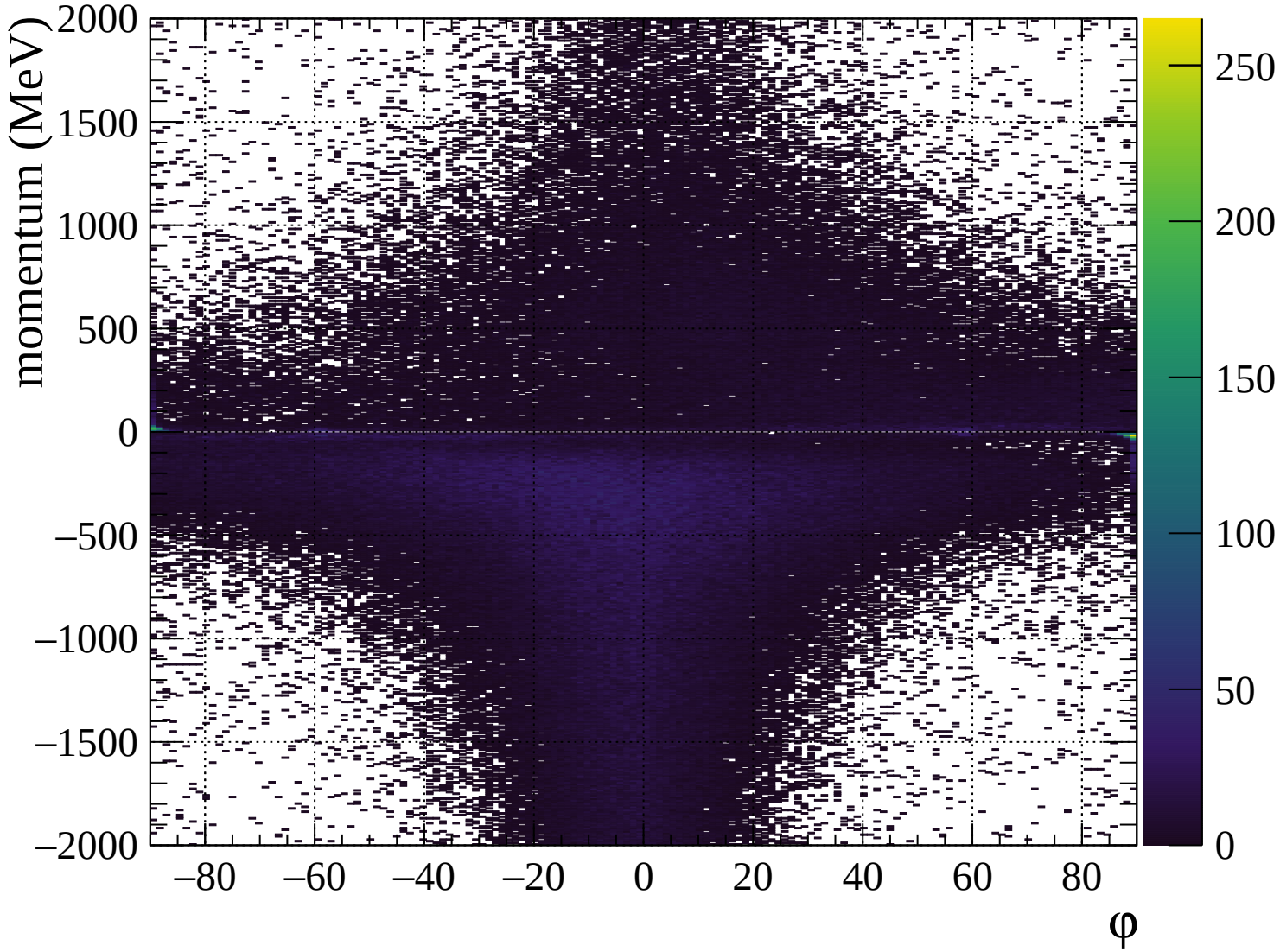


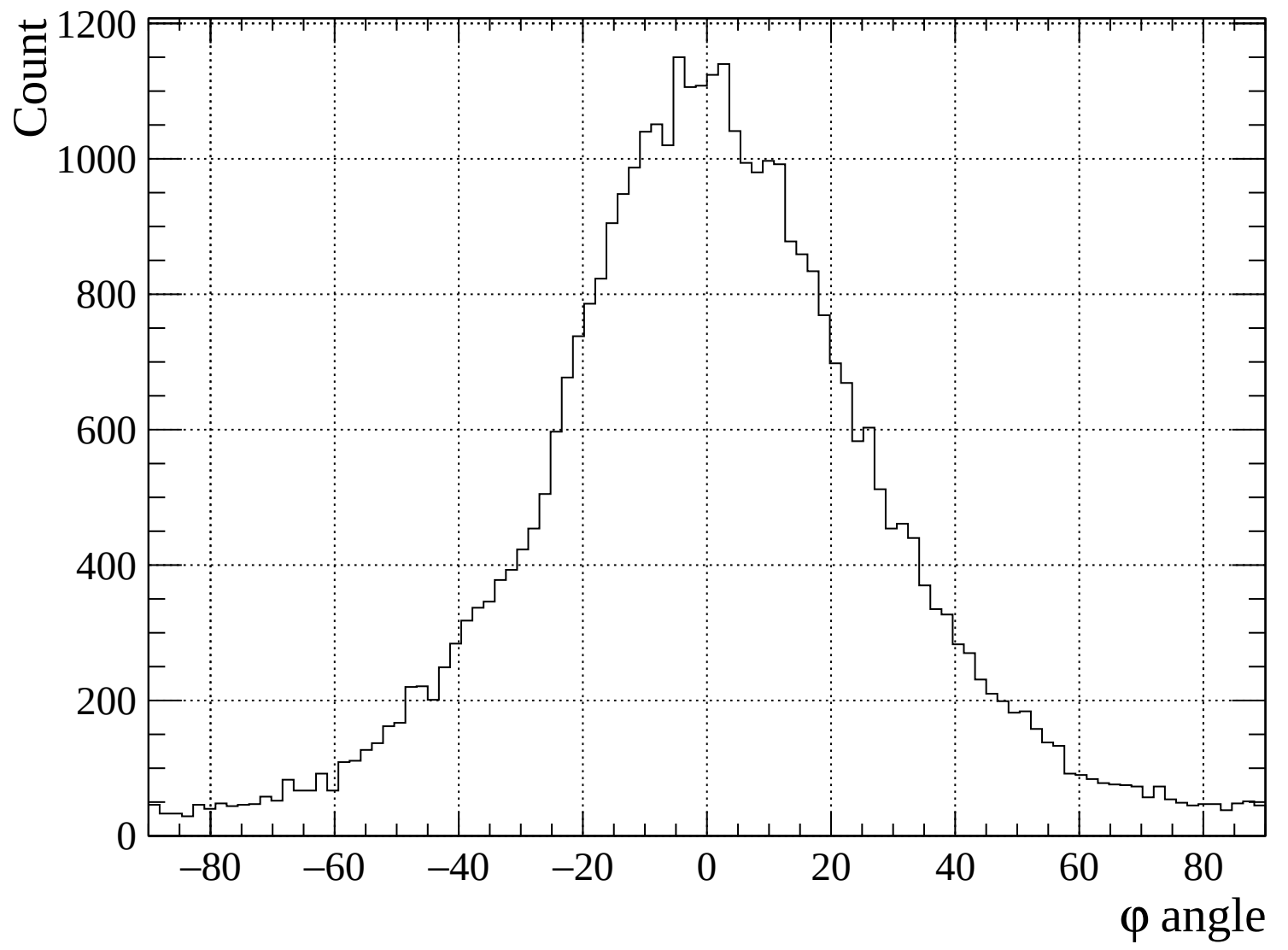


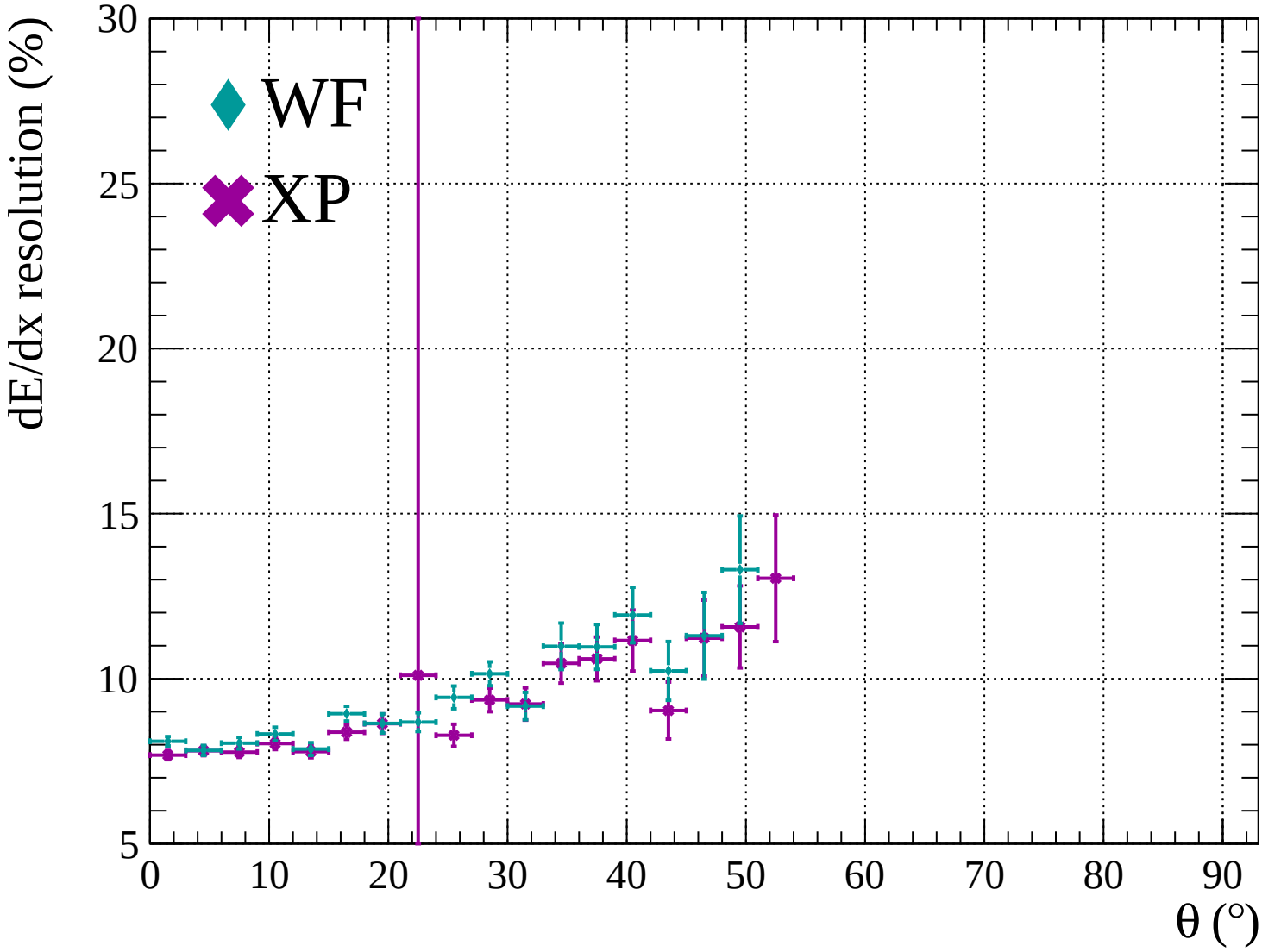


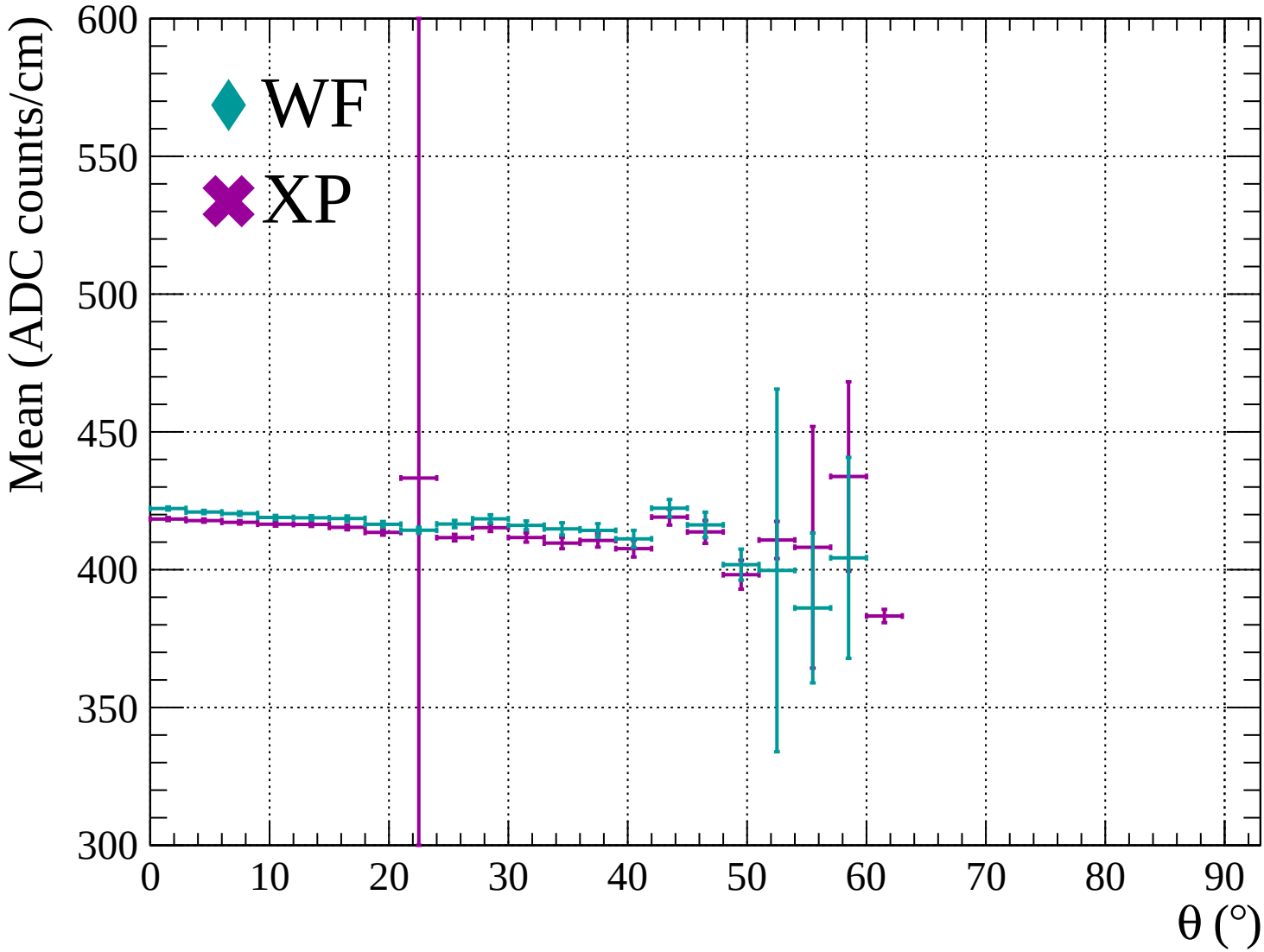


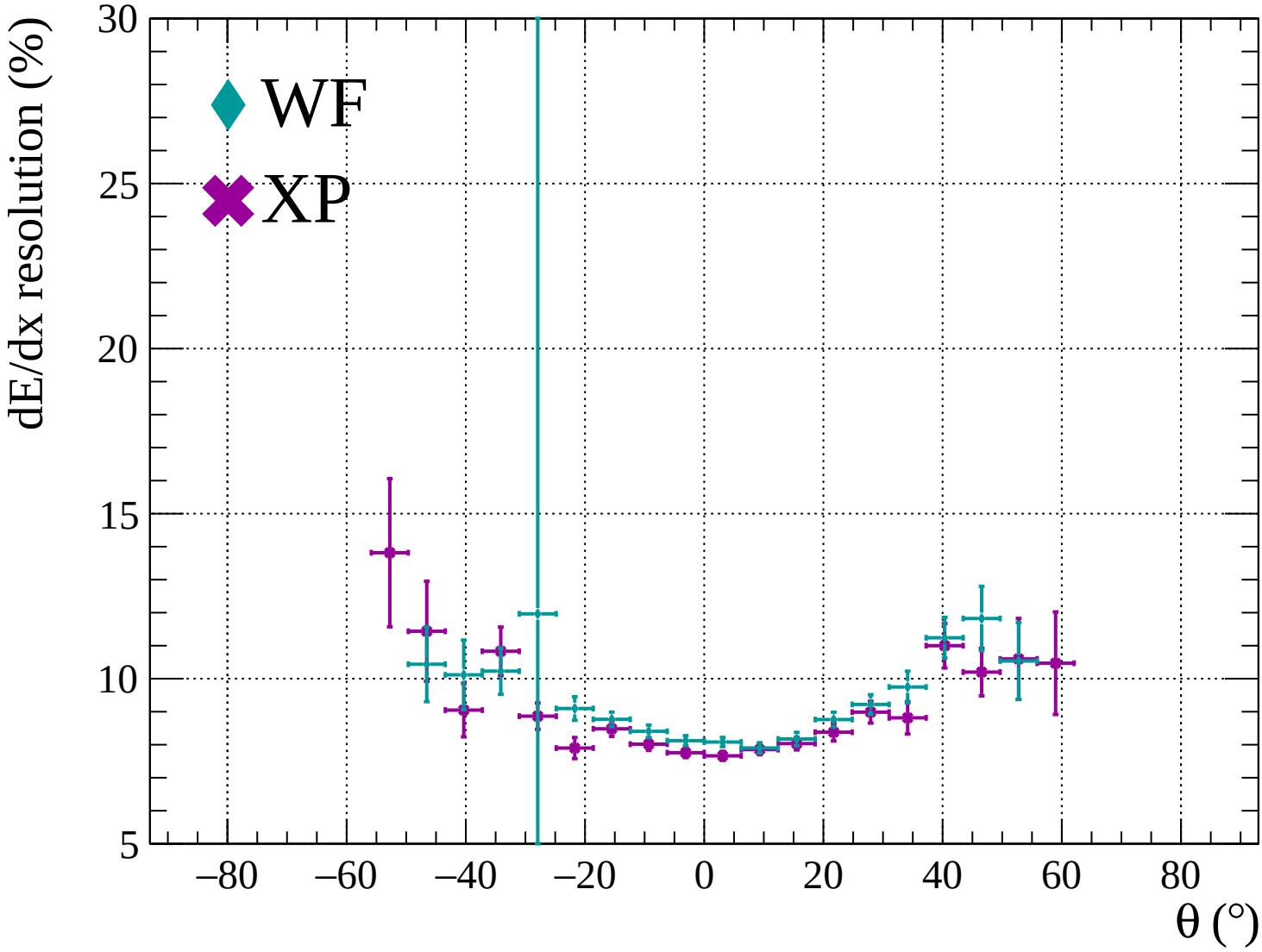


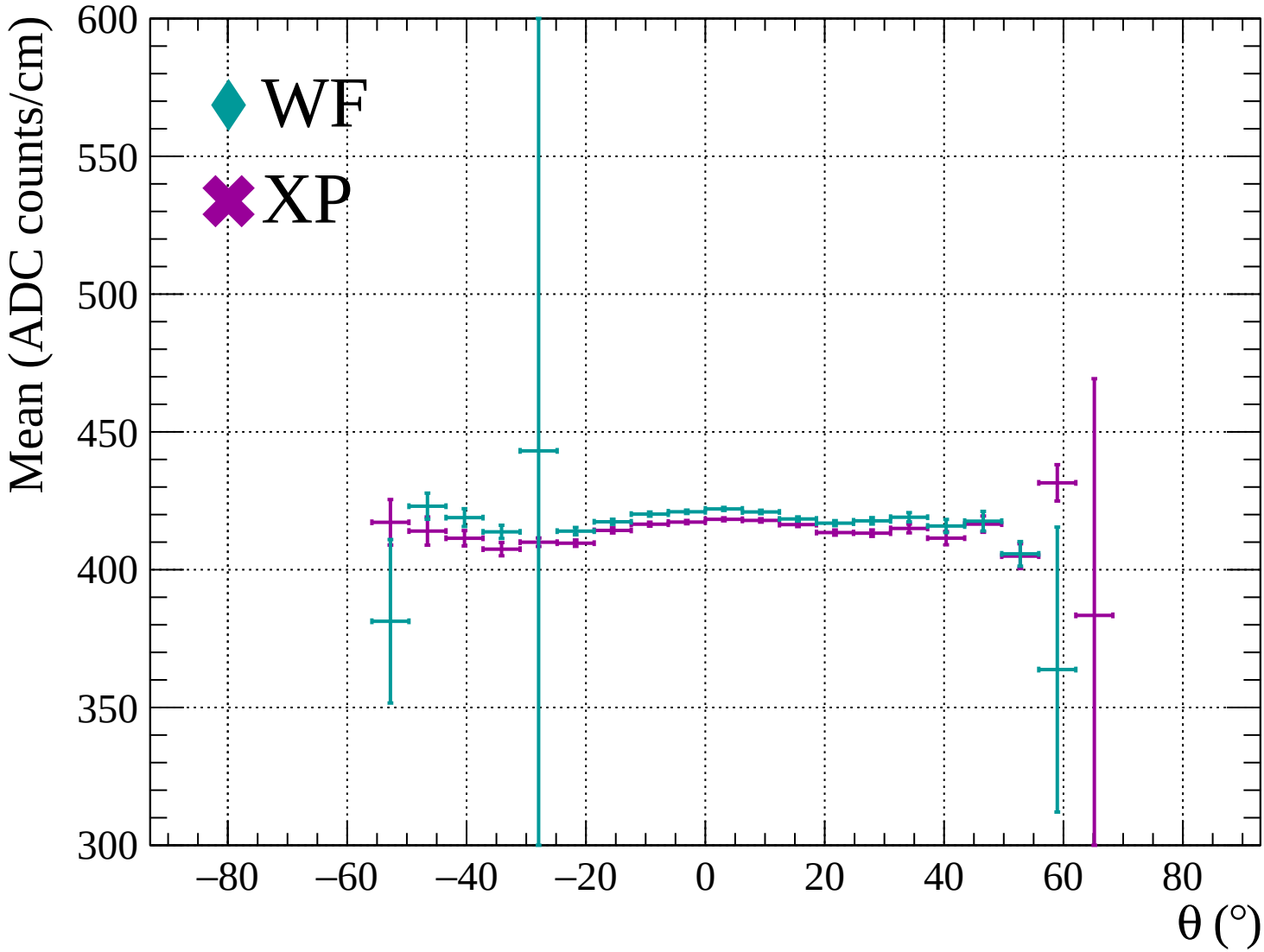


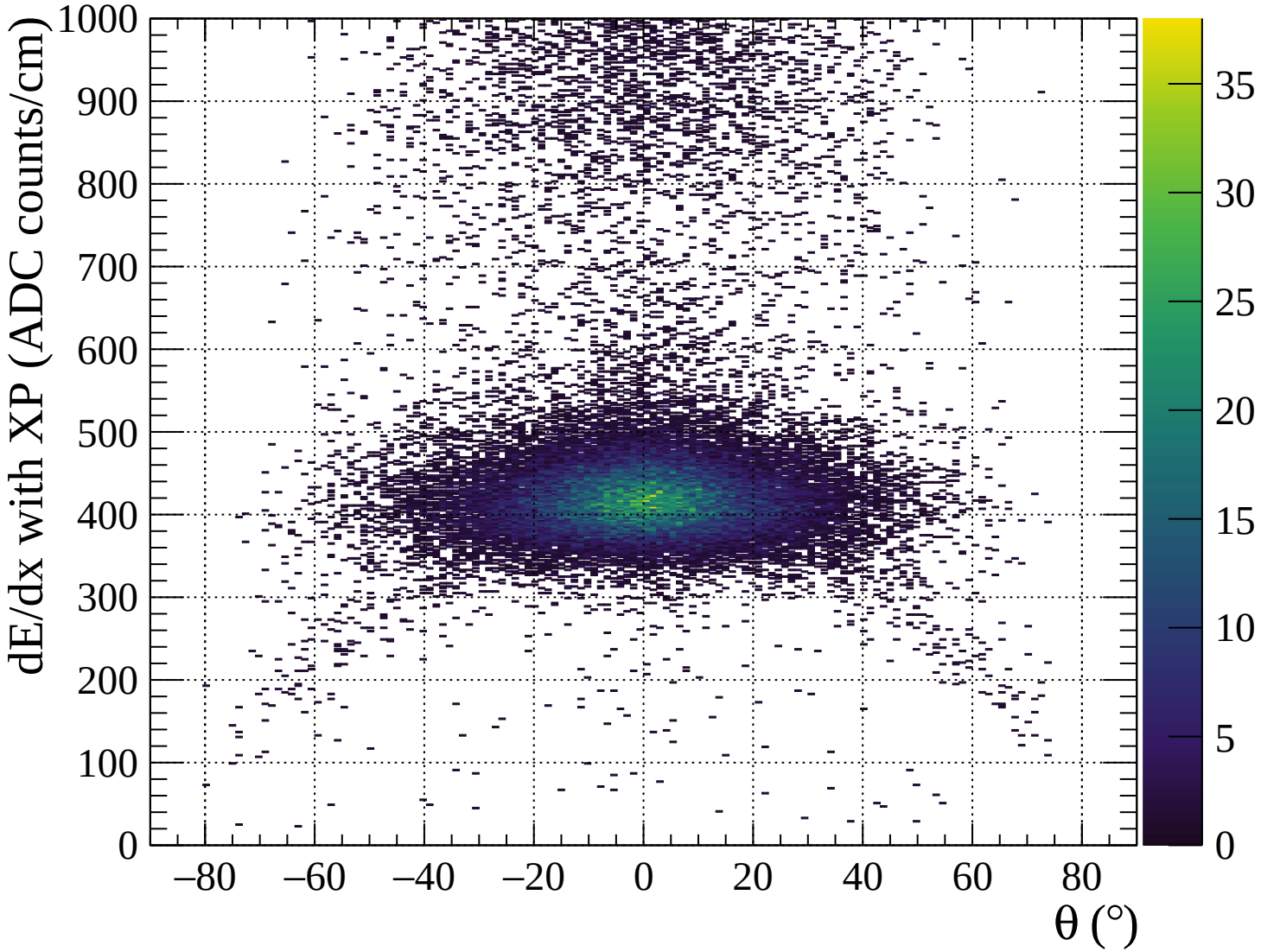


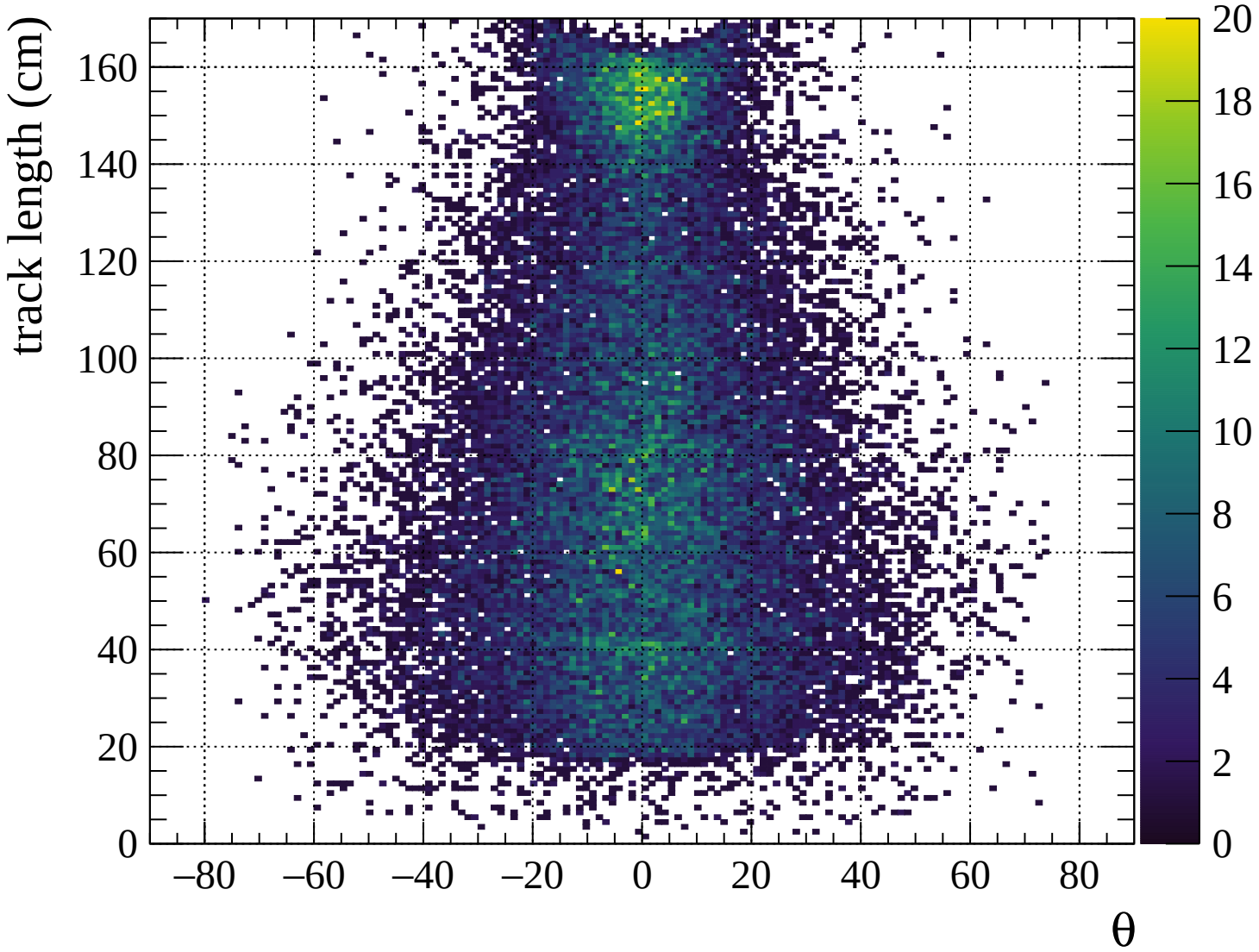


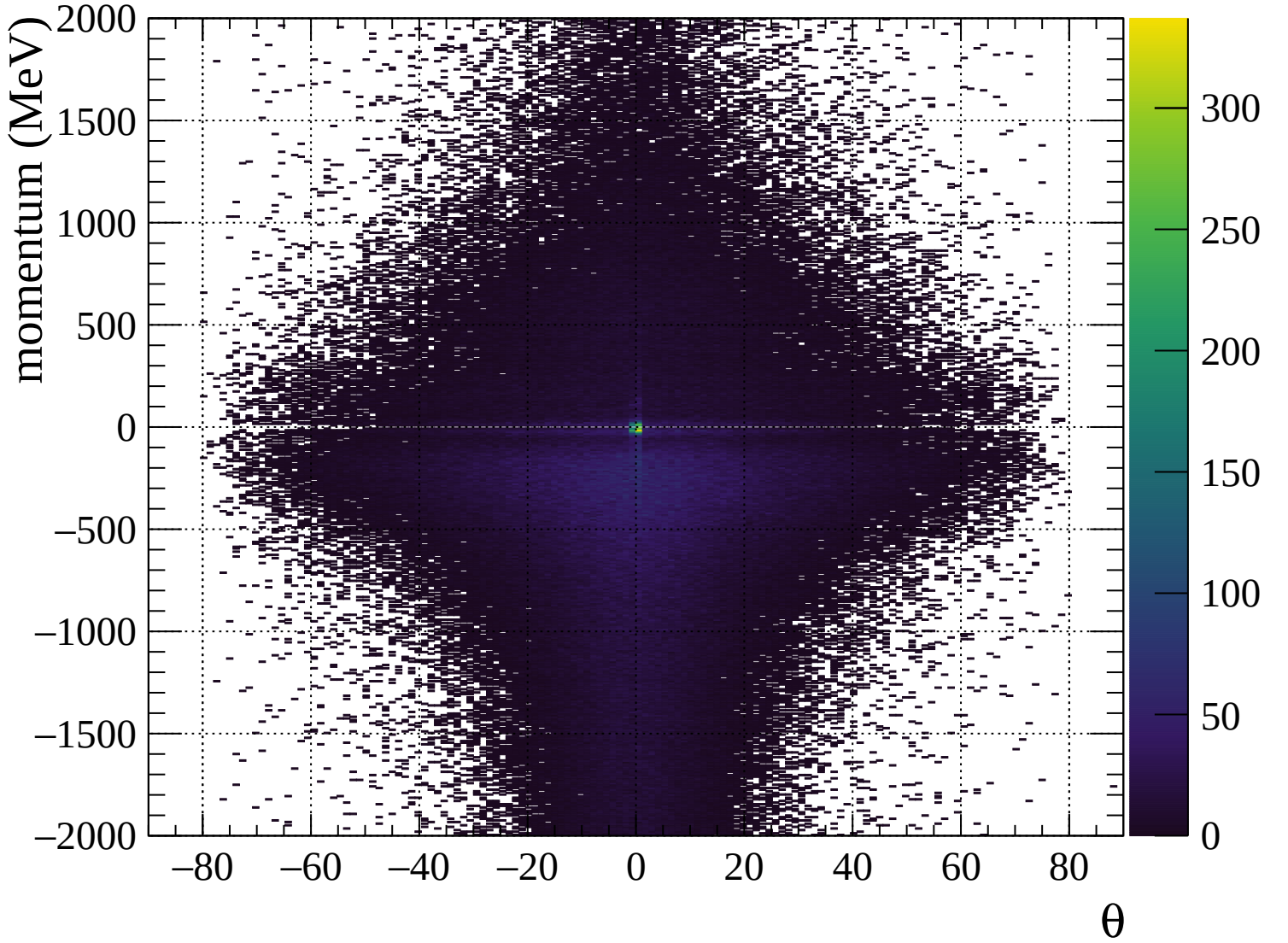


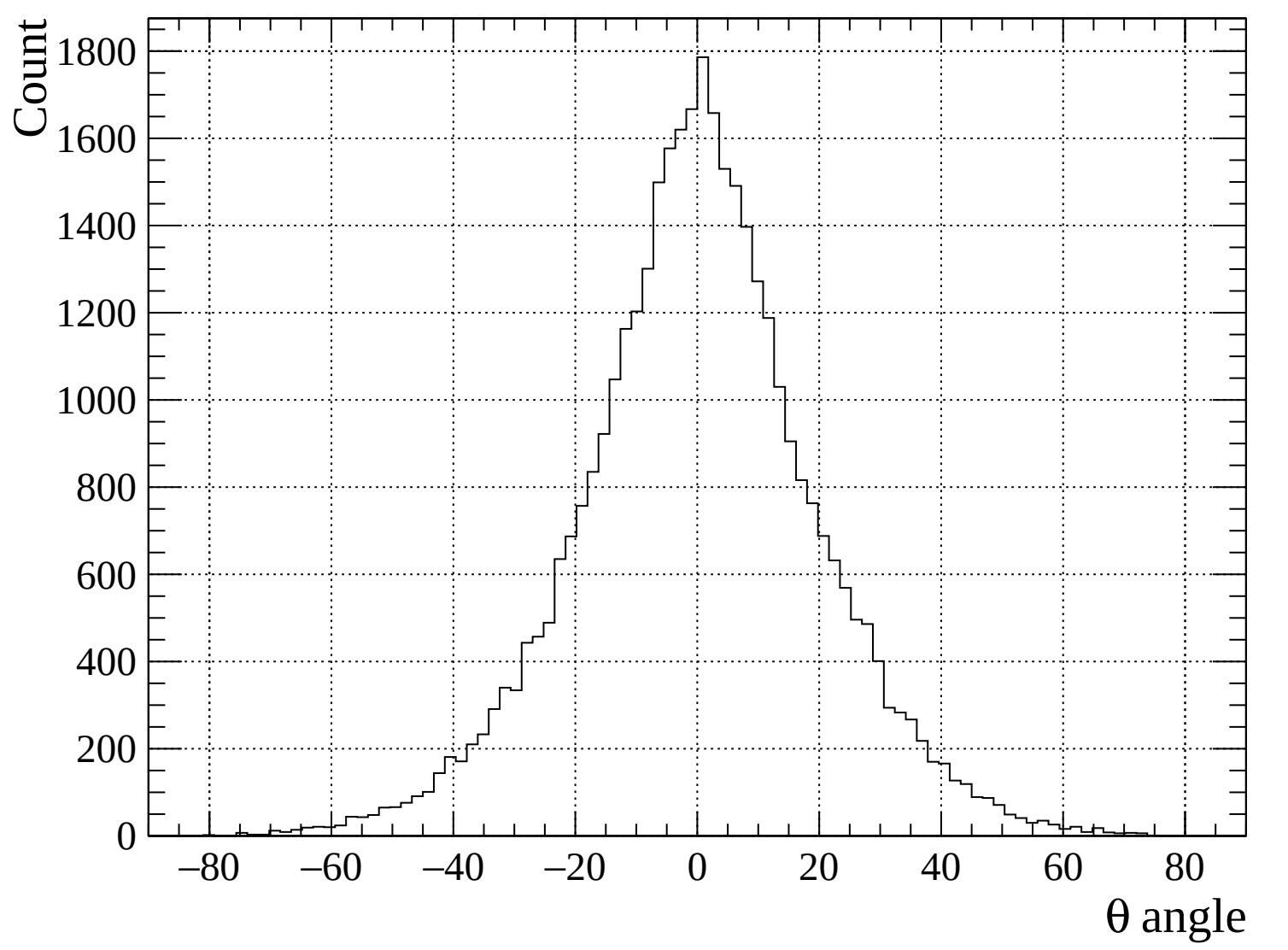


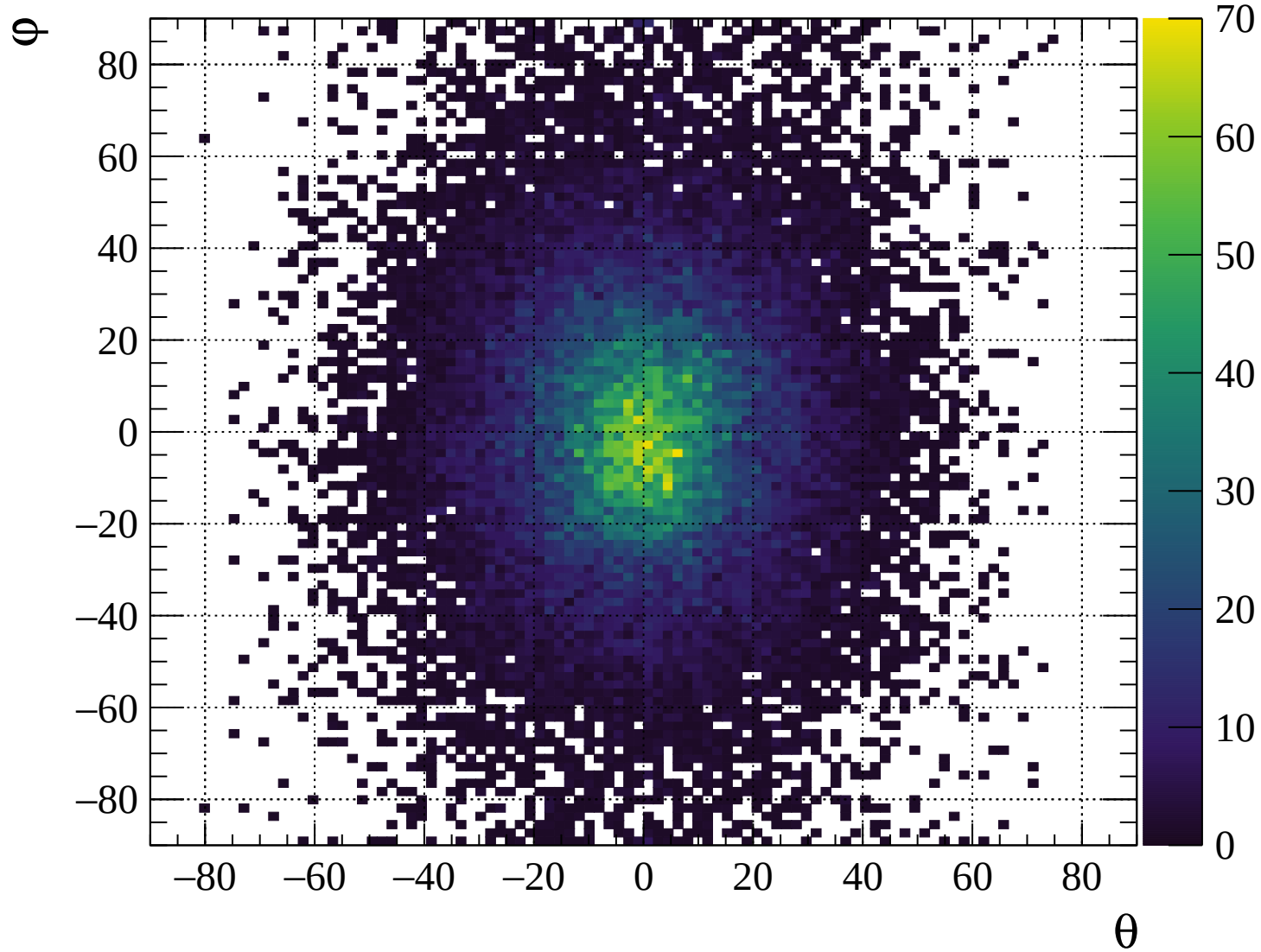


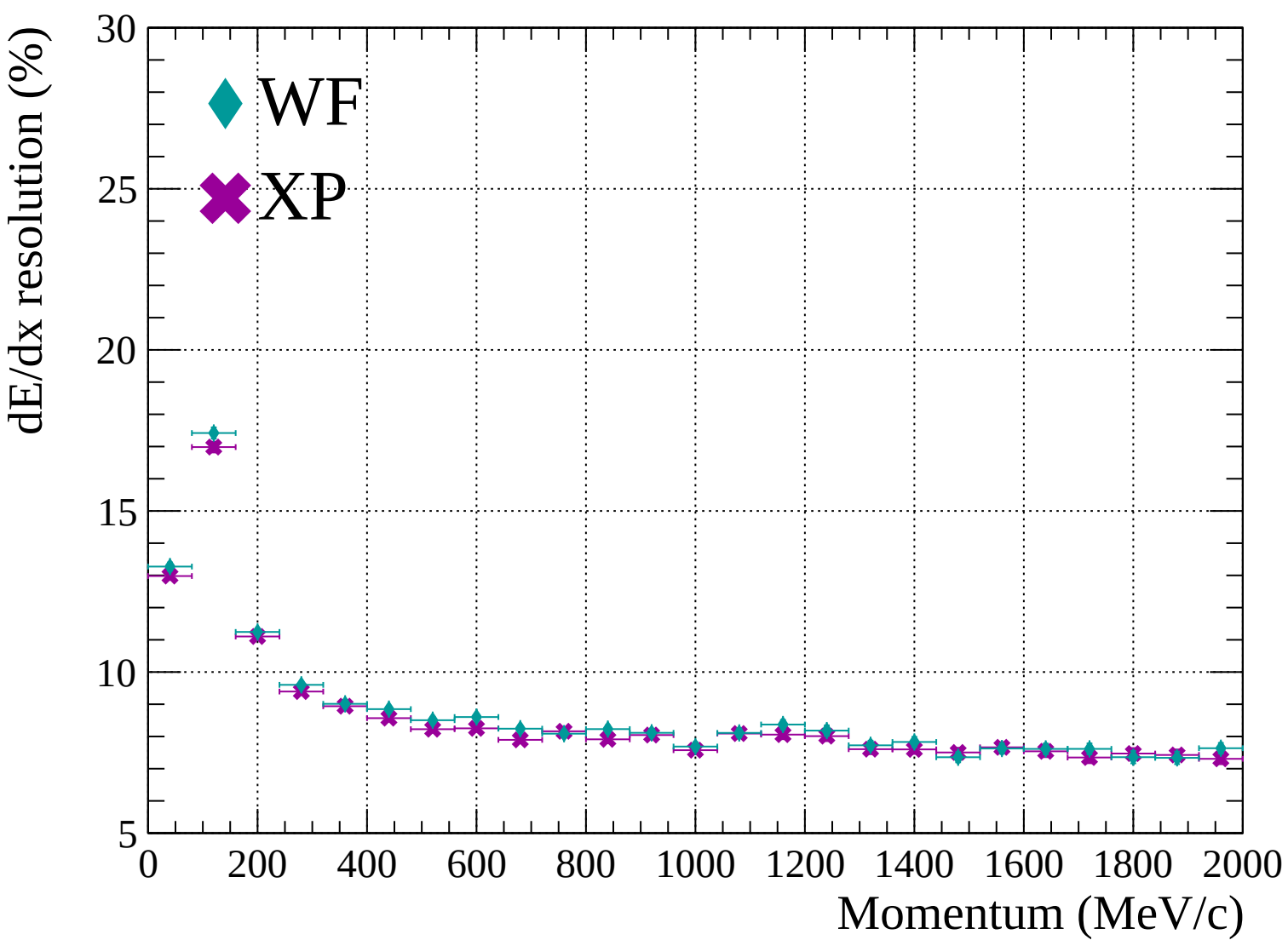


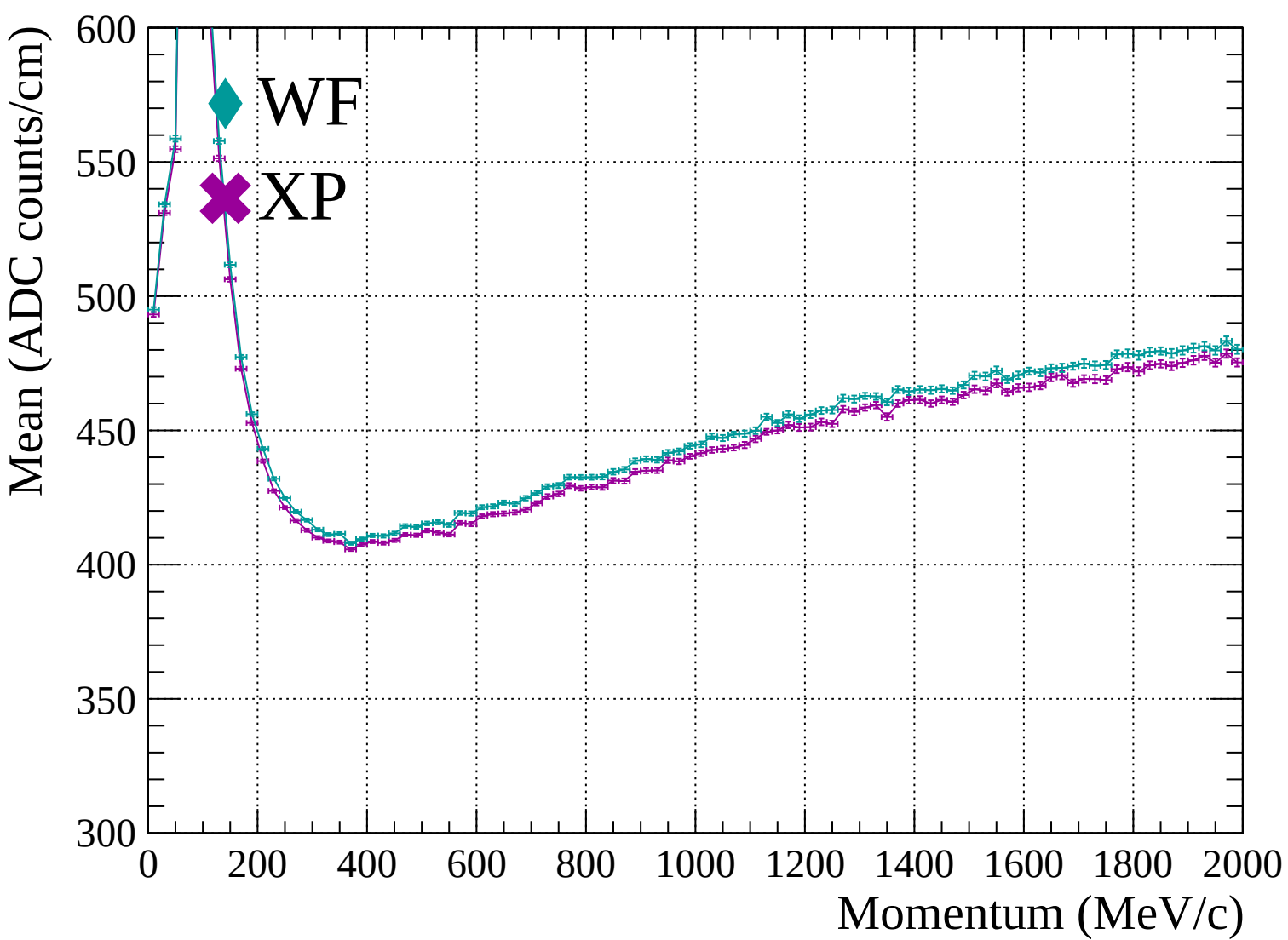


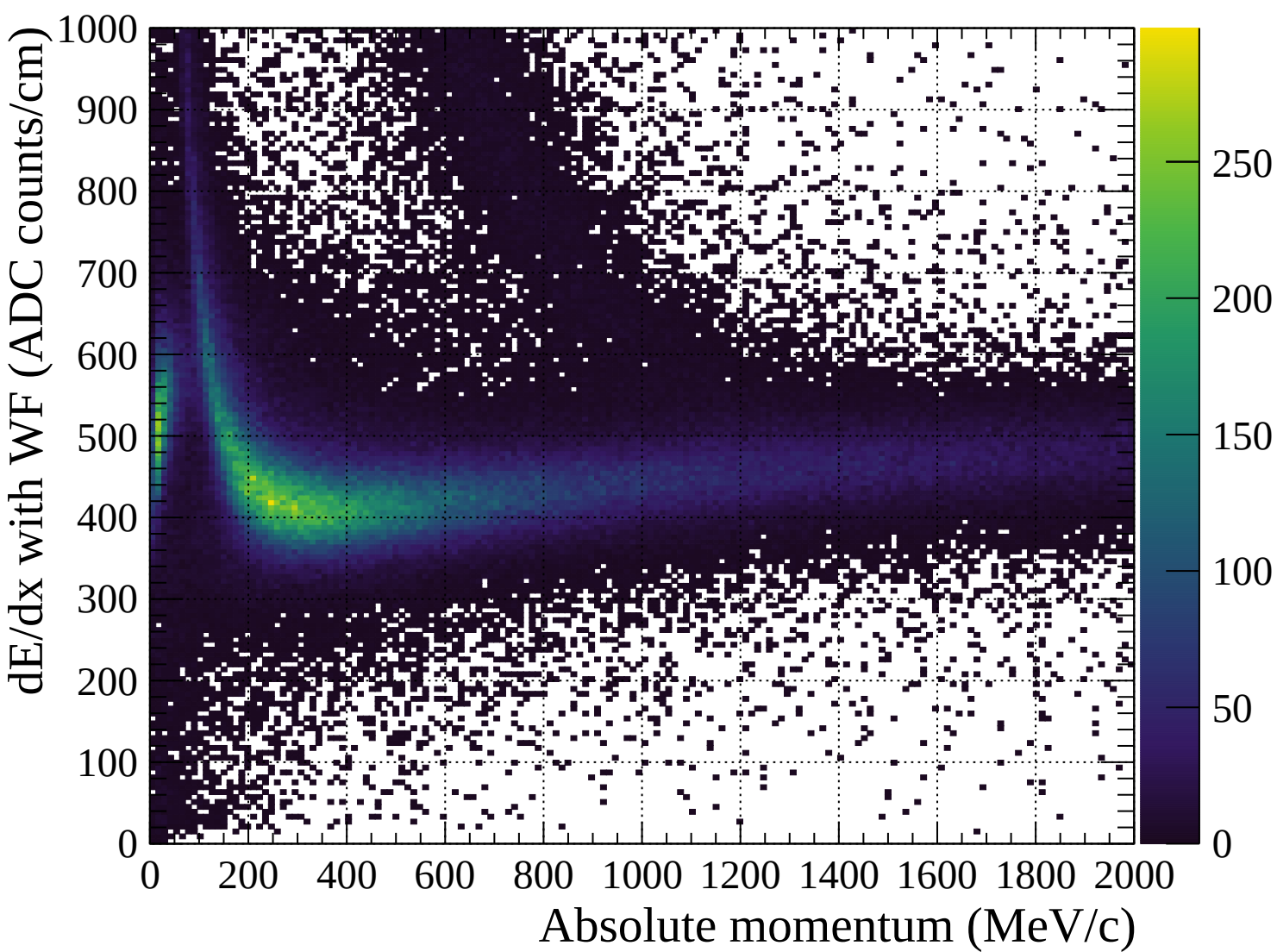


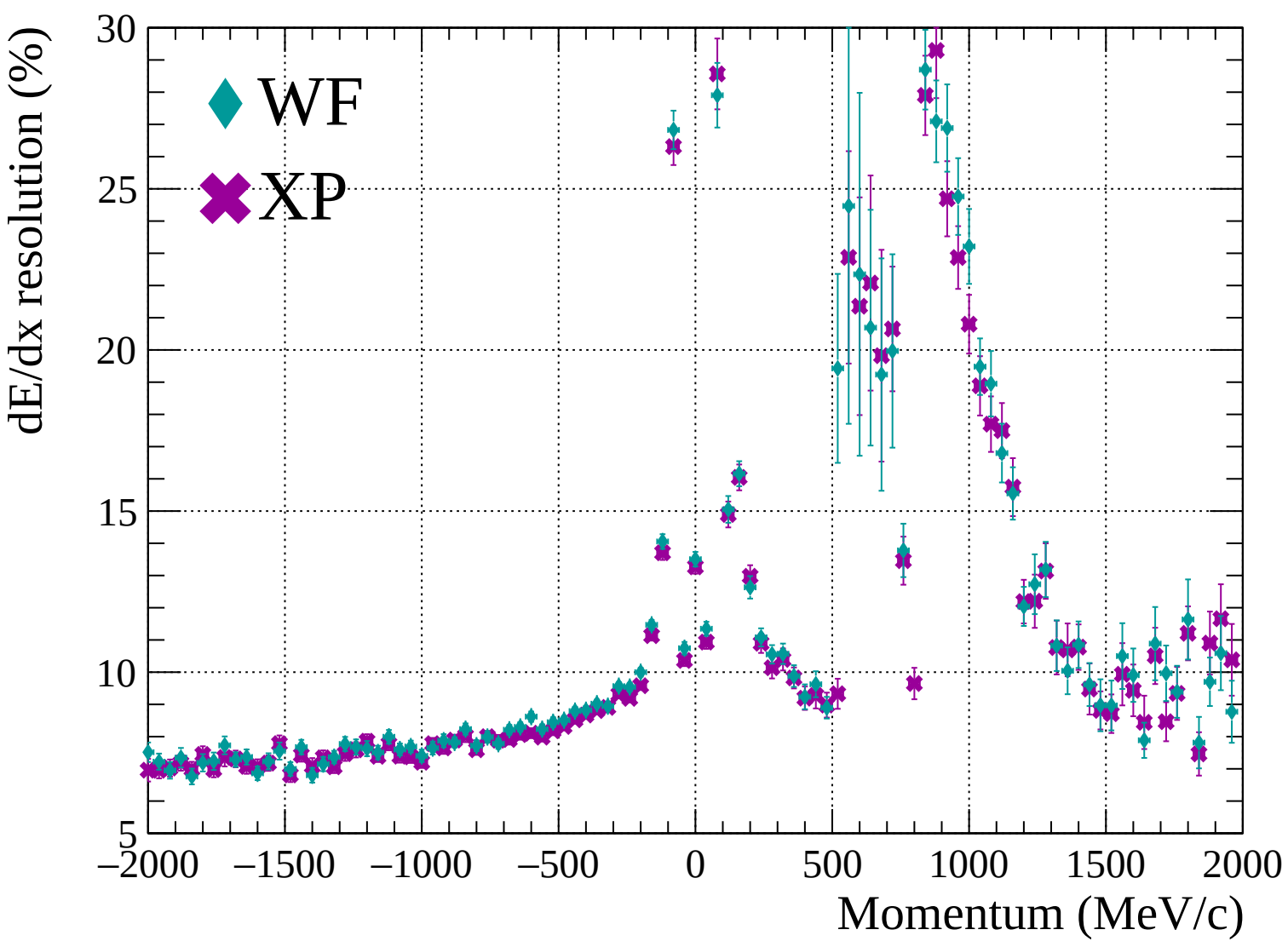


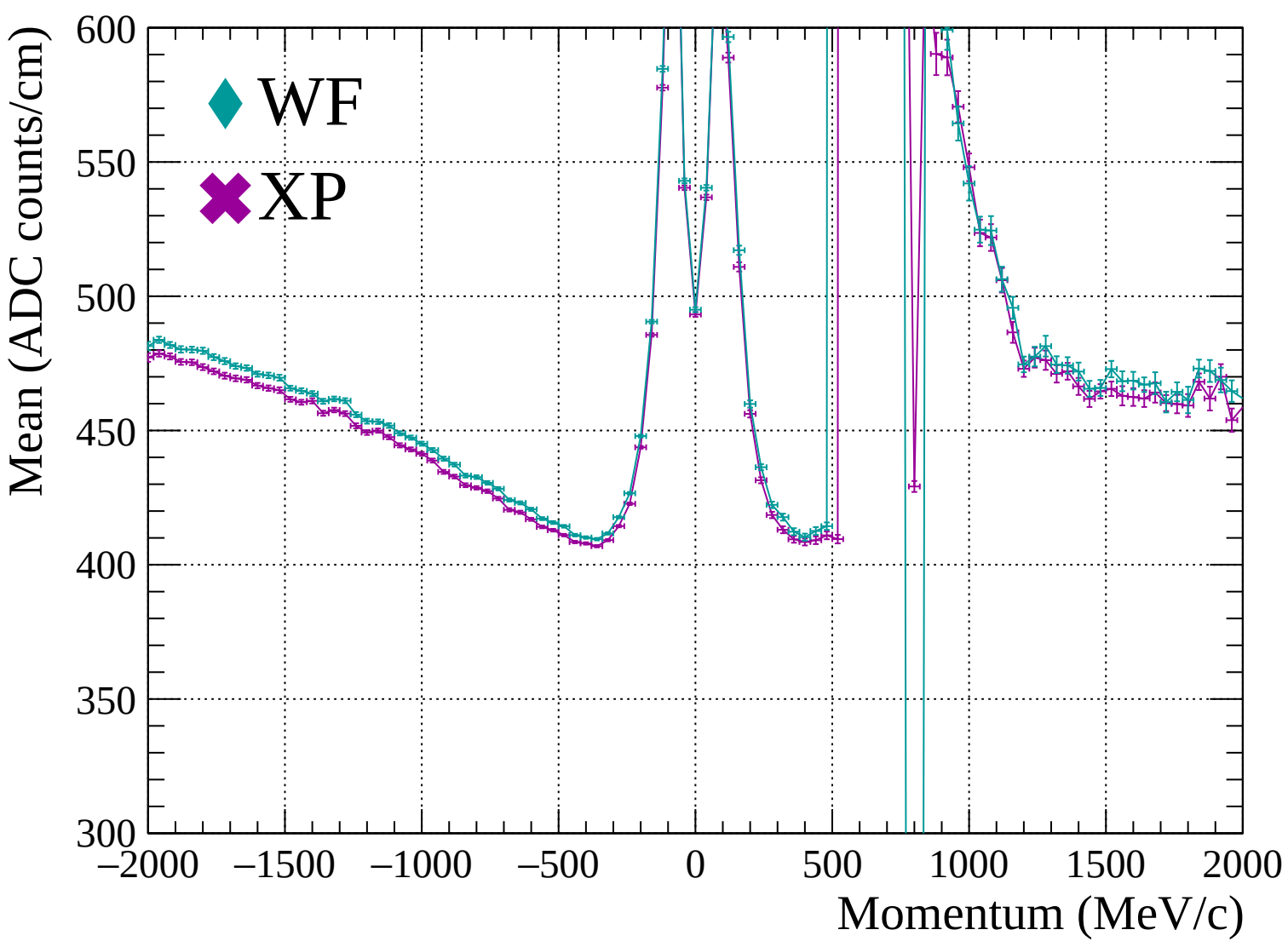


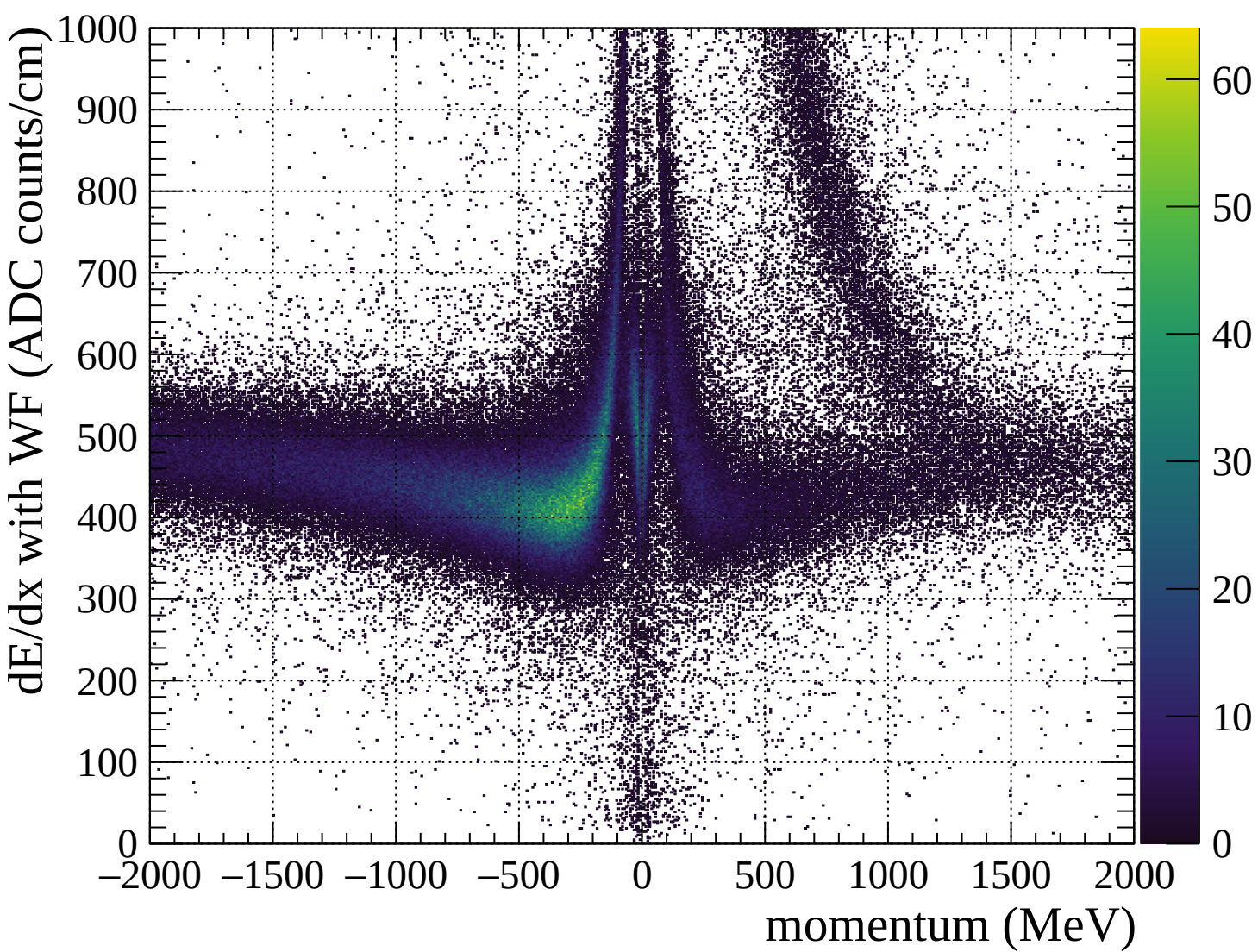


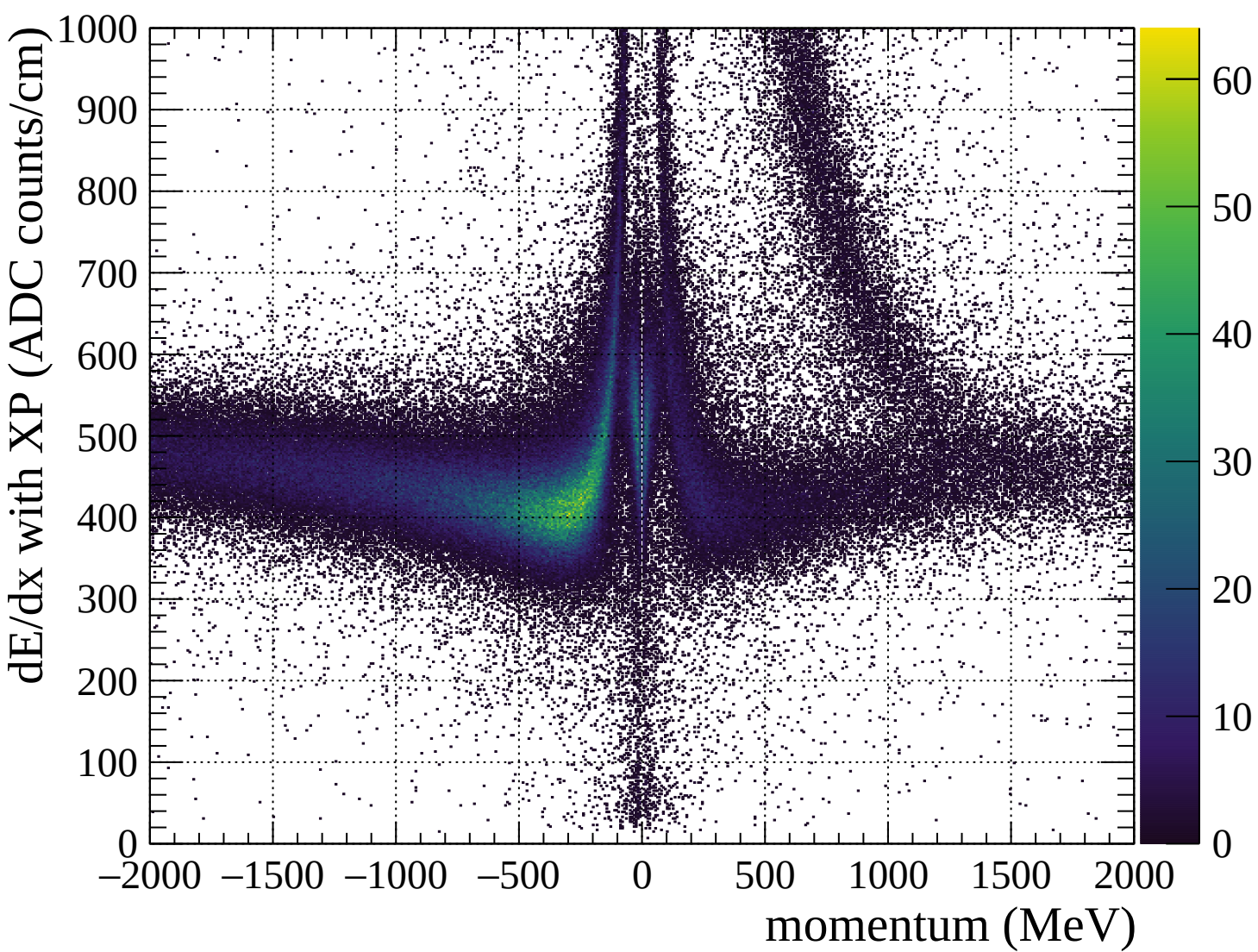


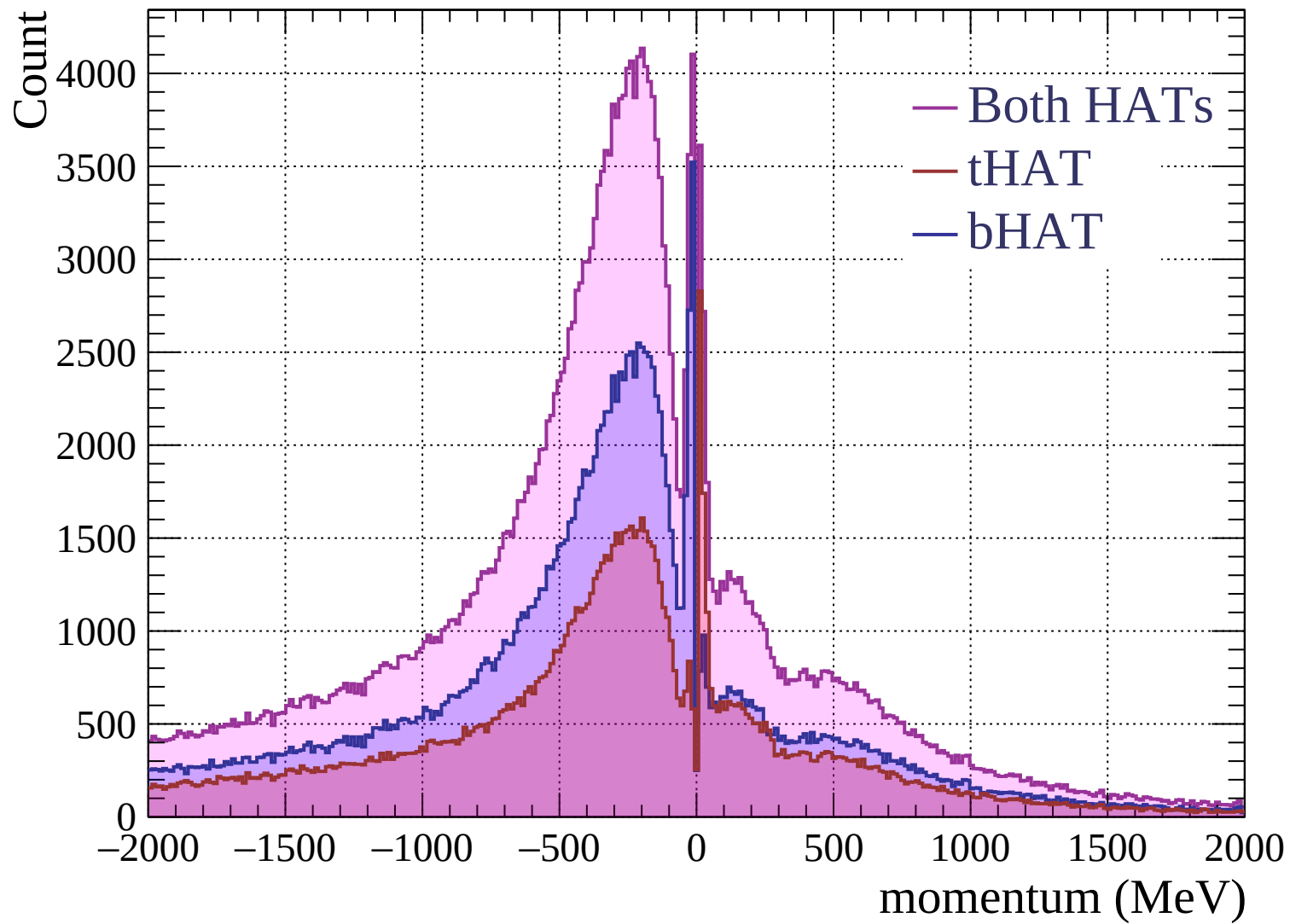


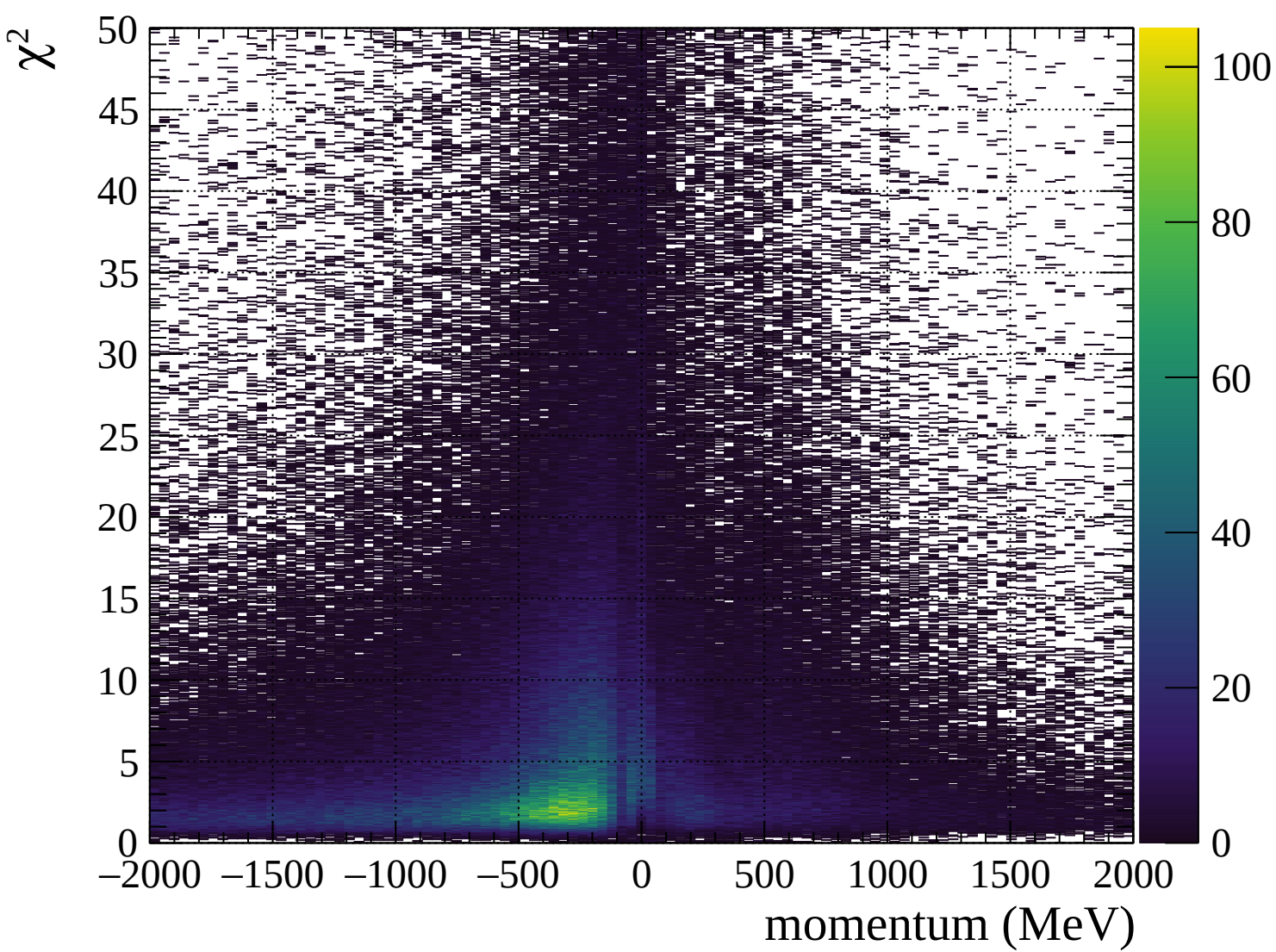




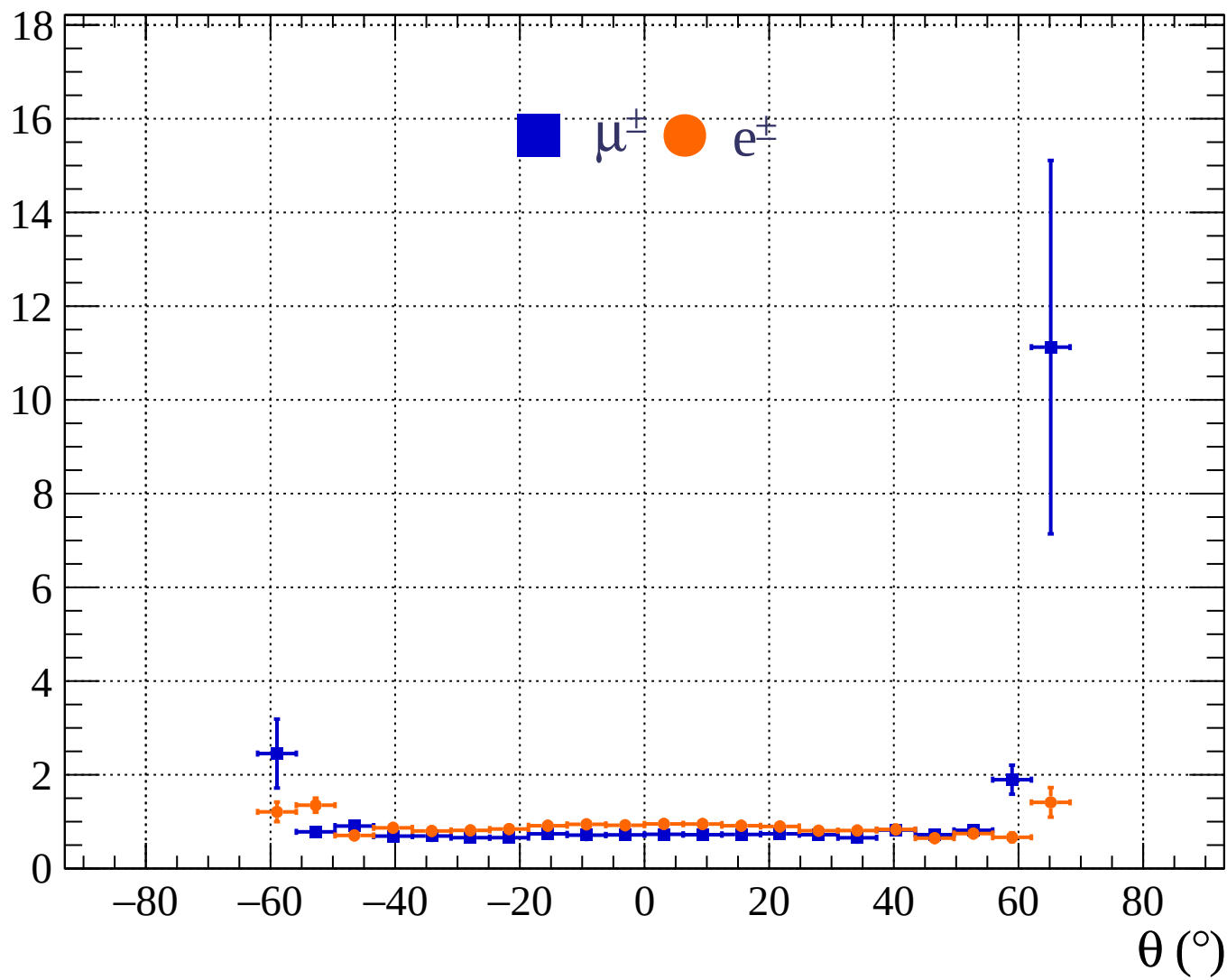


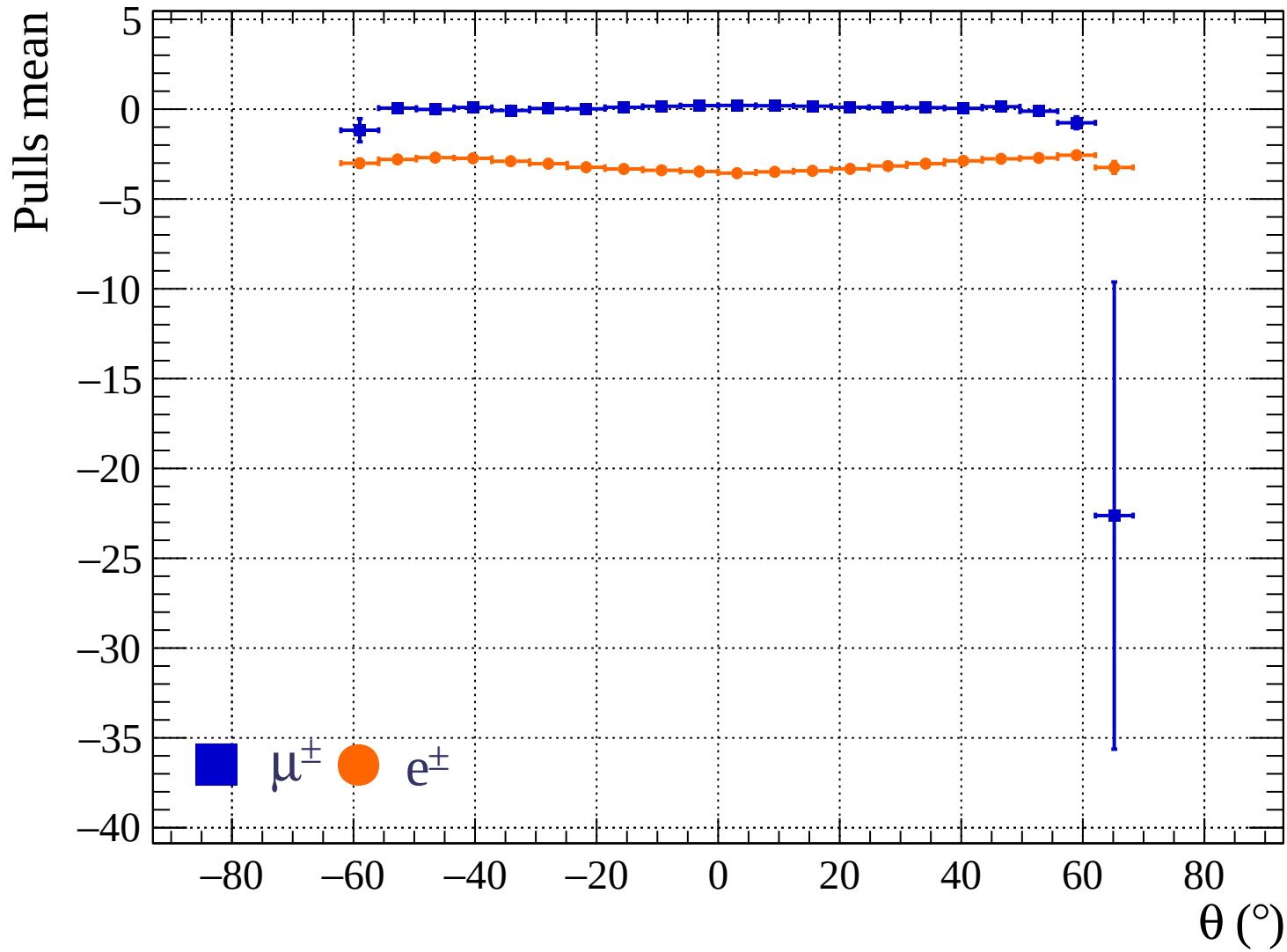




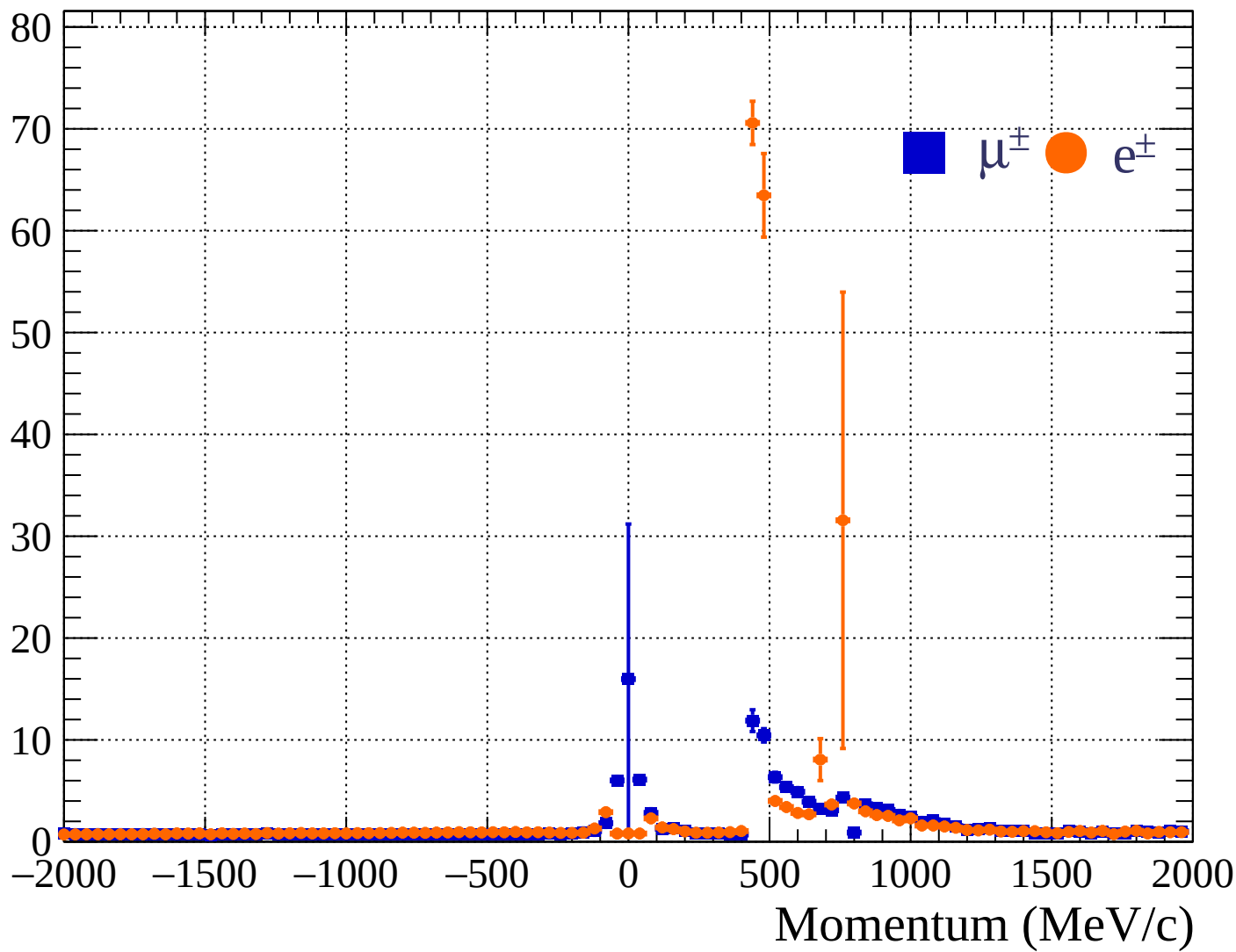


Pulls standard deviation

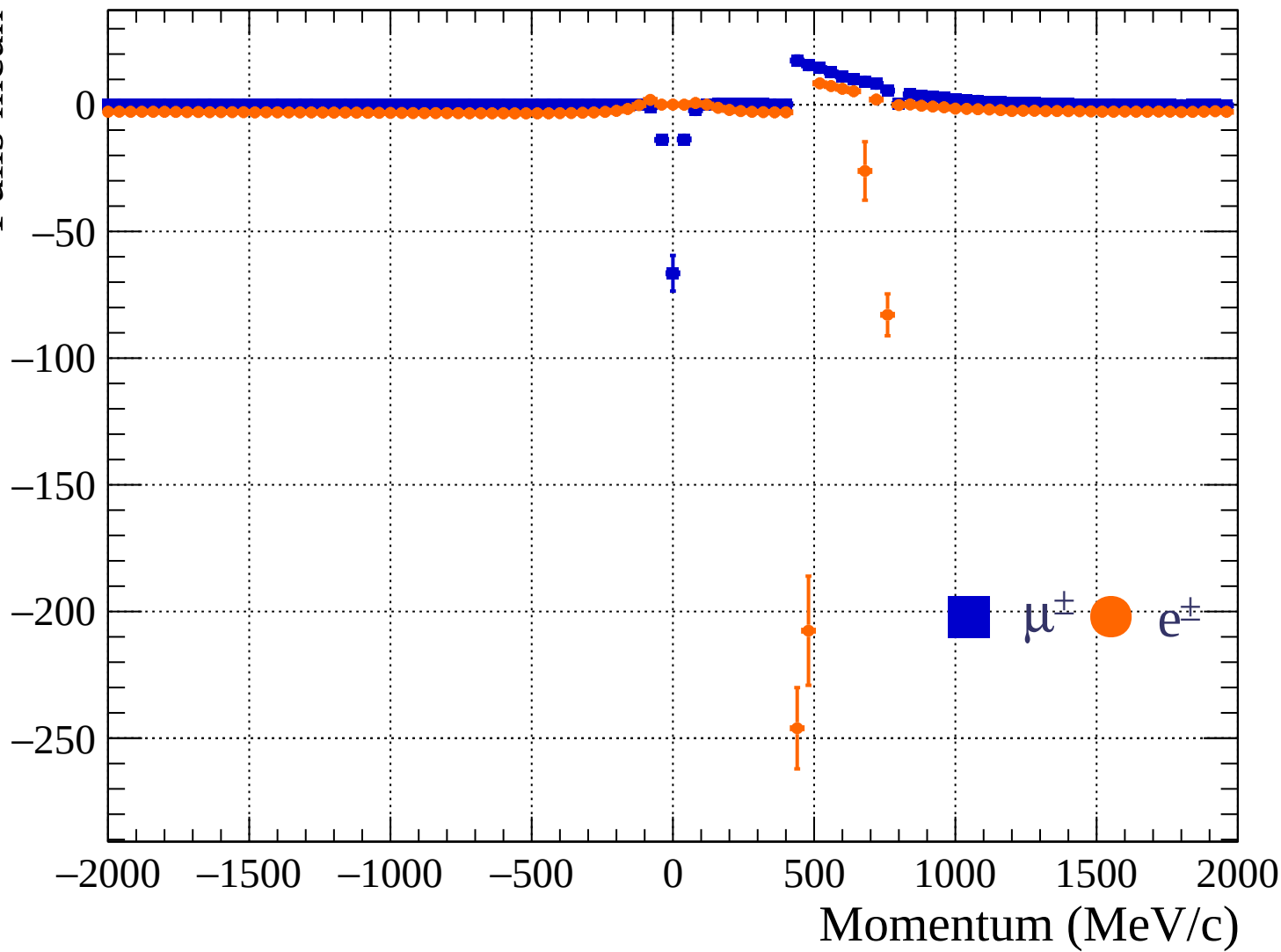


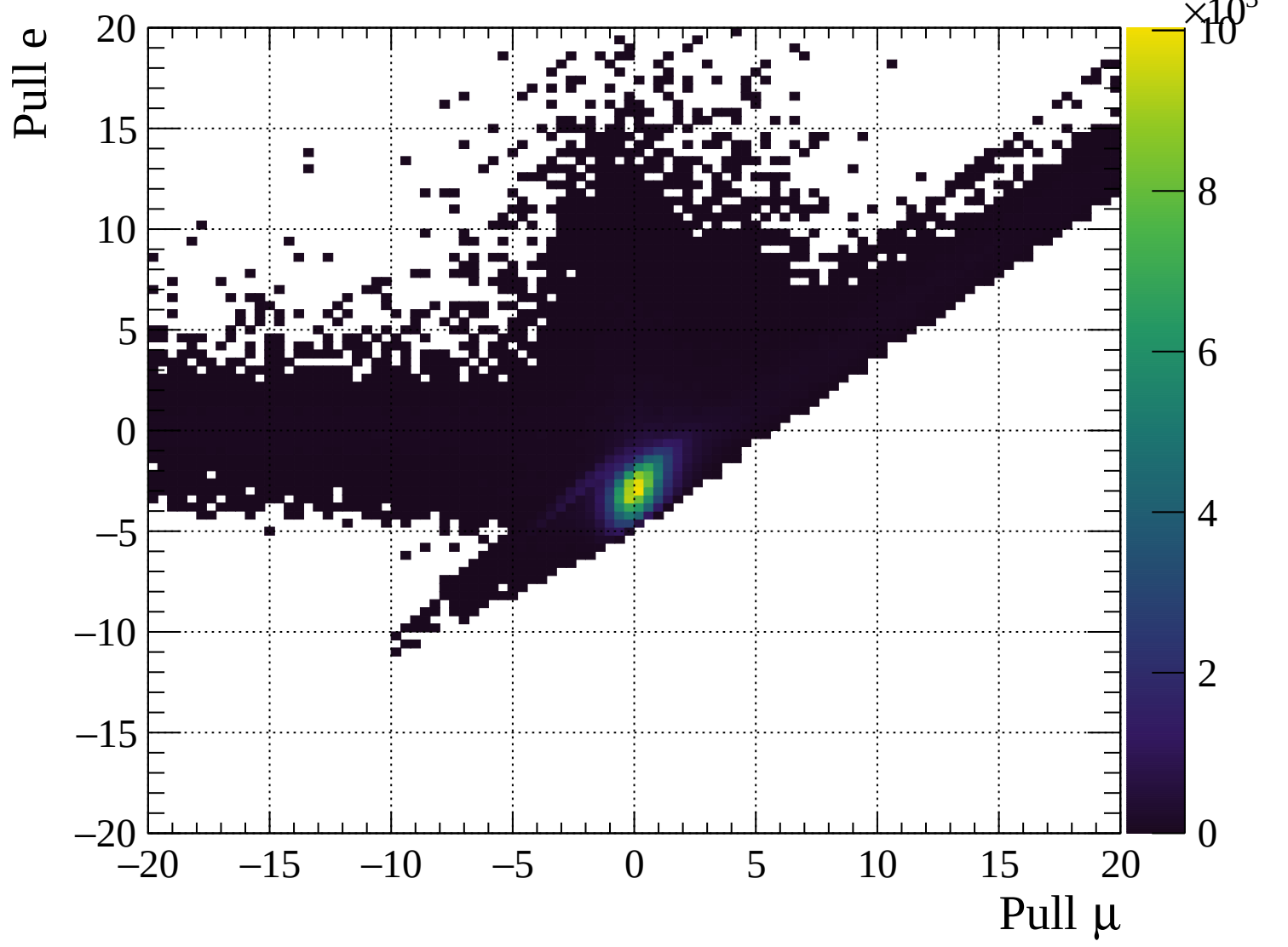


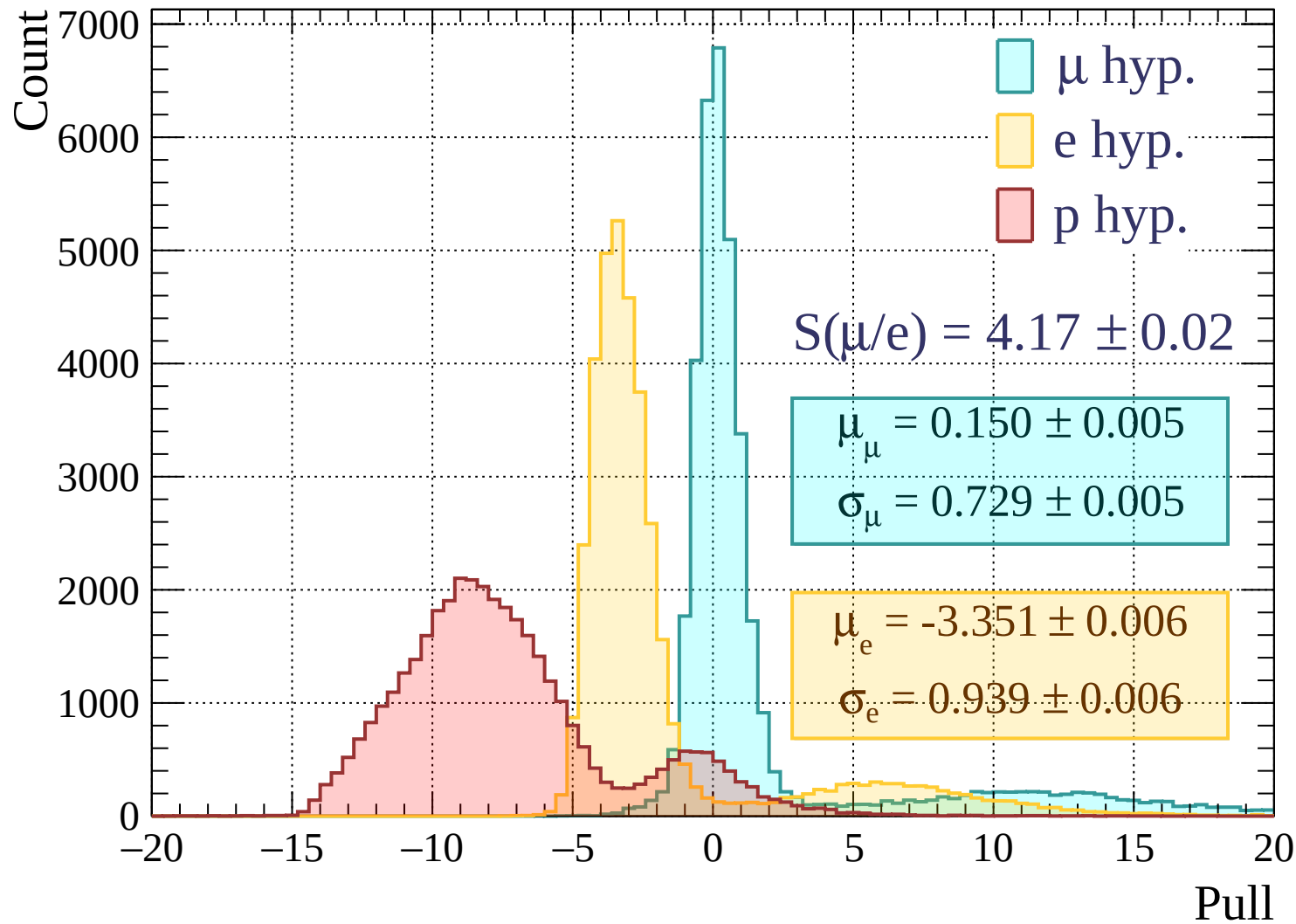
Pulls standard deviation

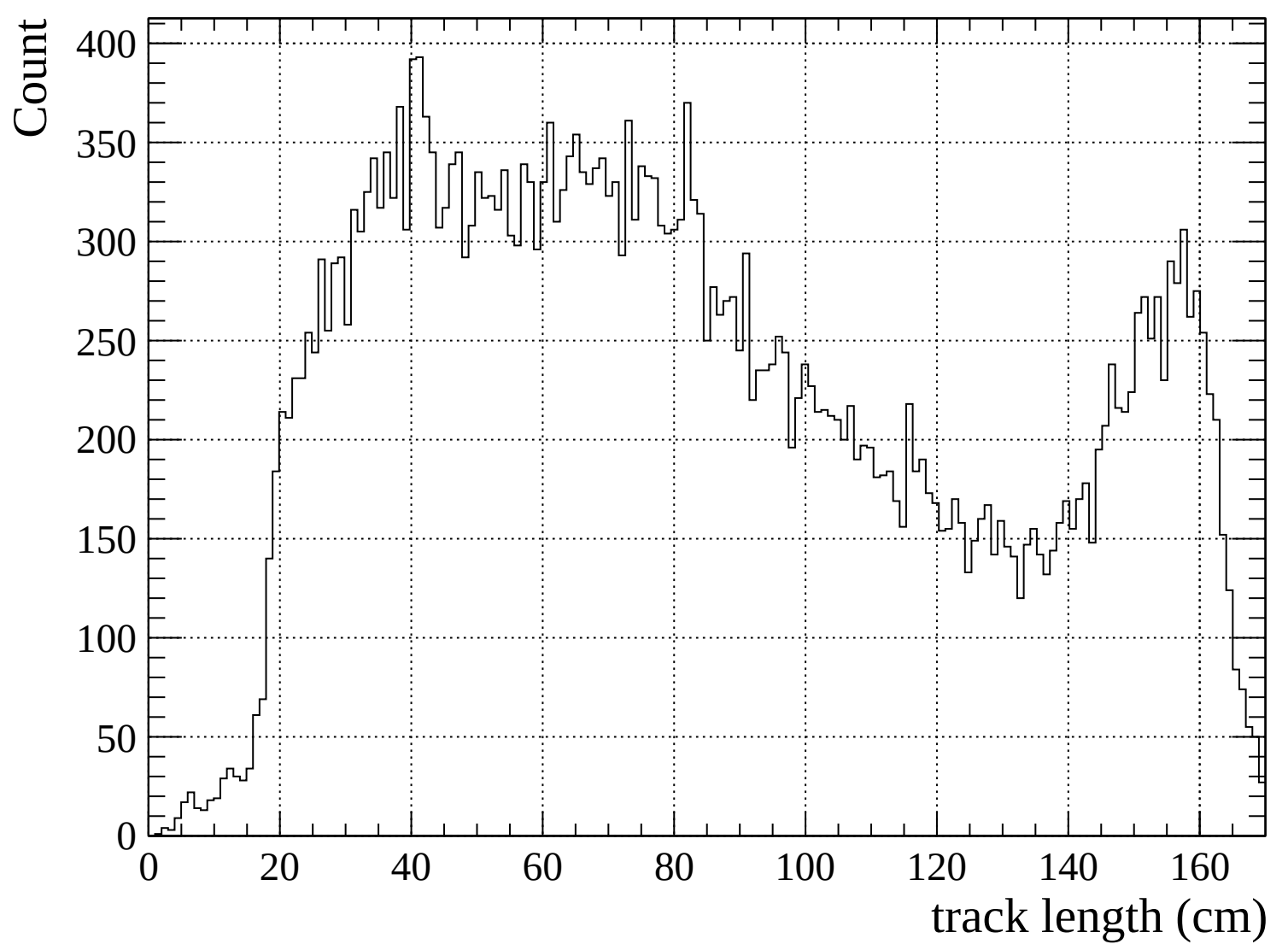


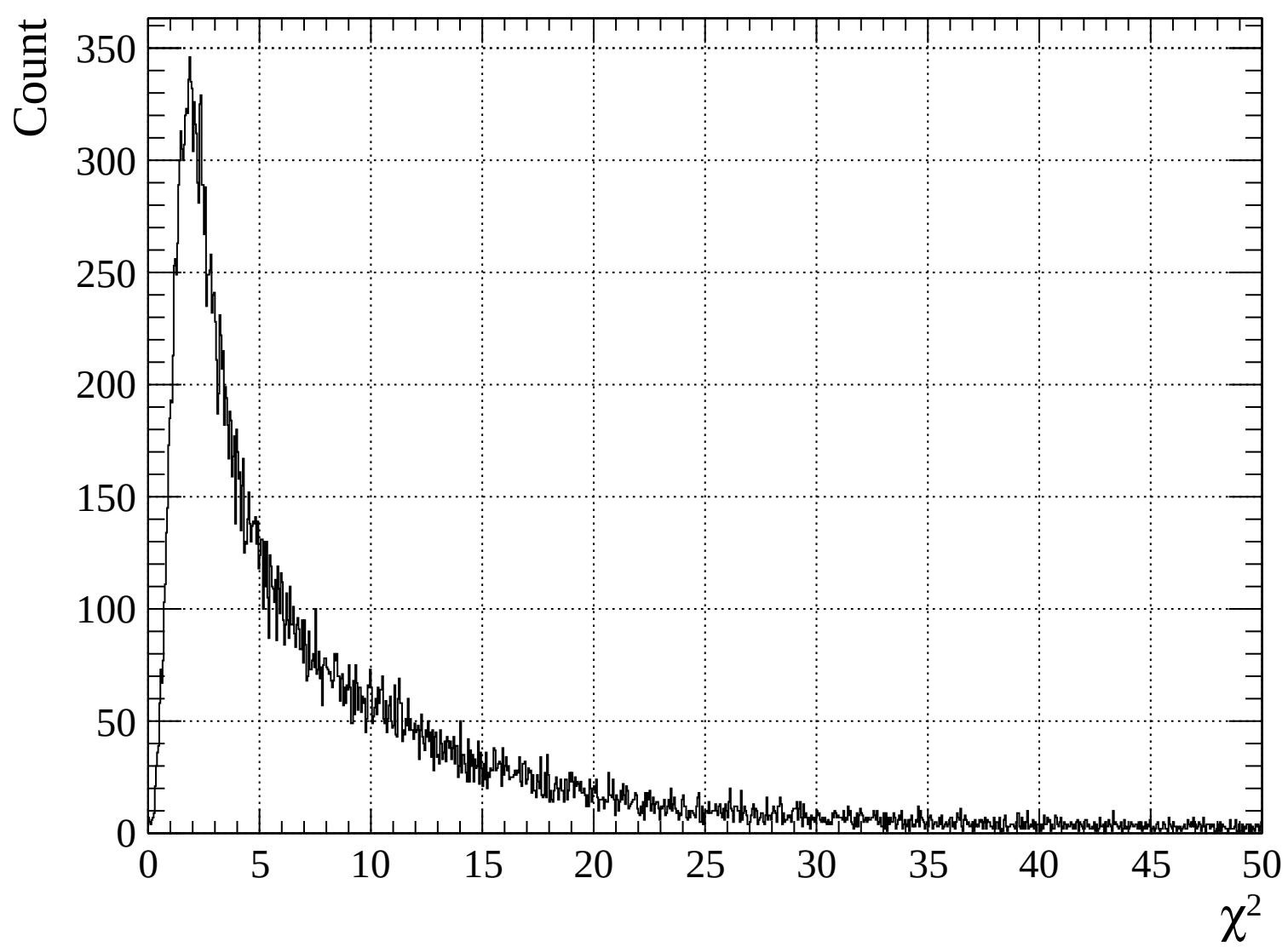
Pulls mean

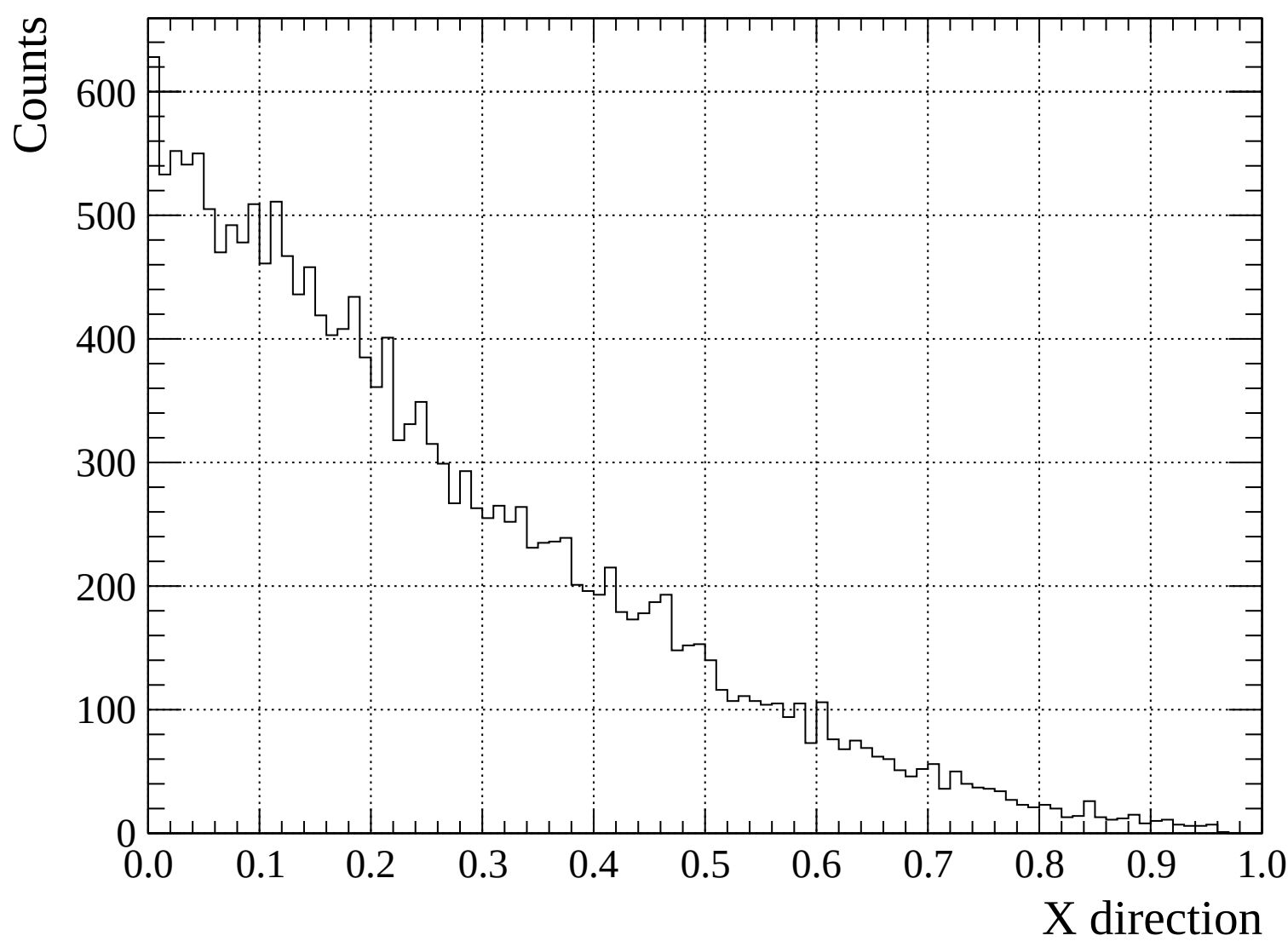


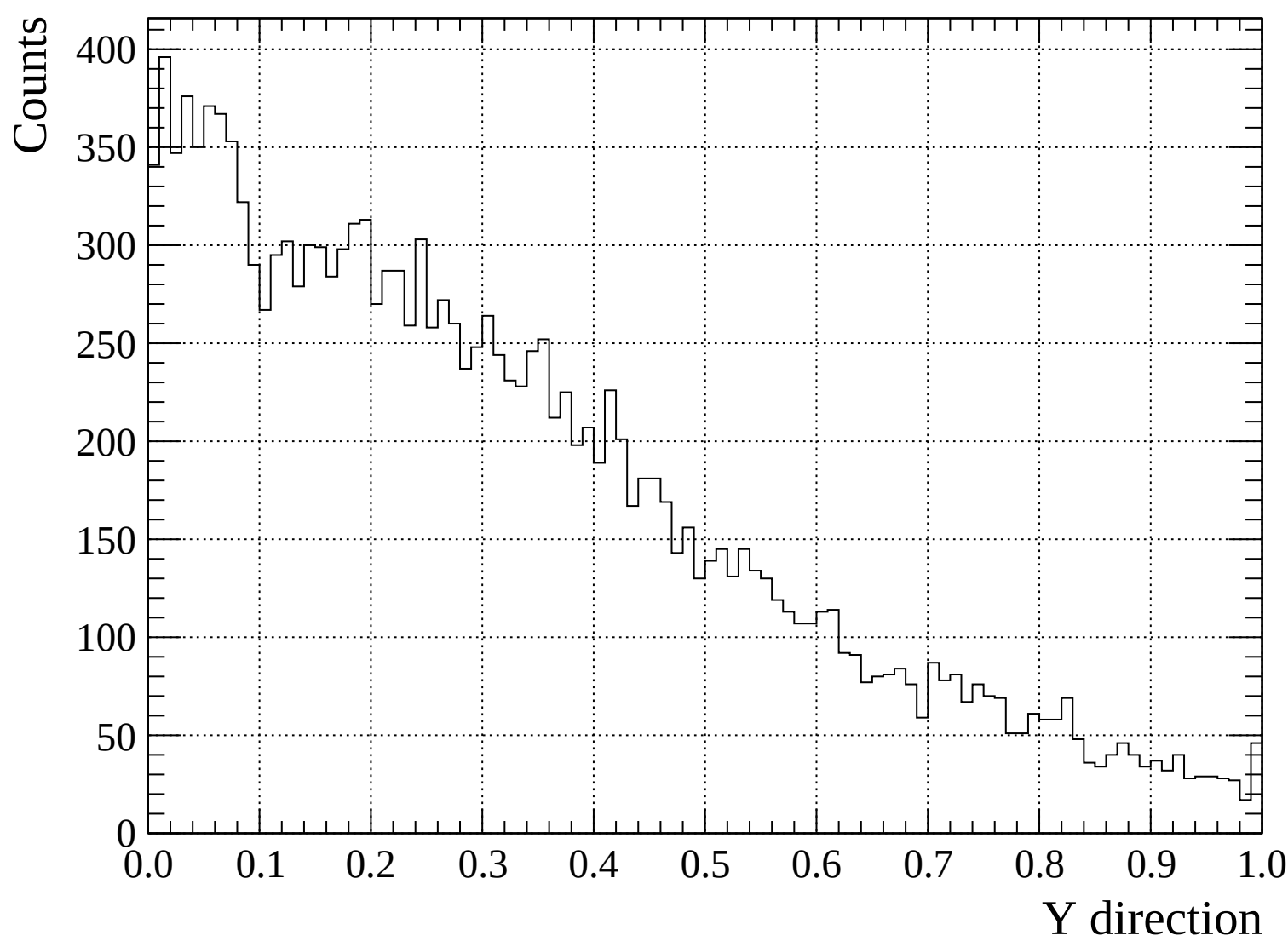


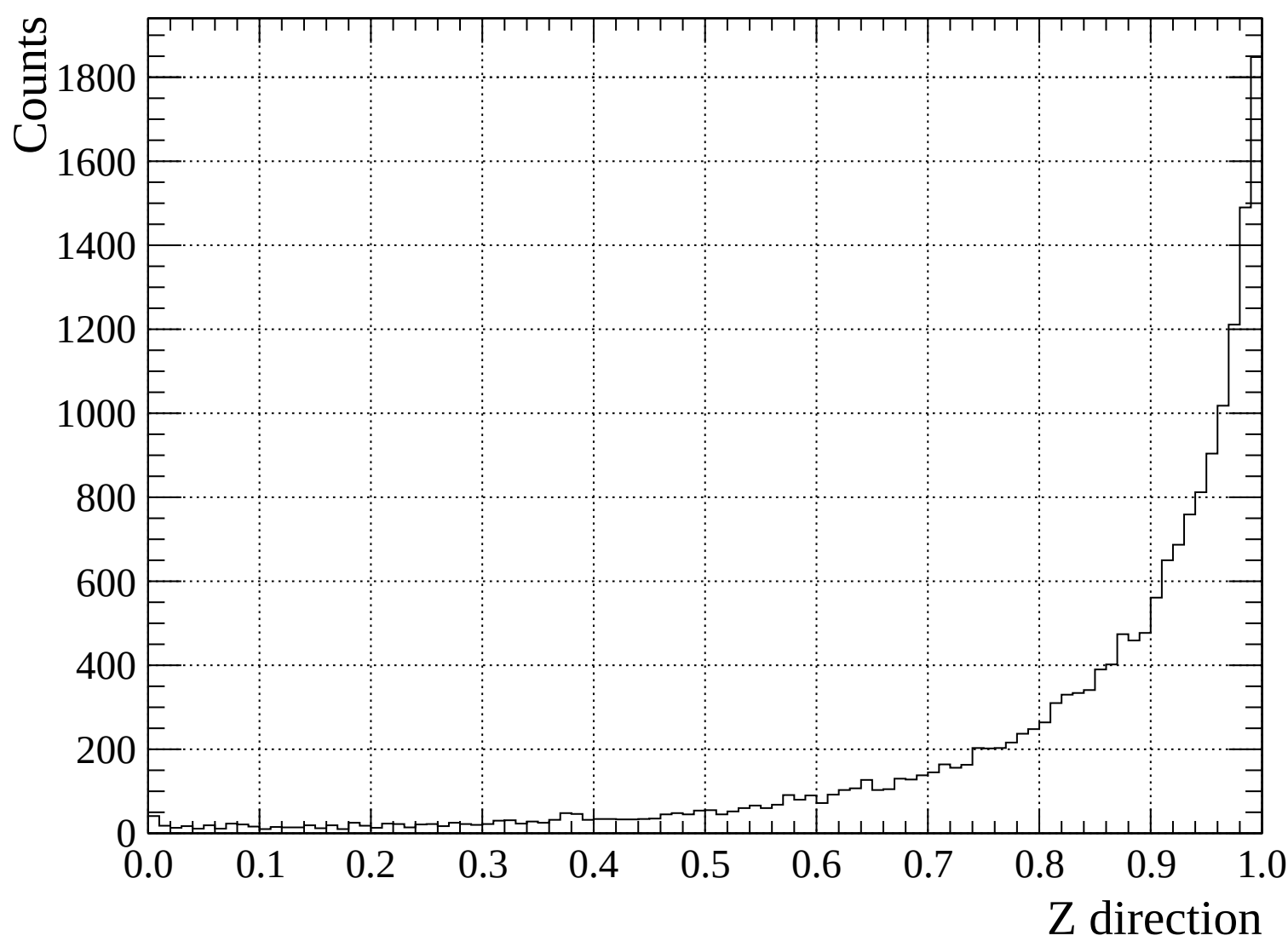


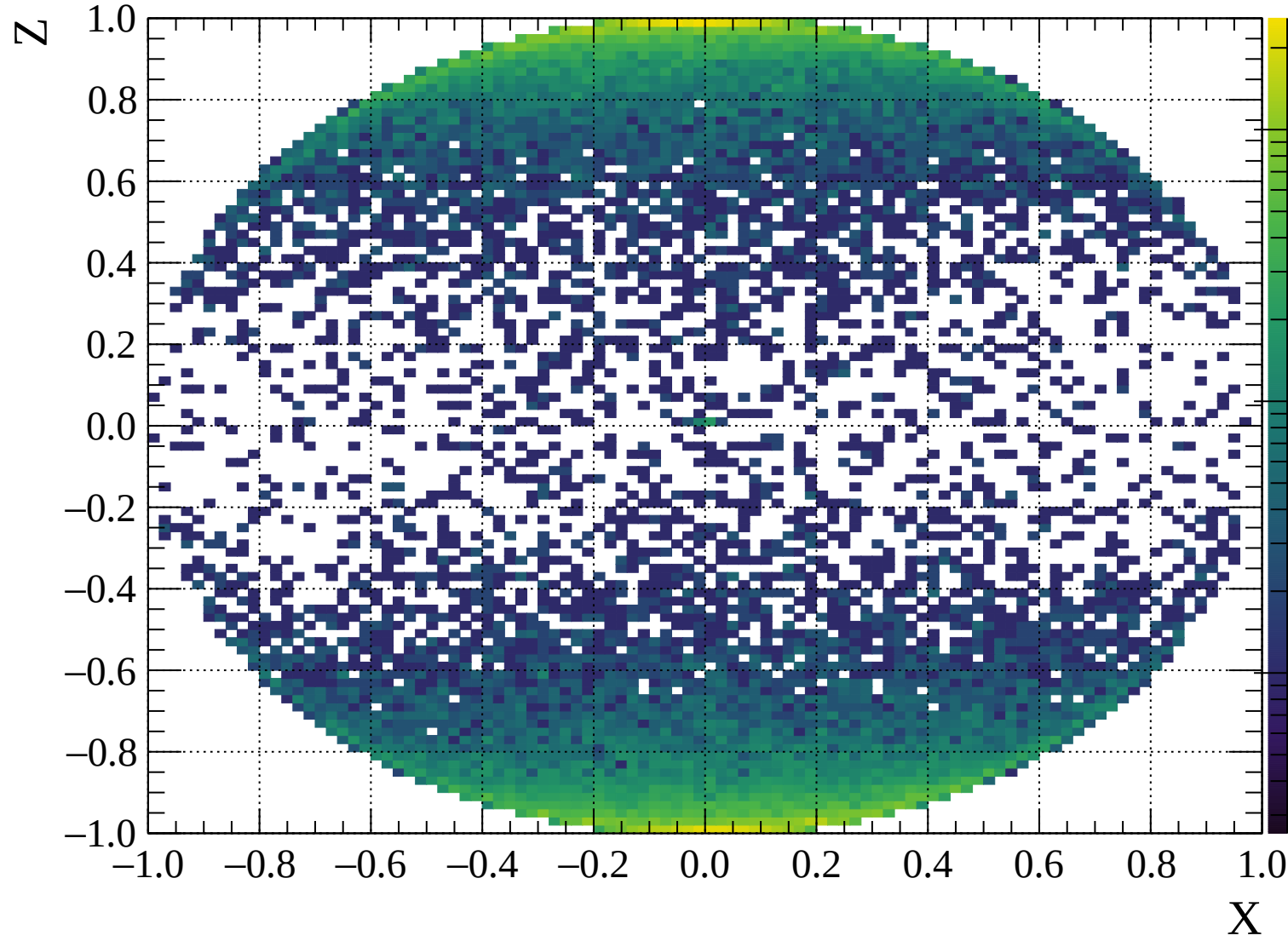


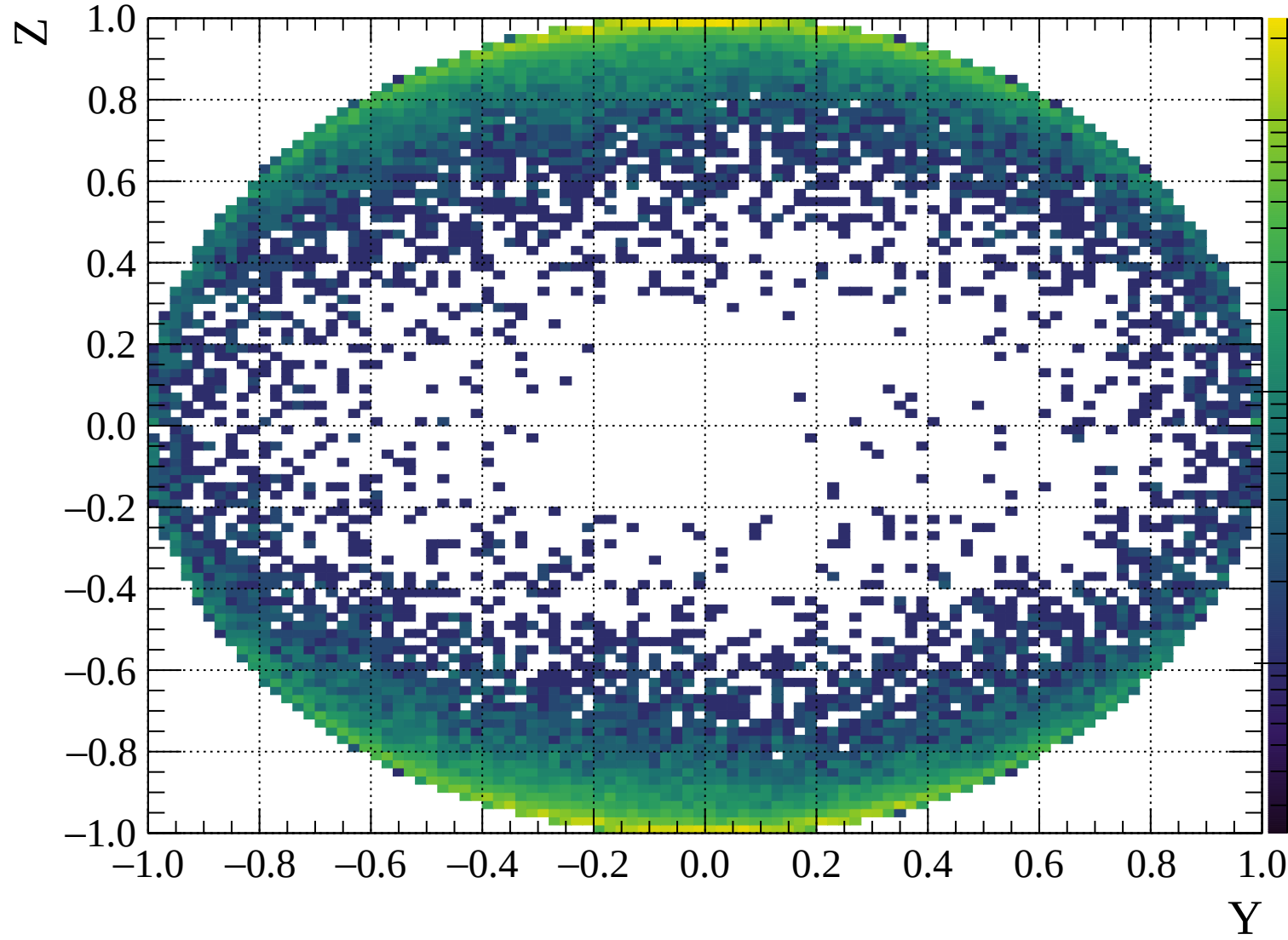


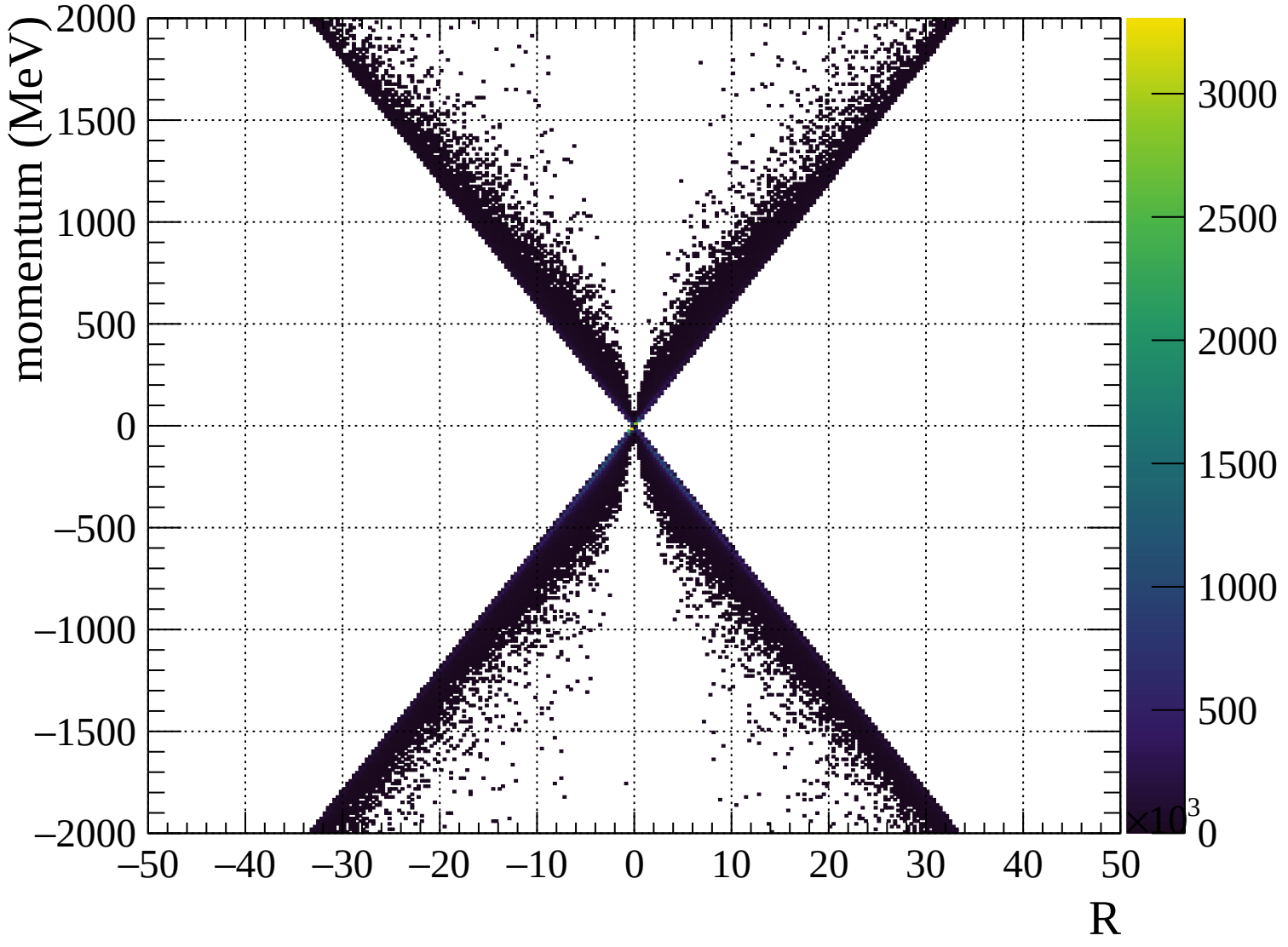


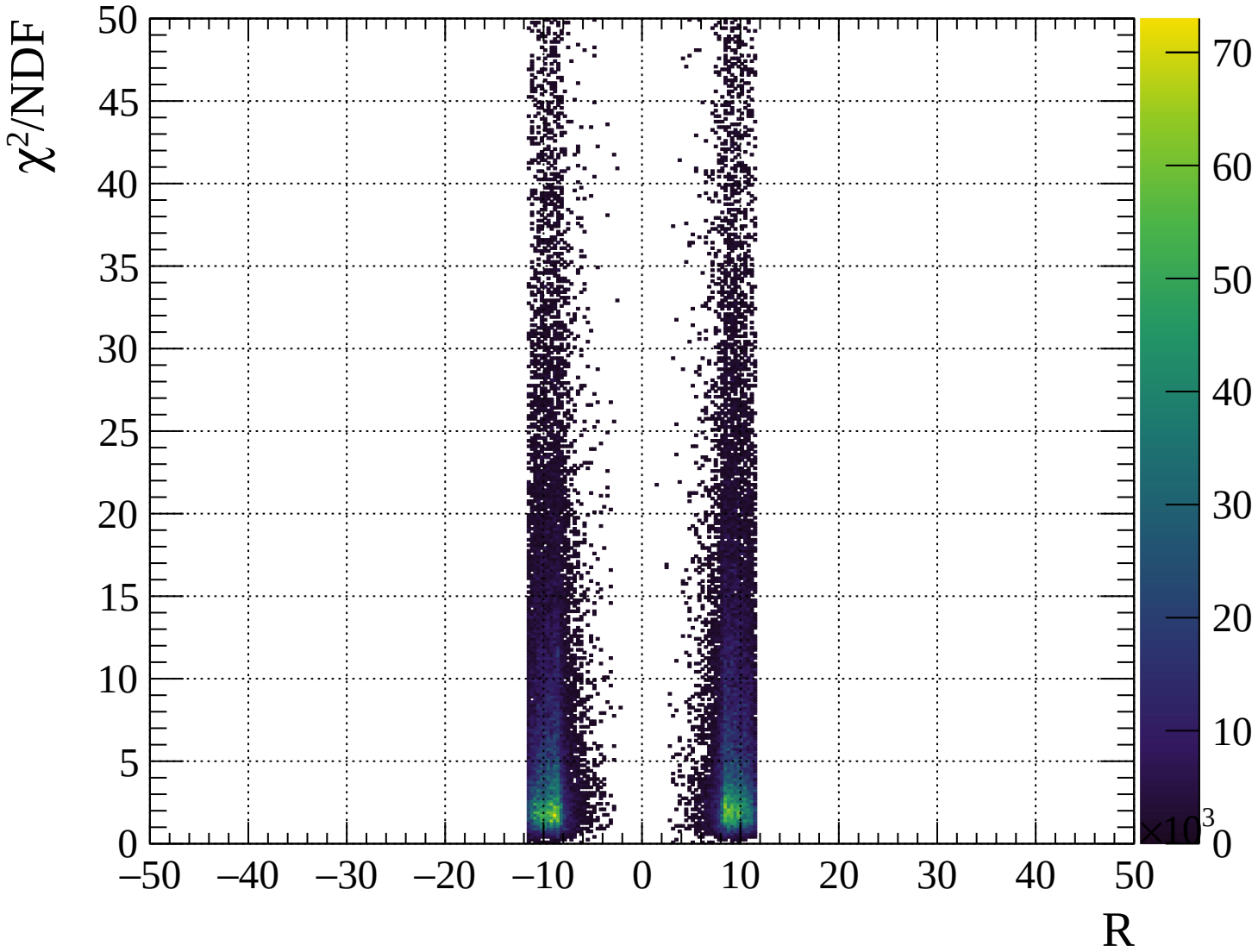




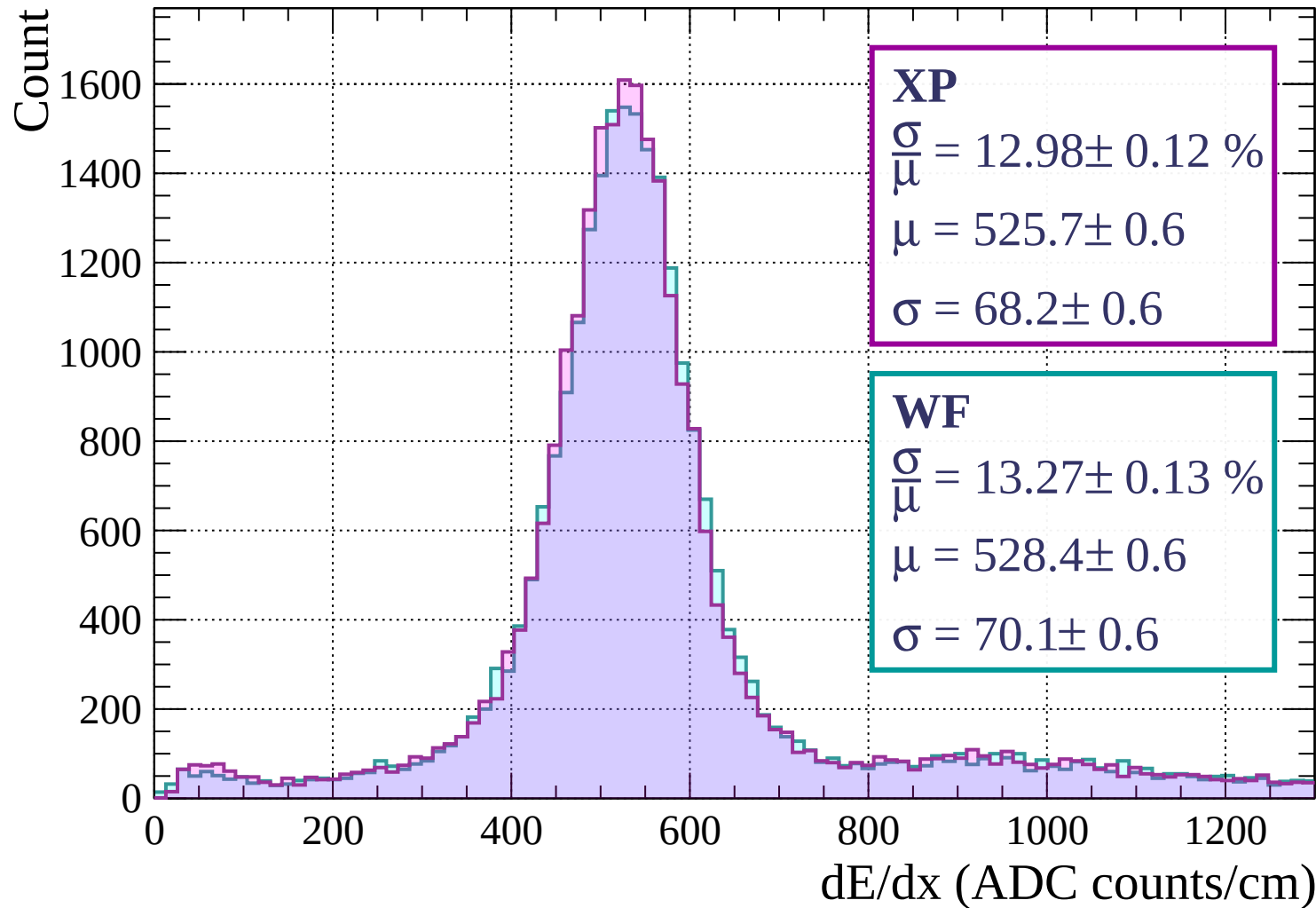




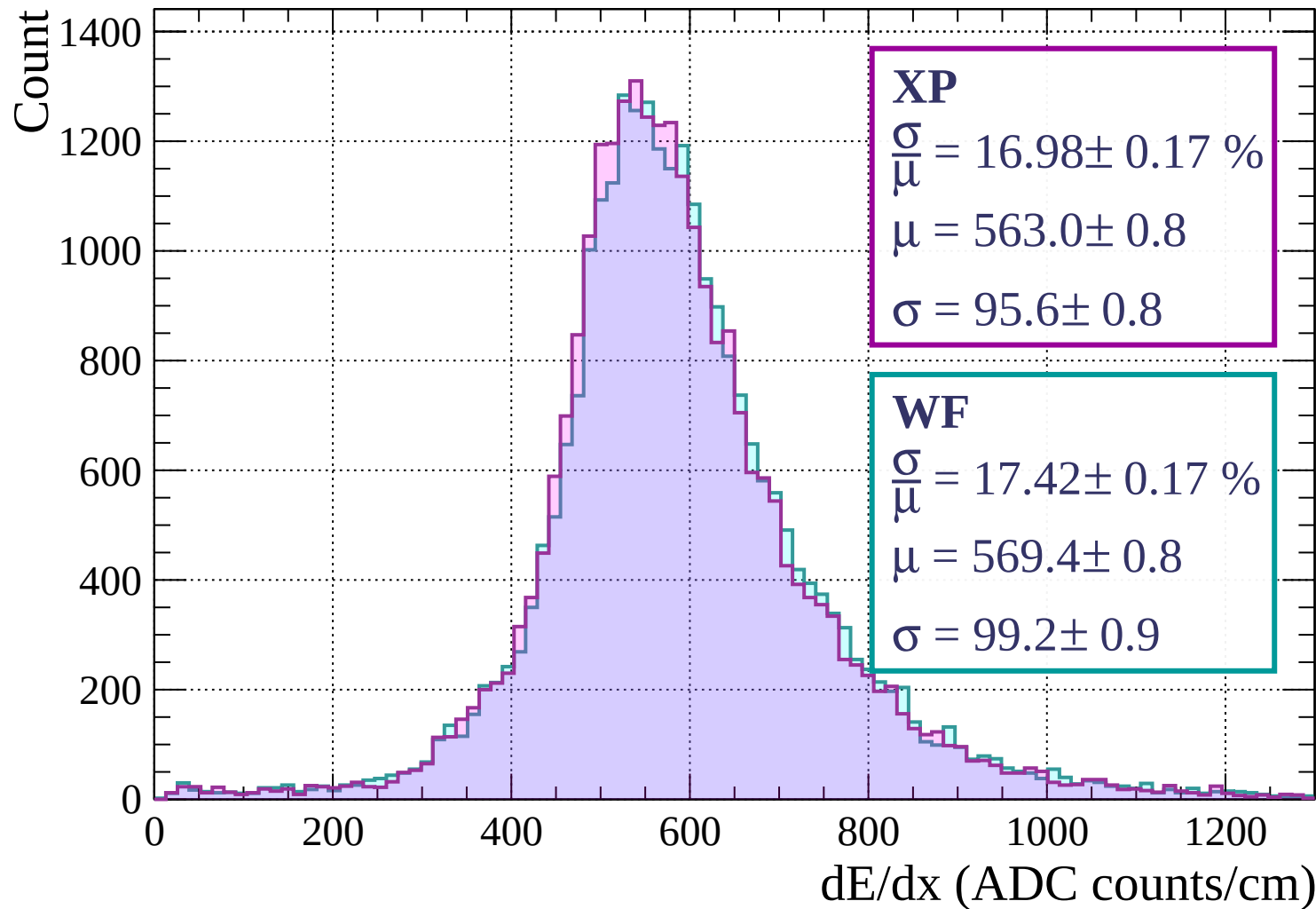




Energy loss | $0 < p < 80$

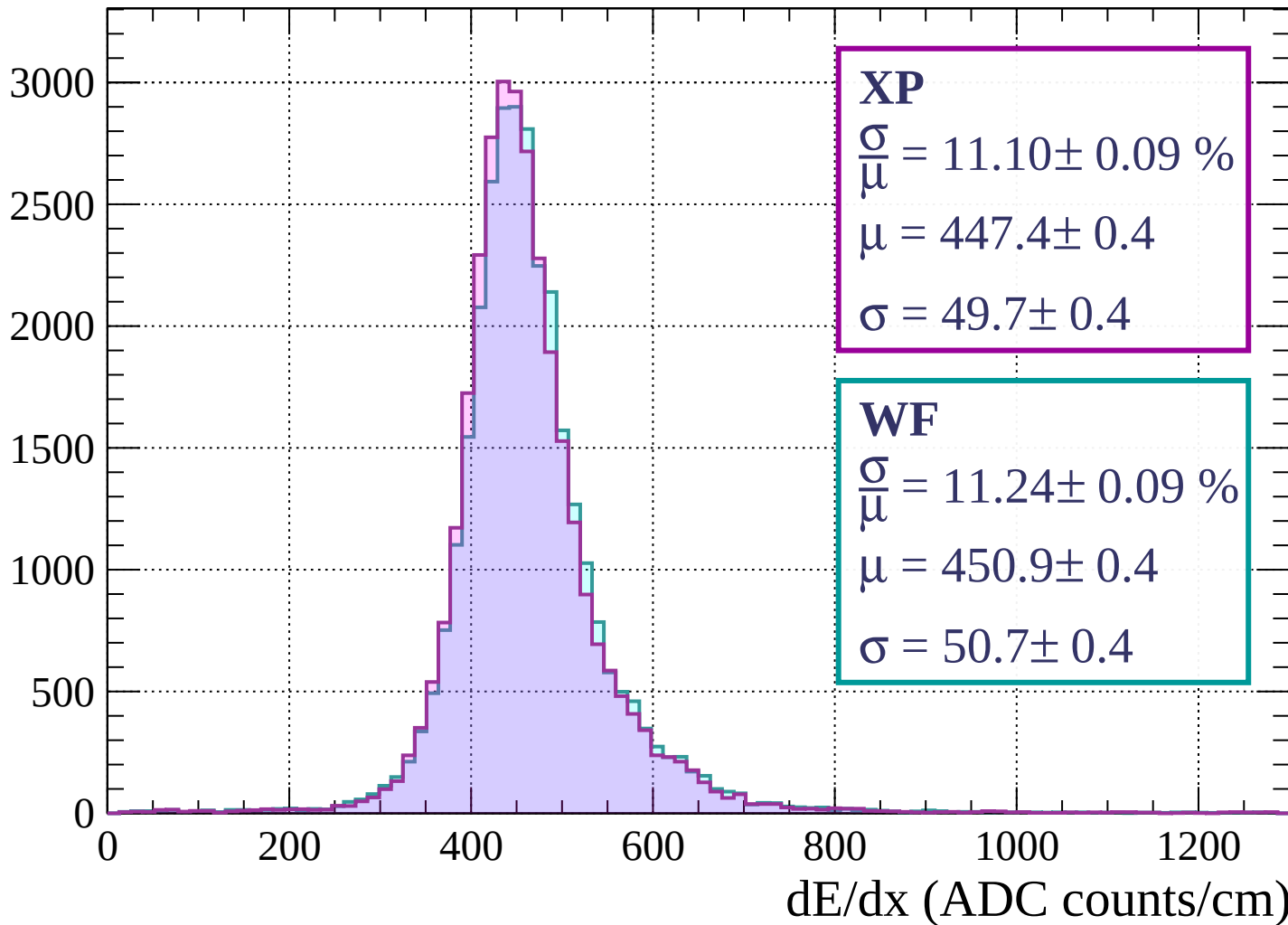


Energy loss | $80 < p < 160$

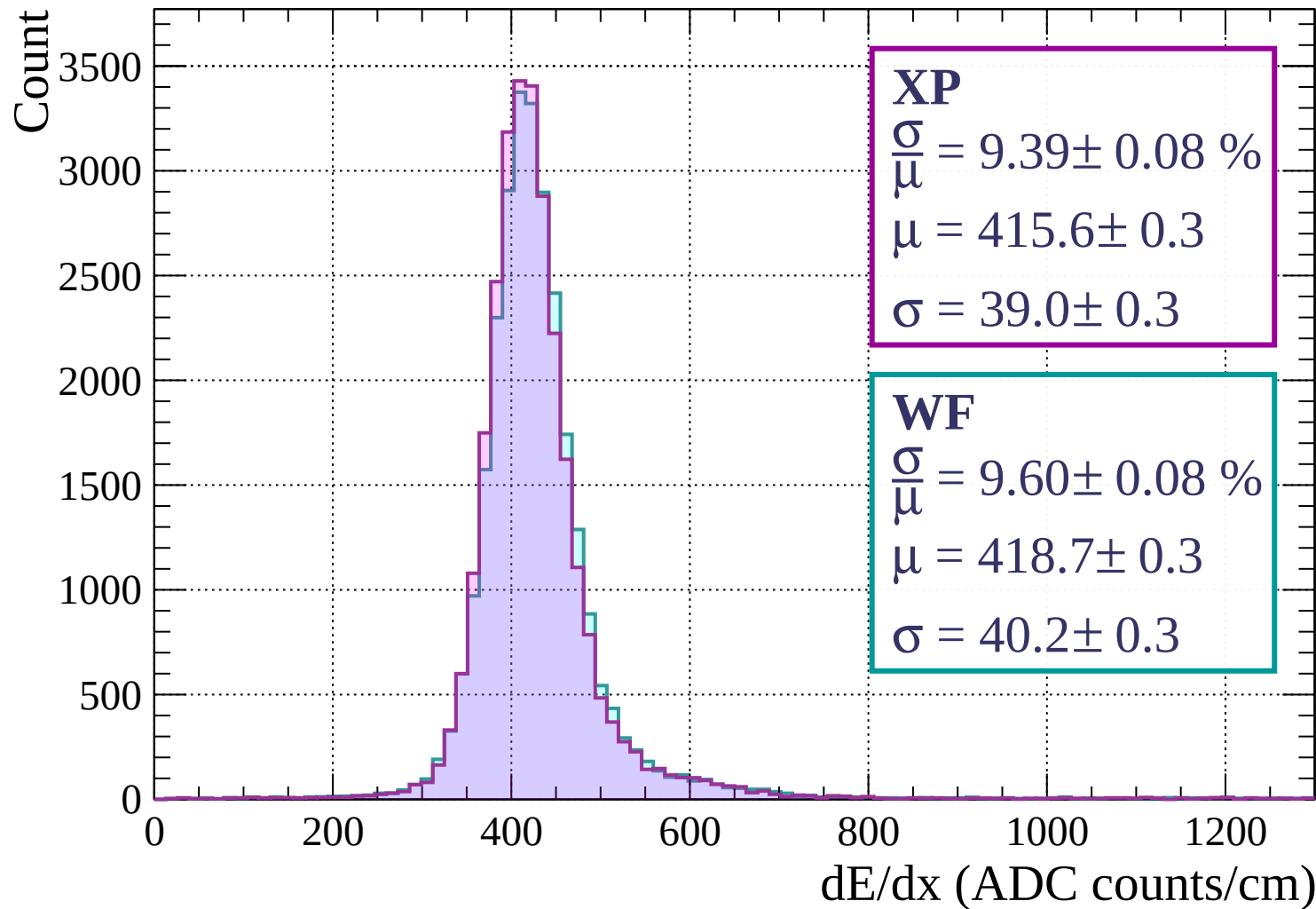


Energy loss | $160 < p < 240$

Count

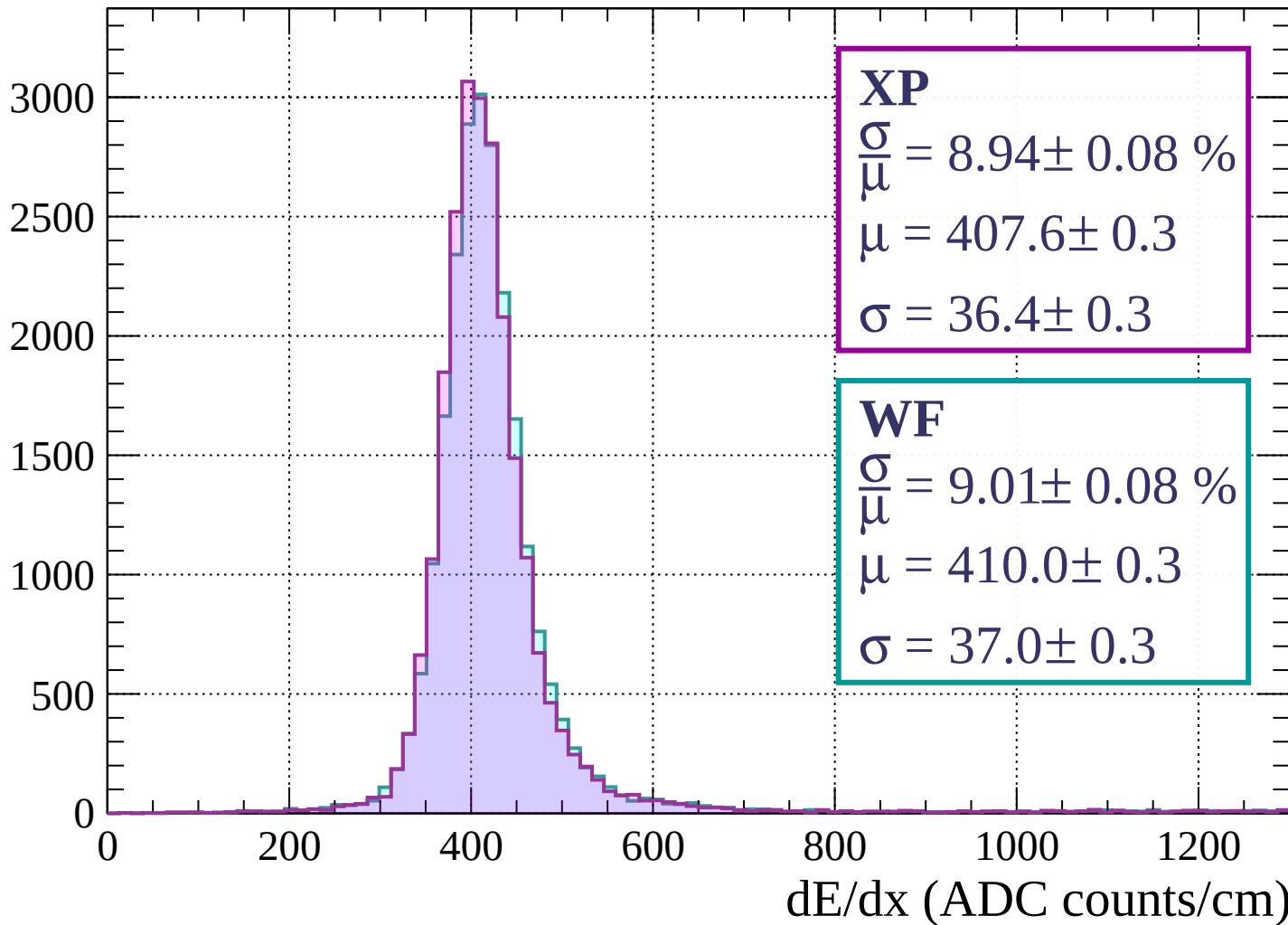


Energy loss | $240 < p < 320$

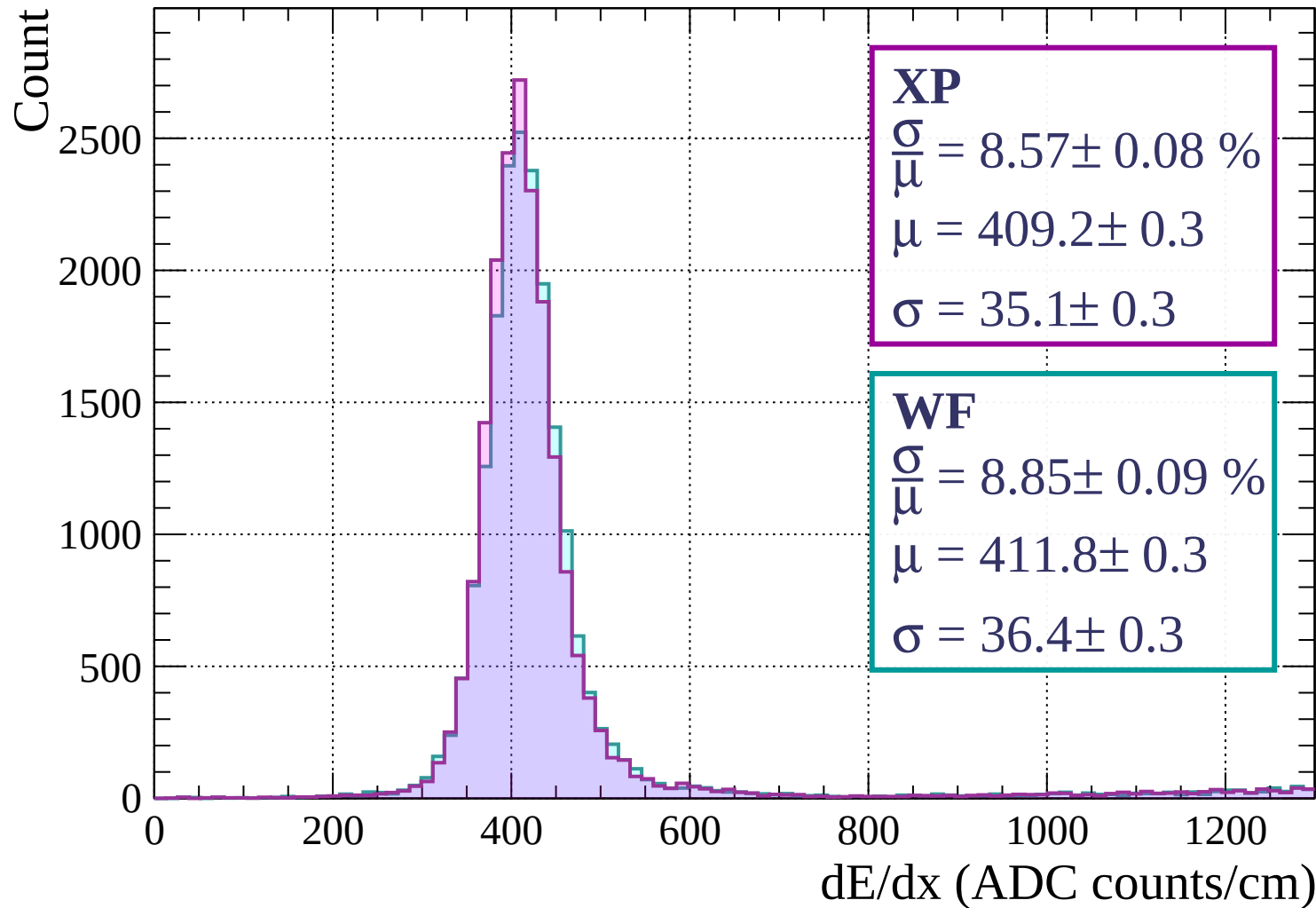


Energy loss | $320 < p < 400$

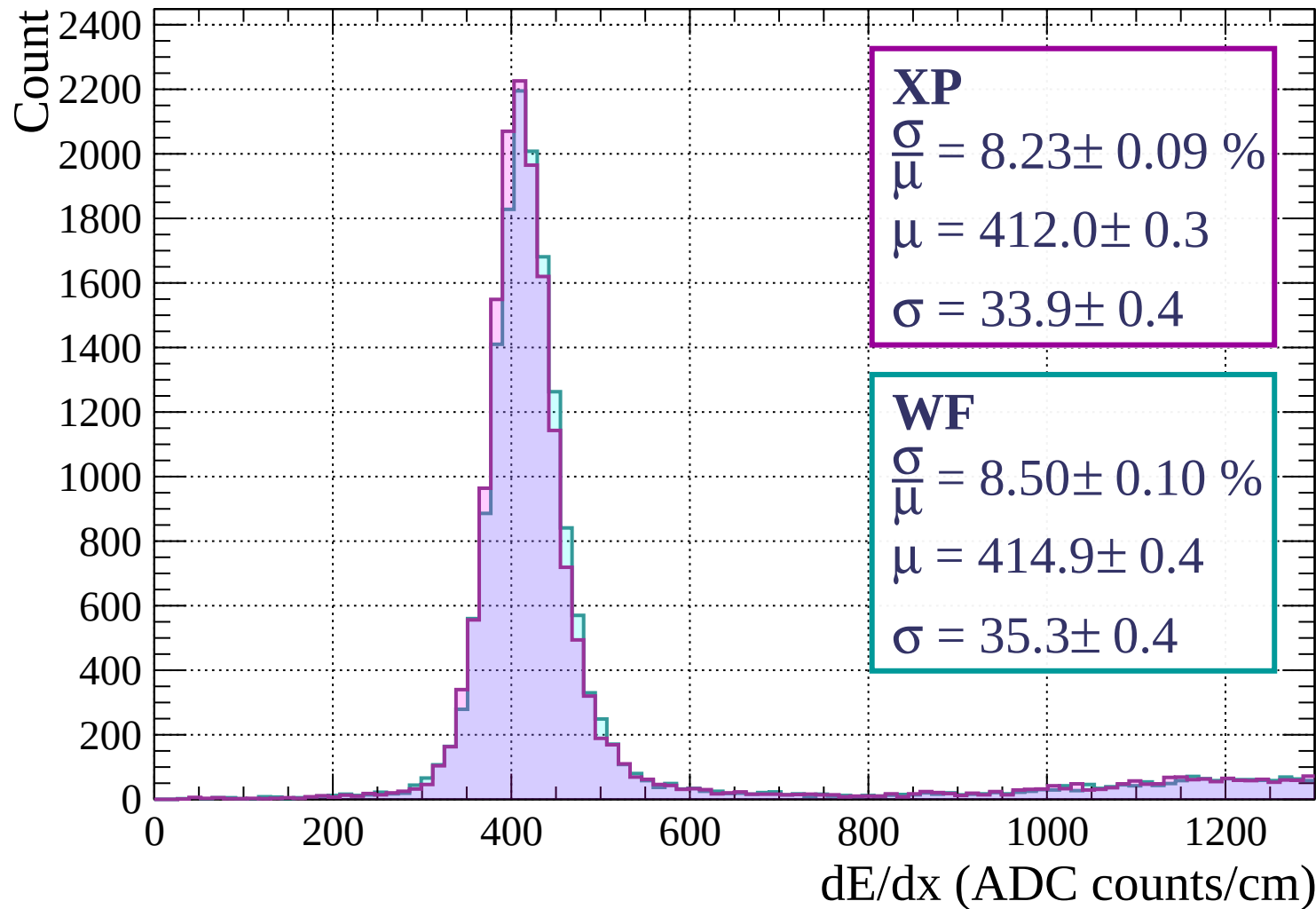
Count



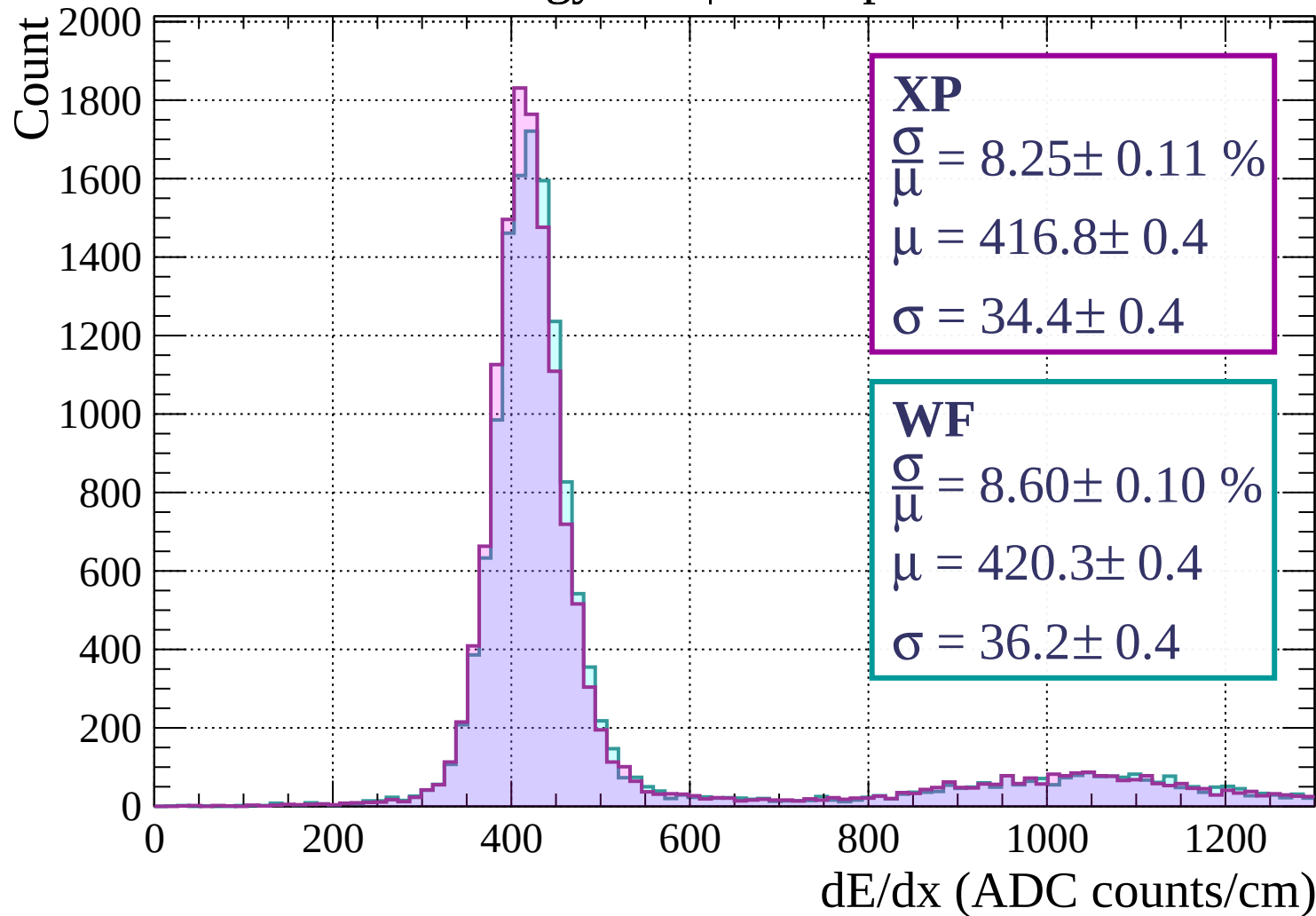
Energy loss | $400 < p < 480$



Energy loss | $480 < p < 560$

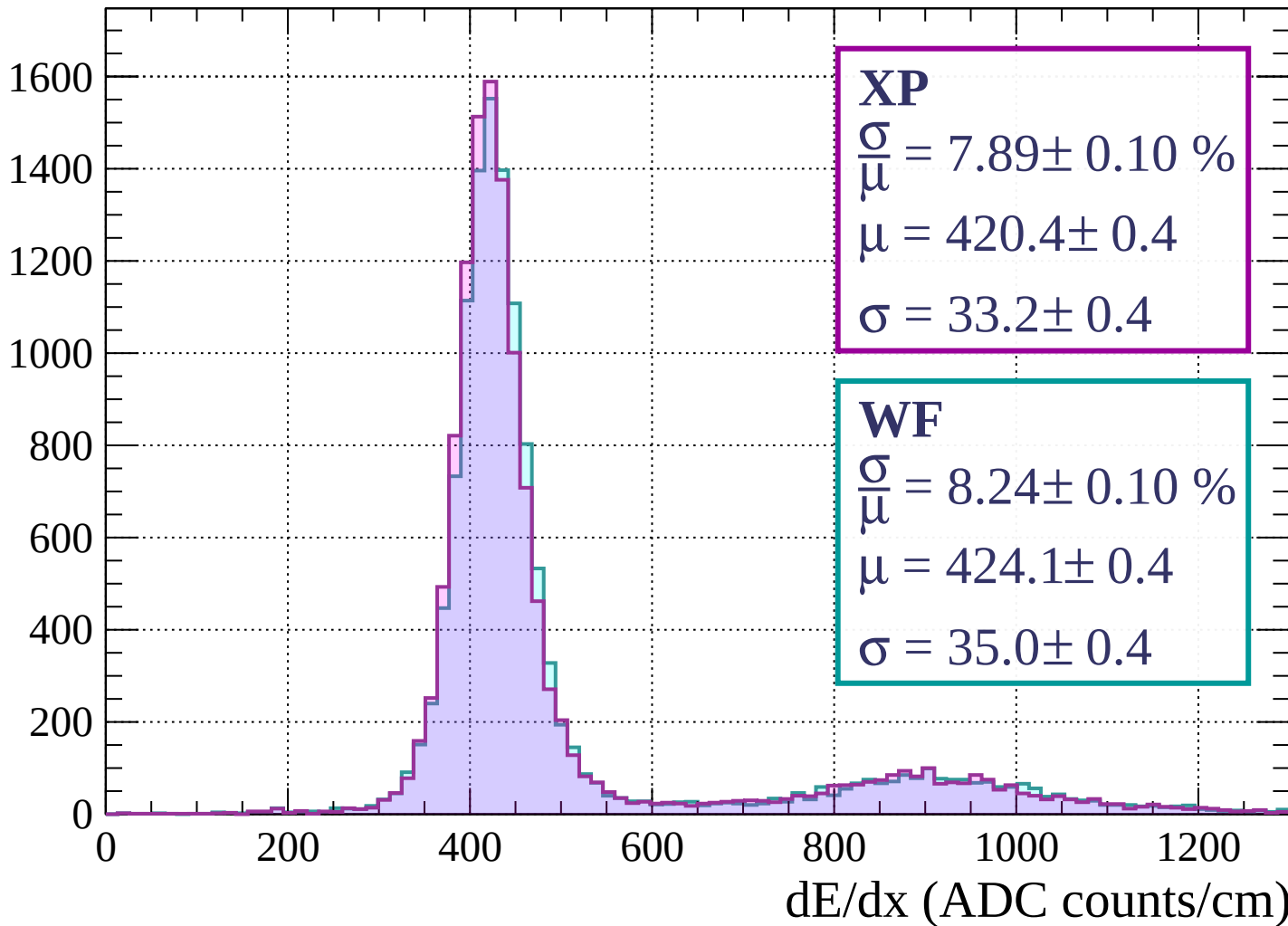


Energy loss | $560 < p < 640$

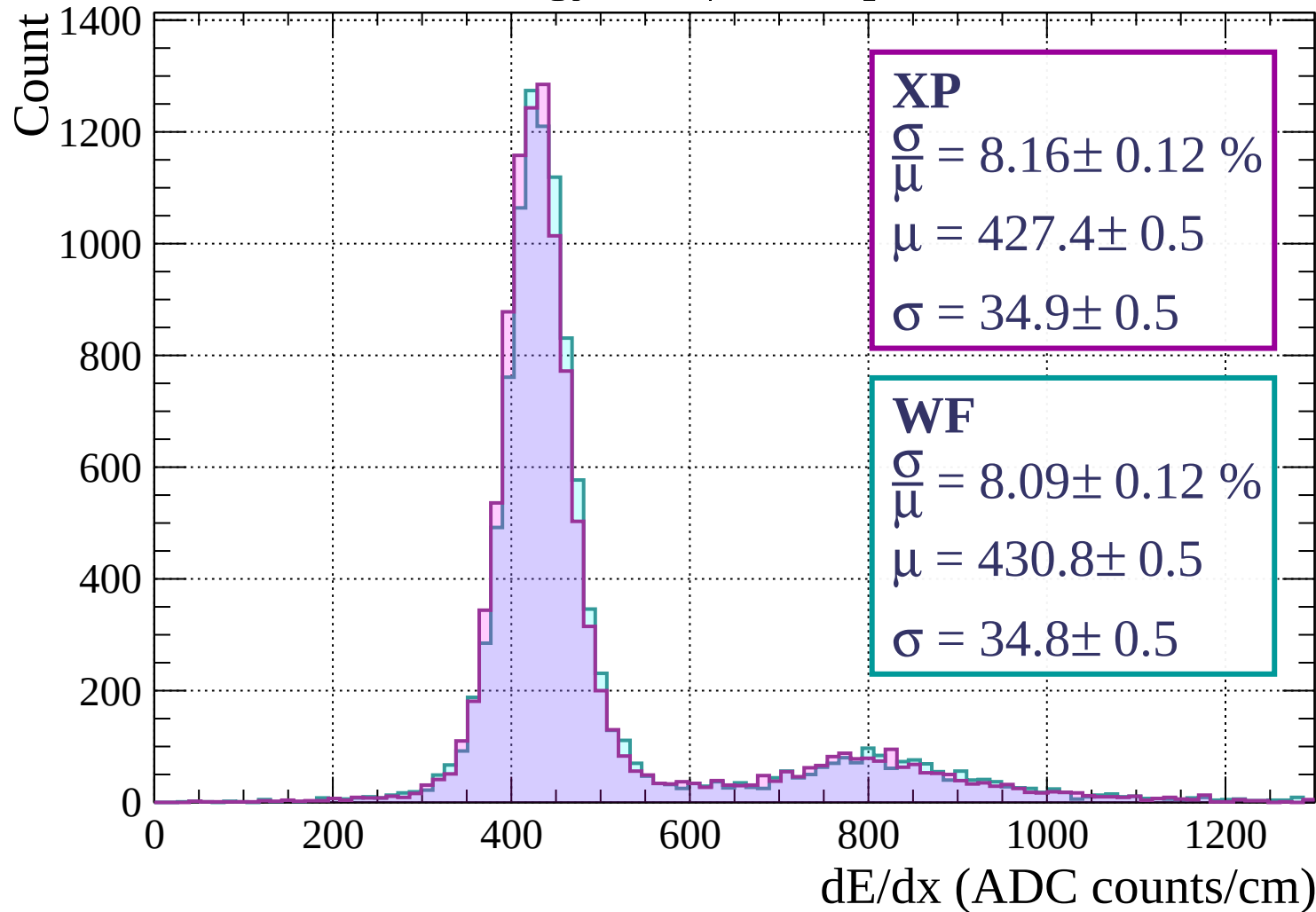


Energy loss | $640 < p < 720$

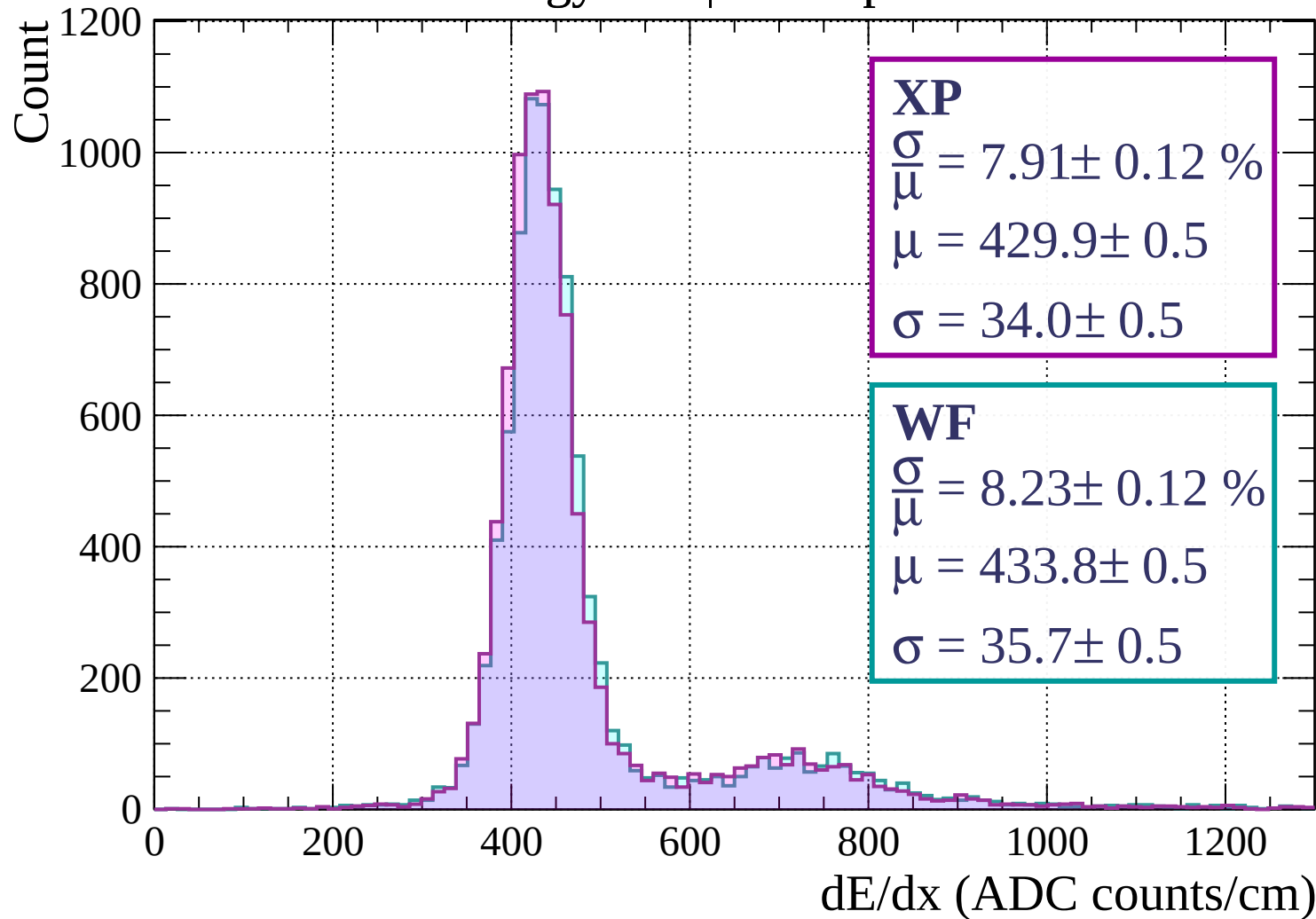
Count



Energy loss | $720 < p < 800$



Energy loss | $800 < p < 880$



Energy loss | $880 < p < 960$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 8.04 \pm 0.12 \%$$

$$\mu = 435.8 \pm 0.5$$

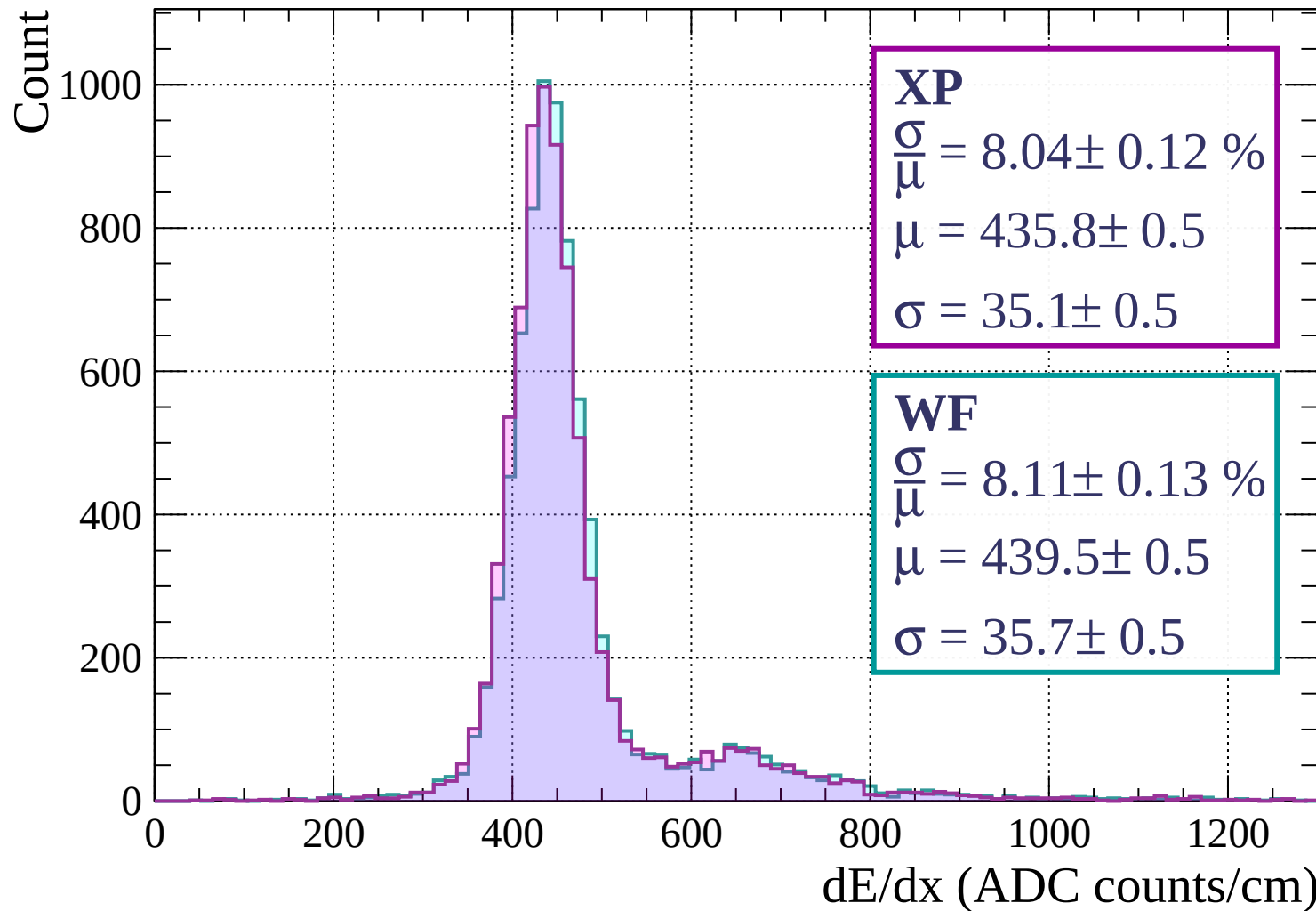
$$\sigma = 35.1 \pm 0.5$$

WF

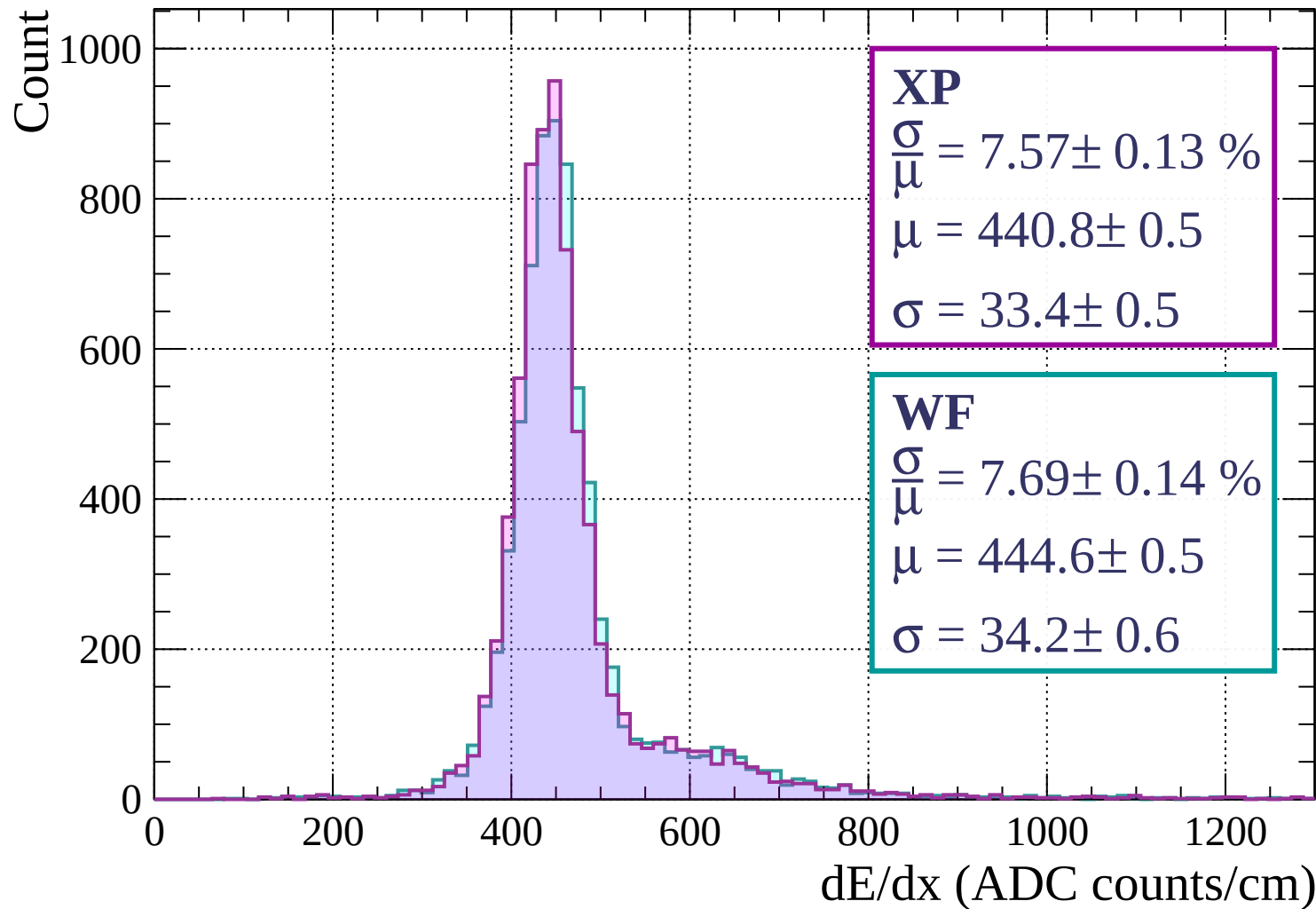
$$\frac{\sigma}{\mu} = 8.11 \pm 0.13 \%$$

$$\mu = 439.5 \pm 0.5$$

$$\sigma = 35.7 \pm 0.5$$

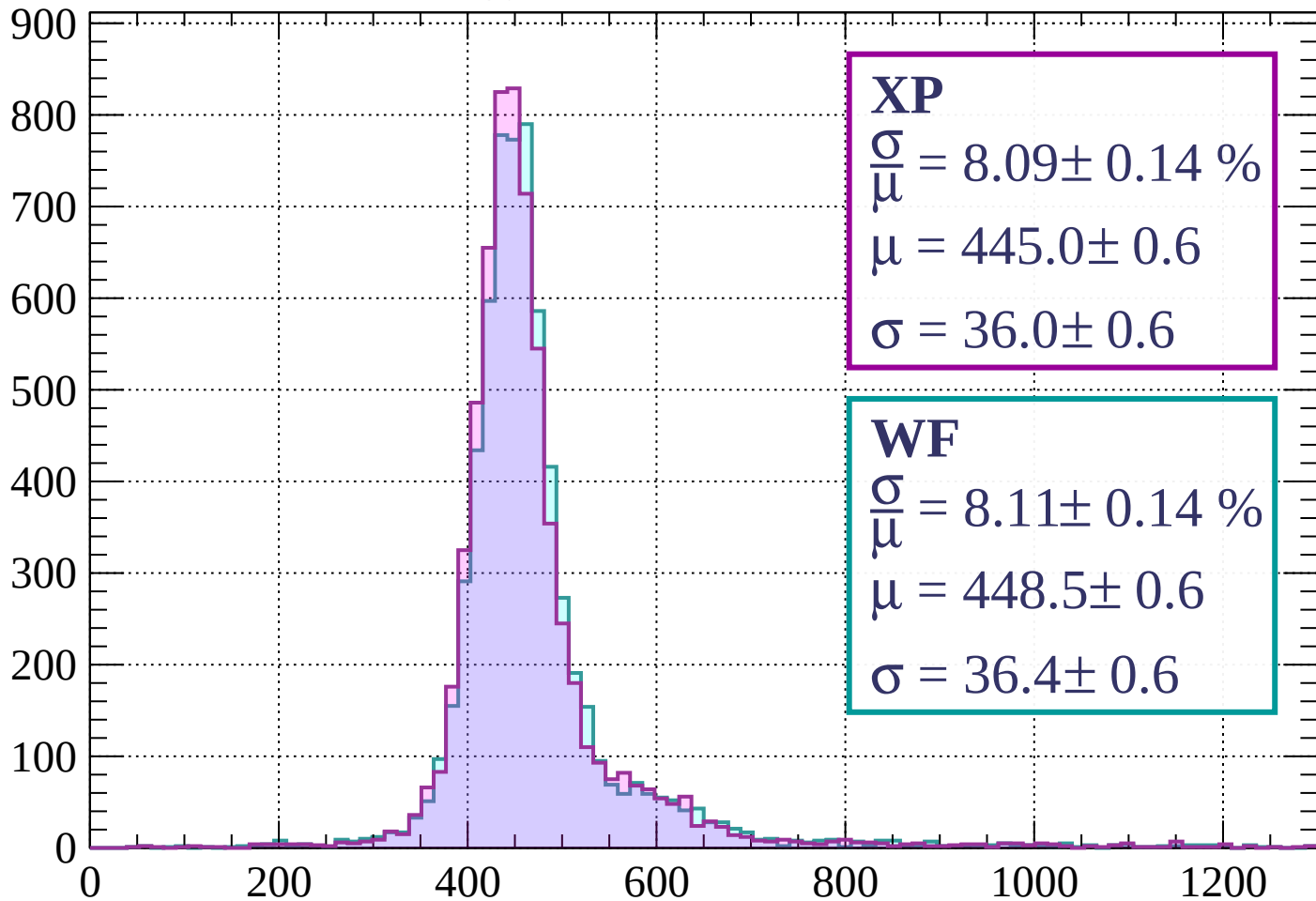


Energy loss | $960 < p < 1040$



Energy loss | $1040 < p < 1120$

Count



XP

$$\frac{\sigma}{\mu} = 8.09 \pm 0.14 \%$$

$$\mu = 445.0 \pm 0.6$$

$$\sigma = 36.0 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.11 \pm 0.14 \%$$

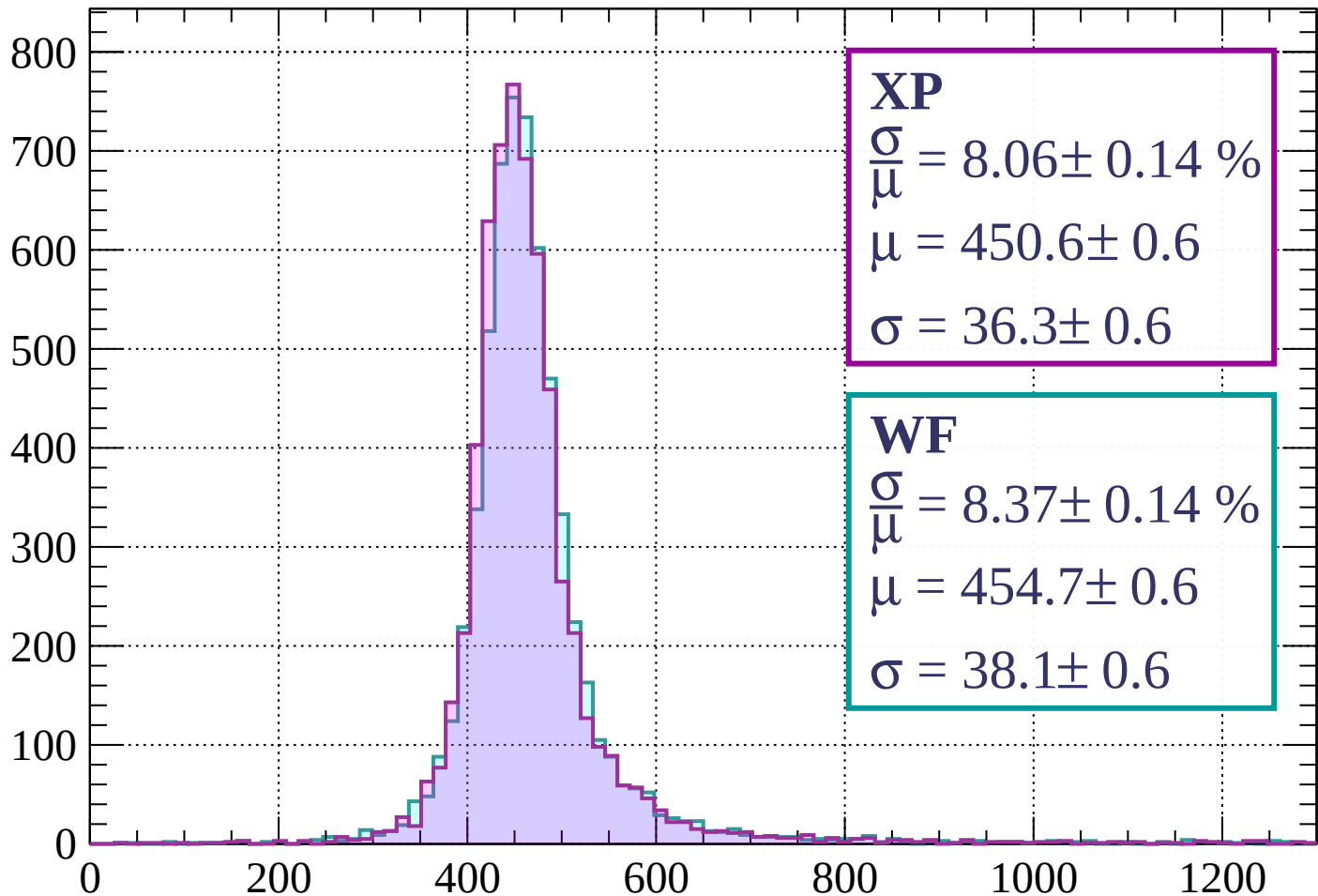
$$\mu = 448.5 \pm 0.6$$

$$\sigma = 36.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1120 < p < 1200$

Count



XP

$$\frac{\sigma}{\mu} = 8.06 \pm 0.14 \%$$

$$\mu = 450.6 \pm 0.6$$

$$\sigma = 36.3 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.37 \pm 0.14 \%$$

$$\mu = 454.7 \pm 0.6$$

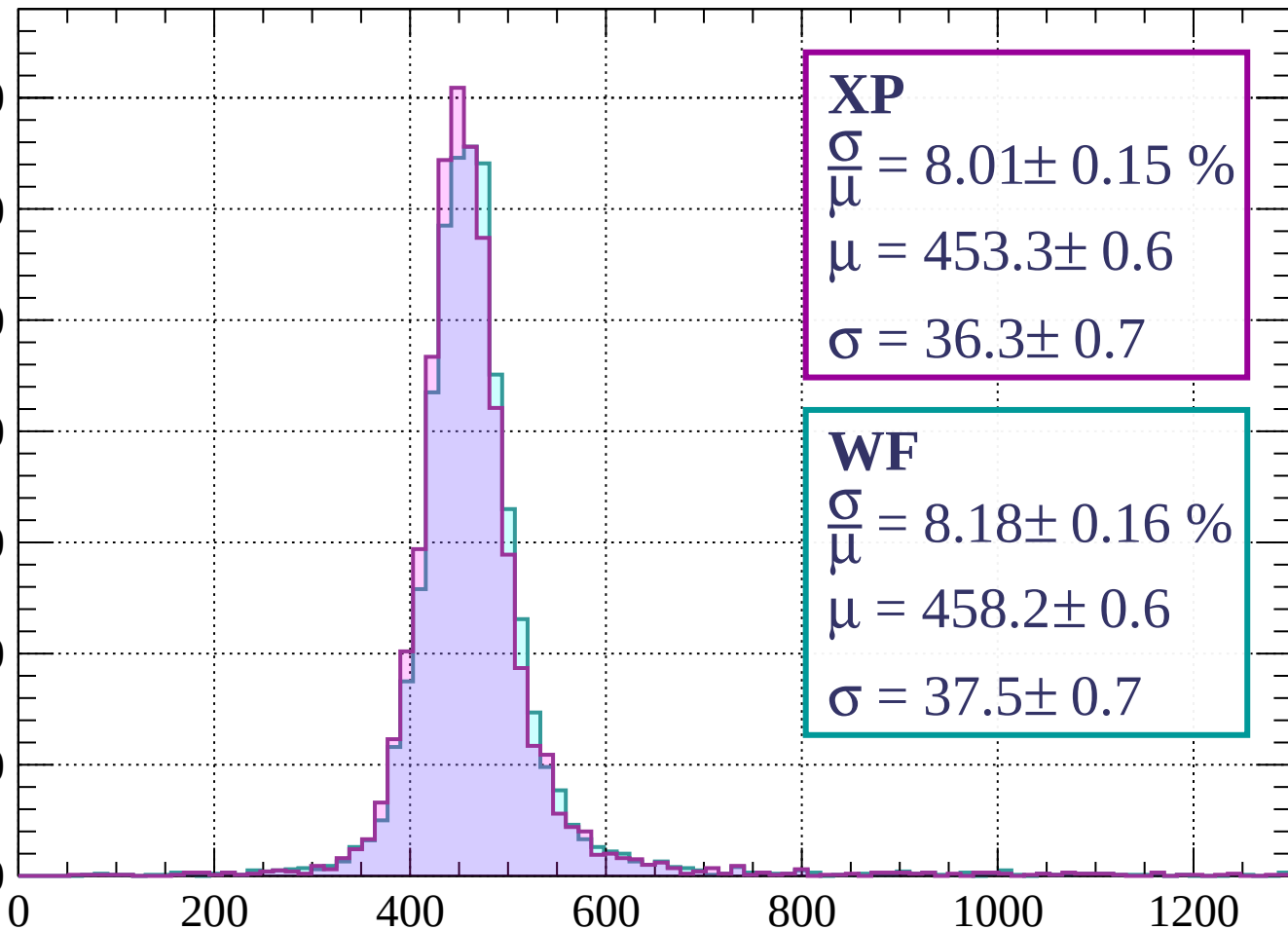
$$\sigma = 38.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1280$

Count

700
600
500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 8.01 \pm 0.15 \%$$

$$\mu = 453.3 \pm 0.6$$

$$\sigma = 36.3 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.18 \pm 0.16 \%$$

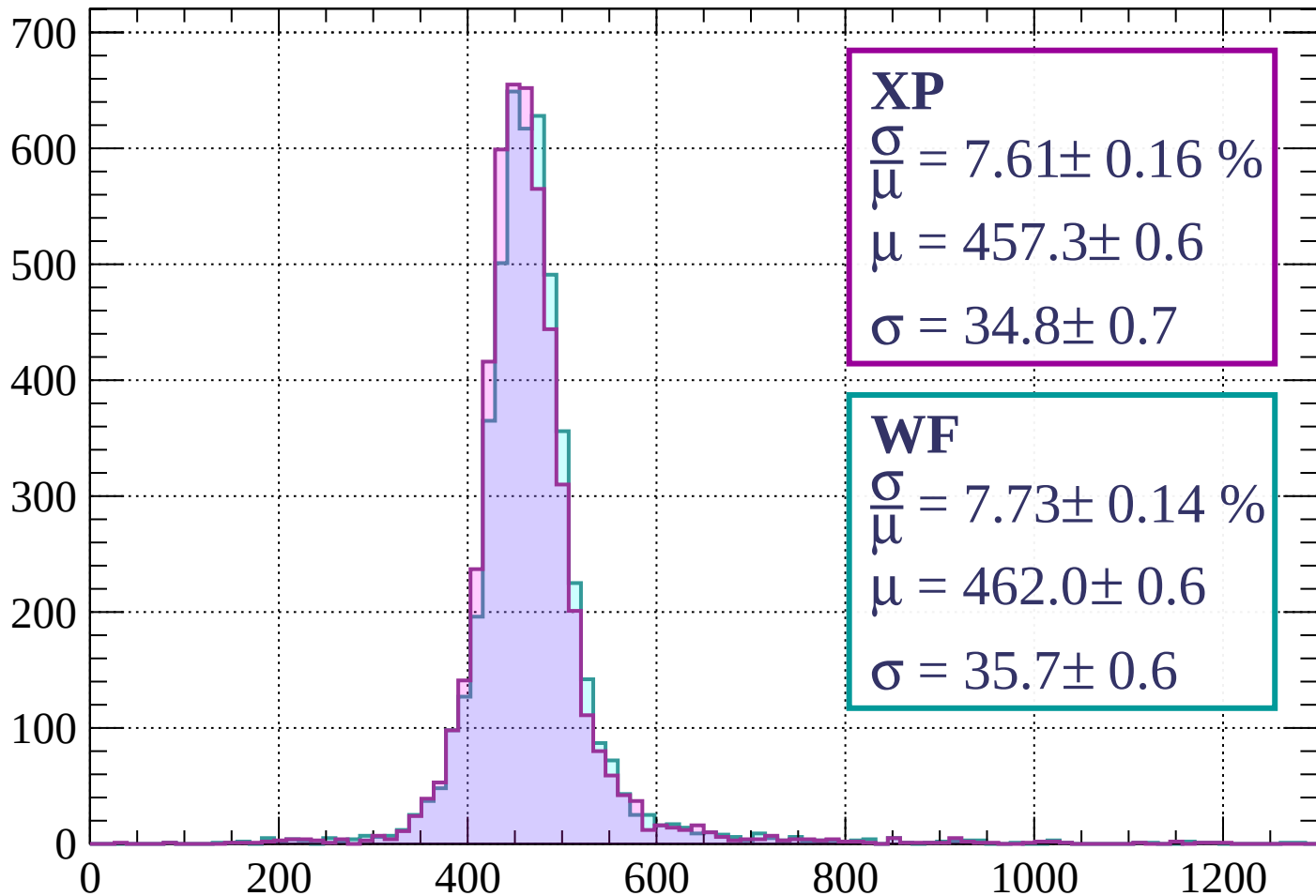
$$\mu = 458.2 \pm 0.6$$

$$\sigma = 37.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1280 < p < 1360$

Count



XP

$$\frac{\sigma}{\mu} = 7.61 \pm 0.16 \%$$

$$\mu = 457.3 \pm 0.6$$

$$\sigma = 34.8 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.73 \pm 0.14 \%$$

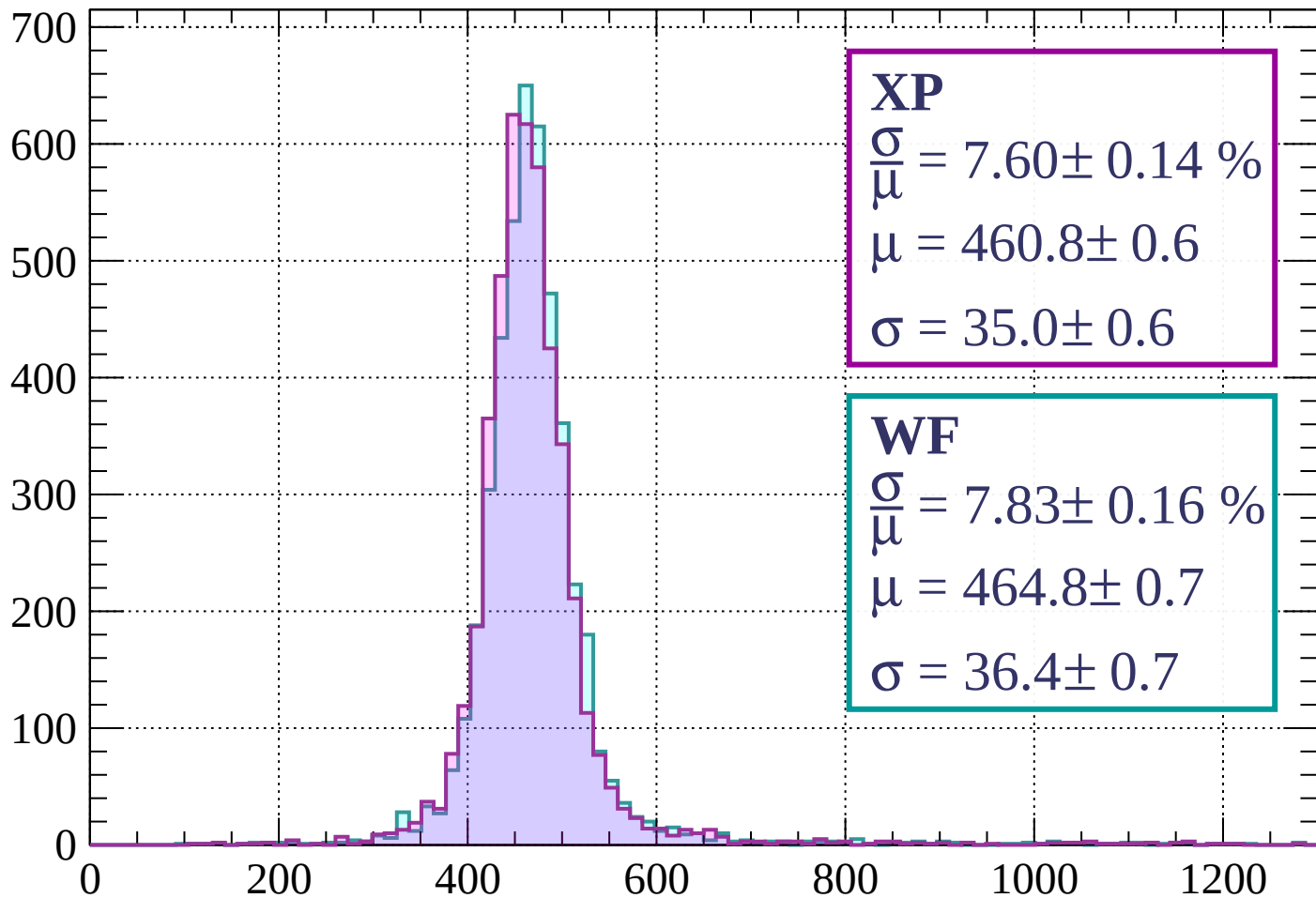
$$\mu = 462.0 \pm 0.6$$

$$\sigma = 35.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1360 < p < 1440$

Count



XP

$$\frac{\sigma}{\mu} = 7.60 \pm 0.14 \%$$

$$\mu = 460.8 \pm 0.6$$

$$\sigma = 35.0 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 7.83 \pm 0.16 \%$$

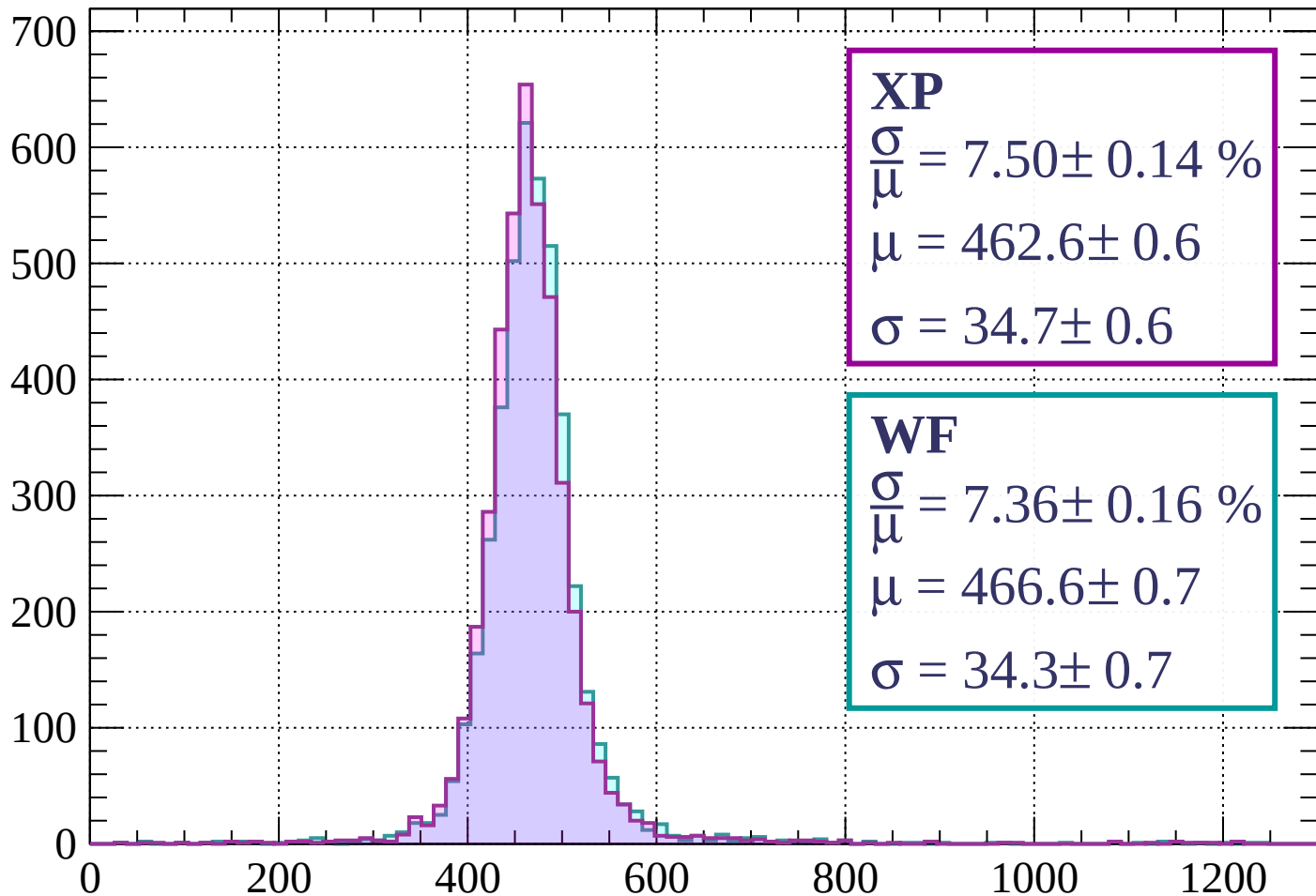
$$\mu = 464.8 \pm 0.7$$

$$\sigma = 36.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1520$

Count



XP

$$\frac{\sigma}{\mu} = 7.50 \pm 0.14 \%$$

$$\mu = 462.6 \pm 0.6$$

$$\sigma = 34.7 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 7.36 \pm 0.16 \%$$

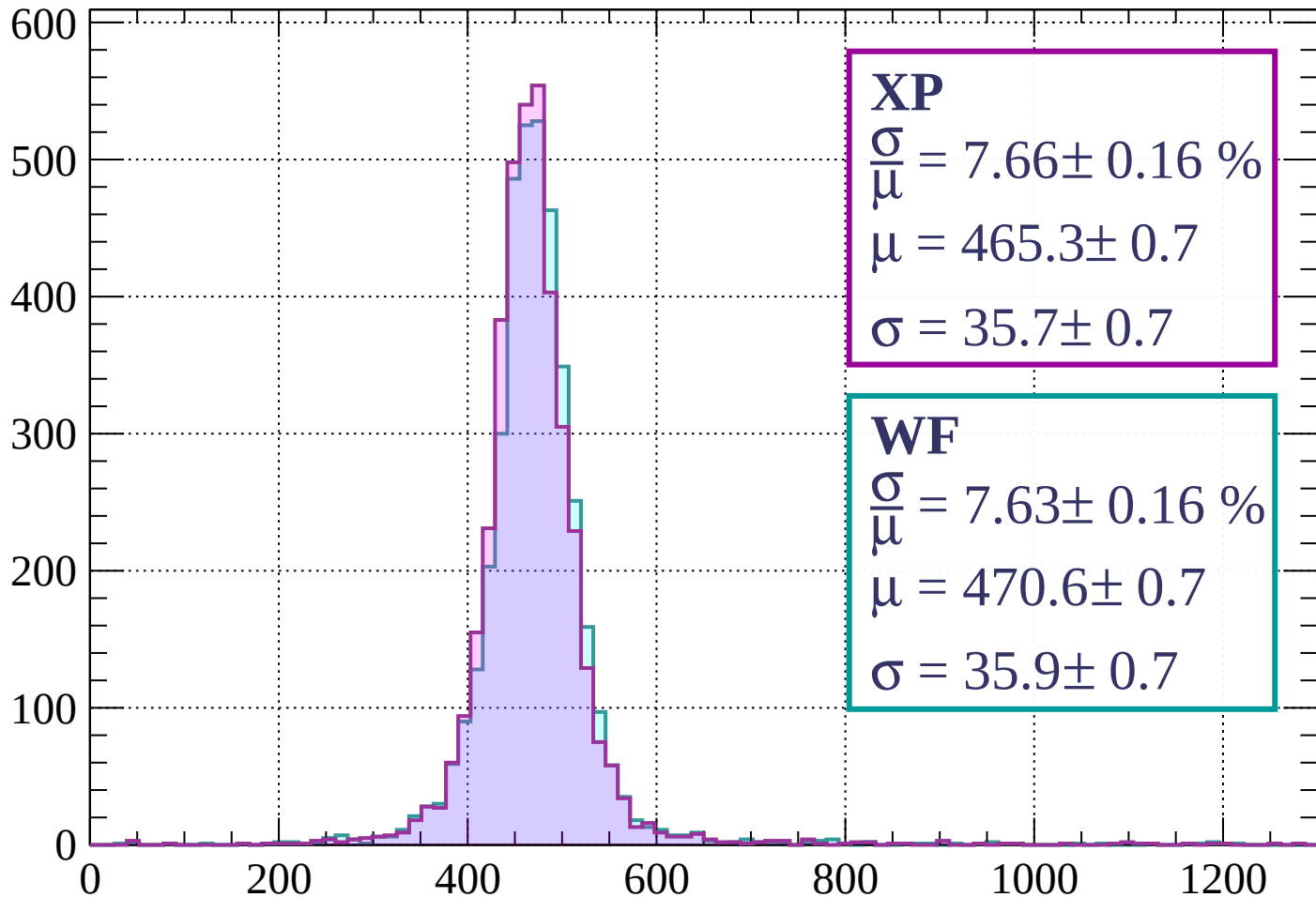
$$\mu = 466.6 \pm 0.7$$

$$\sigma = 34.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1520 < p < 1600$

Count



XP

$$\frac{\sigma}{\mu} = 7.66 \pm 0.16 \%$$

$$\mu = 465.3 \pm 0.7$$

$$\sigma = 35.7 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.63 \pm 0.16 \%$$

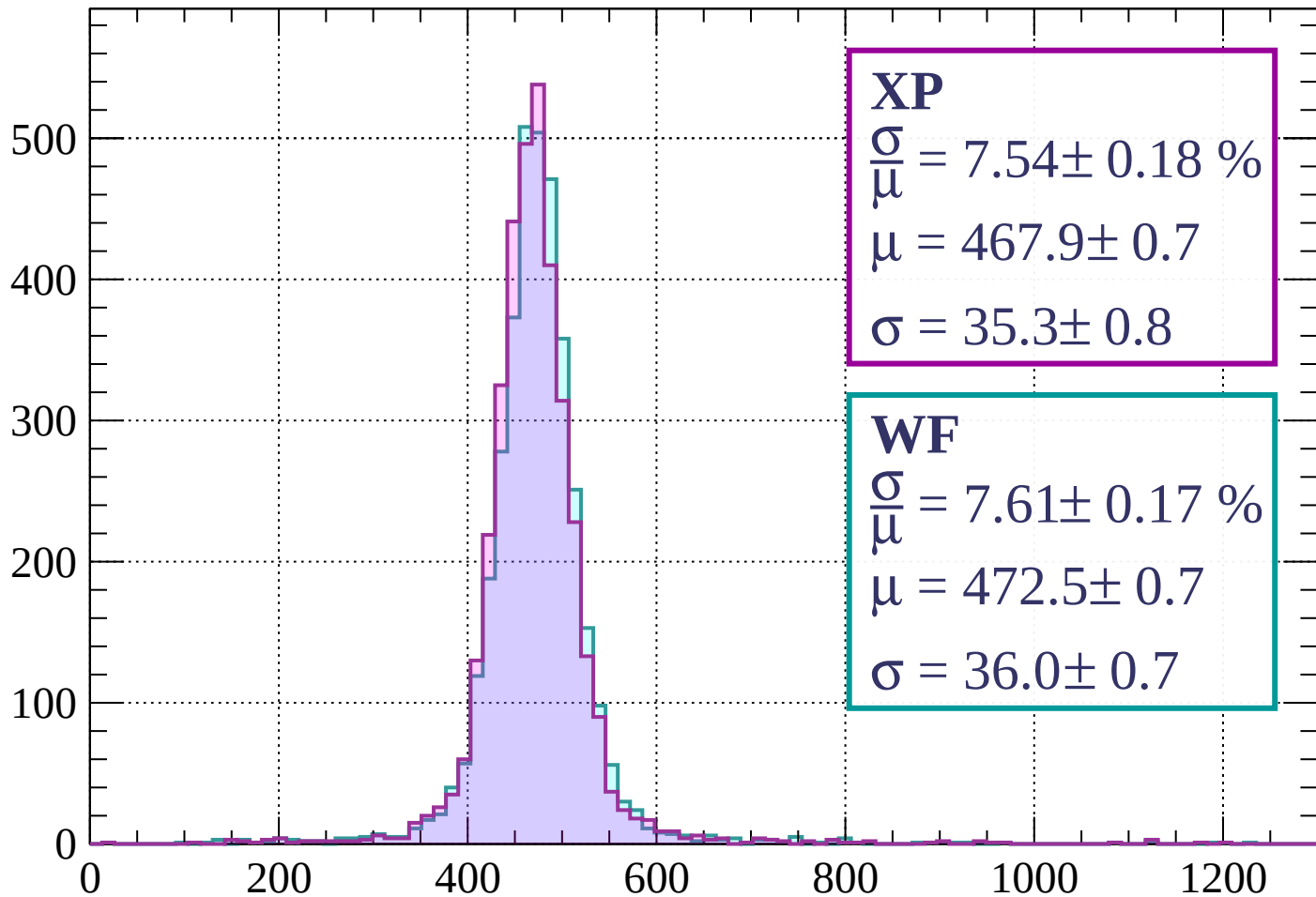
$$\mu = 470.6 \pm 0.7$$

$$\sigma = 35.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1600 < p < 1680$

Count



XP

$$\frac{\sigma}{\mu} = 7.54 \pm 0.18 \%$$

$$\mu = 467.9 \pm 0.7$$

$$\sigma = 35.3 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.61 \pm 0.17 \%$$

$$\mu = 472.5 \pm 0.7$$

$$\sigma = 36.0 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1760$

Count

500

400

300

200

100

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 7.35 \pm 0.18 \%$$

$$\mu = 468.8 \pm 0.7$$

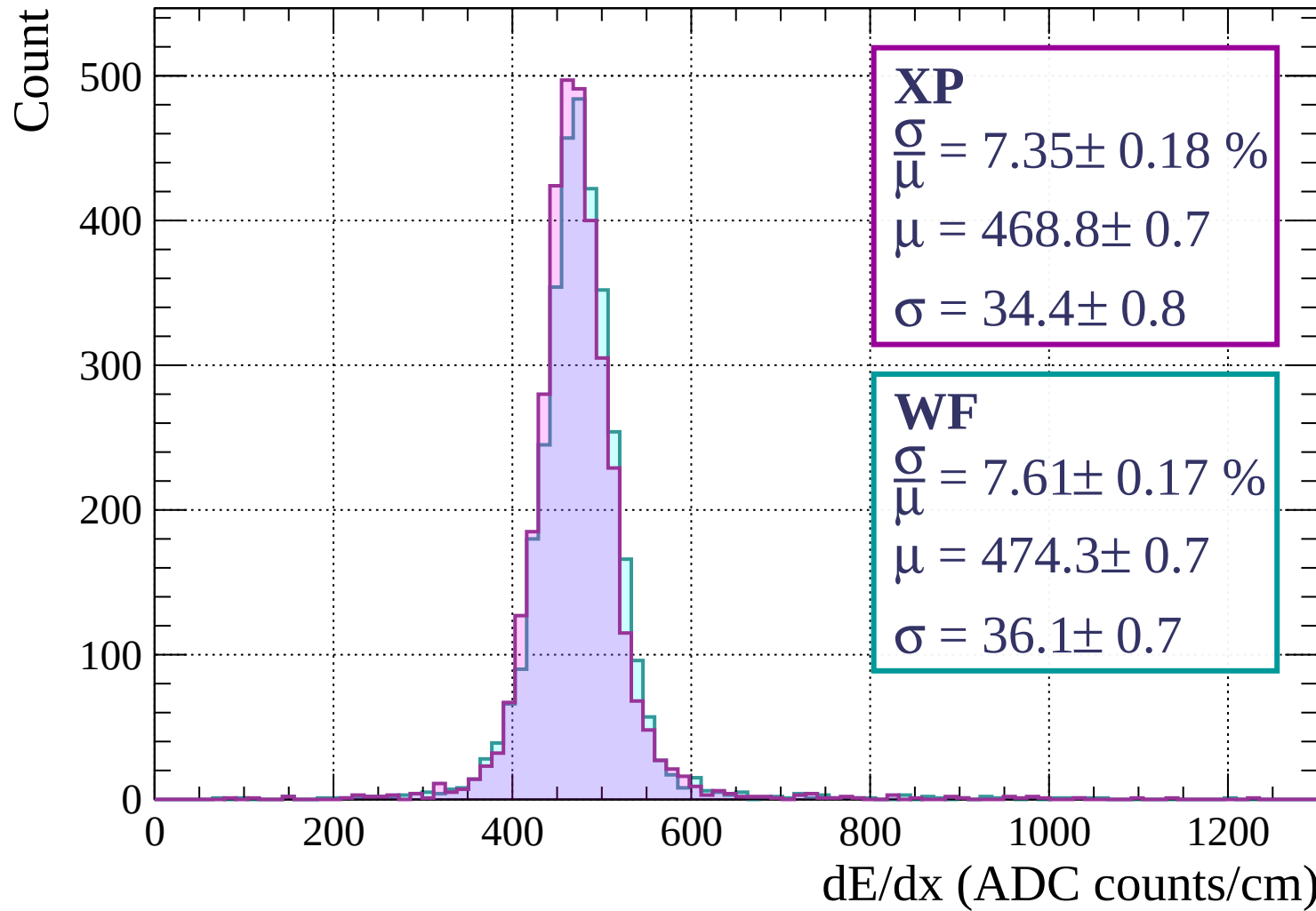
$$\sigma = 34.4 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.61 \pm 0.17 \%$$

$$\mu = 474.3 \pm 0.7$$

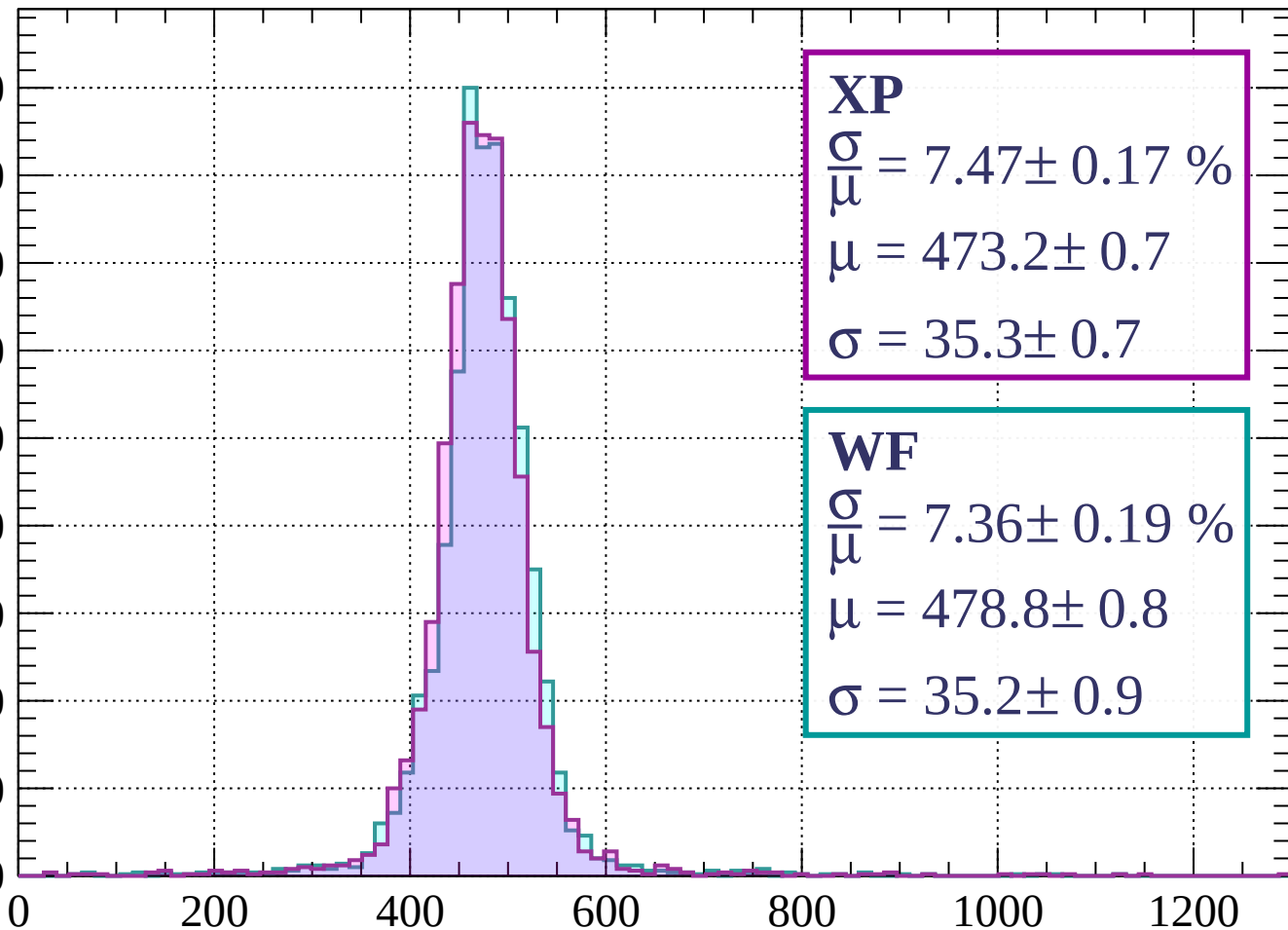
$$\sigma = 36.1 \pm 0.7$$



Energy loss | $1760 < p < 1840$

Count

450
400
350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 7.47 \pm 0.17 \%$$

$$\mu = 473.2 \pm 0.7$$

$$\sigma = 35.3 \pm 0.7$$

WF

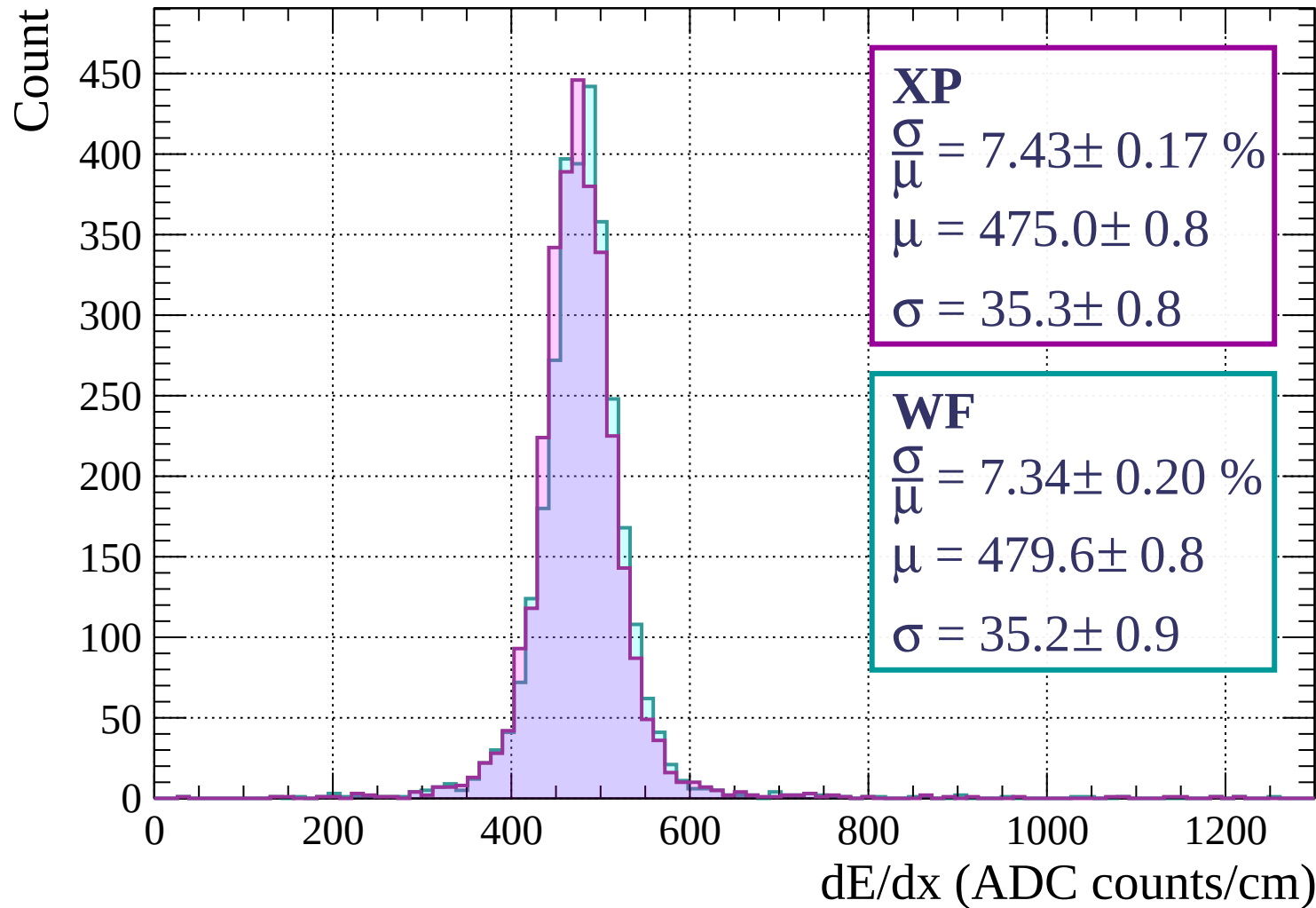
$$\frac{\sigma}{\mu} = 7.36 \pm 0.19 \%$$

$$\mu = 478.8 \pm 0.8$$

$$\sigma = 35.2 \pm 0.9$$

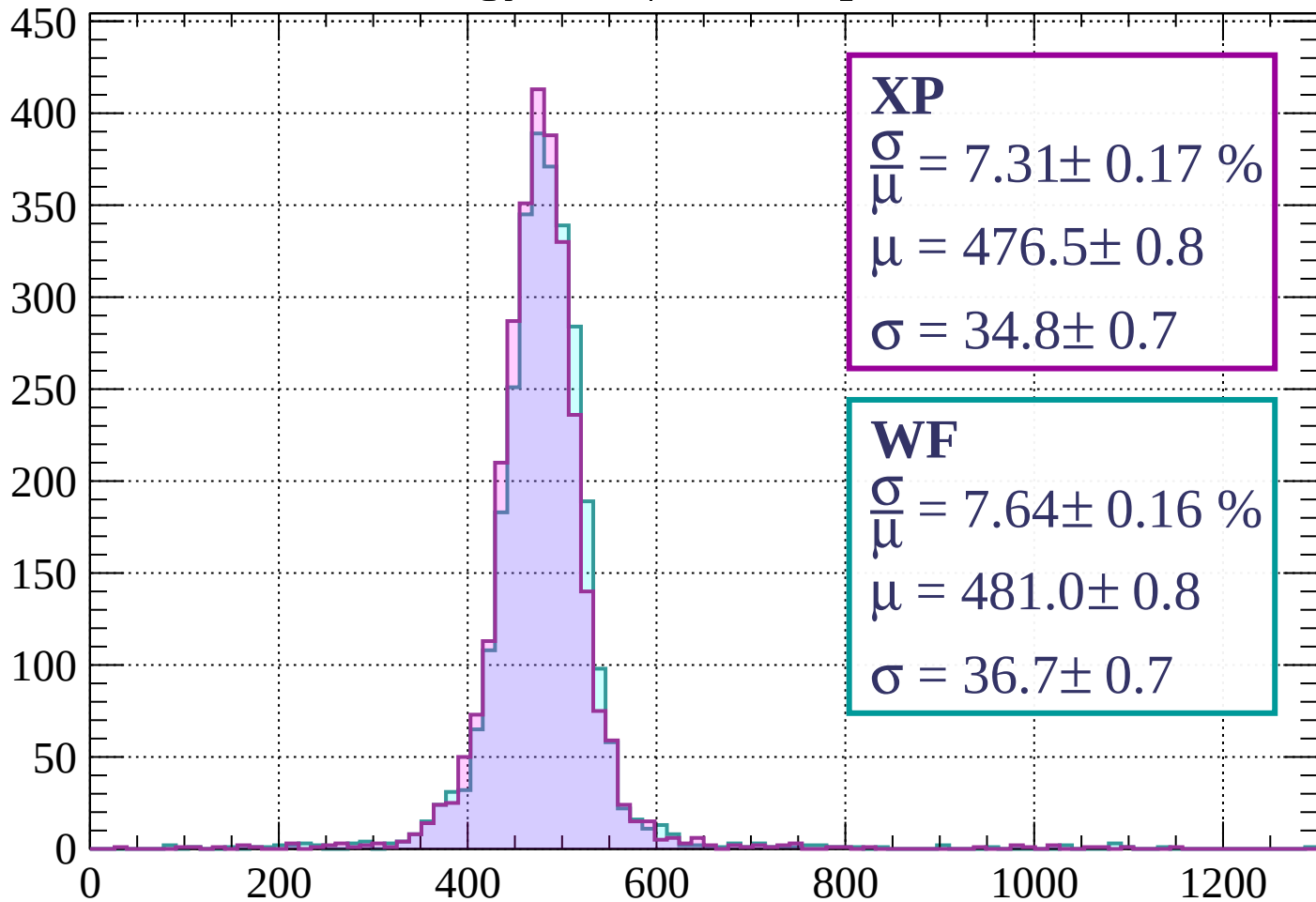
dE/dx (ADC counts/cm)

Energy loss | $1840 < p < 1920$



Energy loss | $1920 < p < 2000$

Count



XP

$$\frac{\sigma}{\mu} = 7.31 \pm 0.17 \%$$

$$\mu = 476.5 \pm 0.8$$

$$\sigma = 34.8 \pm 0.7$$

WF

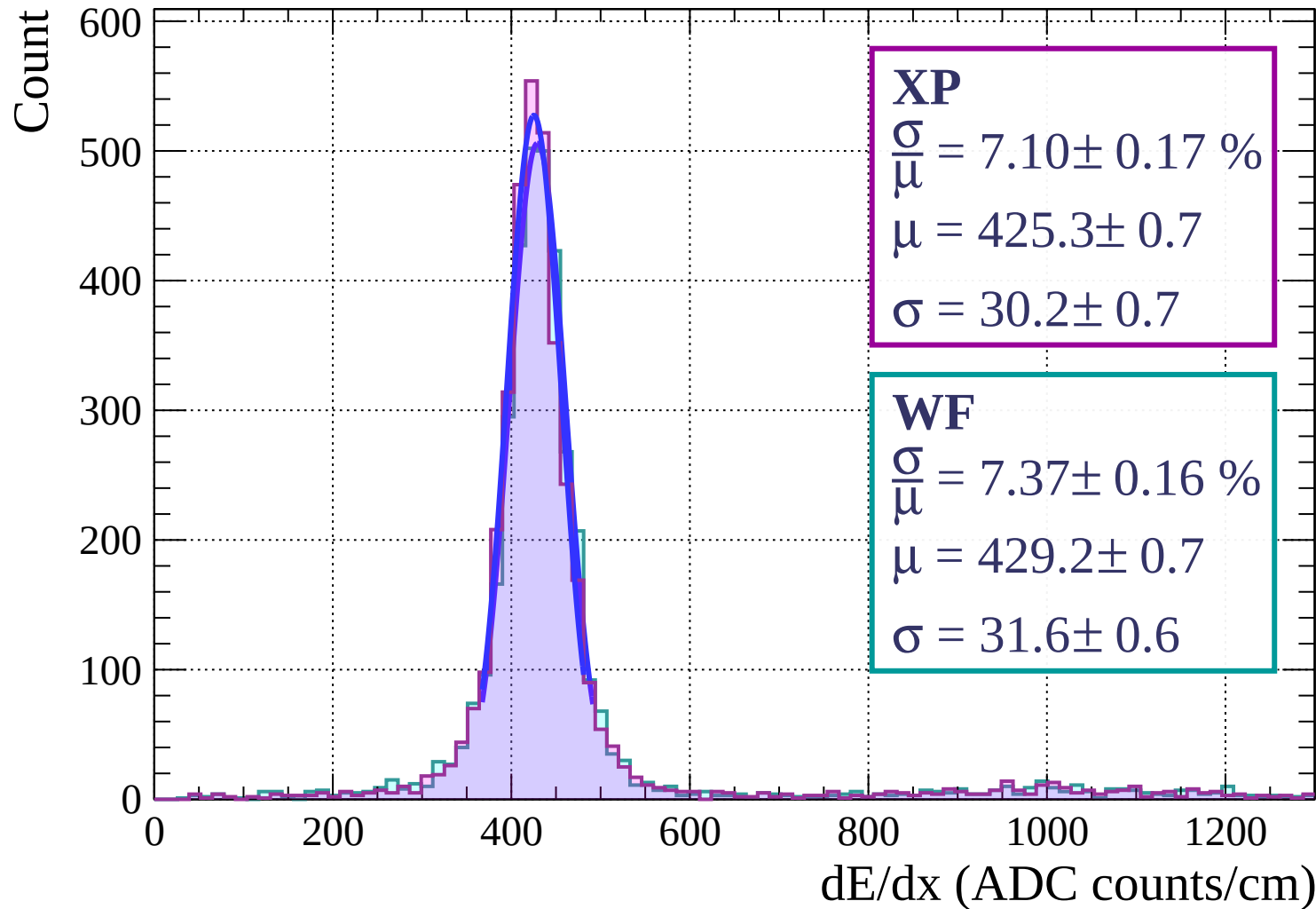
$$\frac{\sigma}{\mu} = 7.64 \pm 0.16 \%$$

$$\mu = 481.0 \pm 0.8$$

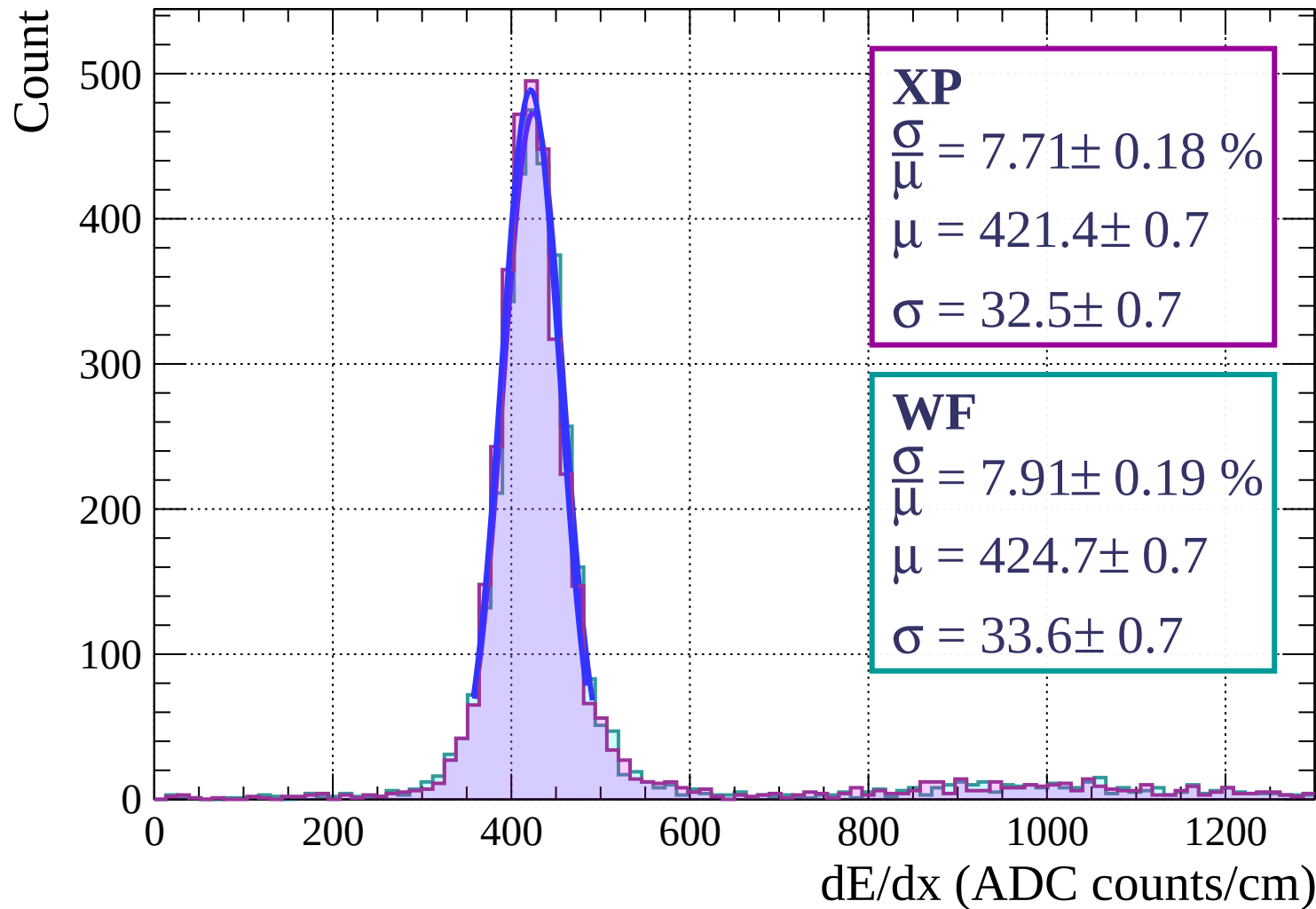
$$\sigma = 36.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $0 < \phi < 3$

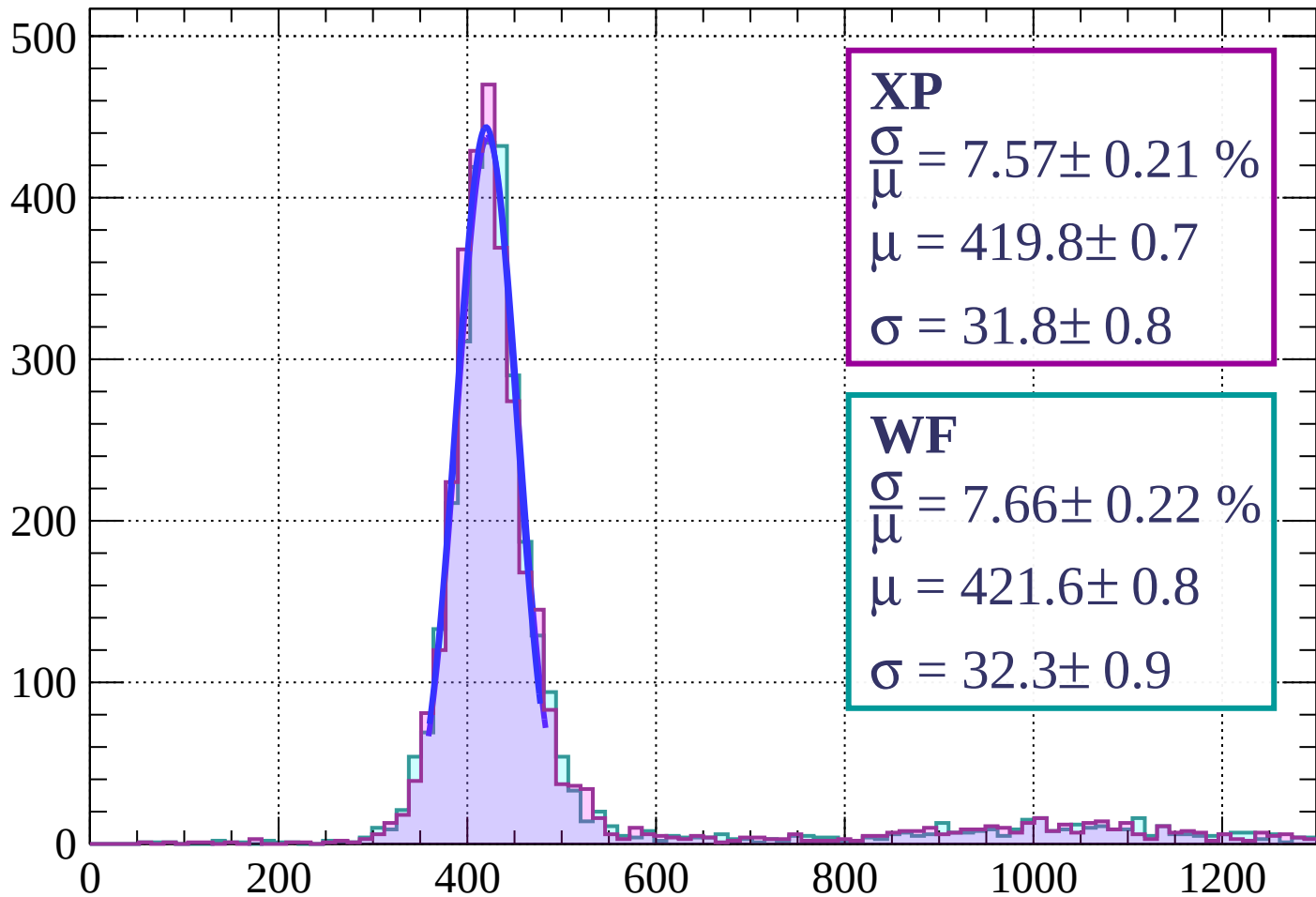


Energy loss | $3 < \phi < 6$



Energy loss | $6 < \phi < 9$

Count



XP

$$\frac{\sigma}{\mu} = 7.57 \pm 0.21 \%$$

$$\mu = 419.8 \pm 0.7$$

$$\sigma = 31.8 \pm 0.8$$

WF

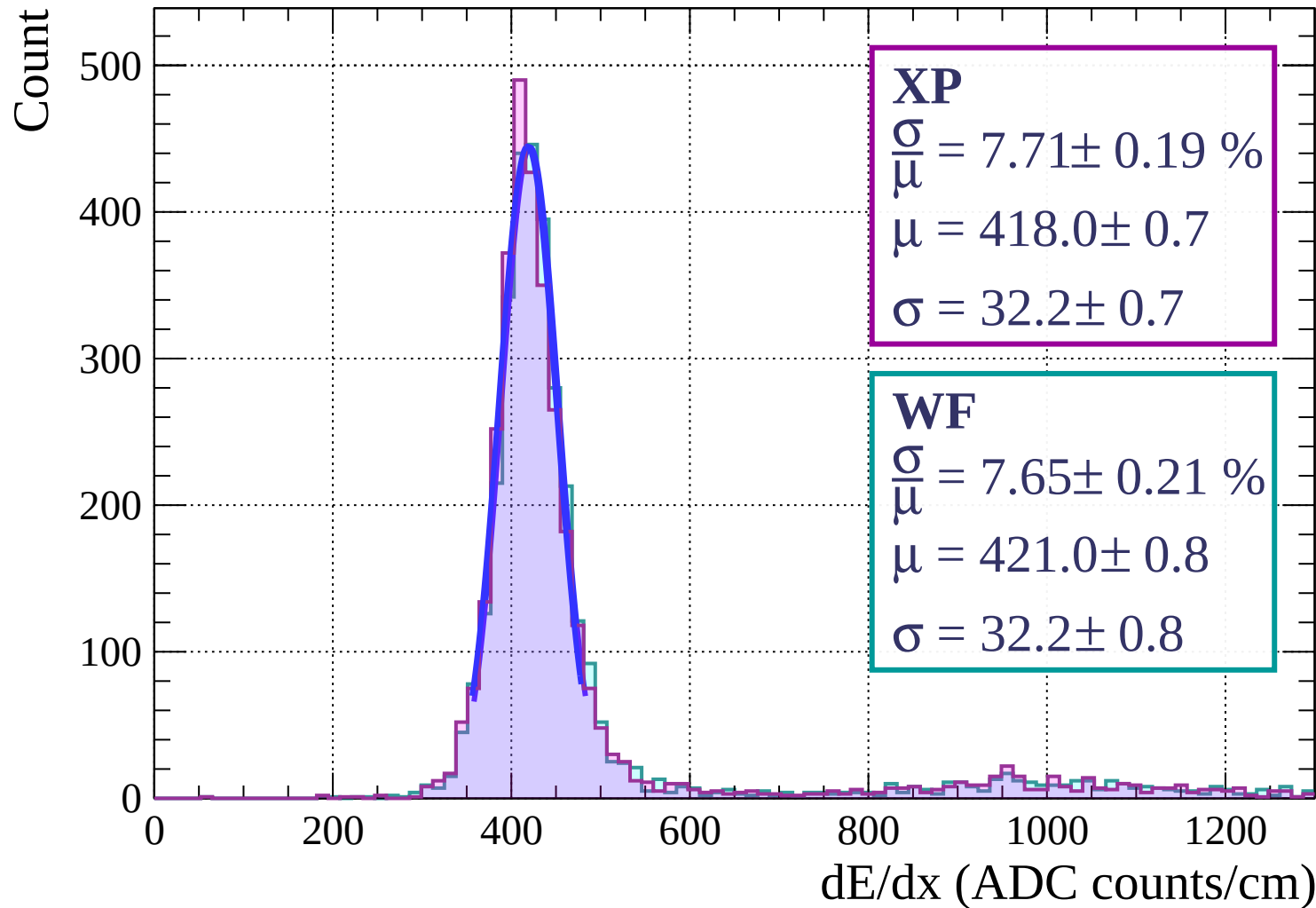
$$\frac{\sigma}{\mu} = 7.66 \pm 0.22 \%$$

$$\mu = 421.6 \pm 0.8$$

$$\sigma = 32.3 \pm 0.9$$

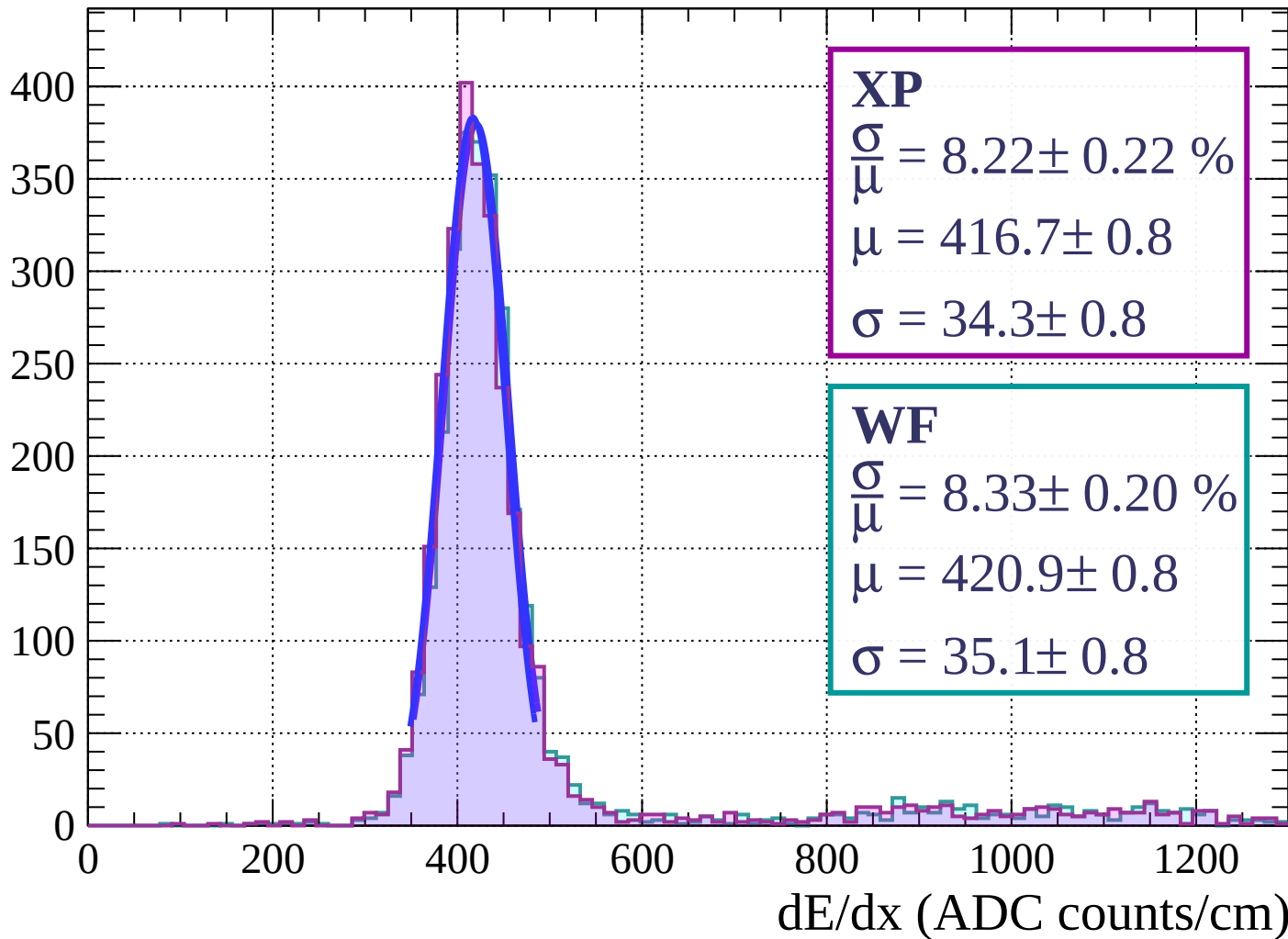
dE/dx (ADC counts/cm)

Energy loss | $9 < \phi < 12$



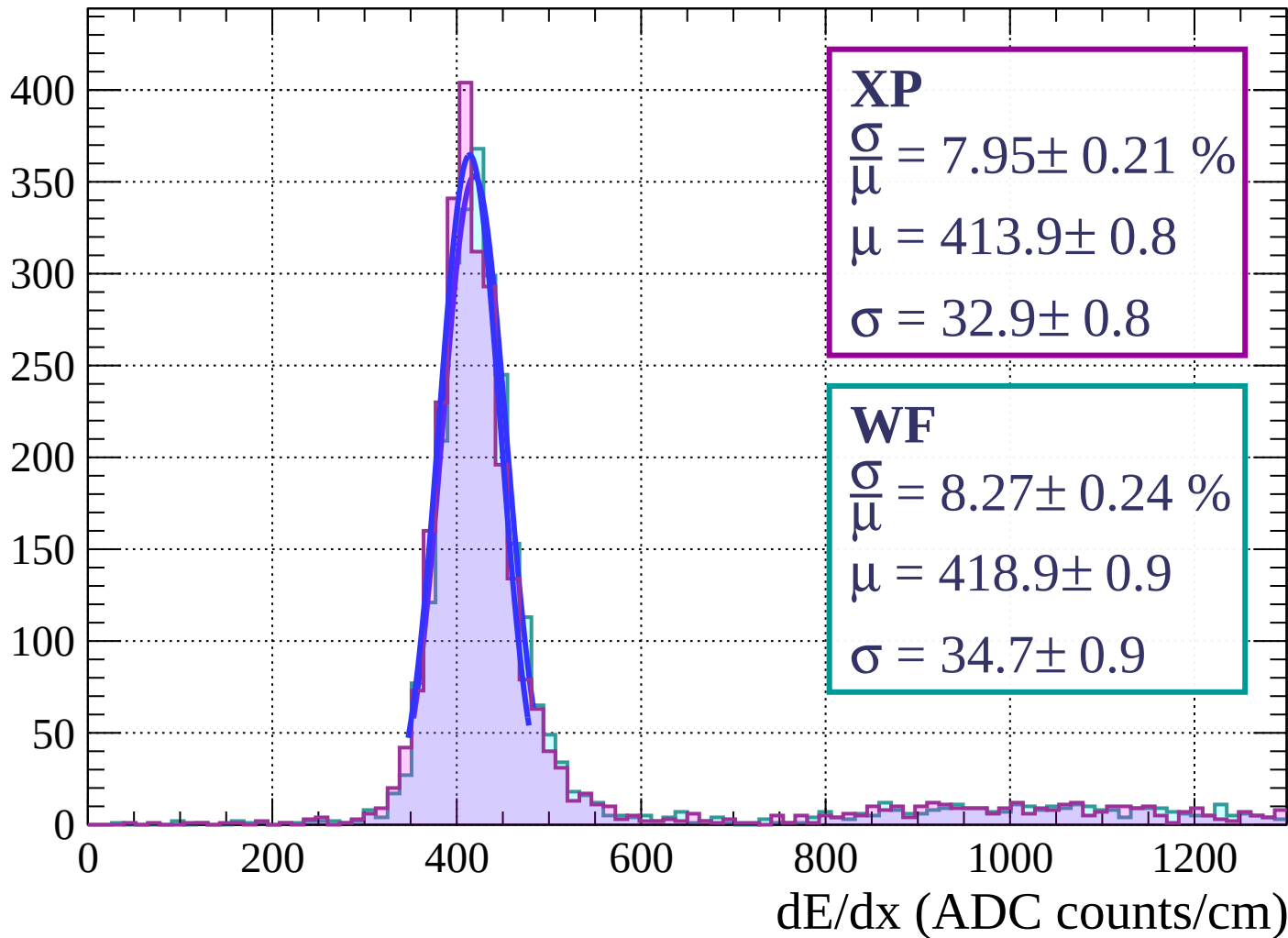
Energy loss | $12 < \phi < 15$

Count

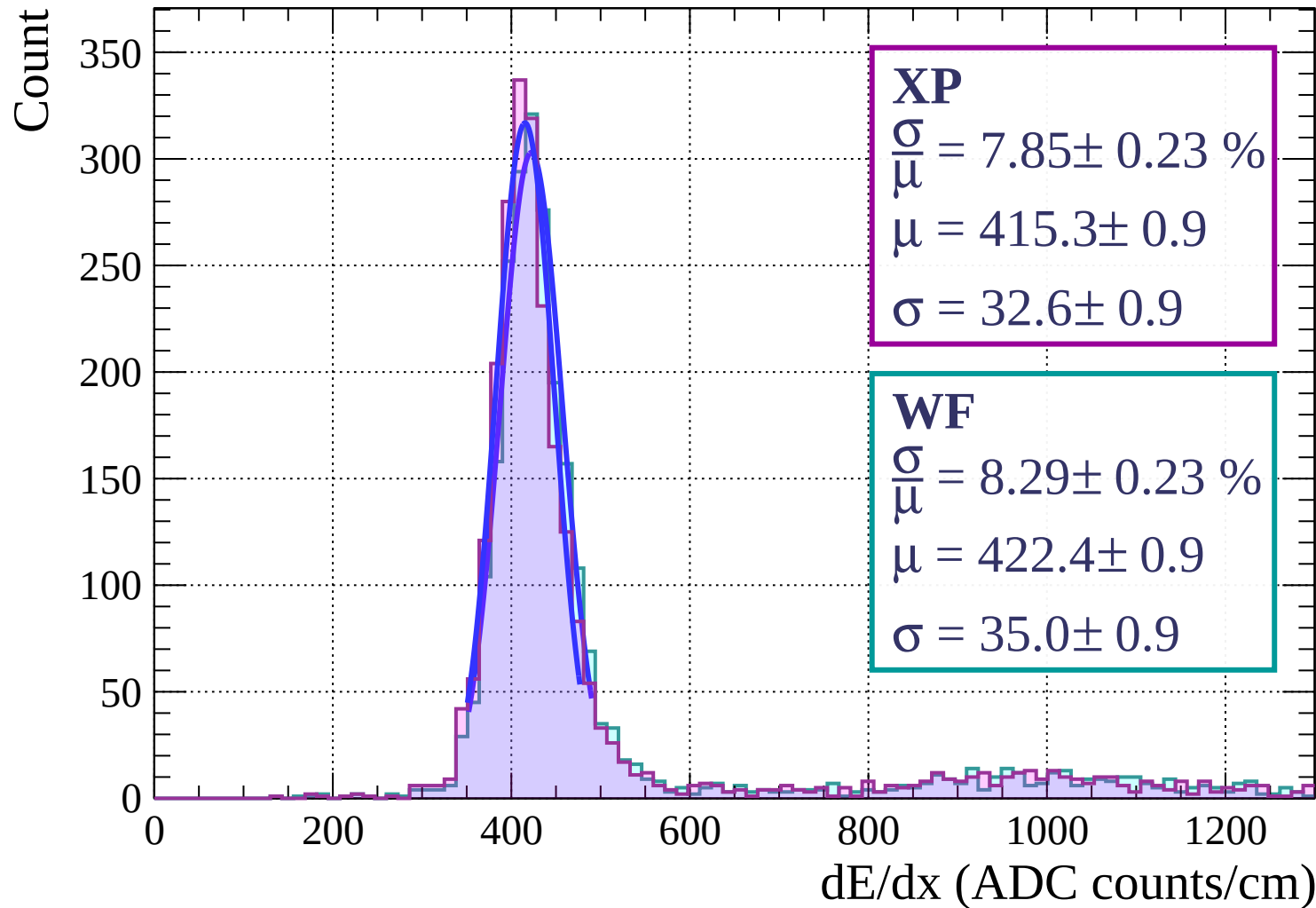


Energy loss | $15 < \phi < 18$

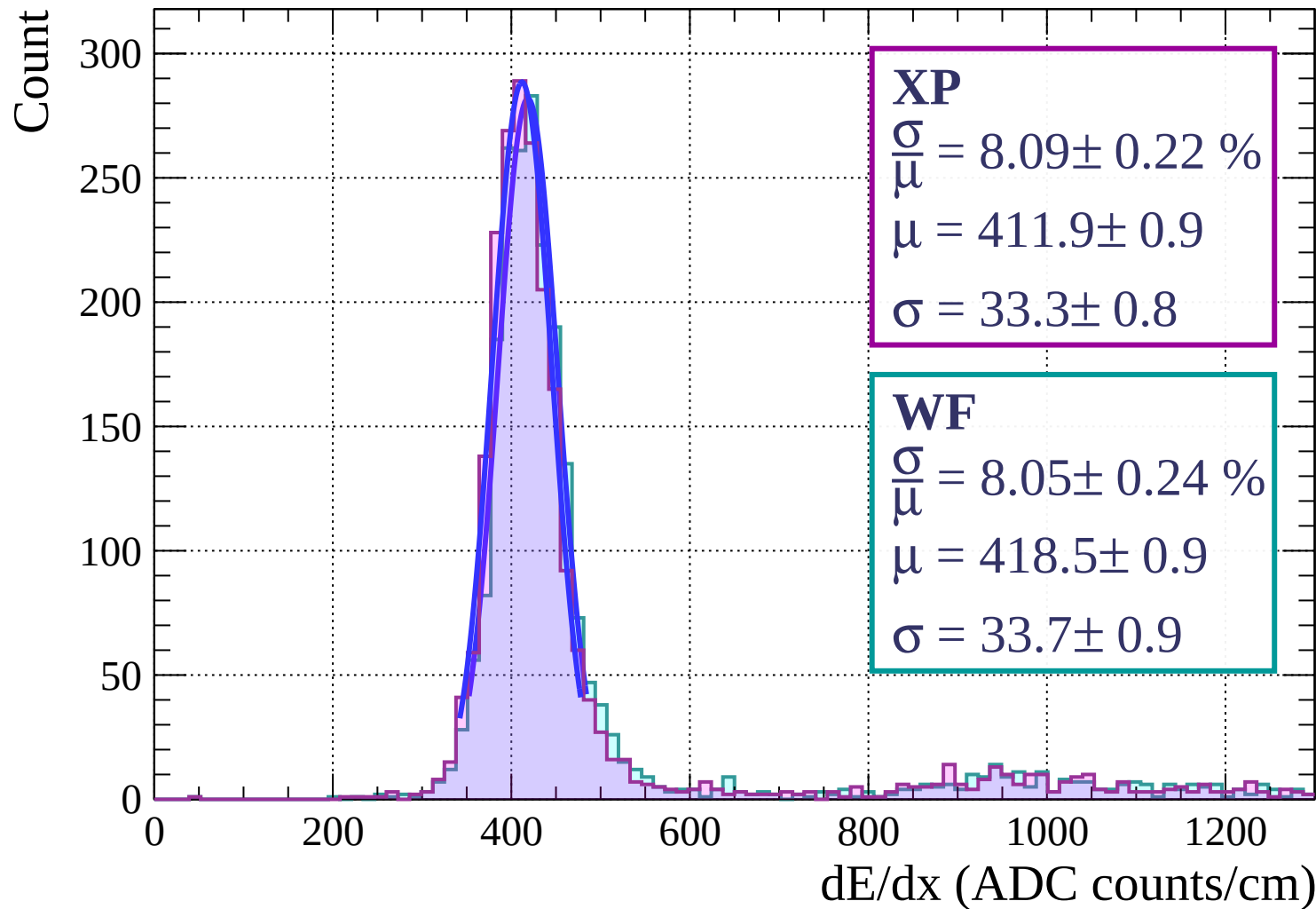
Count



Energy loss | $18 < \phi < 21$

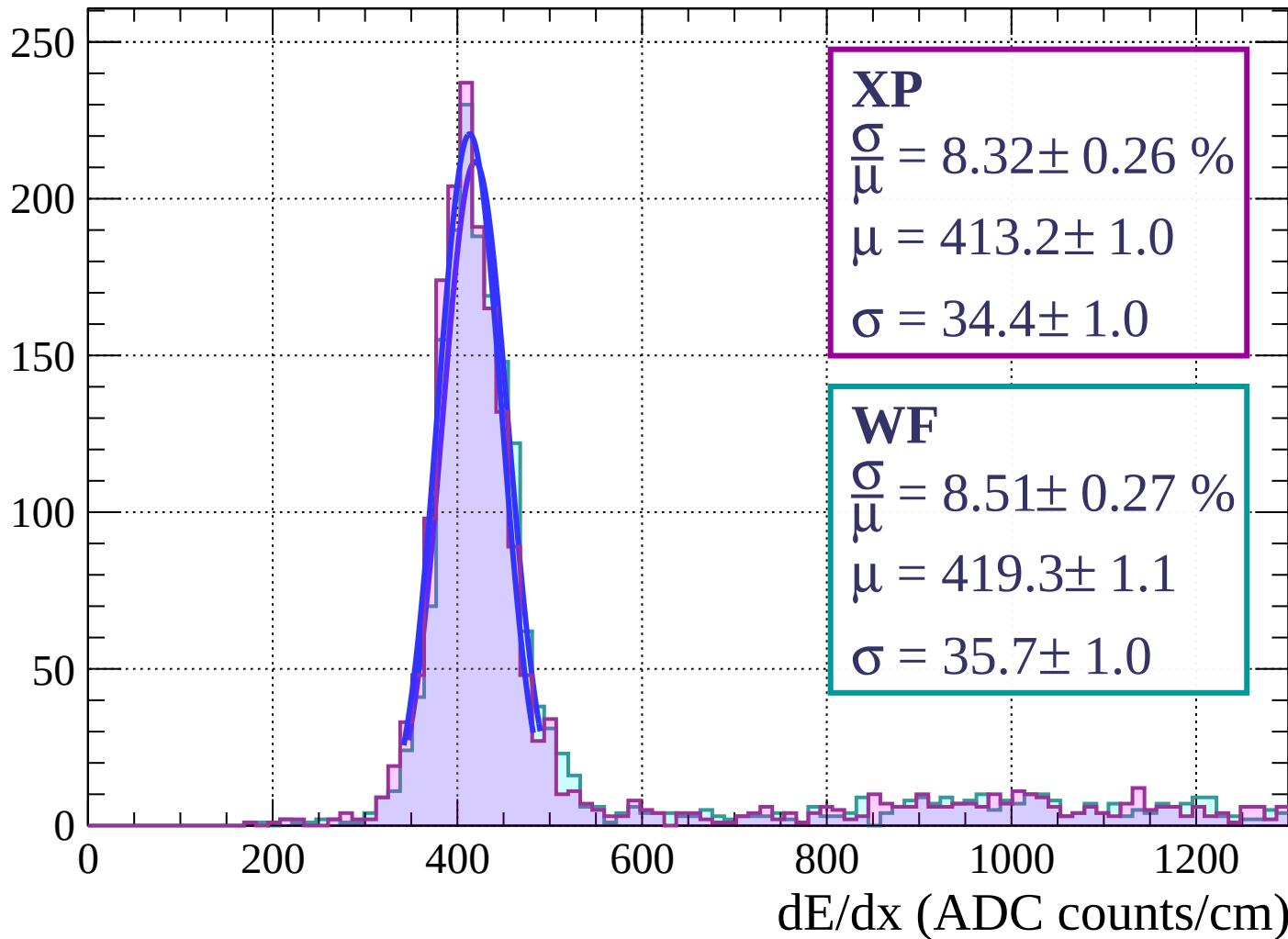


Energy loss | $21 < \phi < 24$

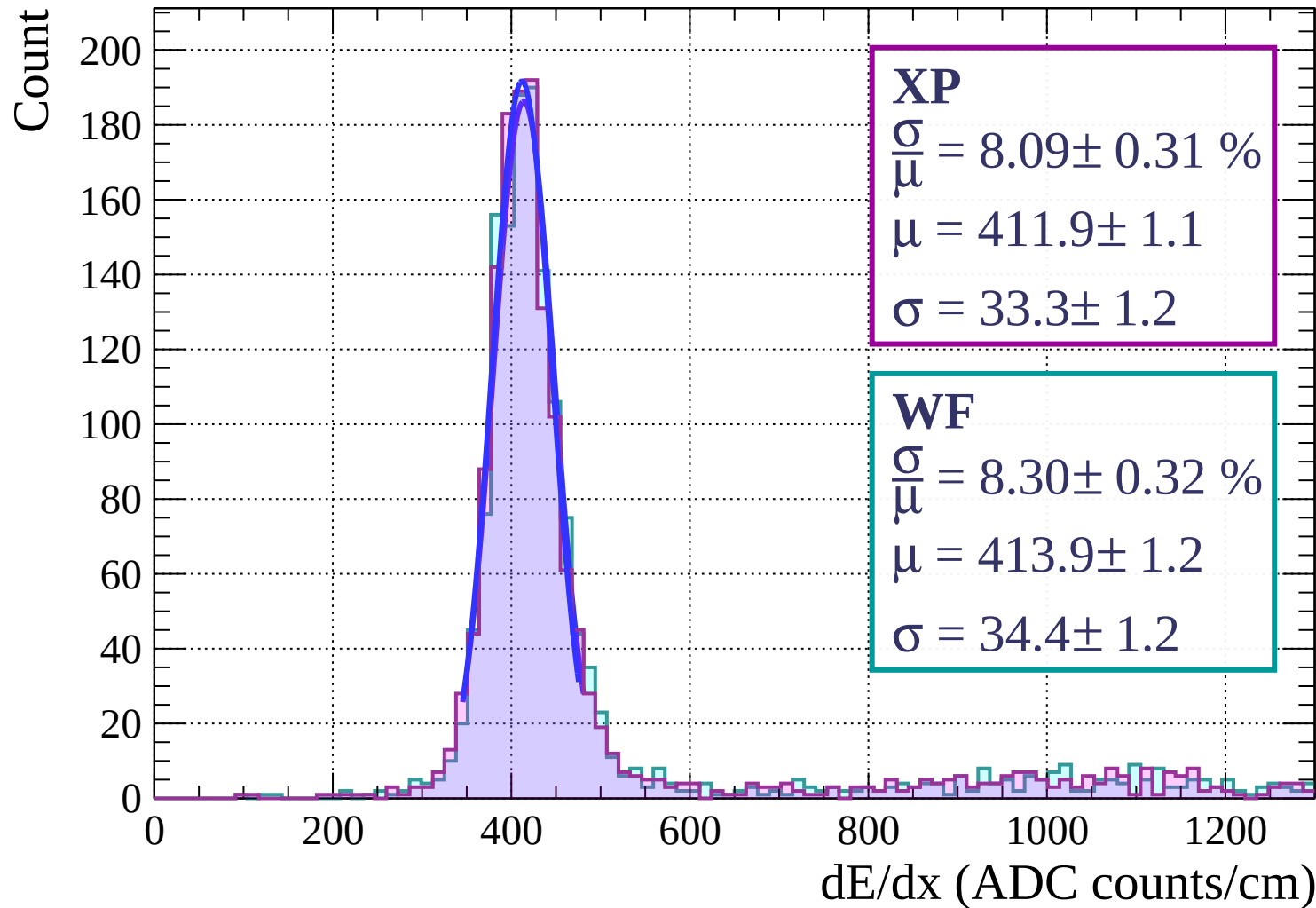


Energy loss | $24 < \phi < 27$

Count

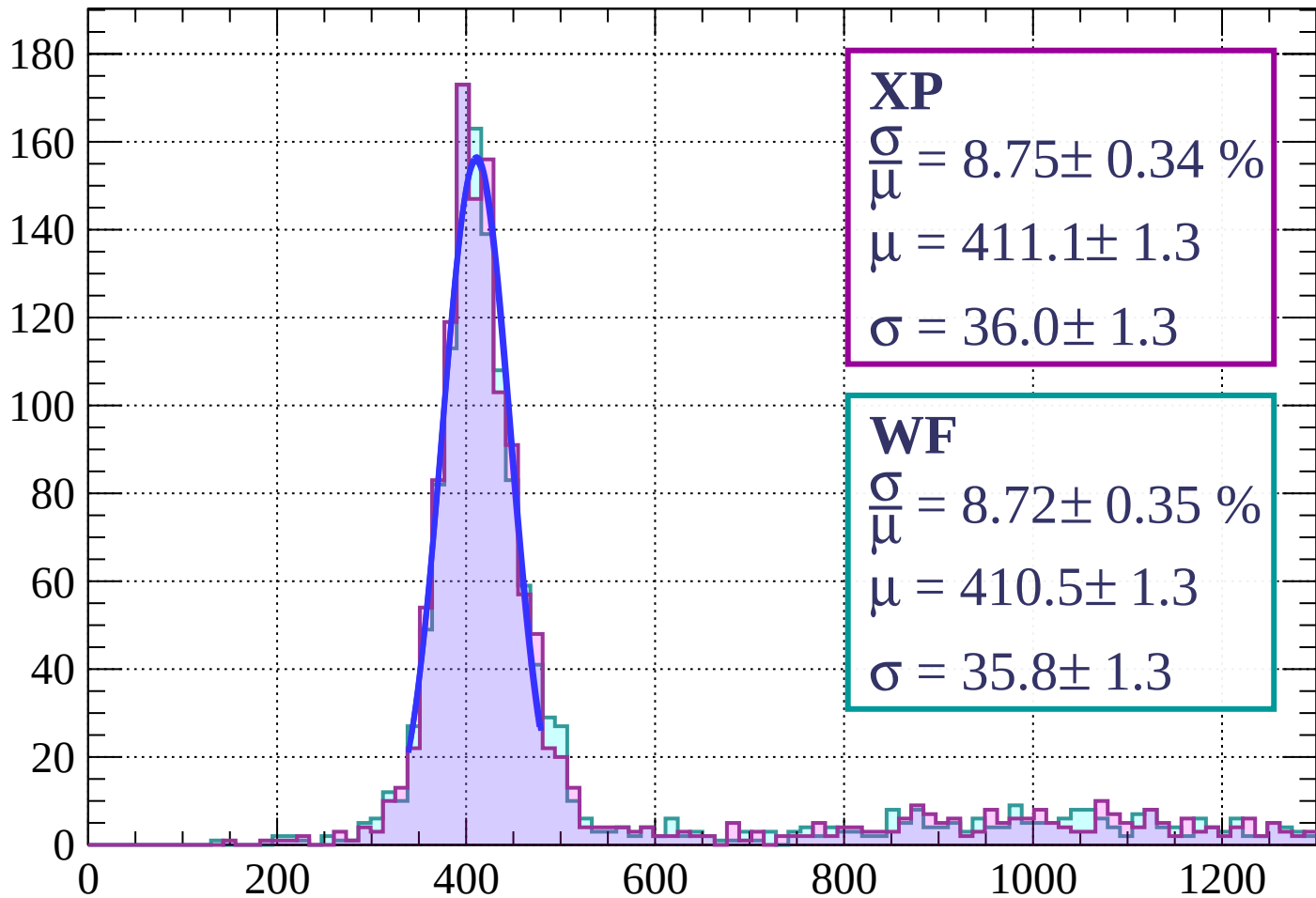


Energy loss | $27 < \phi < 30$



Energy loss | $30 < \phi < 33$

Count



XP

$$\frac{\sigma}{\mu} = 8.75 \pm 0.34 \%$$

$$\mu = 411.1 \pm 1.3$$

$$\sigma = 36.0 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 8.72 \pm 0.35 \%$$

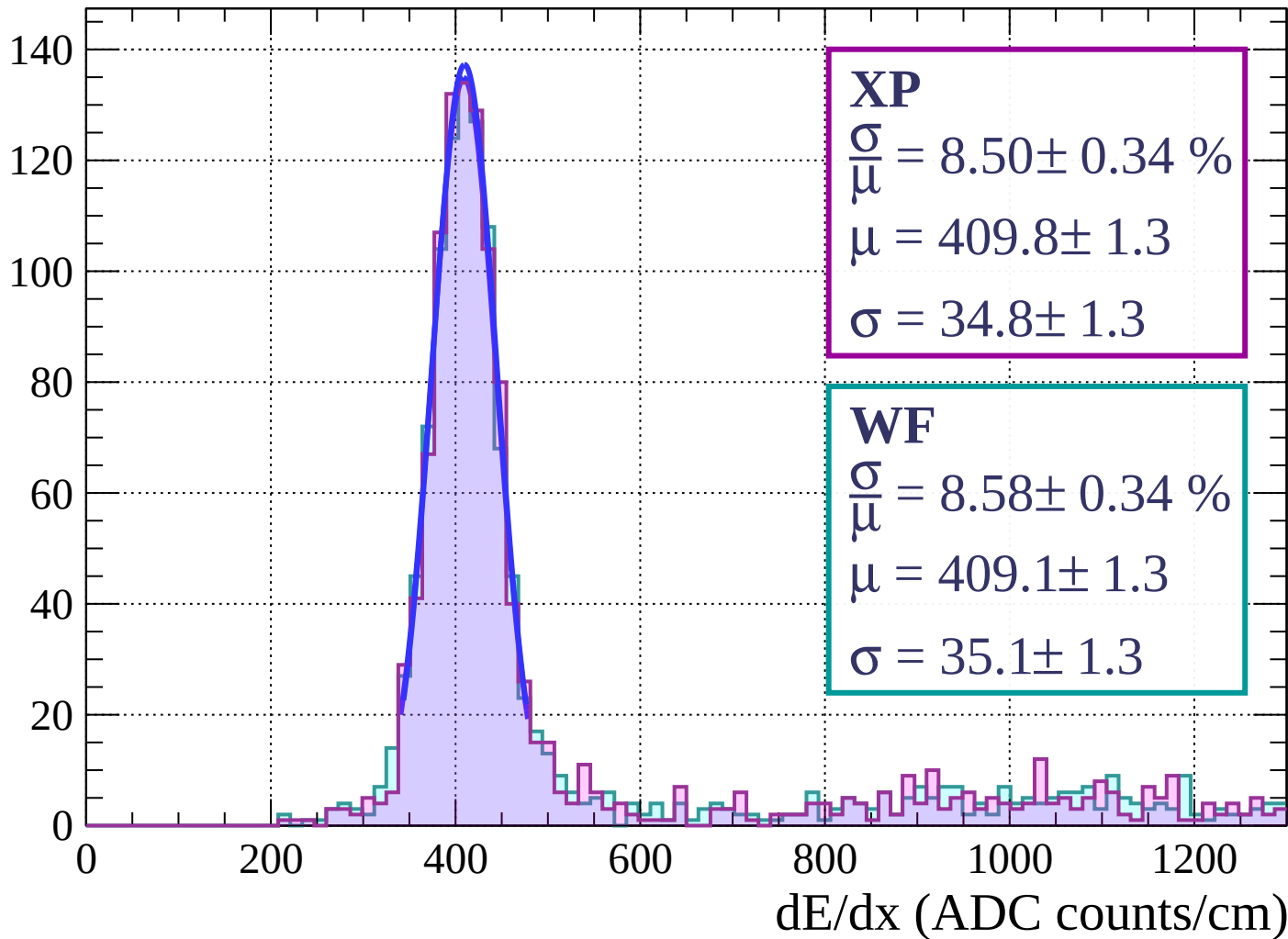
$$\mu = 410.5 \pm 1.3$$

$$\sigma = 35.8 \pm 1.3$$

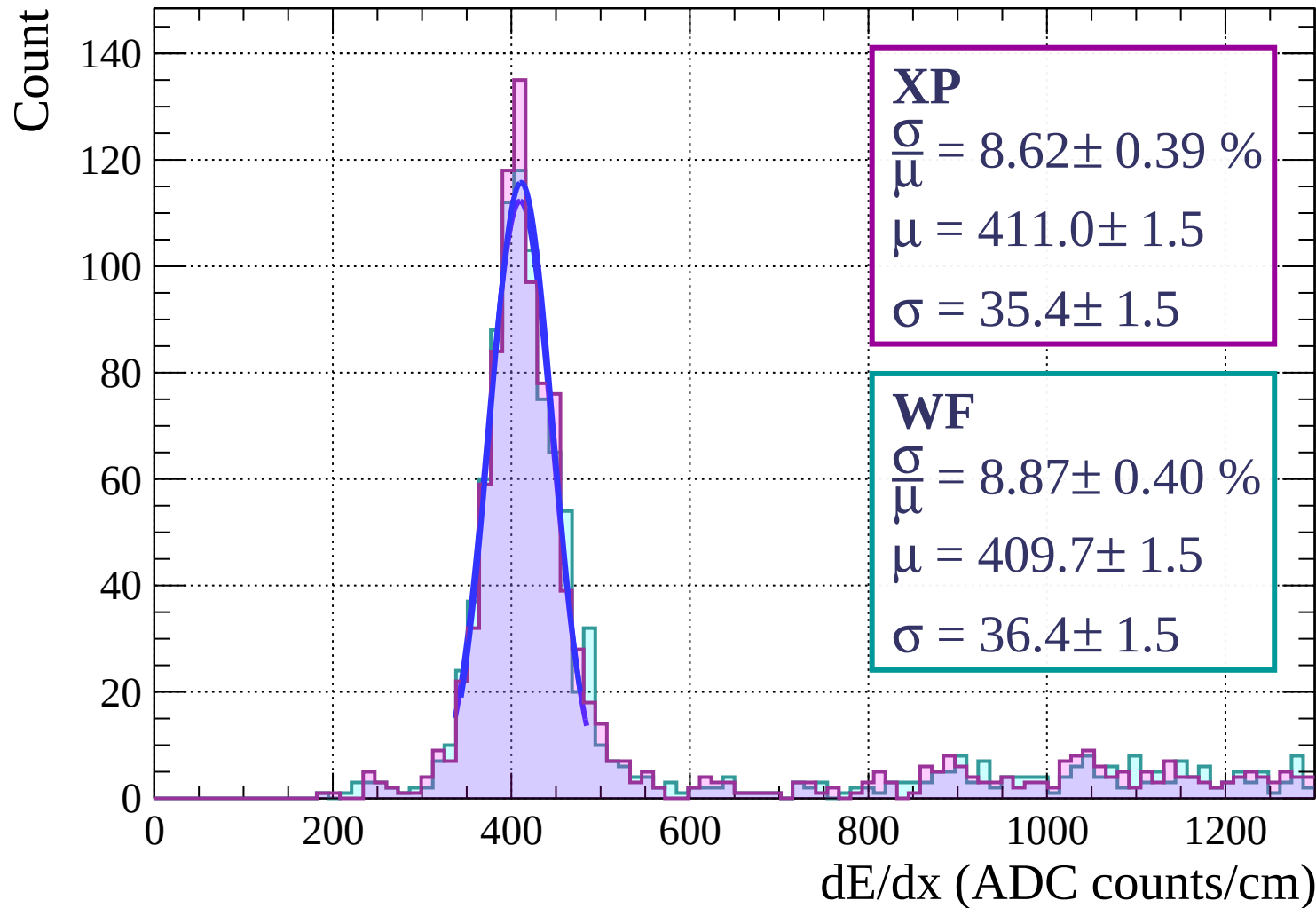
dE/dx (ADC counts/cm)

Energy loss | $33 < \phi < 36$

Count

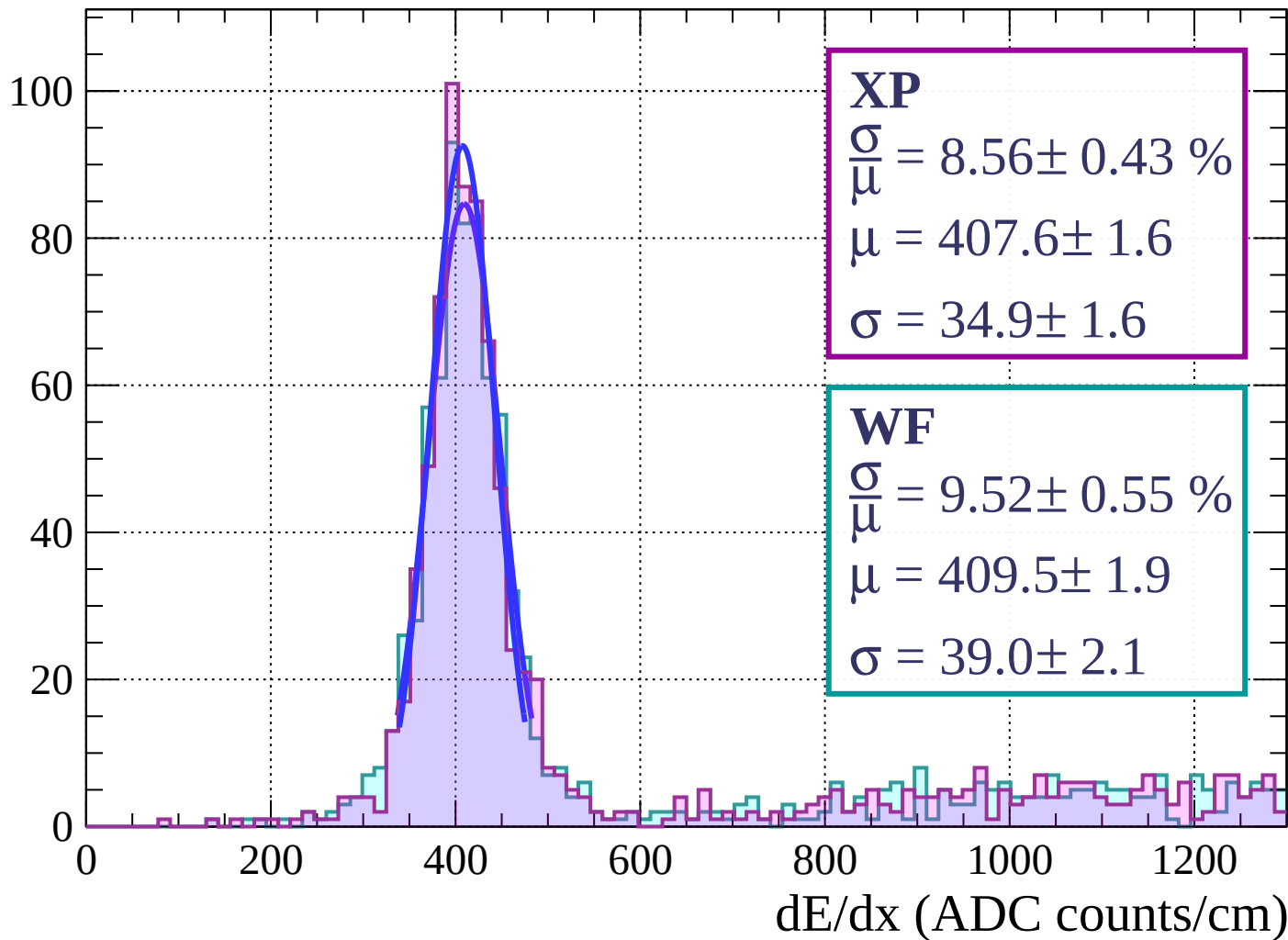


Energy loss | $36 < \phi < 39$



Energy loss | $39 < \phi < 42$

Count



Energy loss | $42 < \phi < 45$

Count

XP

$$\frac{\sigma}{\mu} = 9.96 \pm 0.63 \%$$

$$\mu = 409.2 \pm 2.3$$

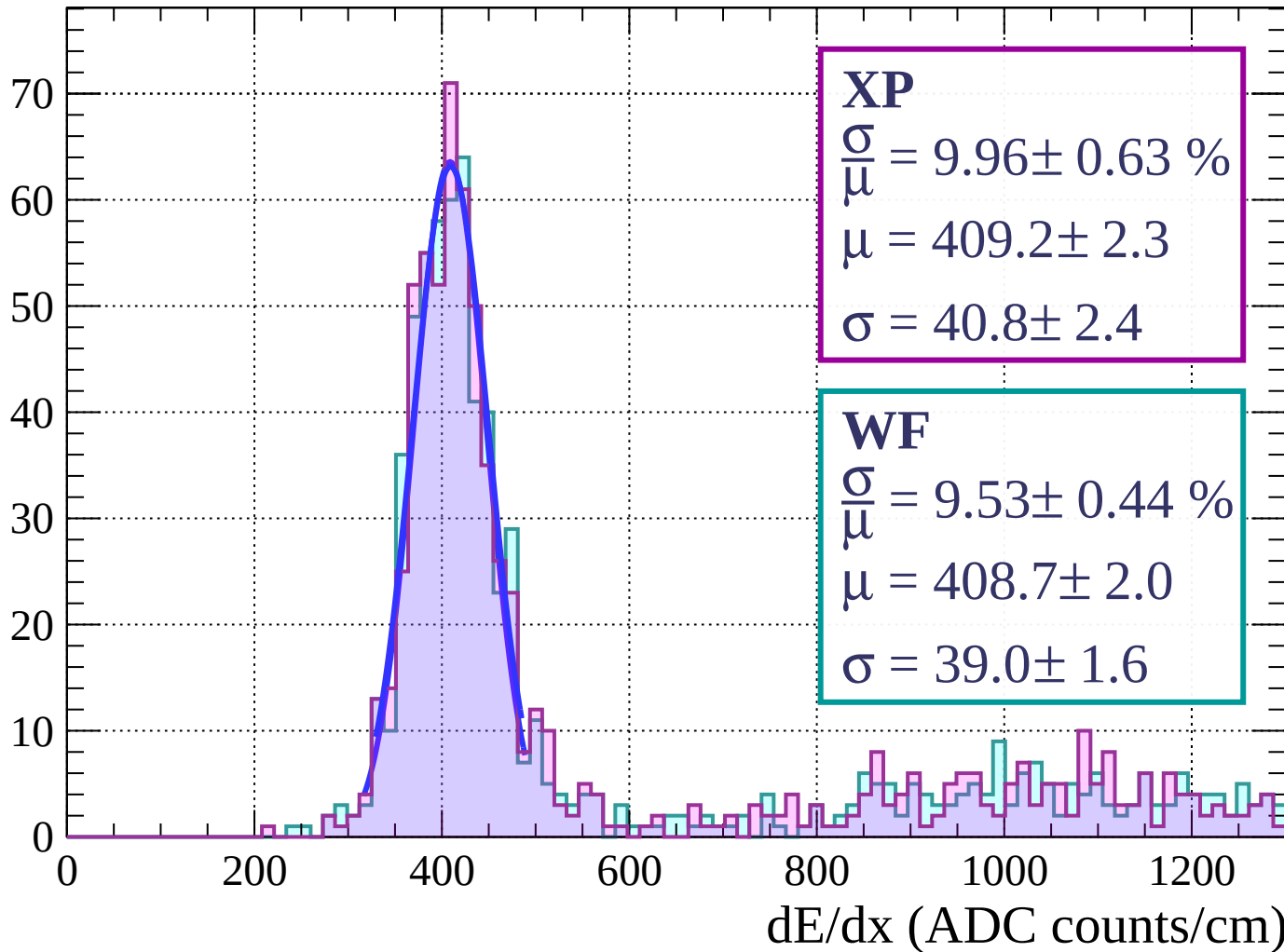
$$\sigma = 40.8 \pm 2.4$$

WF

$$\frac{\sigma}{\mu} = 9.53 \pm 0.44 \%$$

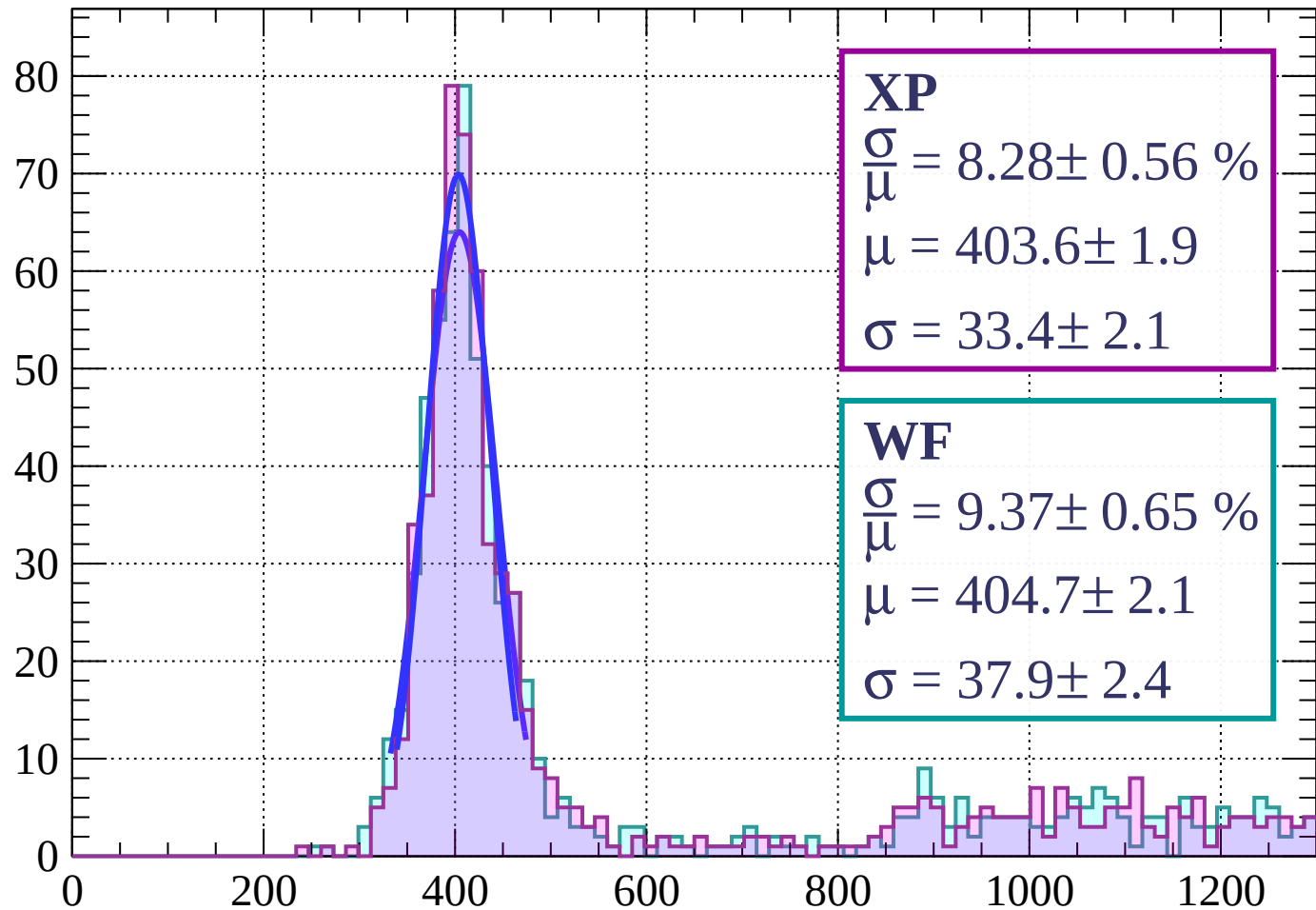
$$\mu = 408.7 \pm 2.0$$

$$\sigma = 39.0 \pm 1.6$$



Energy loss | $45 < \phi < 48$

Count



XP

$$\frac{\sigma}{\mu} = 8.28 \pm 0.56 \%$$

$$\mu = 403.6 \pm 1.9$$

$$\sigma = 33.4 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 9.37 \pm 0.65 \%$$

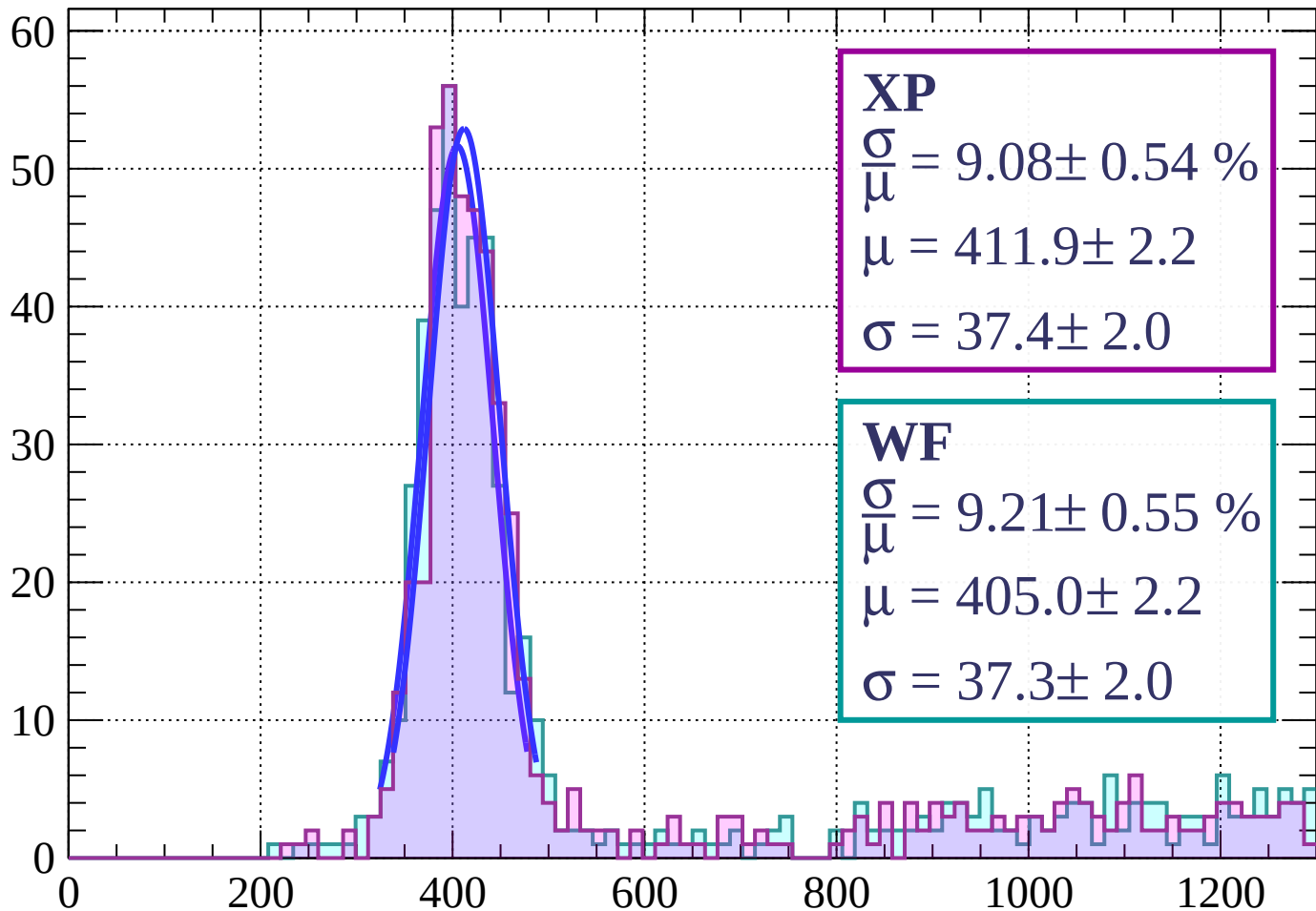
$$\mu = 404.7 \pm 2.1$$

$$\sigma = 37.9 \pm 2.4$$

dE/dx (ADC counts/cm)

Energy loss | $48 < \phi < 51$

Count



XP

$$\frac{\sigma}{\mu} = 9.08 \pm 0.54 \%$$

$$\mu = 411.9 \pm 2.2$$

$$\sigma = 37.4 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 9.21 \pm 0.55 \%$$

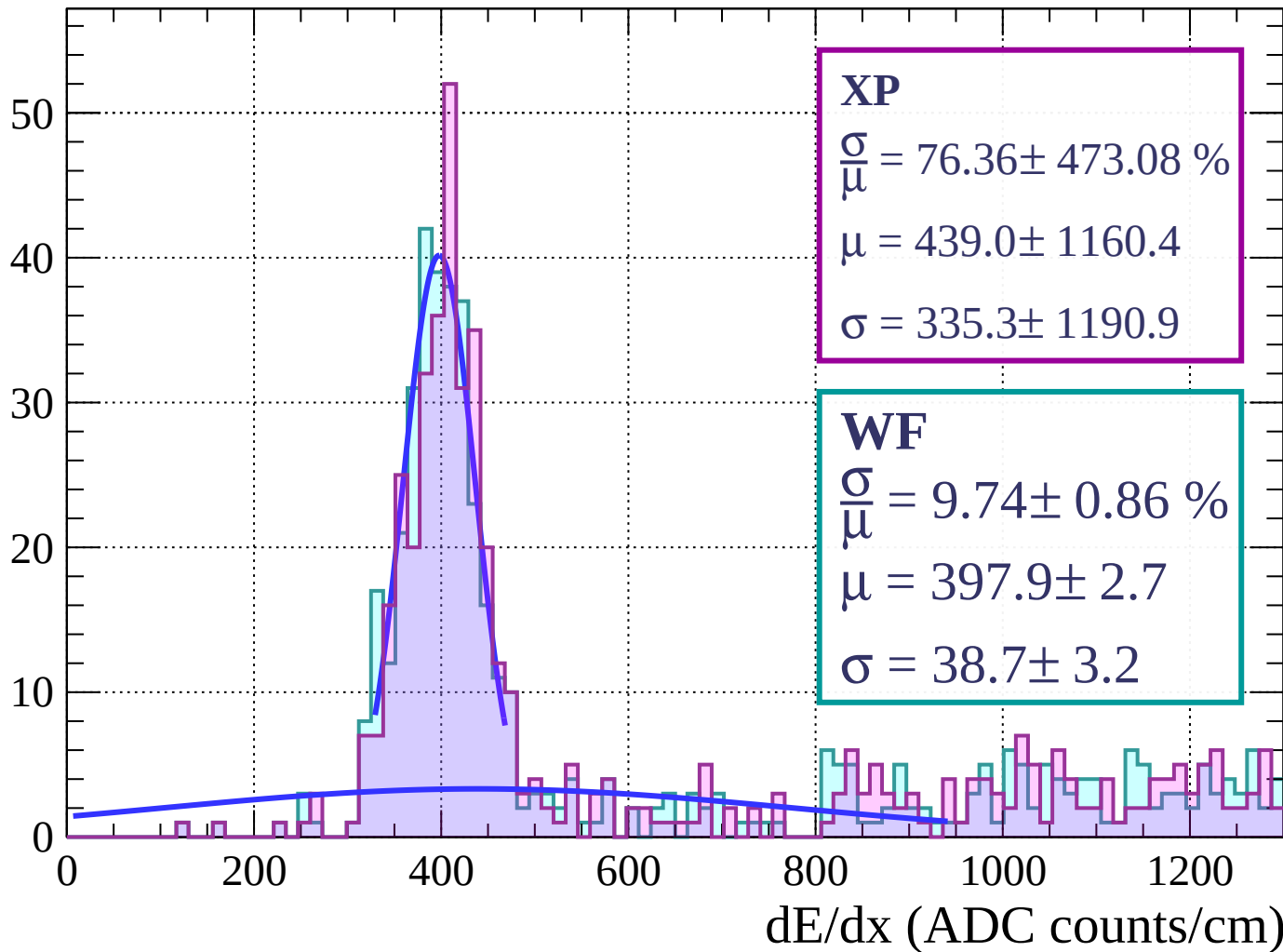
$$\mu = 405.0 \pm 2.2$$

$$\sigma = 37.3 \pm 2.0$$

dE/dx (ADC counts/cm)

Energy loss | $51 < \phi < 54$

Count



Energy loss | $54 < \phi < 57$

Count

XP

$$\frac{\sigma}{\mu} = 6.99 \pm 0.70 \%$$

$$\mu = 404.2 \pm 2.4$$

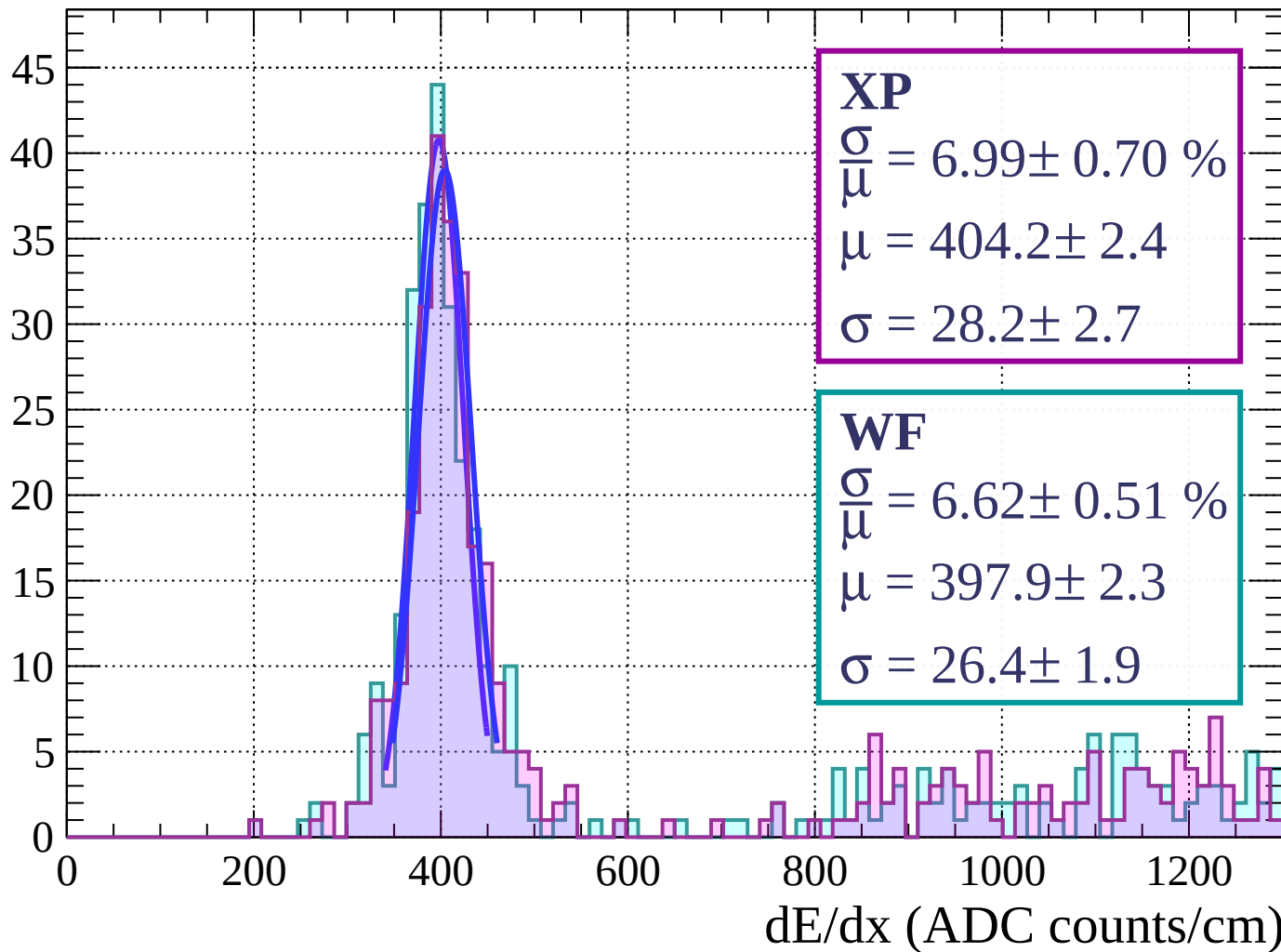
$$\sigma = 28.2 \pm 2.7$$

WF

$$\frac{\sigma}{\mu} = 6.62 \pm 0.51 \%$$

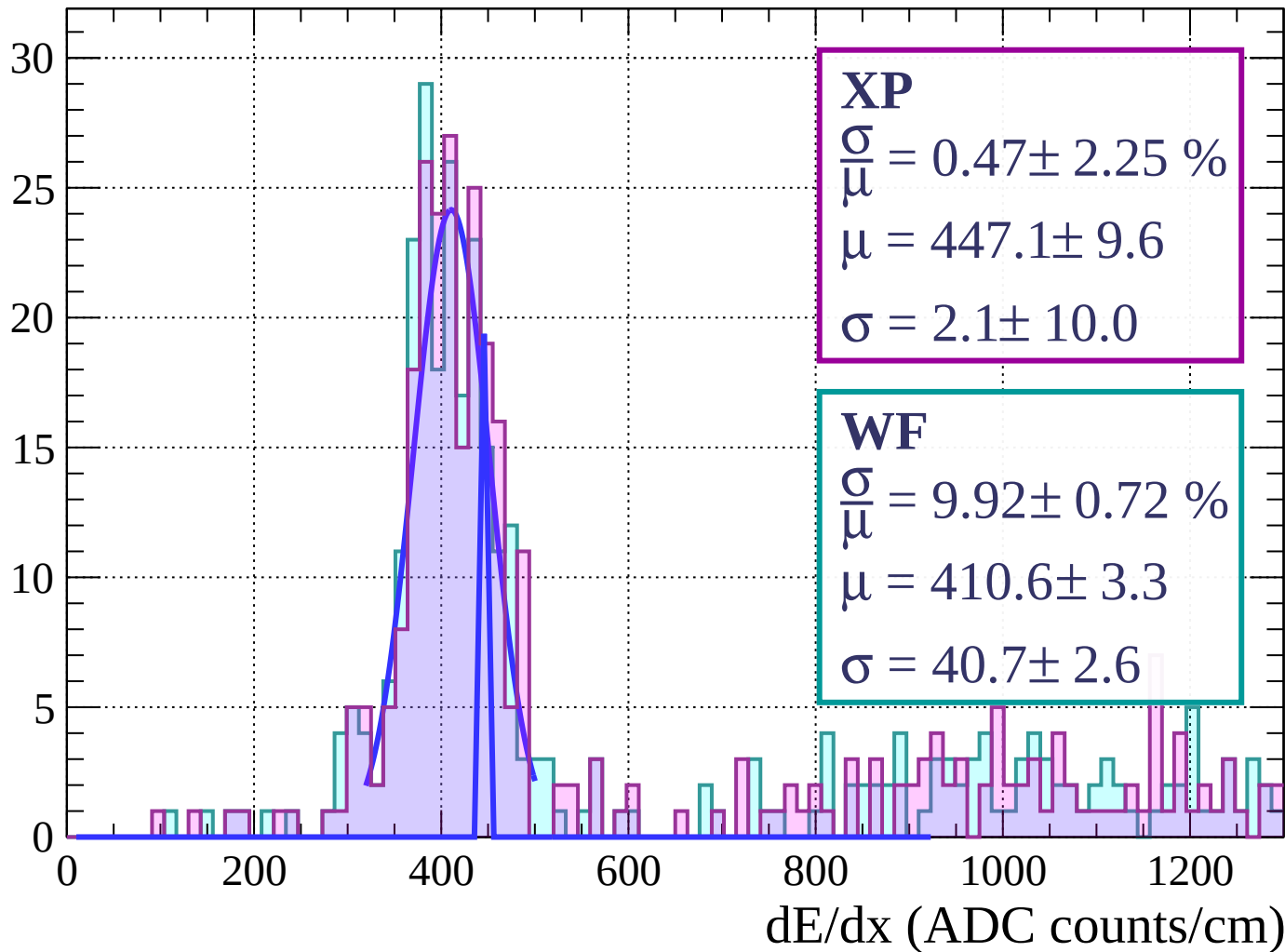
$$\mu = 397.9 \pm 2.3$$

$$\sigma = 26.4 \pm 1.9$$



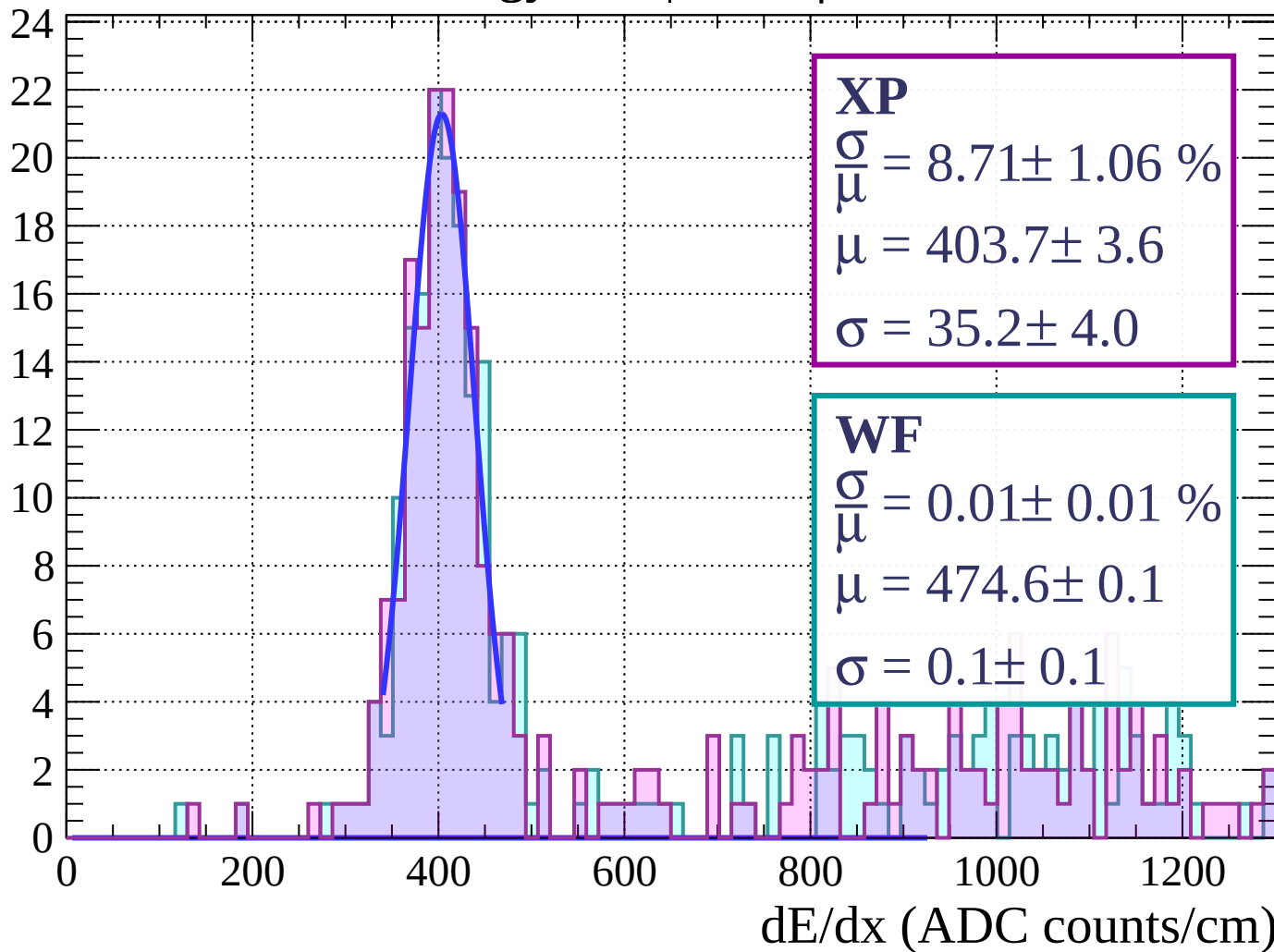
Energy loss | $57 < \phi < 60$

Count



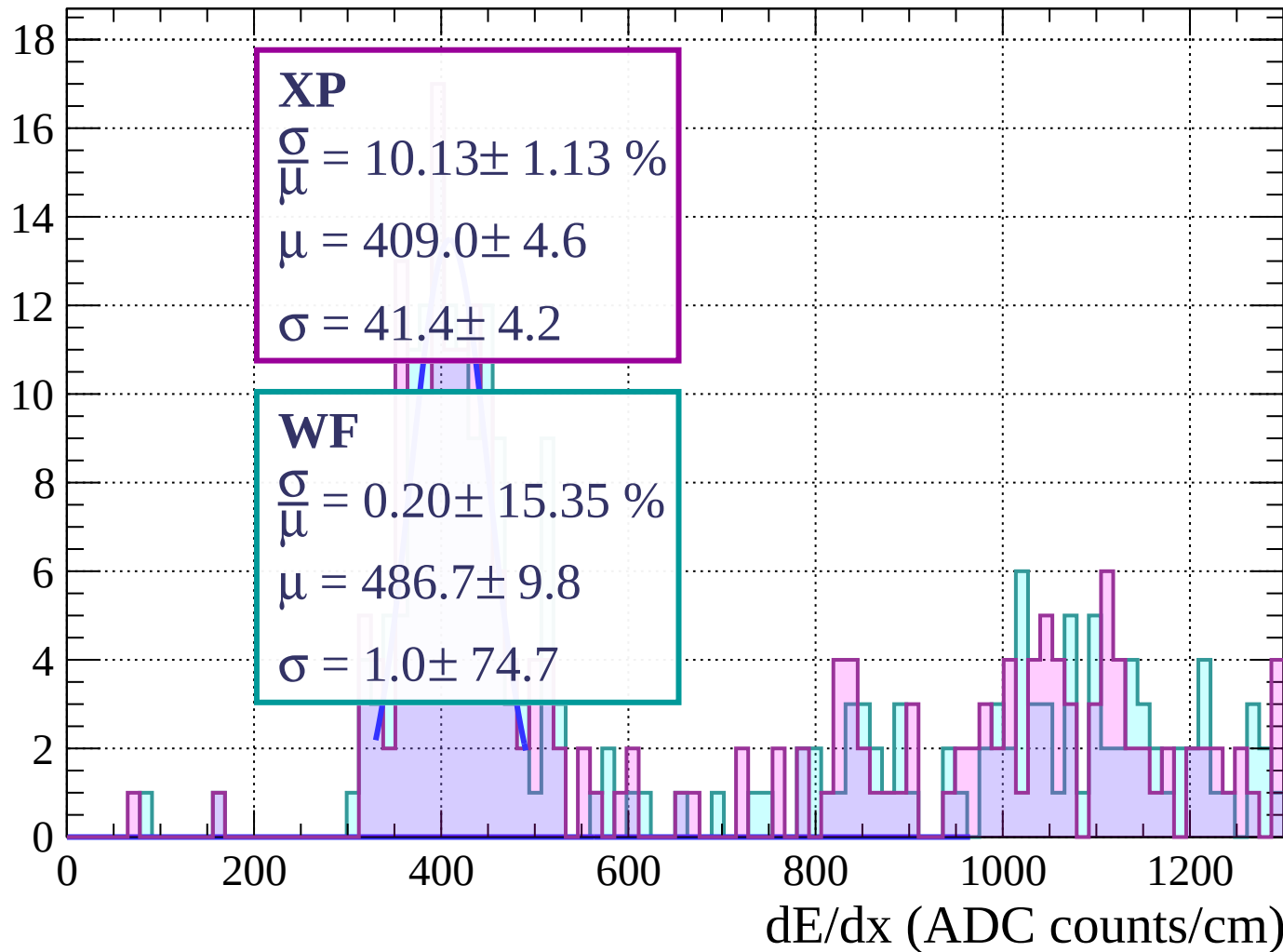
Energy loss | $60 < \phi < 63$

Count



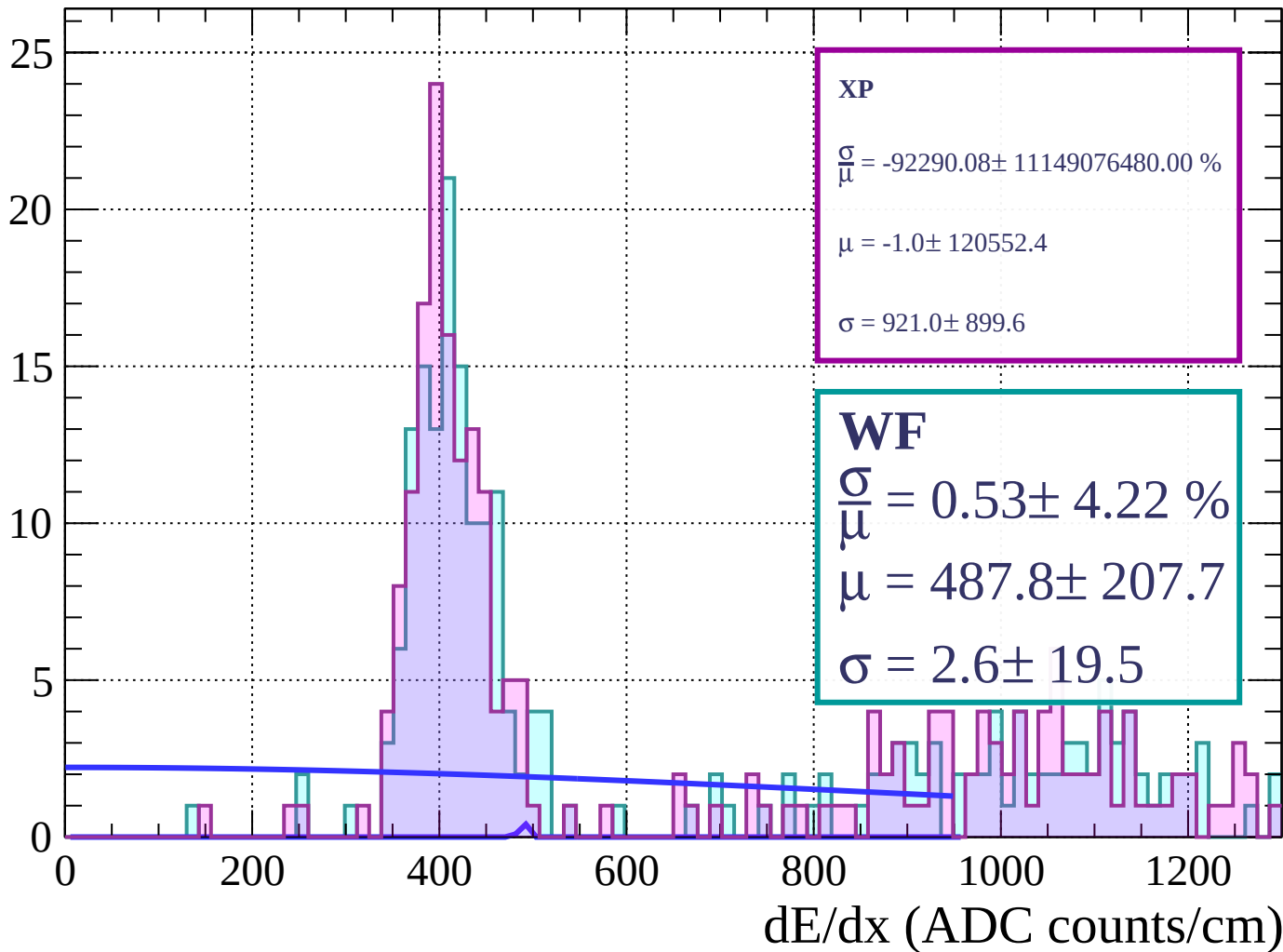
Energy loss | $63 < \phi < 66$

Count



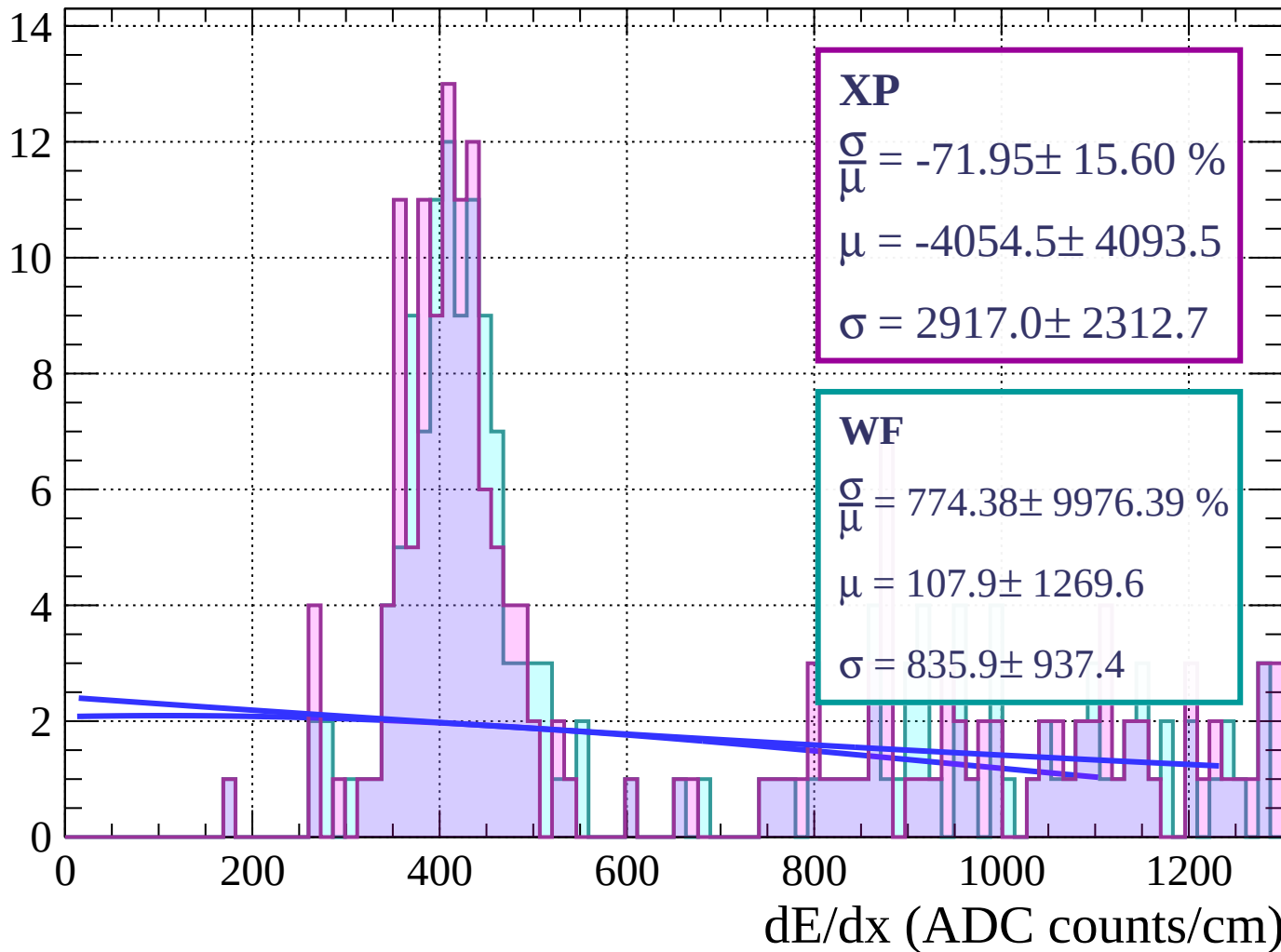
Energy loss | $66 < \phi < 69$

Count



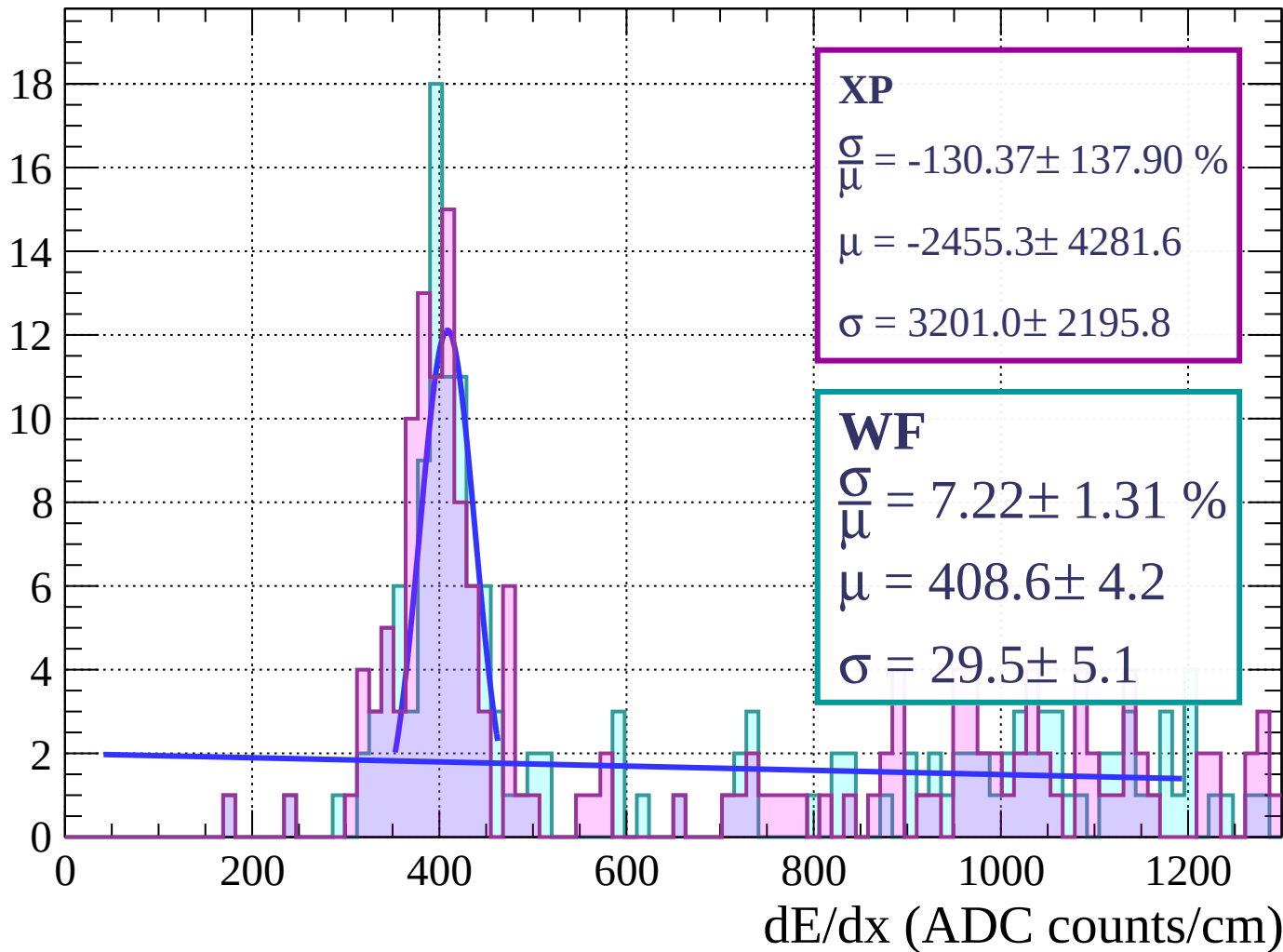
Energy loss | $69 < \phi < 72$

Count



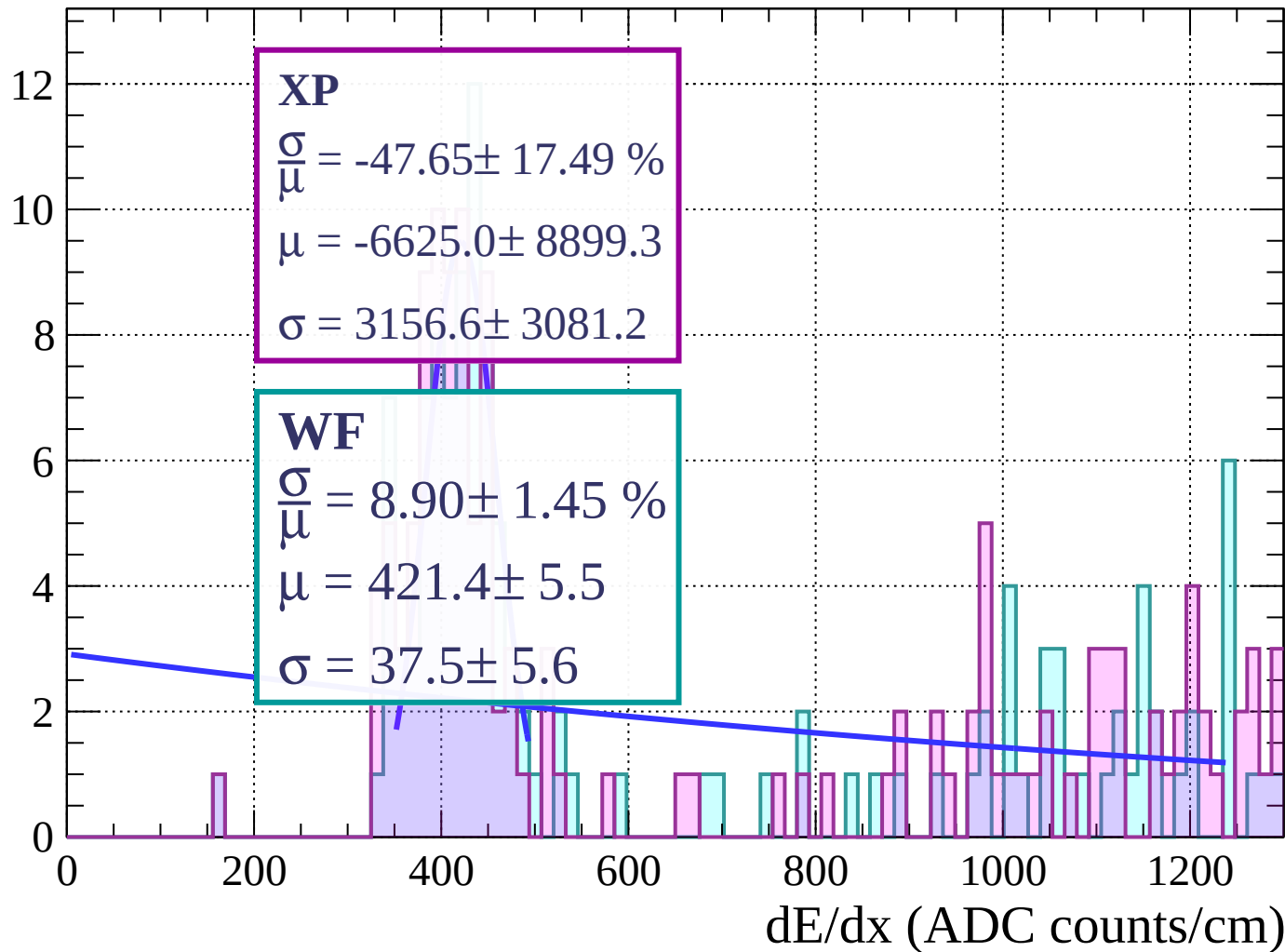
Energy loss | $72 < \phi < 75$

Count



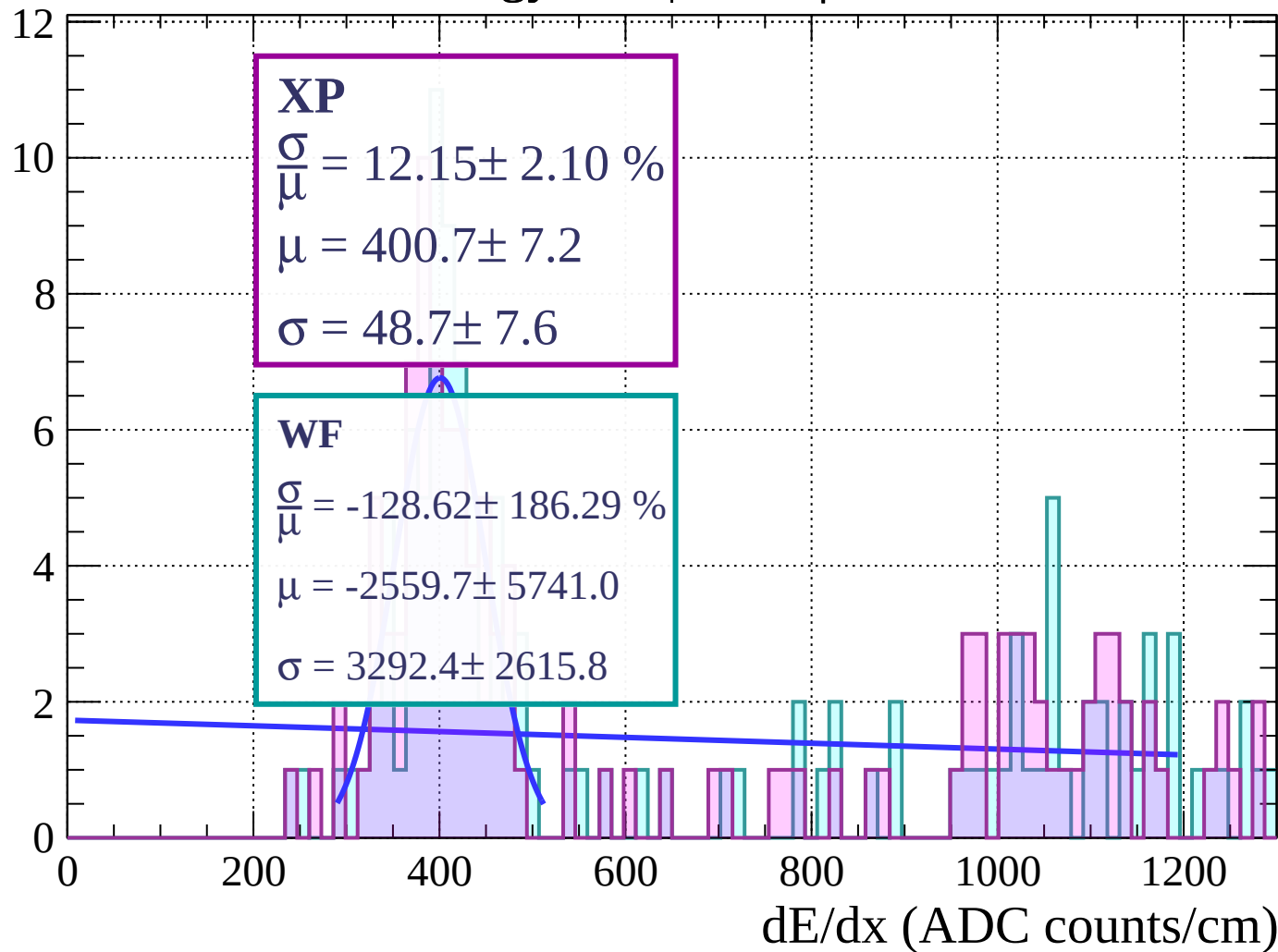
Energy loss | $75 < \phi < 78$

Count



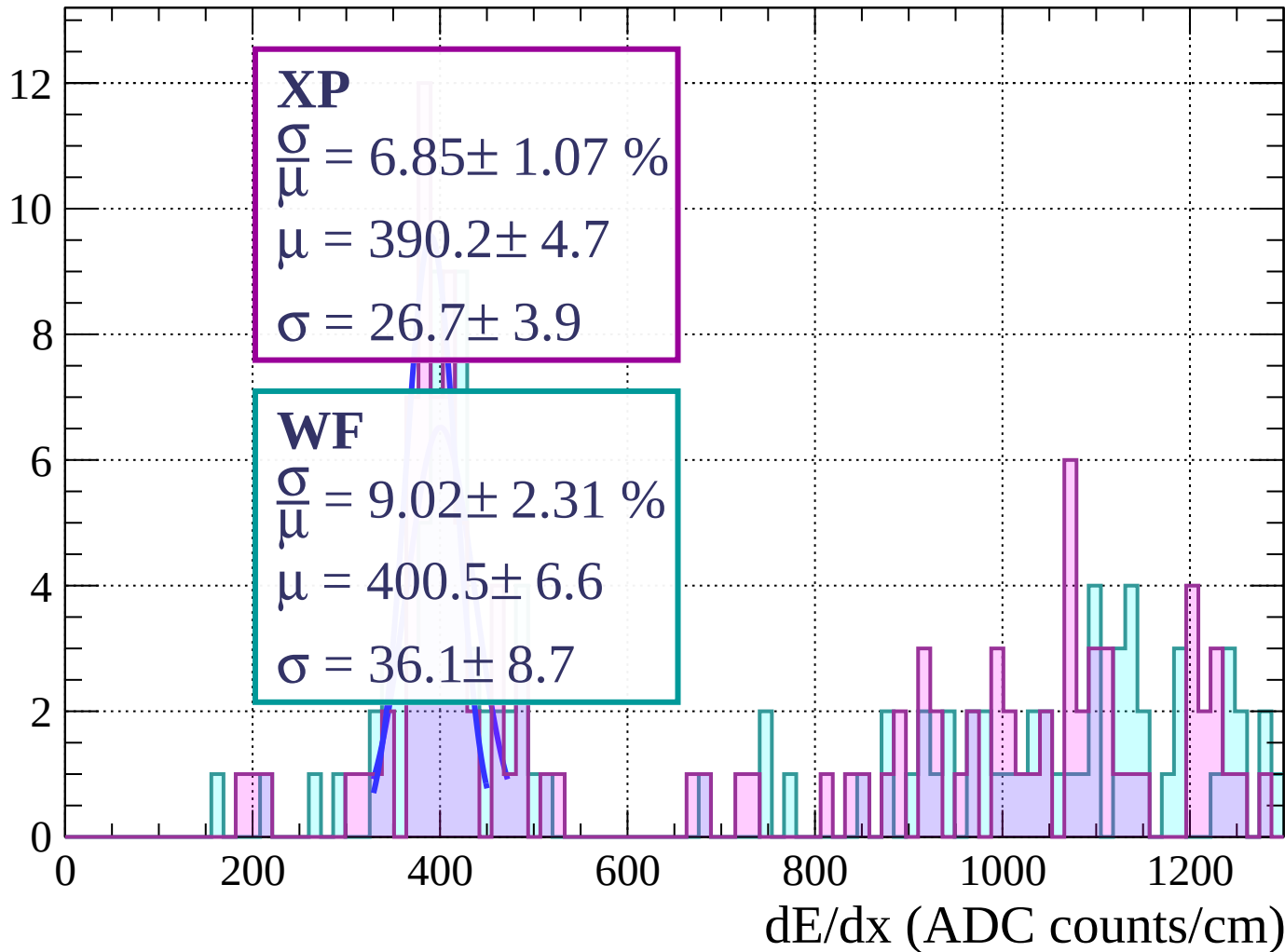
Energy loss | $78 < \phi < 81$

Count



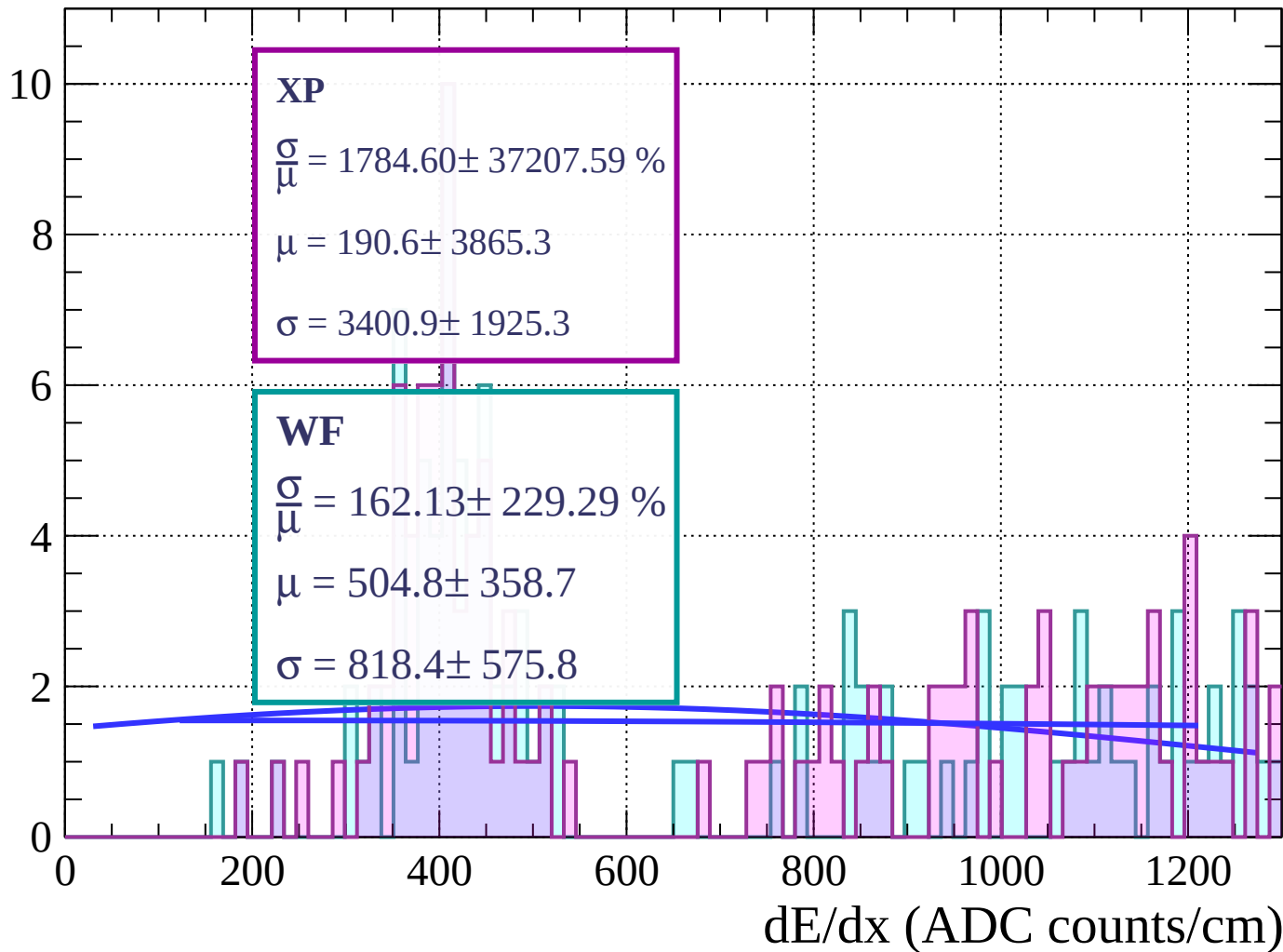
Energy loss | $81 < \phi < 84$

Count



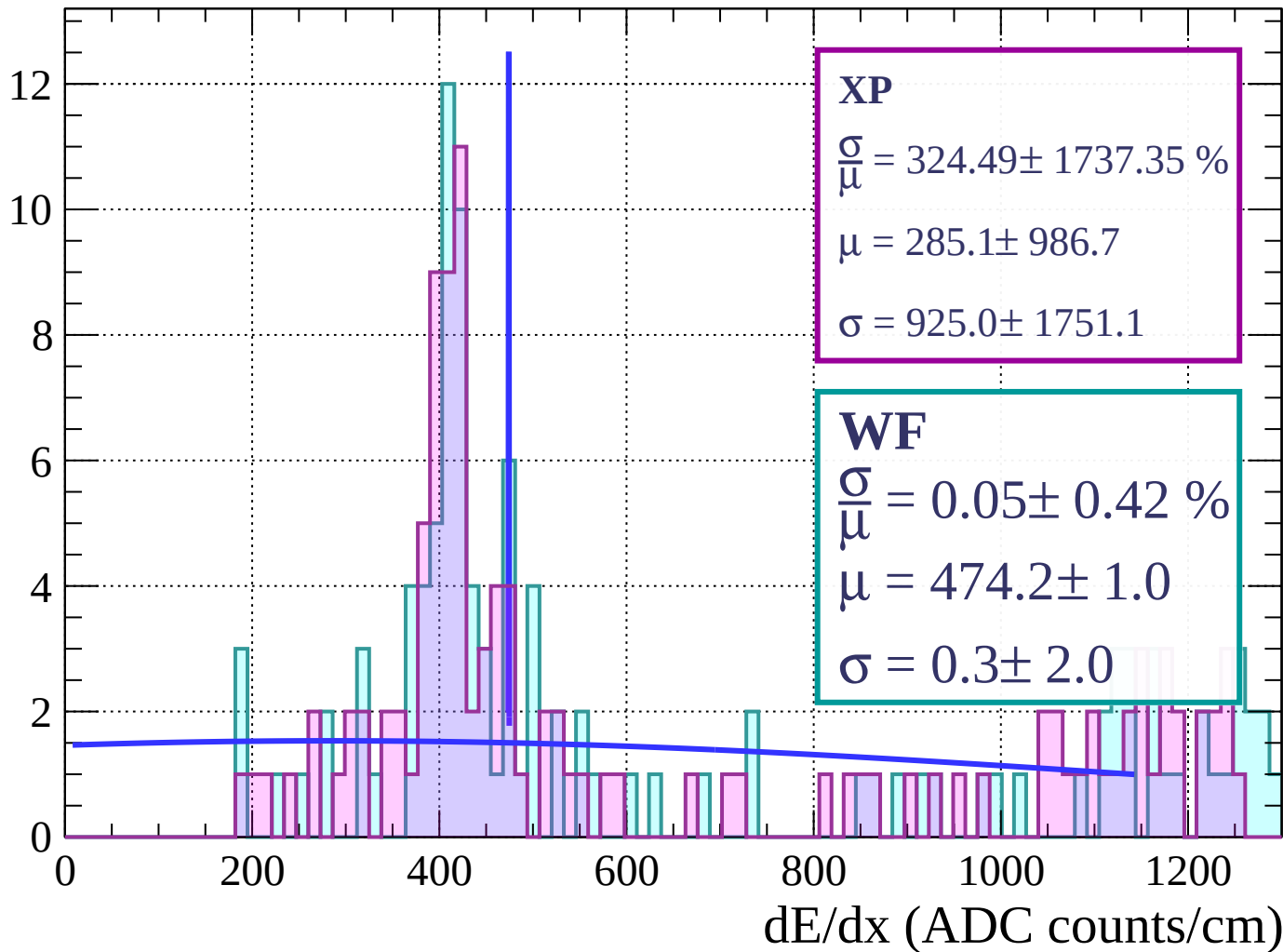
Energy loss | $84 < \phi < 87$

Count



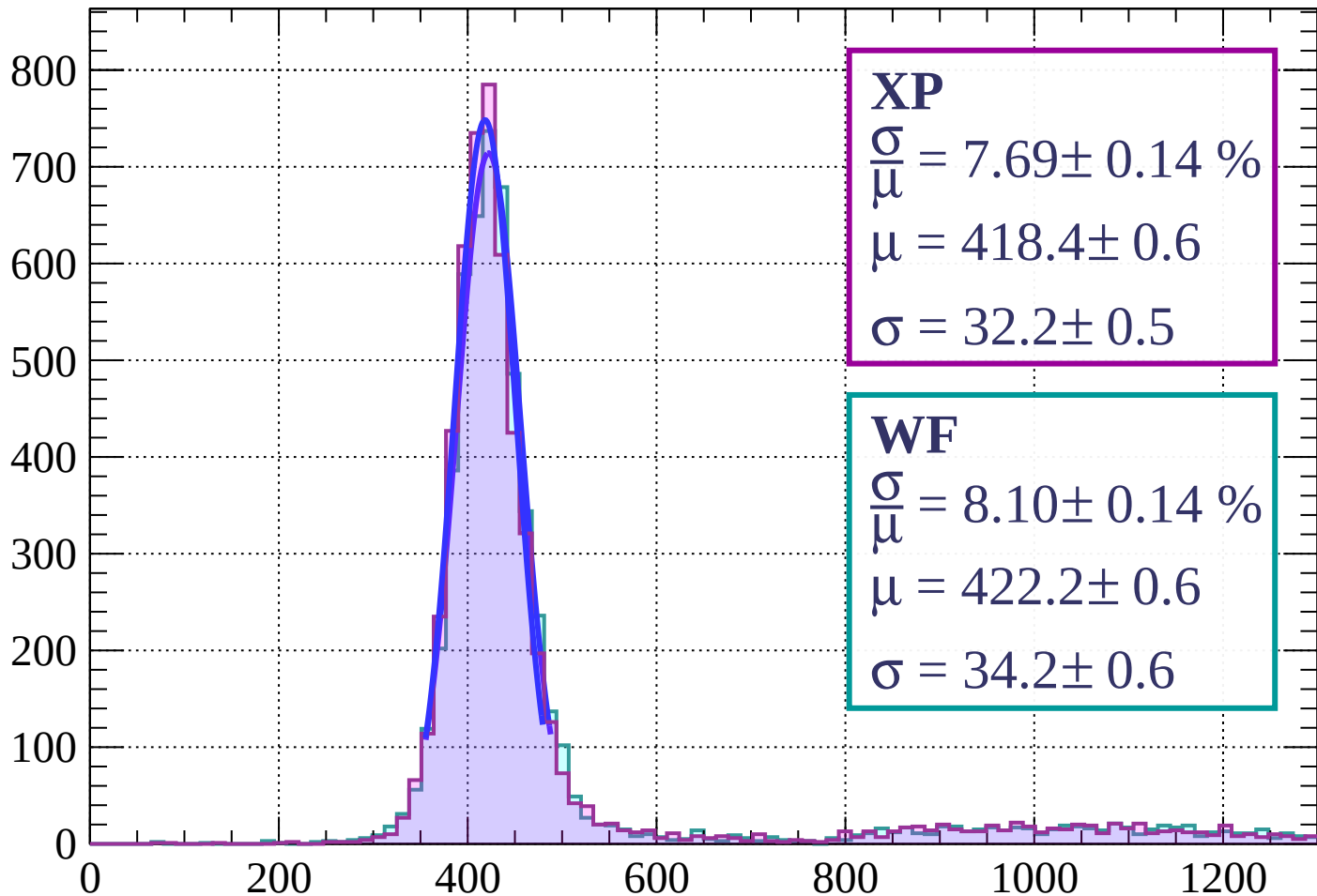
Energy loss | $87 < \phi < 90$

Count



Energy loss | $0 < \theta < 3$

Count



XP

$$\frac{\sigma}{\mu} = 7.69 \pm 0.14 \%$$

$$\mu = 418.4 \pm 0.6$$

$$\sigma = 32.2 \pm 0.5$$

WF

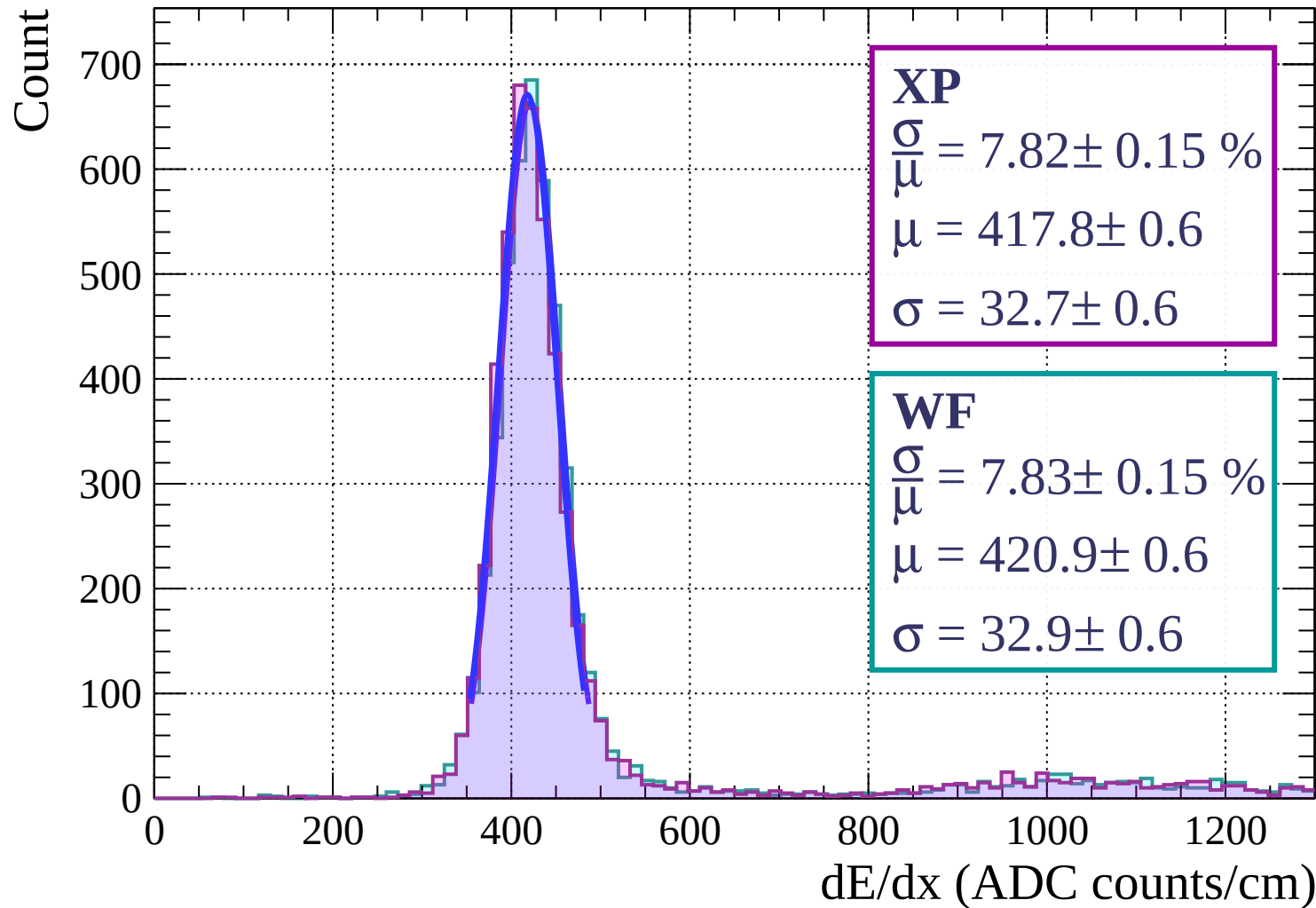
$$\frac{\sigma}{\mu} = 8.10 \pm 0.14 \%$$

$$\mu = 422.2 \pm 0.6$$

$$\sigma = 34.2 \pm 0.6$$

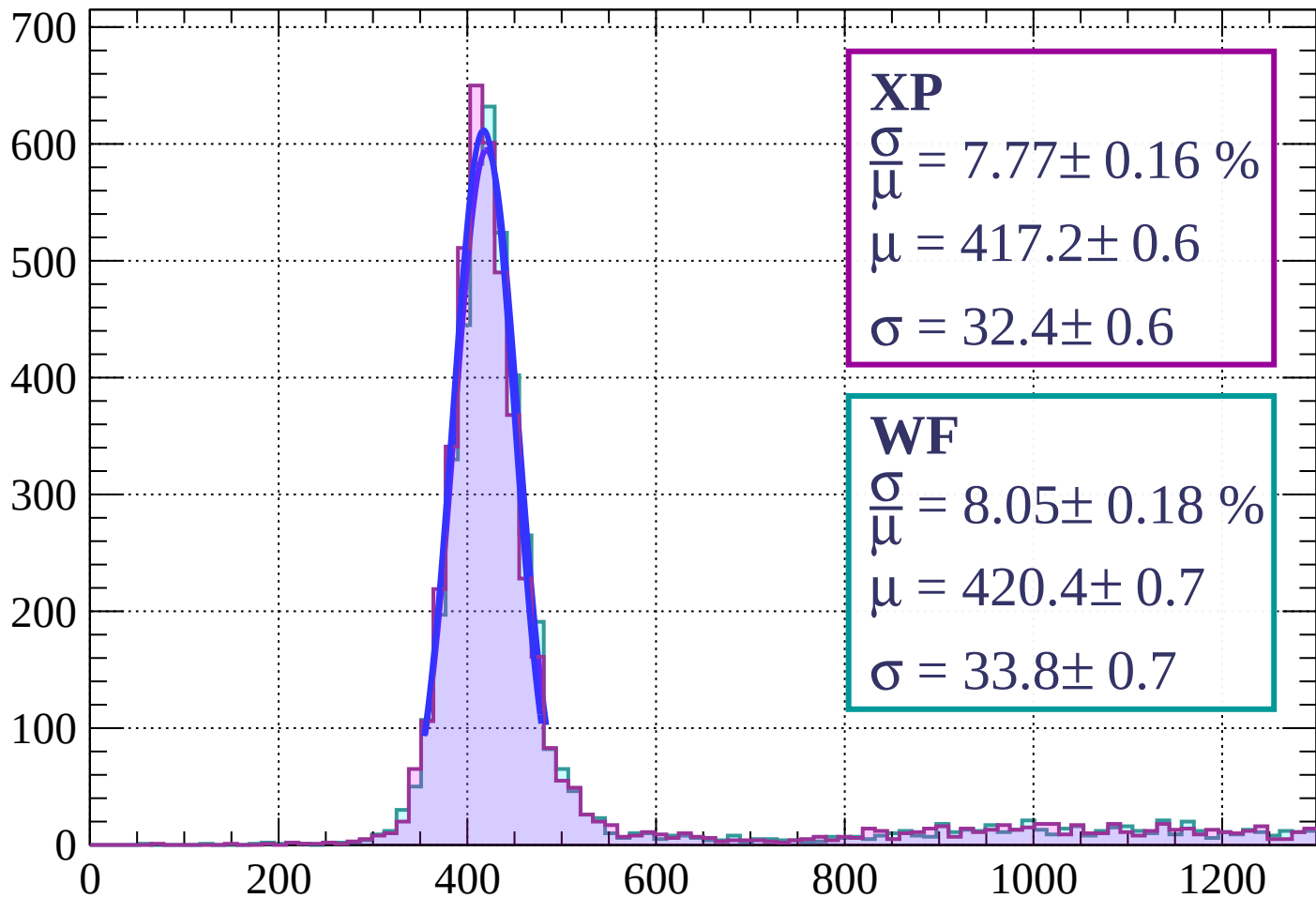
dE/dx (ADC counts/cm)

Energy loss | $3 < \theta < 6$



Energy loss | $6 < \theta < 9$

Count



XP

$$\frac{\sigma}{\mu} = 7.77 \pm 0.16 \%$$

$$\mu = 417.2 \pm 0.6$$

$$\sigma = 32.4 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.05 \pm 0.18 \%$$

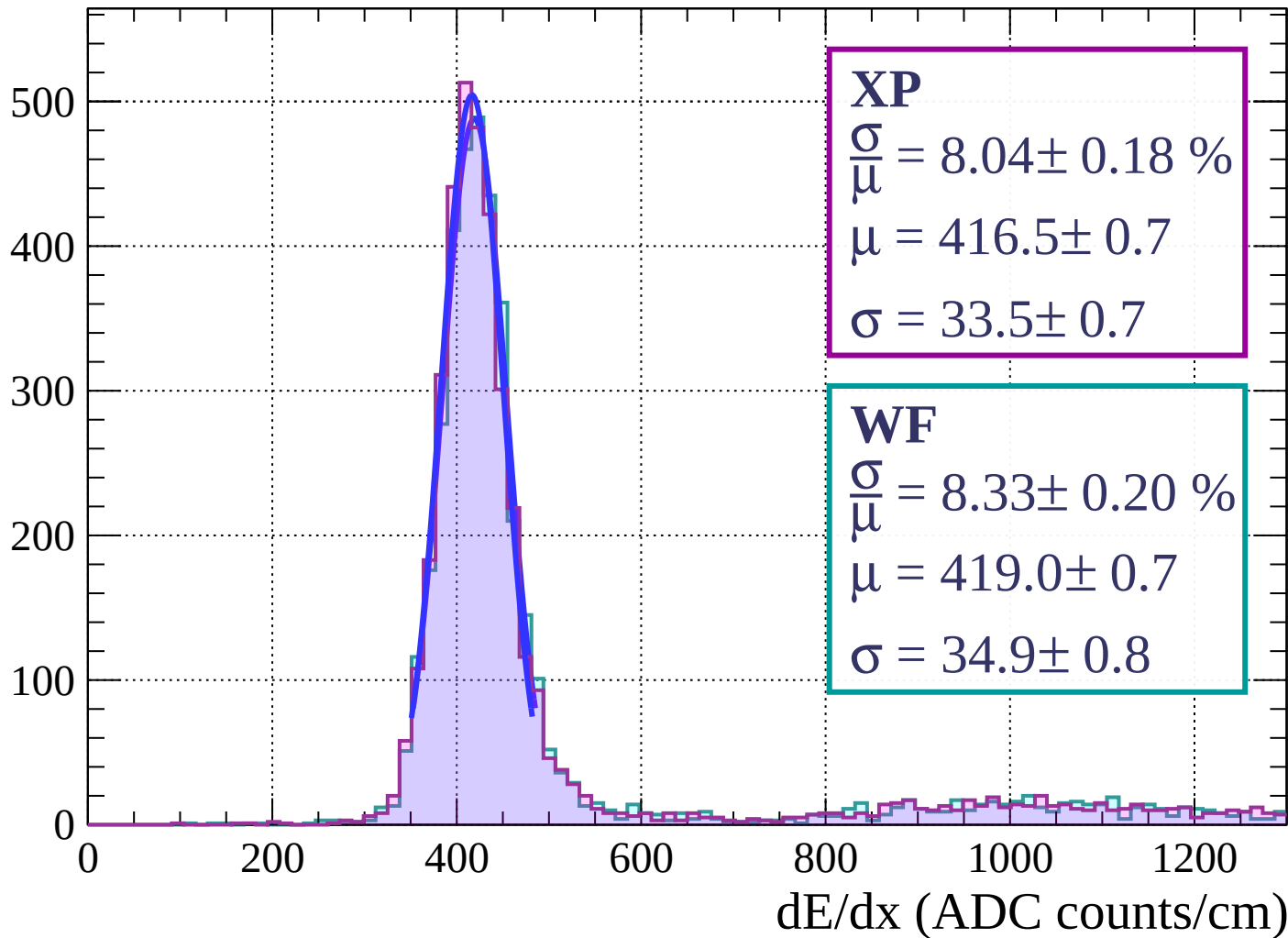
$$\mu = 420.4 \pm 0.7$$

$$\sigma = 33.8 \pm 0.7$$

dE/dx (ADC counts/cm)

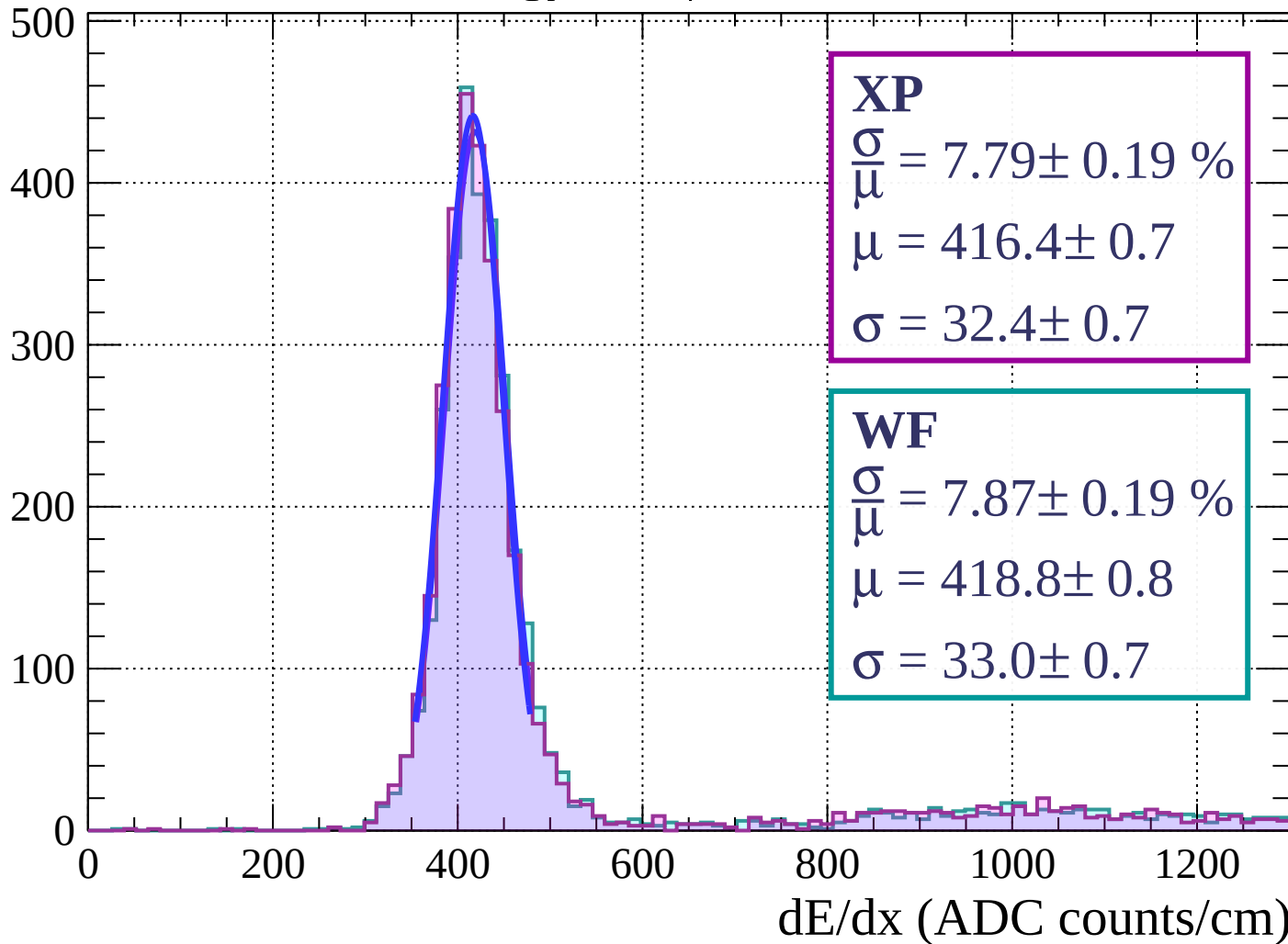
Energy loss | $9 < \theta < 12$

Count

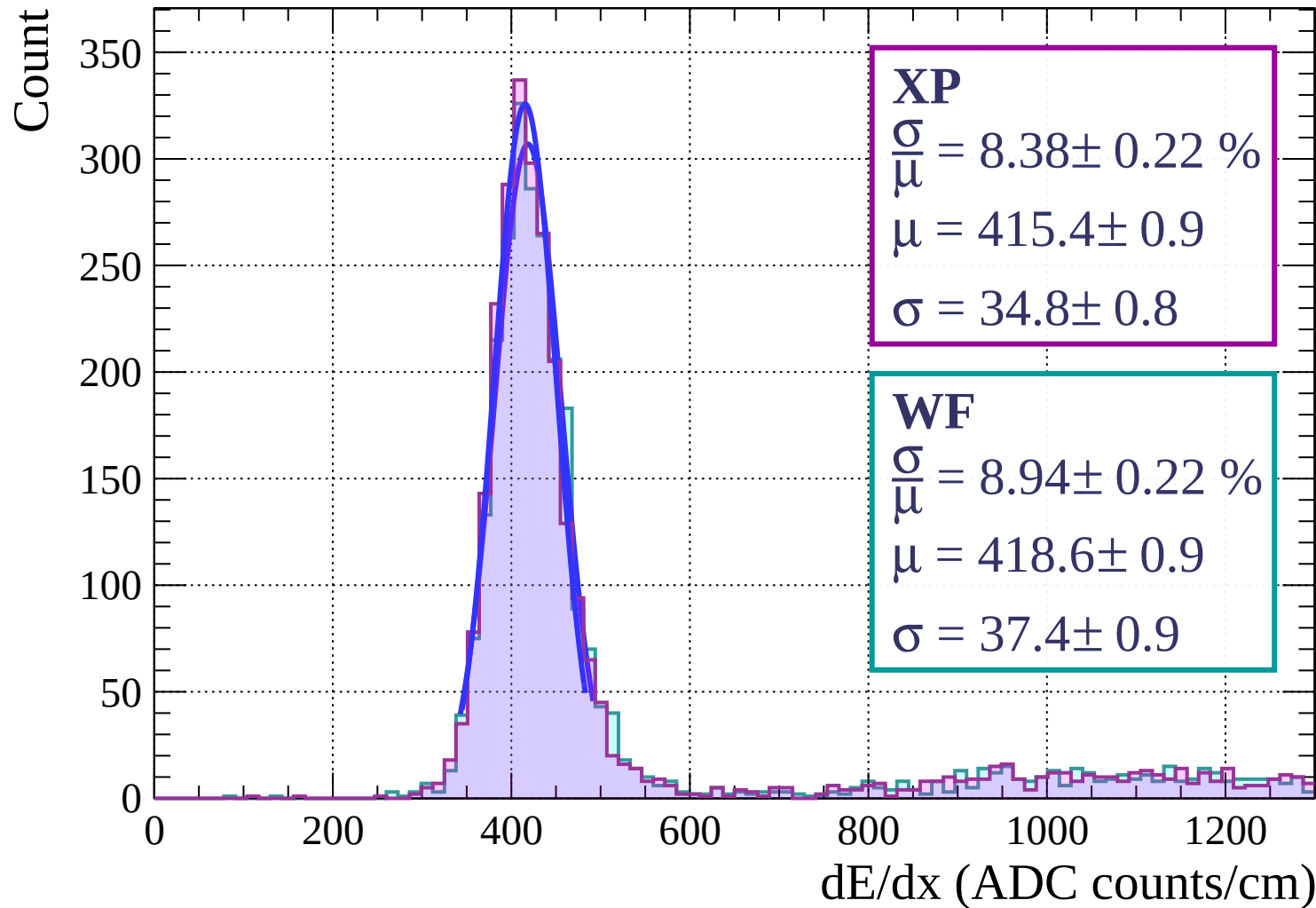


Energy loss | $12 < \theta < 15$

Count

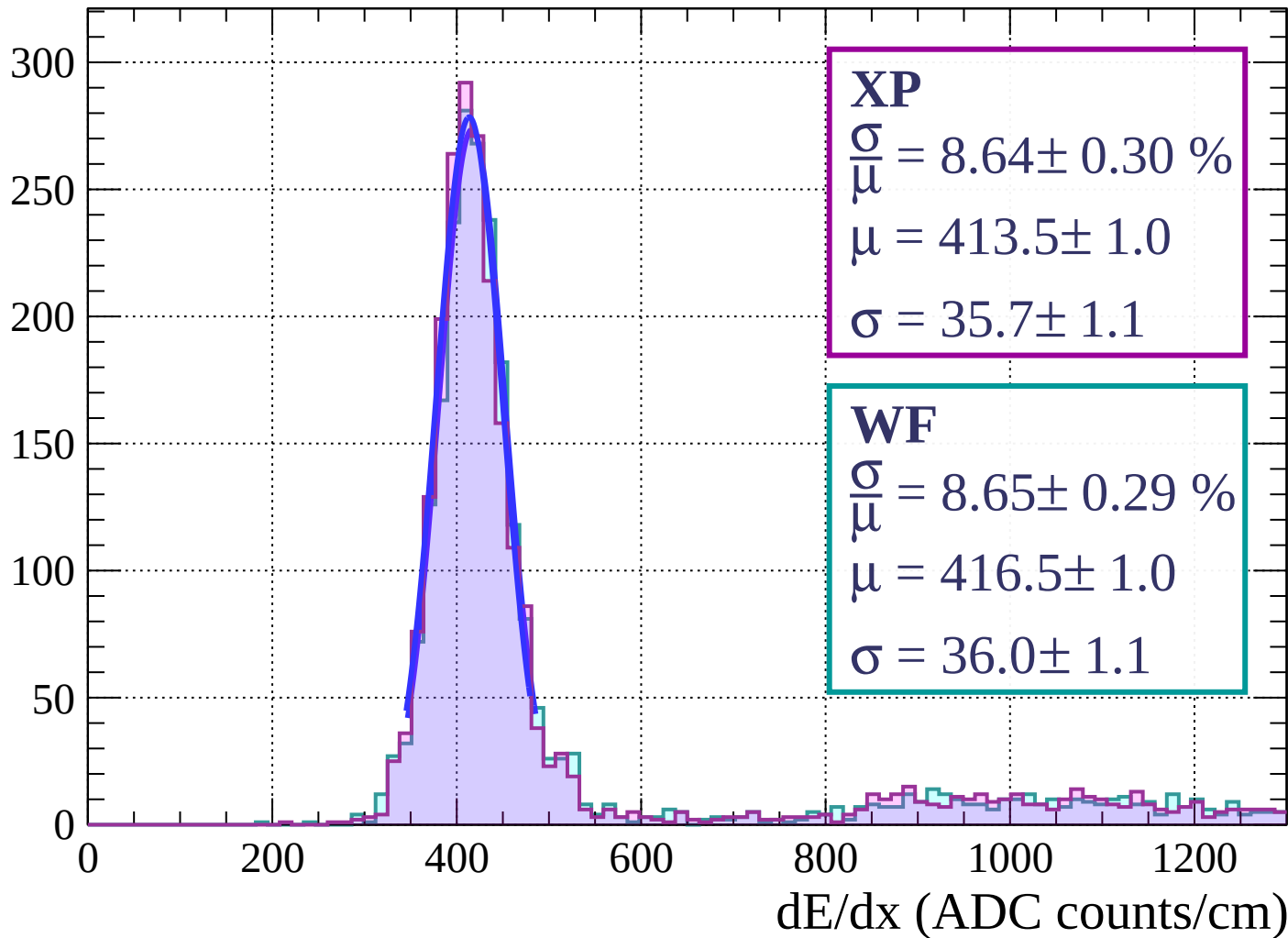


Energy loss | $15 < \theta < 18$



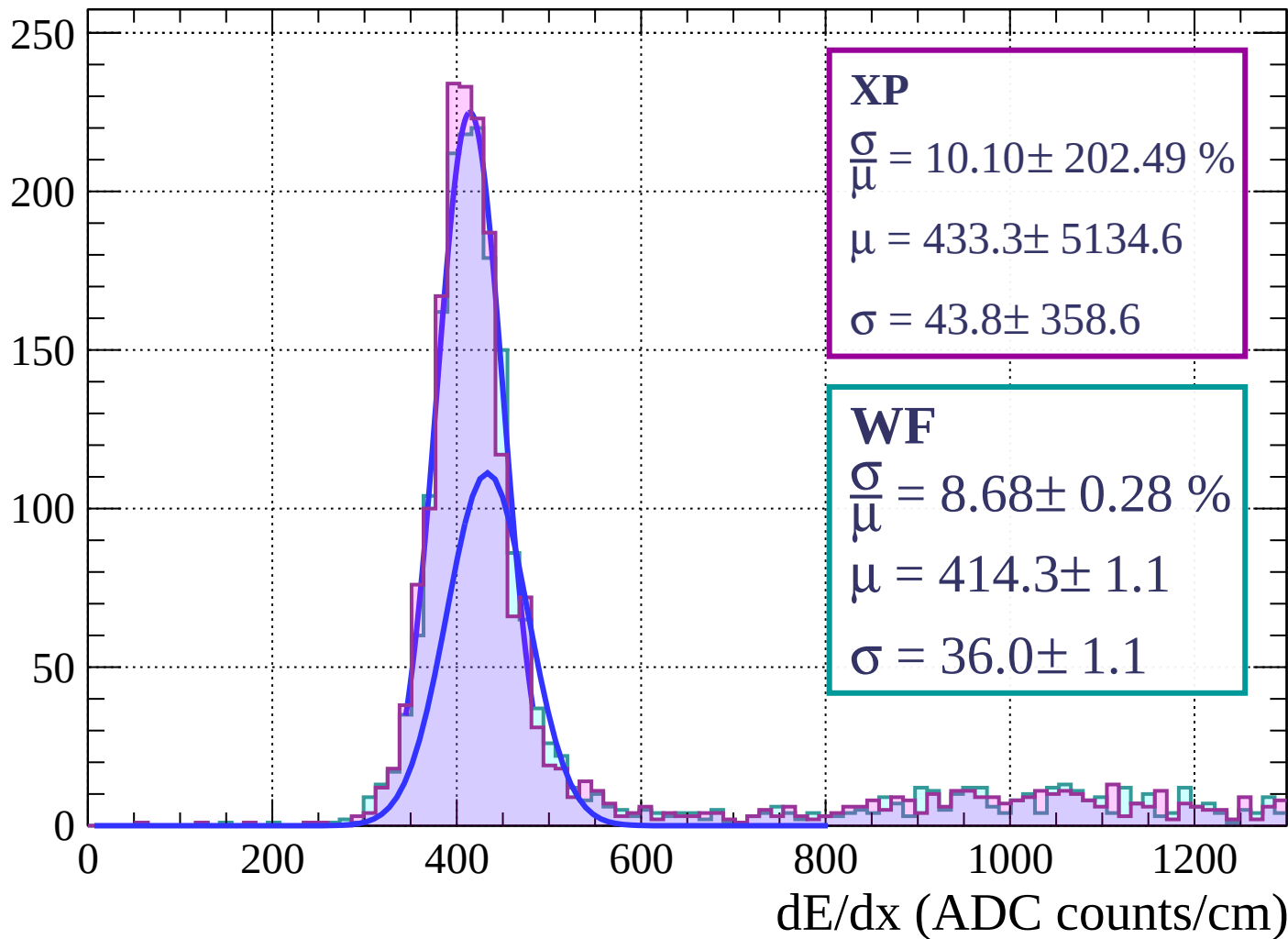
Energy loss | $18 < \theta < 21$

Count



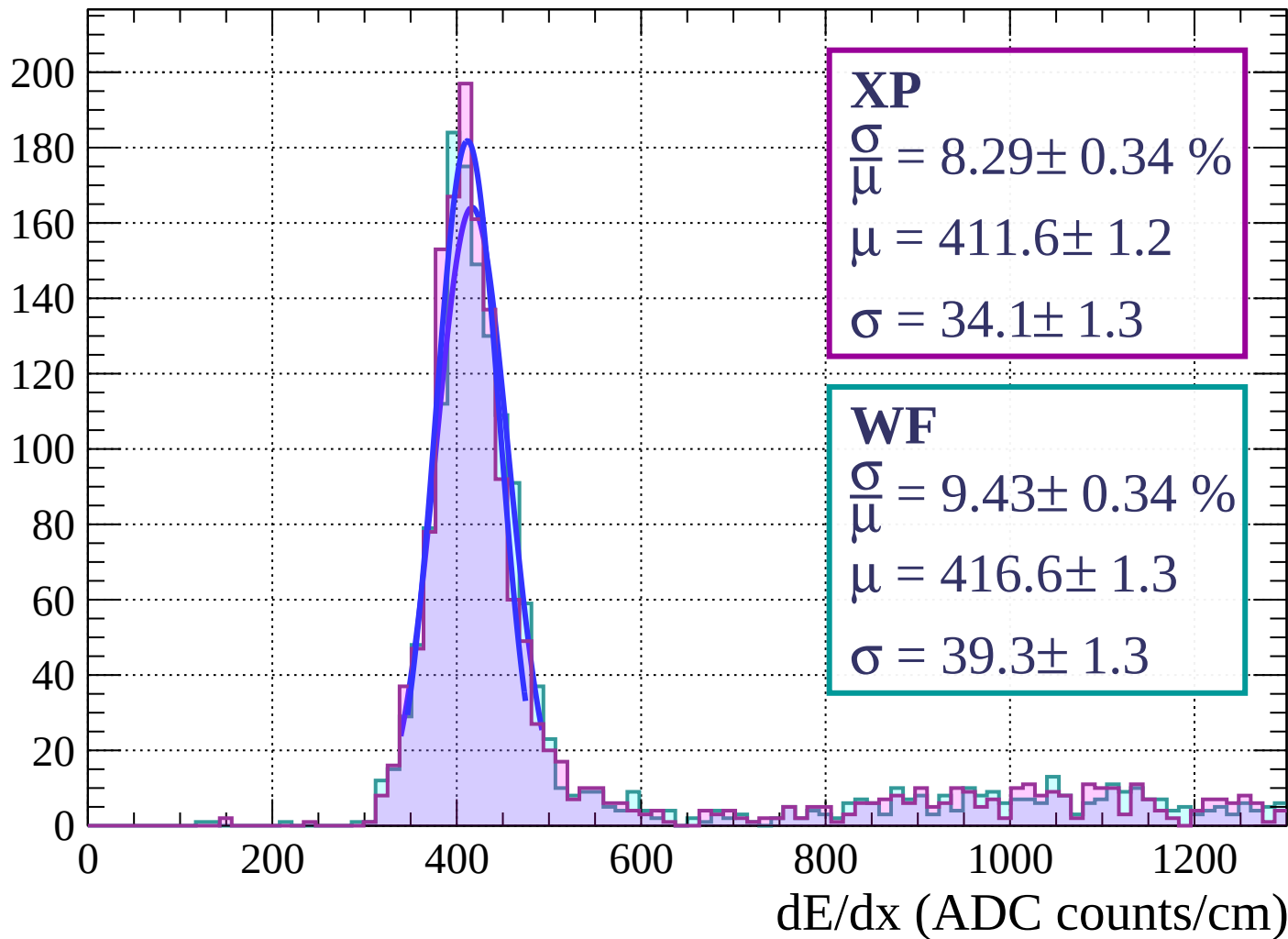
Energy loss | $21 < \theta < 24$

Count



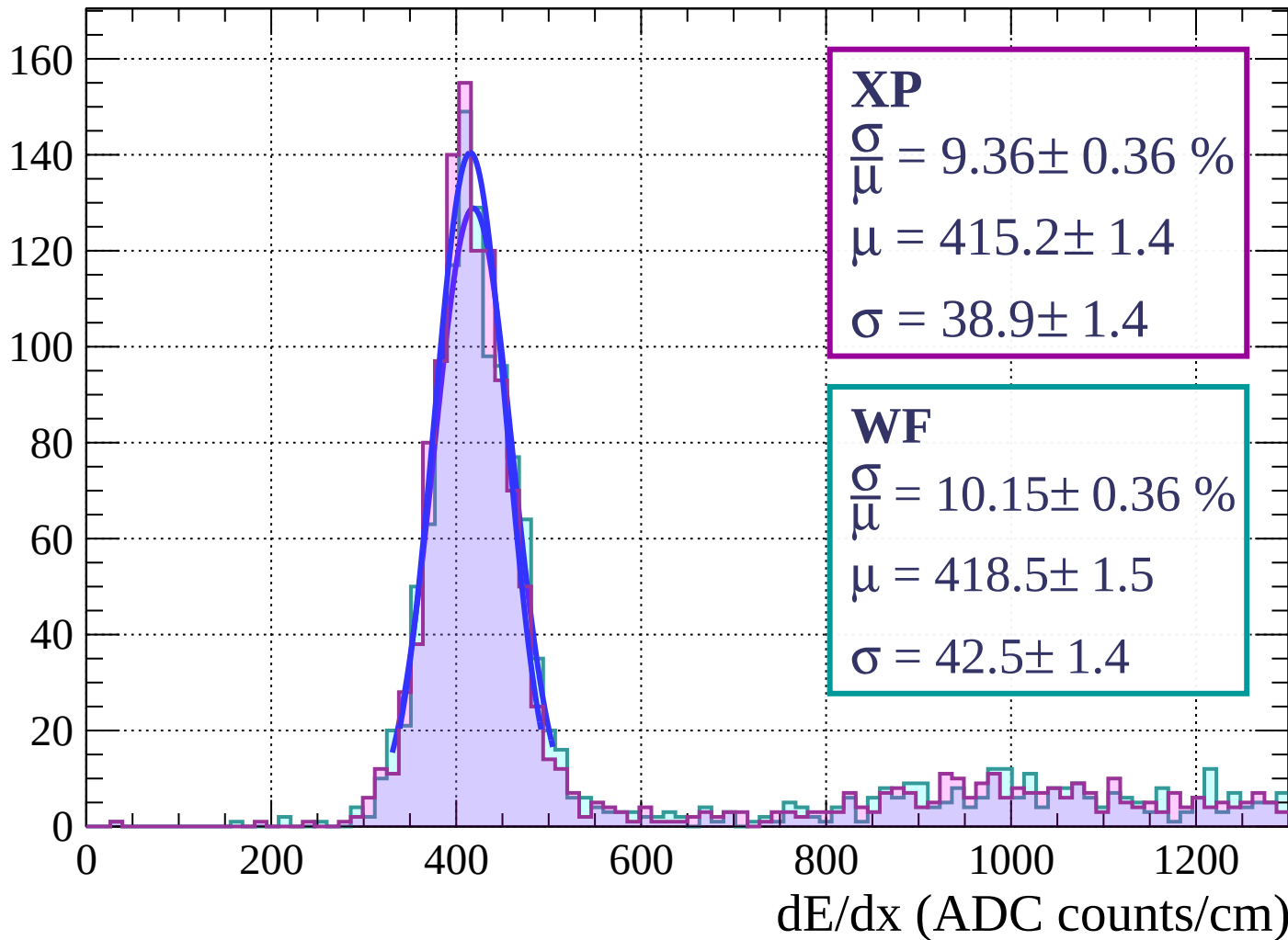
Energy loss | $24 < \theta < 27$

Count



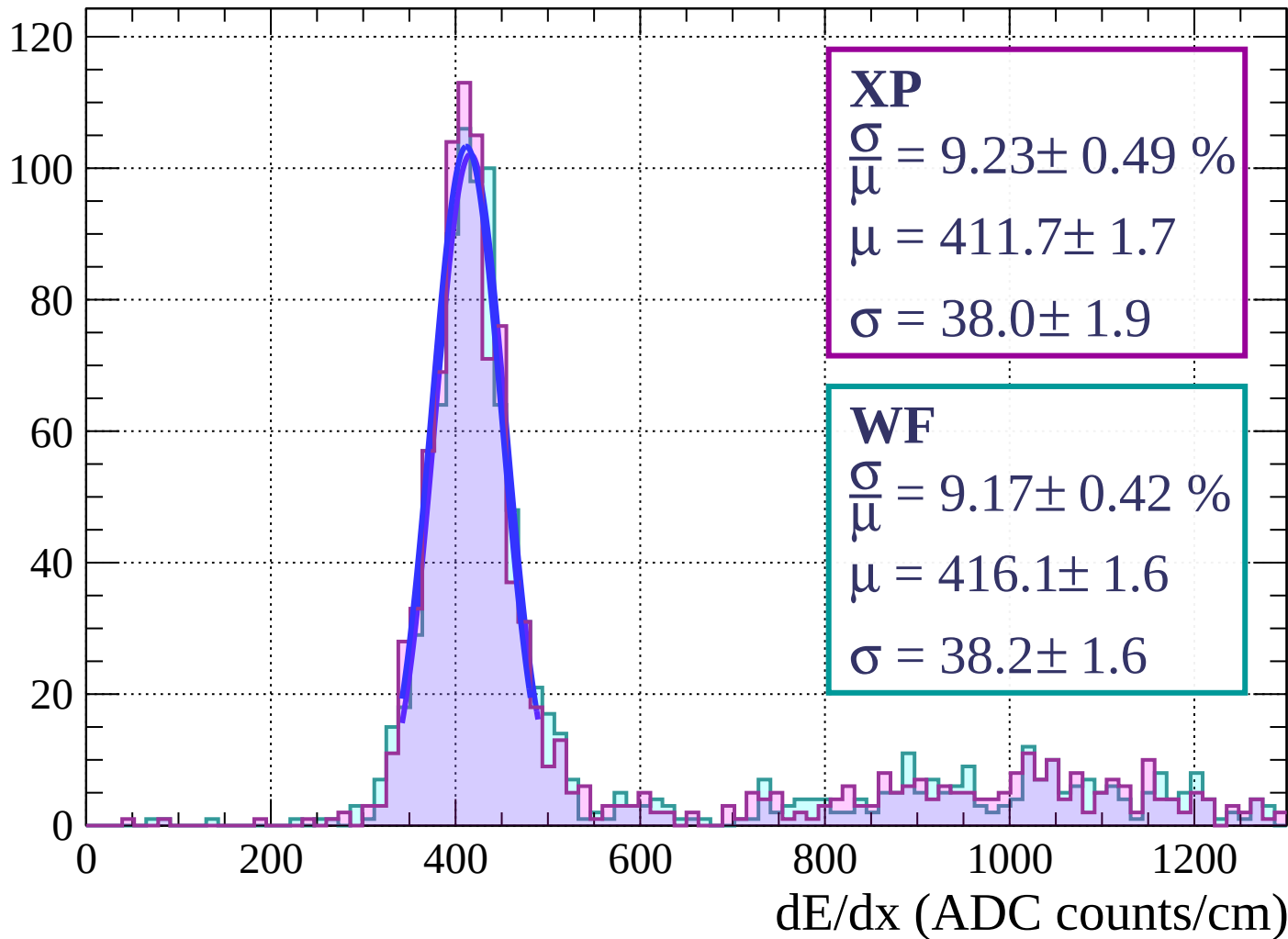
Energy loss | $27 < \theta < 30$

Count



Energy loss | $30 < \theta < 33$

Count



Energy loss | $33 < \theta < 36$

Count

XP

$$\frac{\sigma}{\mu} = 10.47 \pm 0.60 \%$$

$$\mu = 409.7 \pm 2.0$$

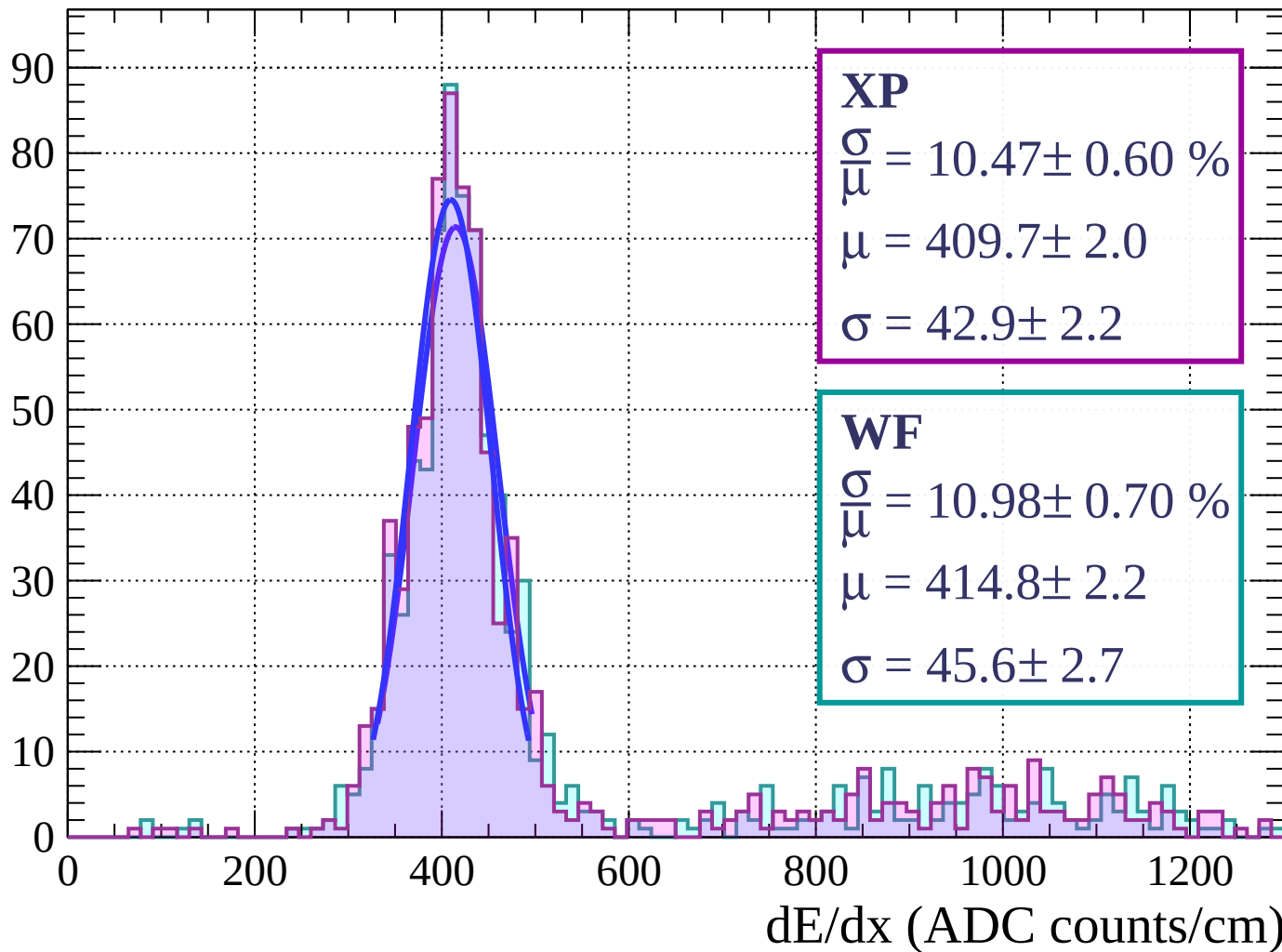
$$\sigma = 42.9 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 10.98 \pm 0.70 \%$$

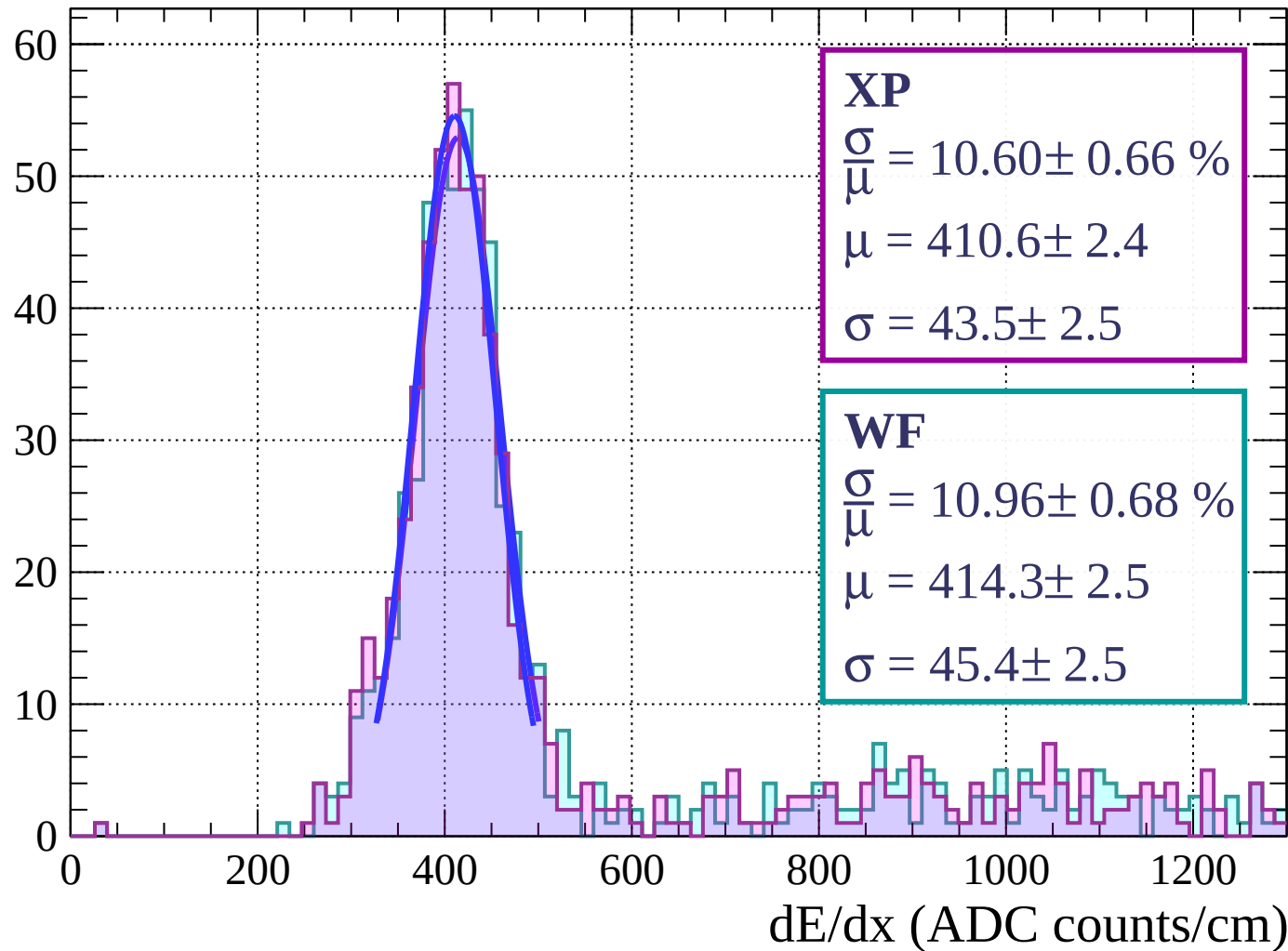
$$\mu = 414.8 \pm 2.2$$

$$\sigma = 45.6 \pm 2.7$$



Energy loss | $36 < \theta < 39$

Count



Energy loss | $39 < \theta < 42$

Count

XP

$$\frac{\sigma}{\mu} = 11.16 \pm 0.93 \%$$

$$\mu = 407.6 \pm 3.0$$

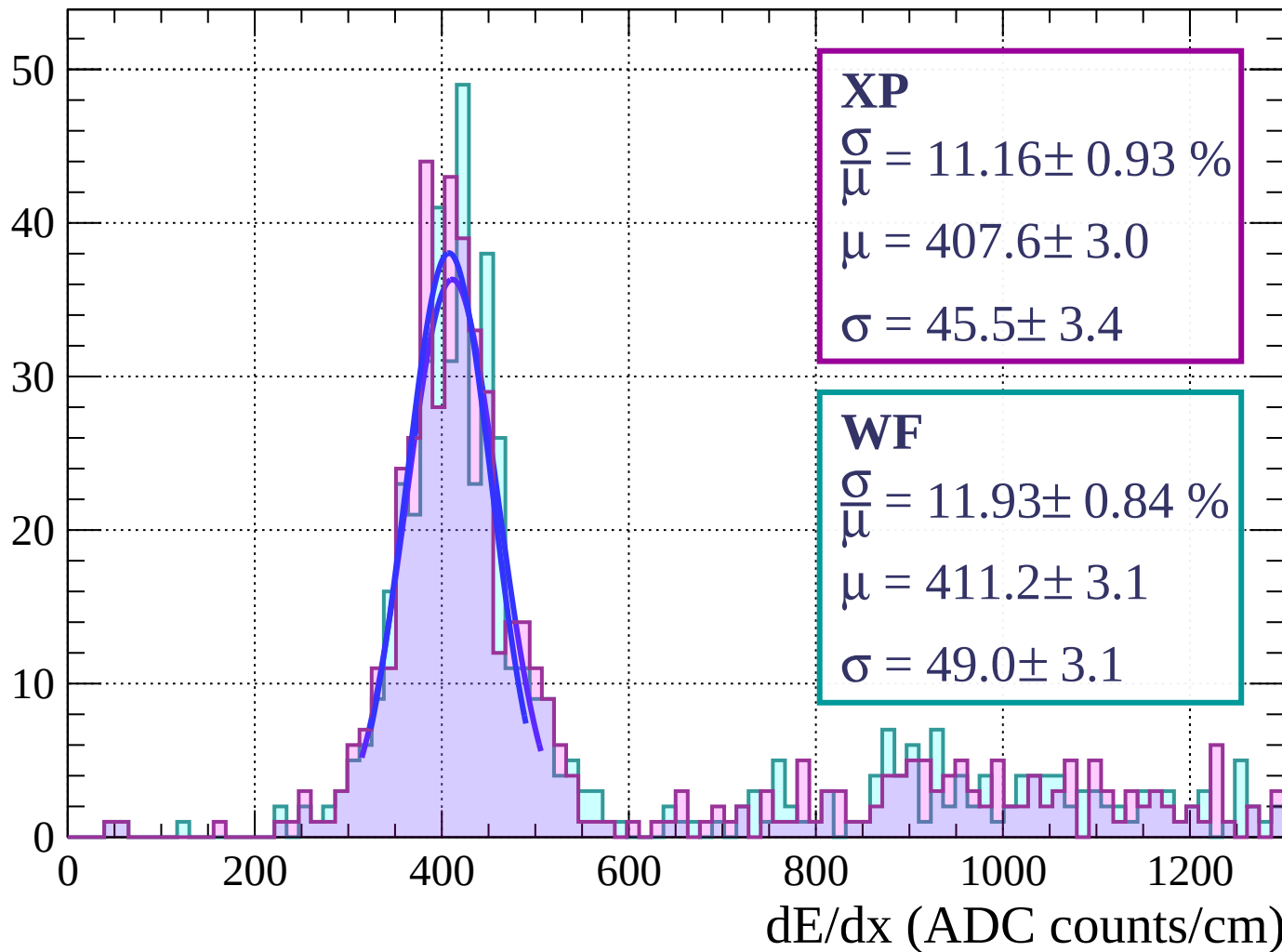
$$\sigma = 45.5 \pm 3.4$$

WF

$$\frac{\sigma}{\mu} = 11.93 \pm 0.84 \%$$

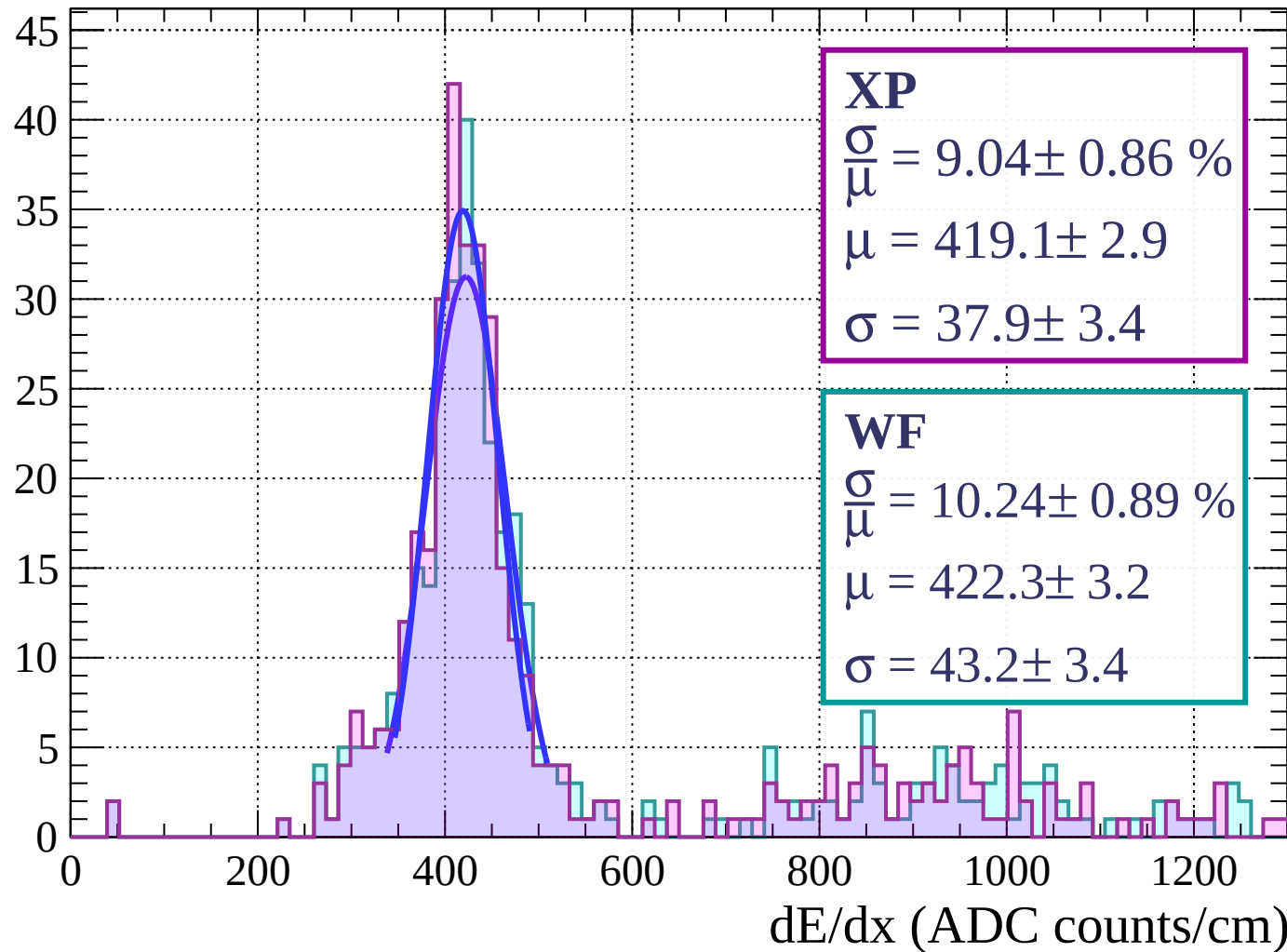
$$\mu = 411.2 \pm 3.1$$

$$\sigma = 49.0 \pm 3.1$$



Energy loss | $42 < \theta < 45$

Count



Energy loss | $45 < \theta < 48$

Count

XP

$$\frac{\sigma}{\mu} = 11.23 \pm 1.15 \%$$

$$\mu = 413.7 \pm 4.2$$

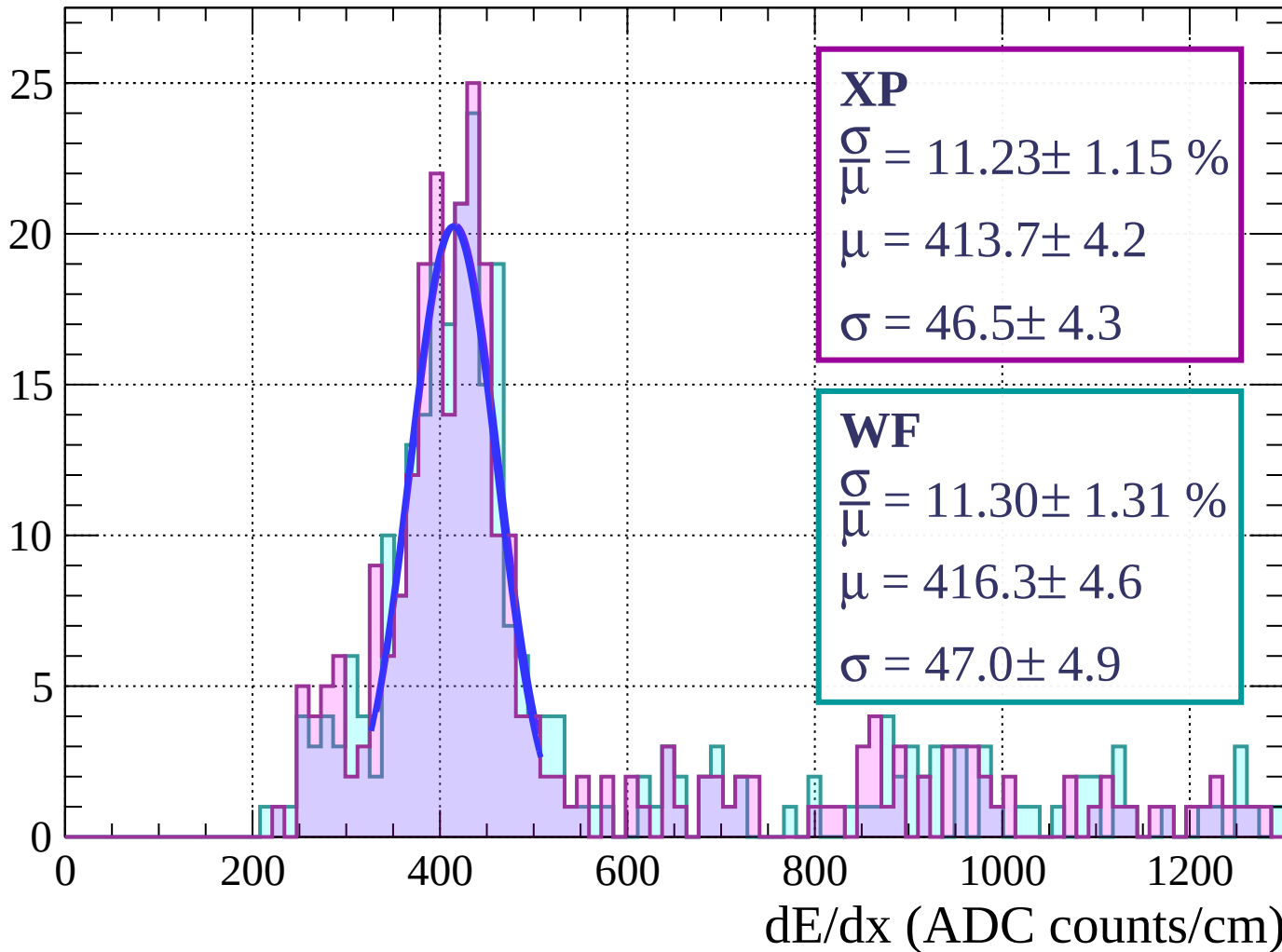
$$\sigma = 46.5 \pm 4.3$$

WF

$$\frac{\sigma}{\mu} = 11.30 \pm 1.31 \%$$

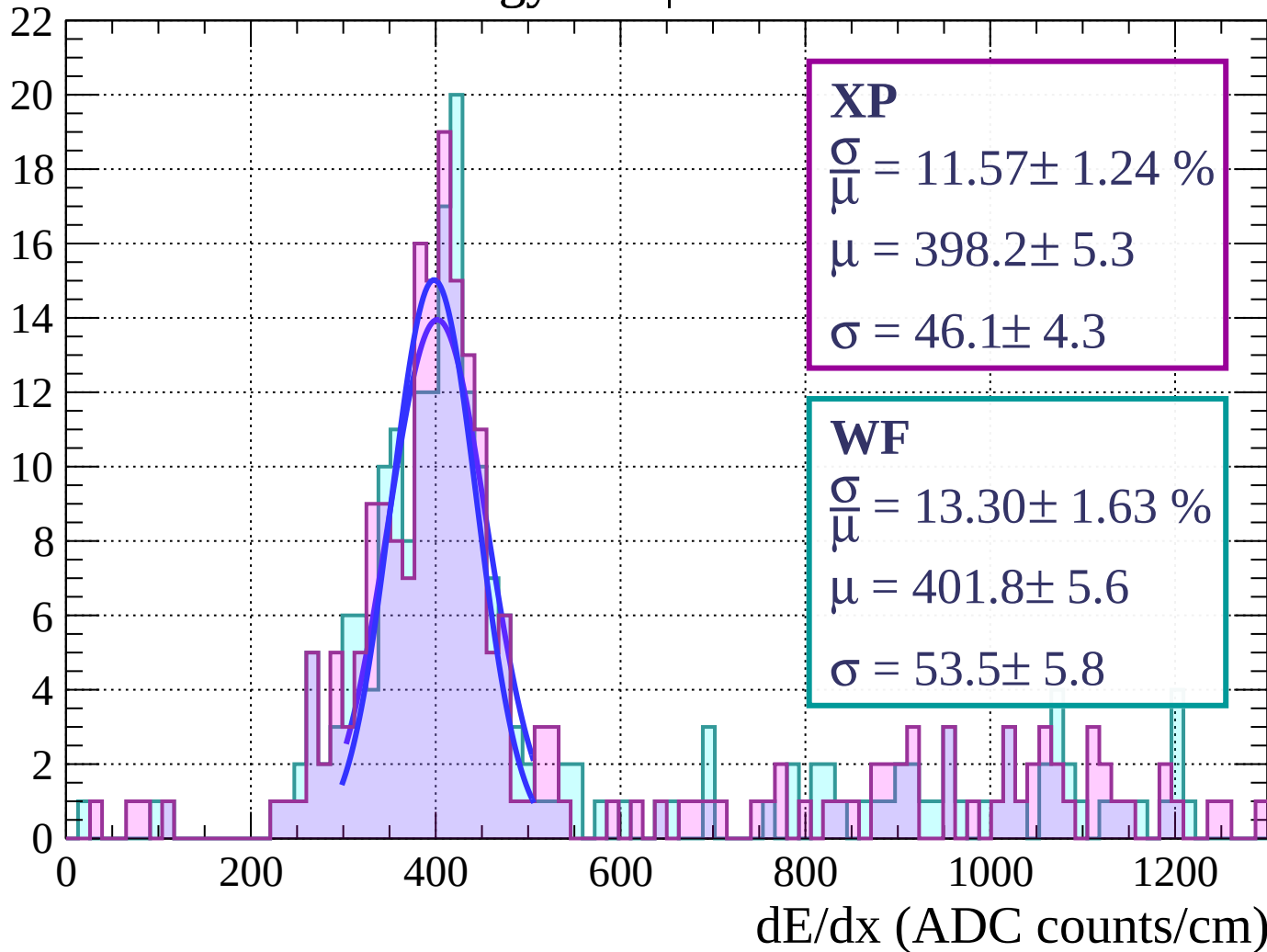
$$\mu = 416.3 \pm 4.6$$

$$\sigma = 47.0 \pm 4.9$$



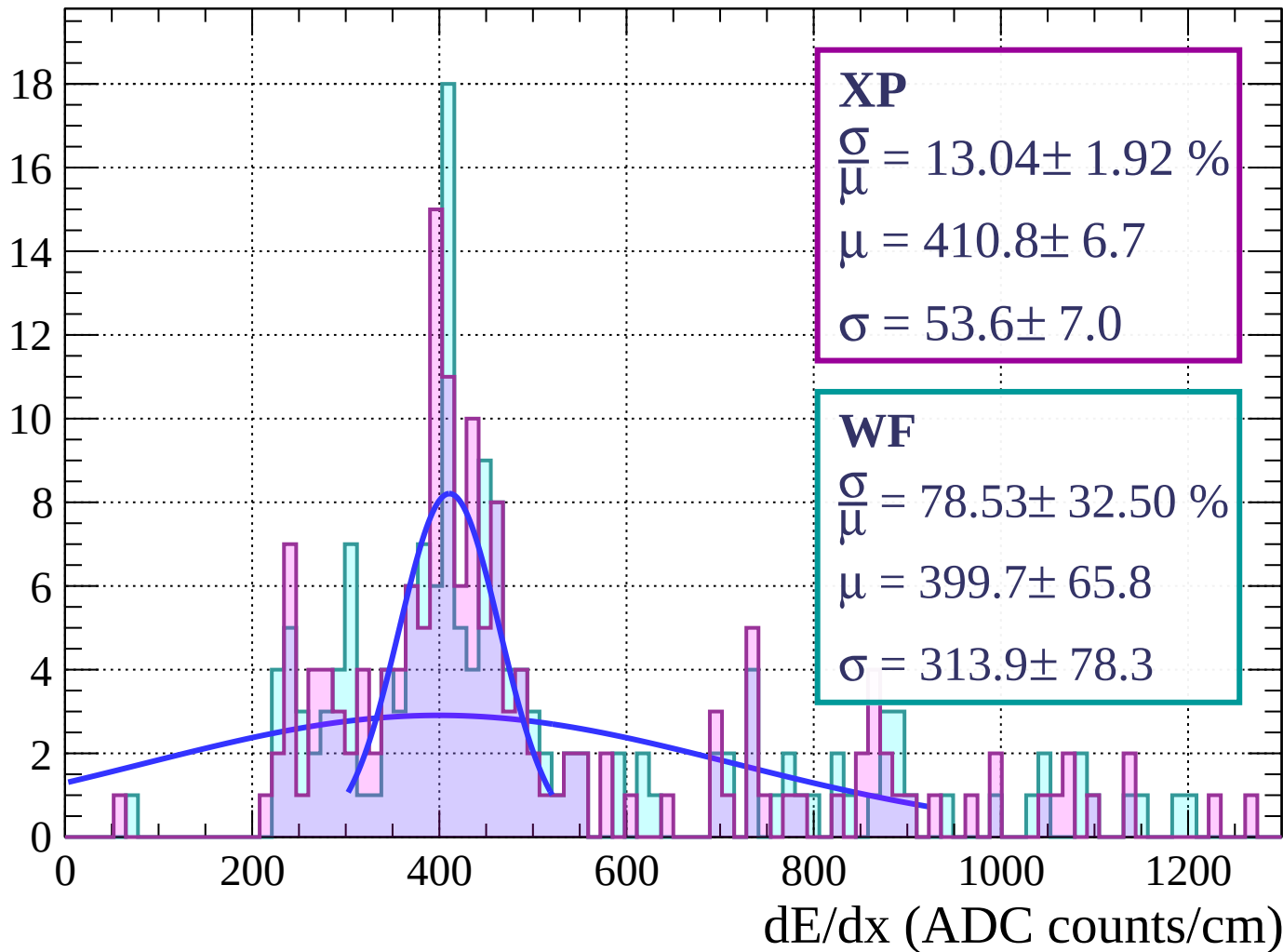
Energy loss | $48 < \theta < 51$

Count



Energy loss | $51 < \theta < 54$

Count



Energy loss | $54 < \theta < 57$

Count

XP

$$\frac{\sigma}{\mu} = 53.93 \pm 22.01 \%$$

$$\mu = 408.1 \pm 43.9$$

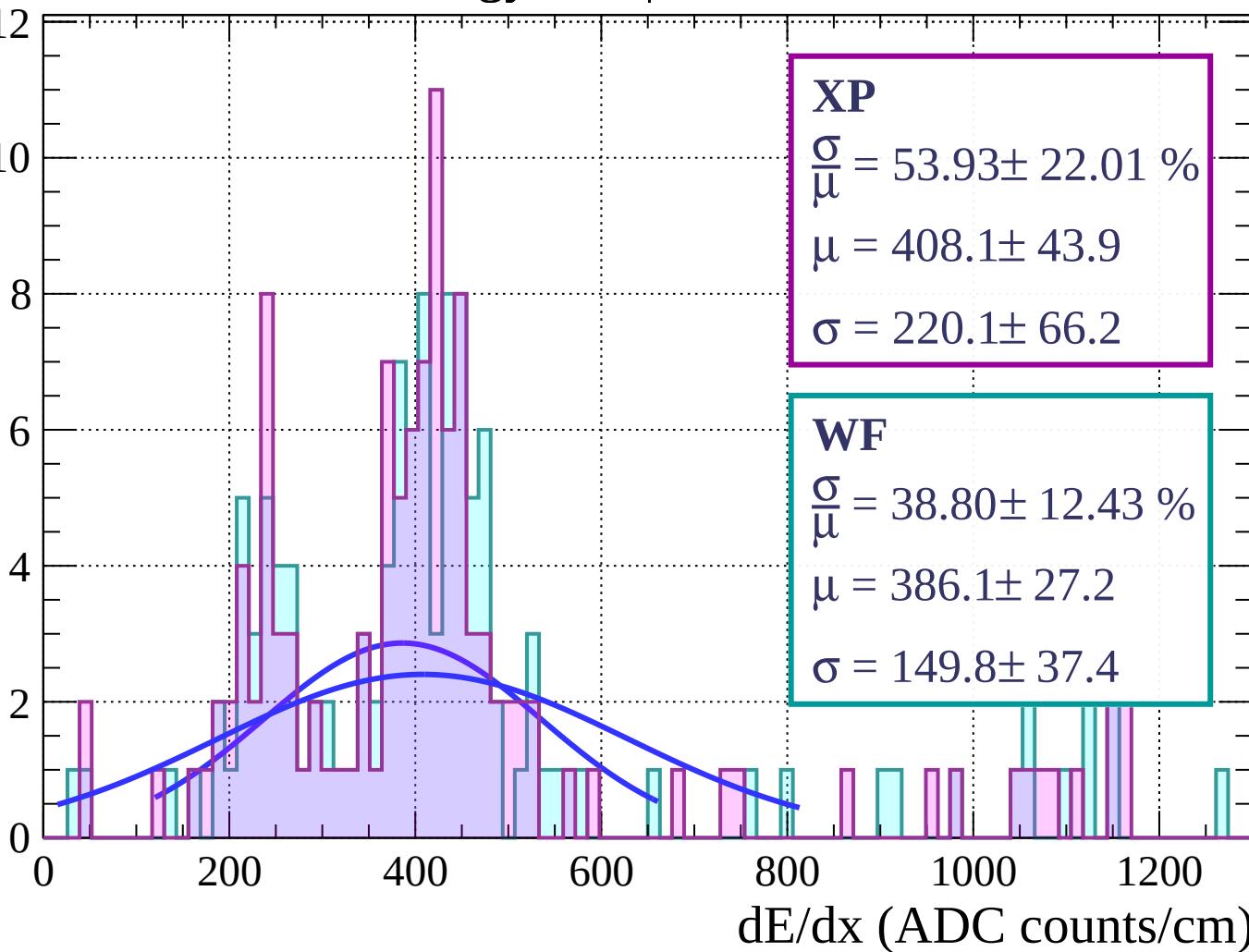
$$\sigma = 220.1 \pm 66.2$$

WF

$$\frac{\sigma}{\mu} = 38.80 \pm 12.43 \%$$

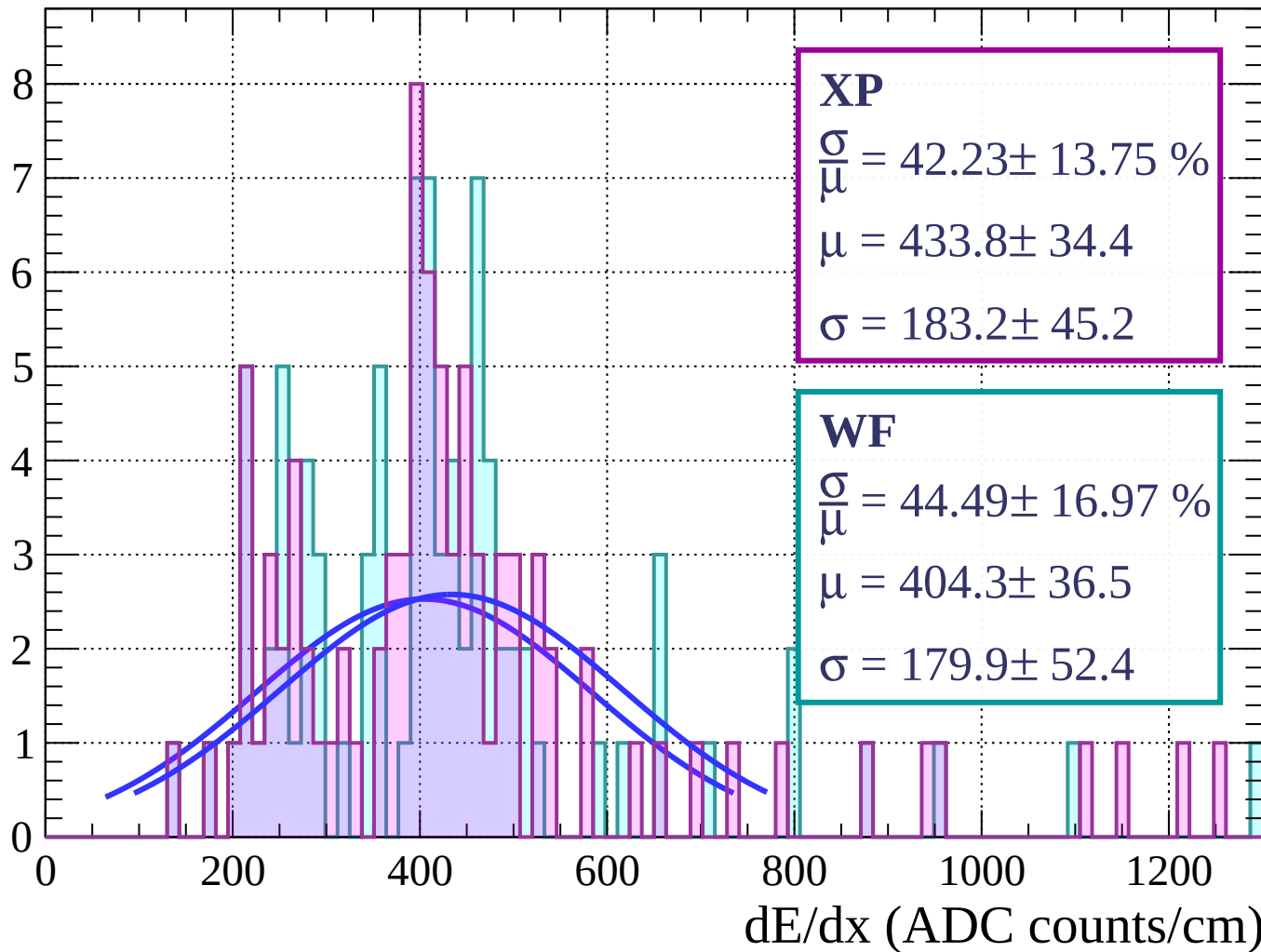
$$\mu = 386.1 \pm 27.2$$

$$\sigma = 149.8 \pm 37.4$$



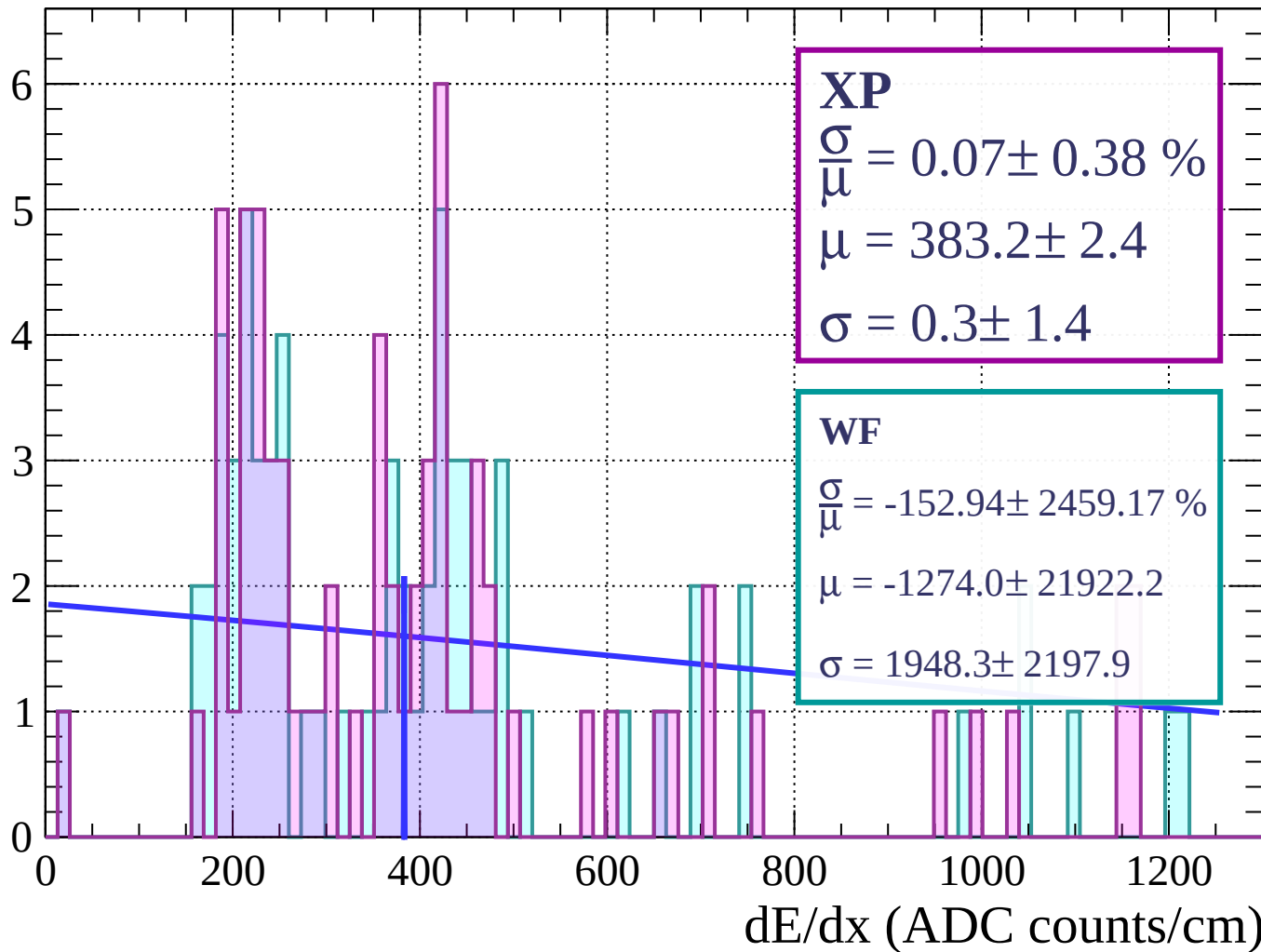
Energy loss | $57 < \theta < 60$

Count



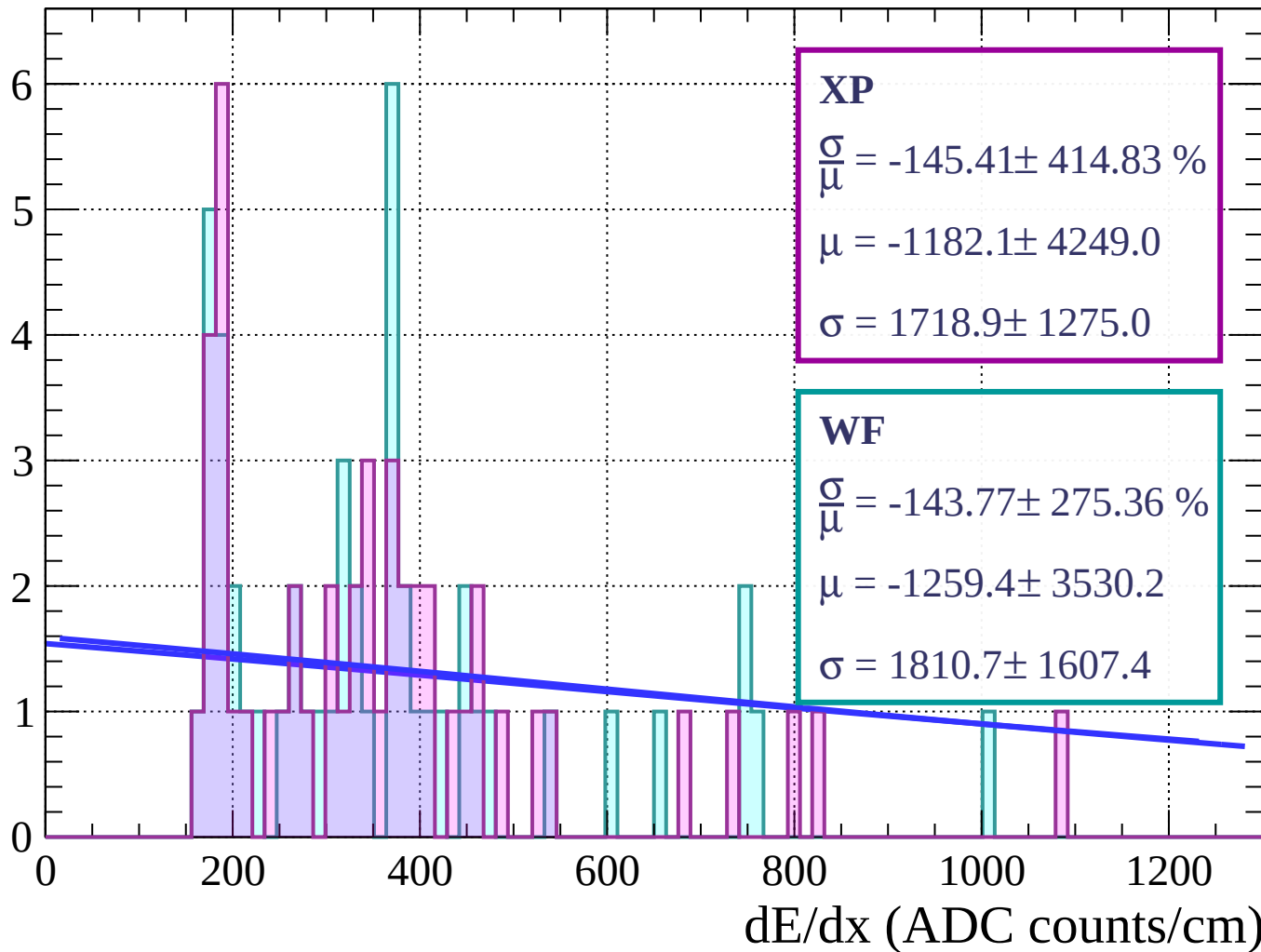
Energy loss | $60 < \theta < 63$

Count



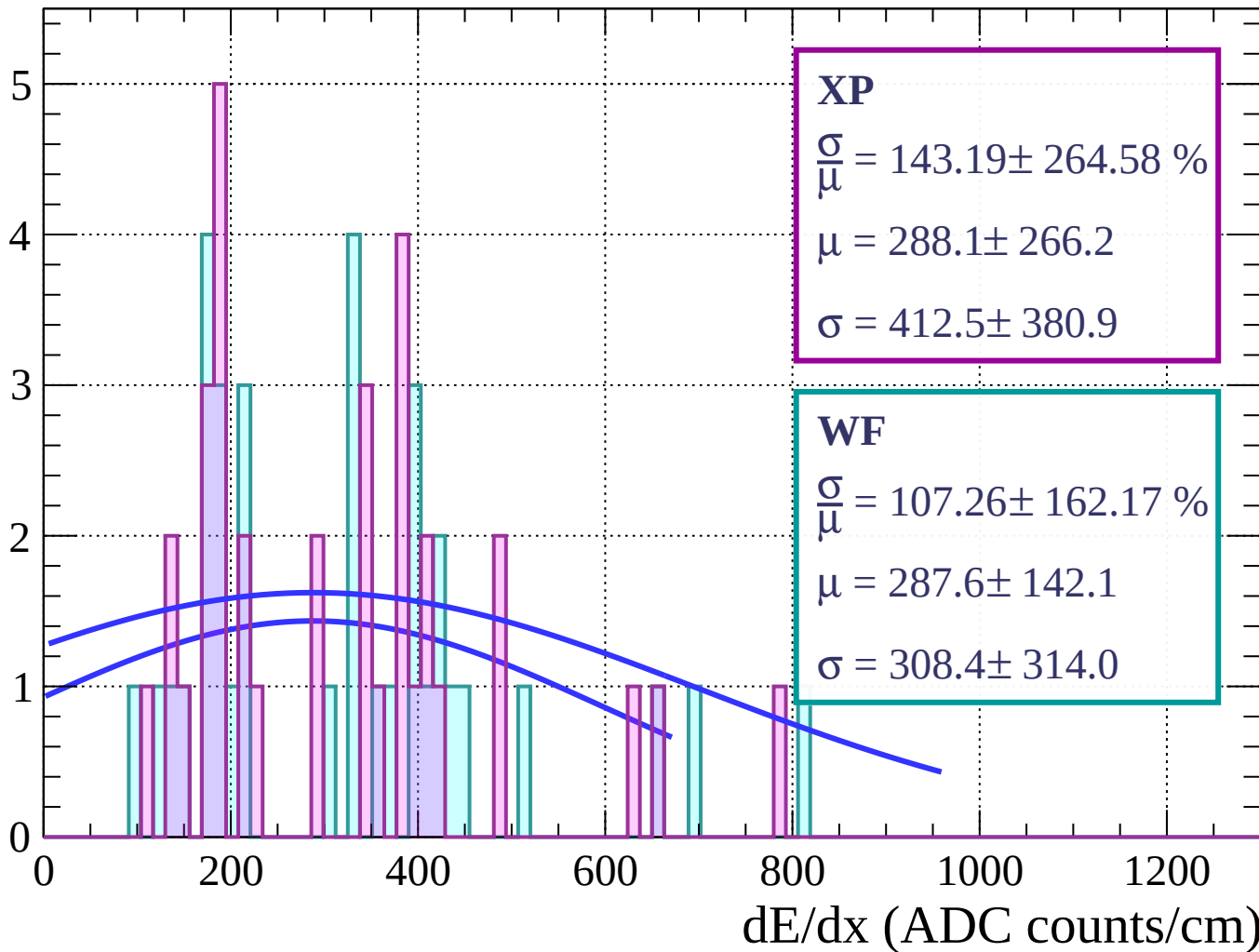
Energy loss | $63 < \theta < 66$

Count



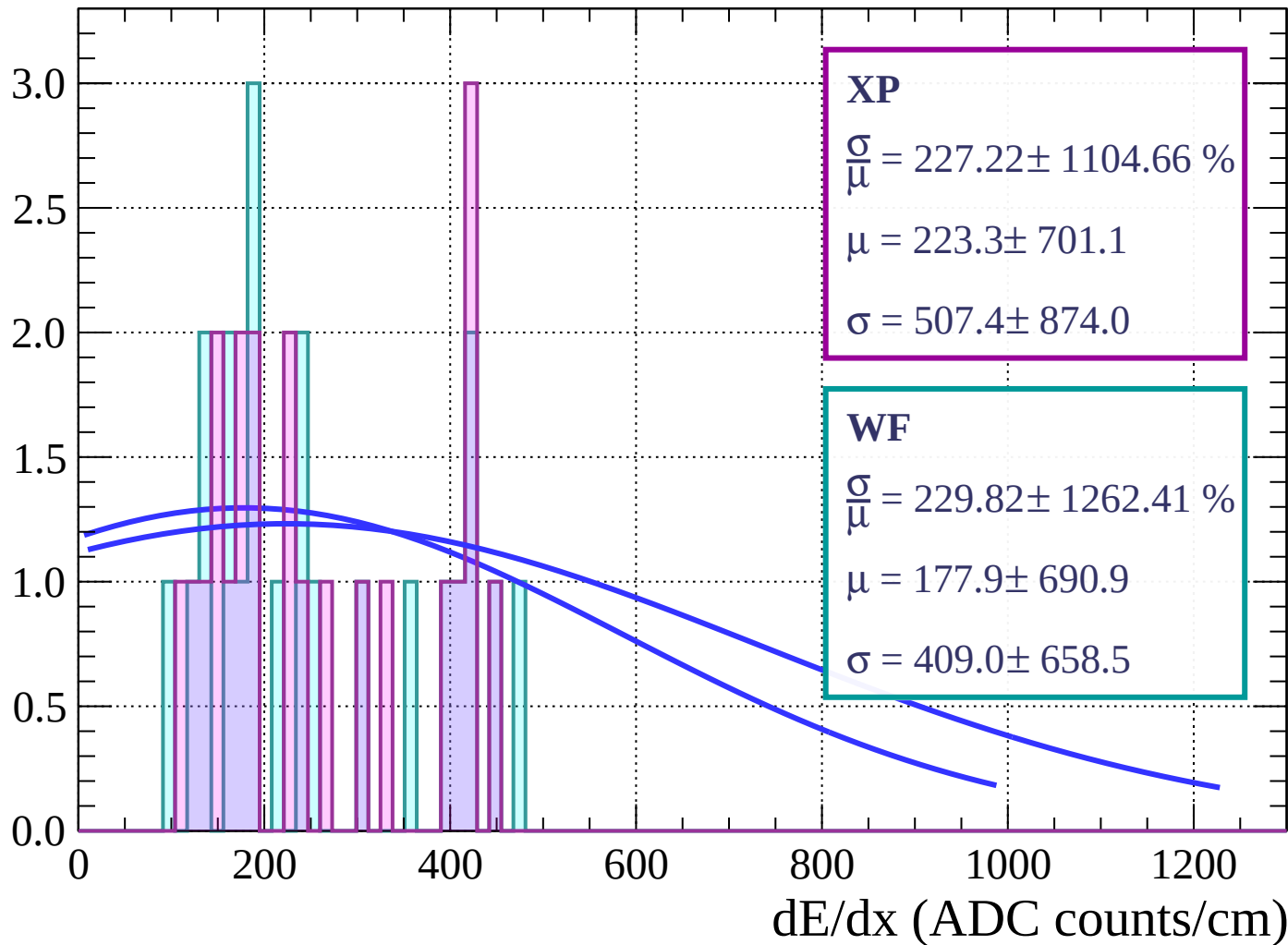
Energy loss | $66 < \theta < 69$

Count



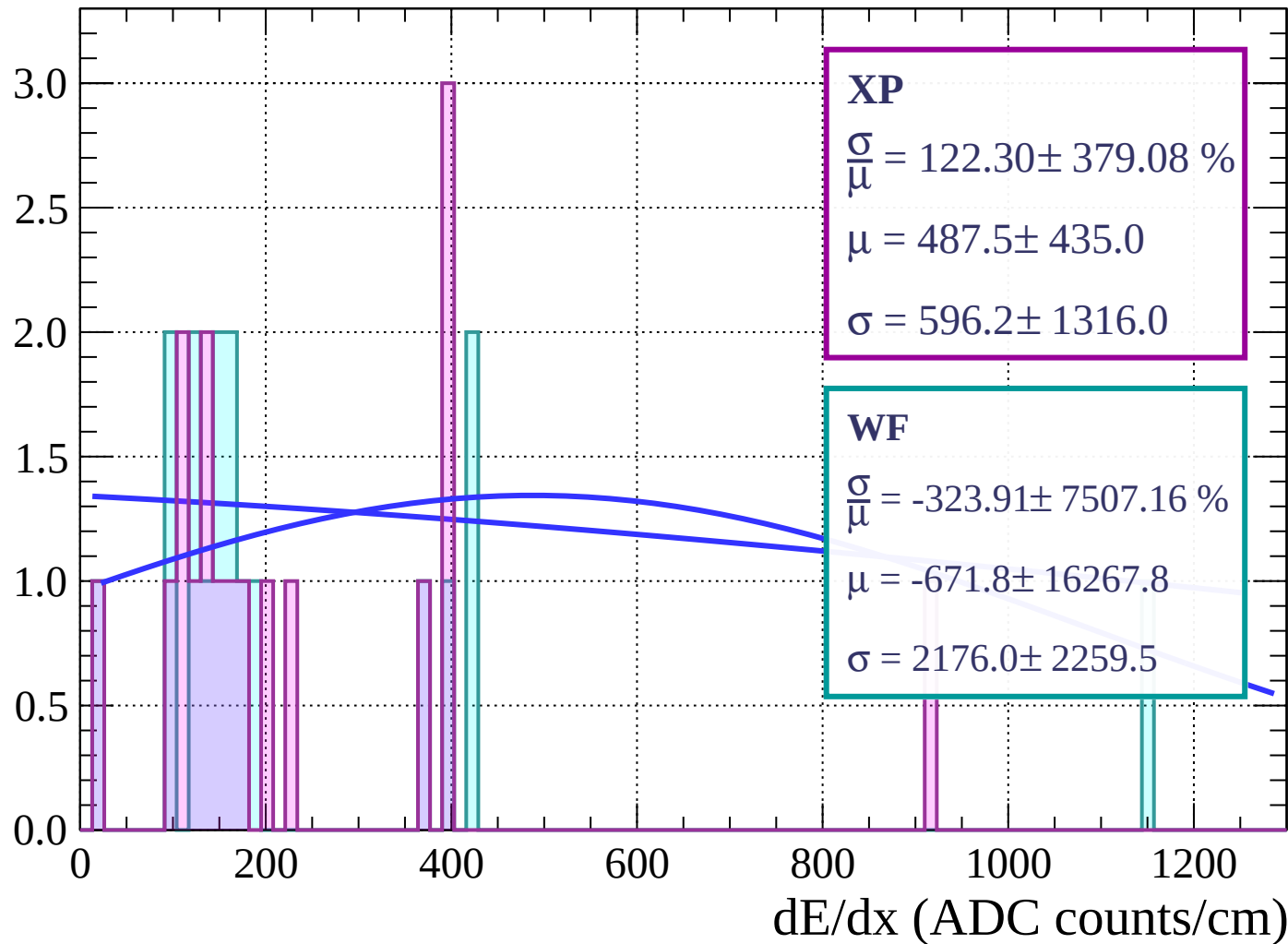
Energy loss | $69 < \theta < 72$

Count



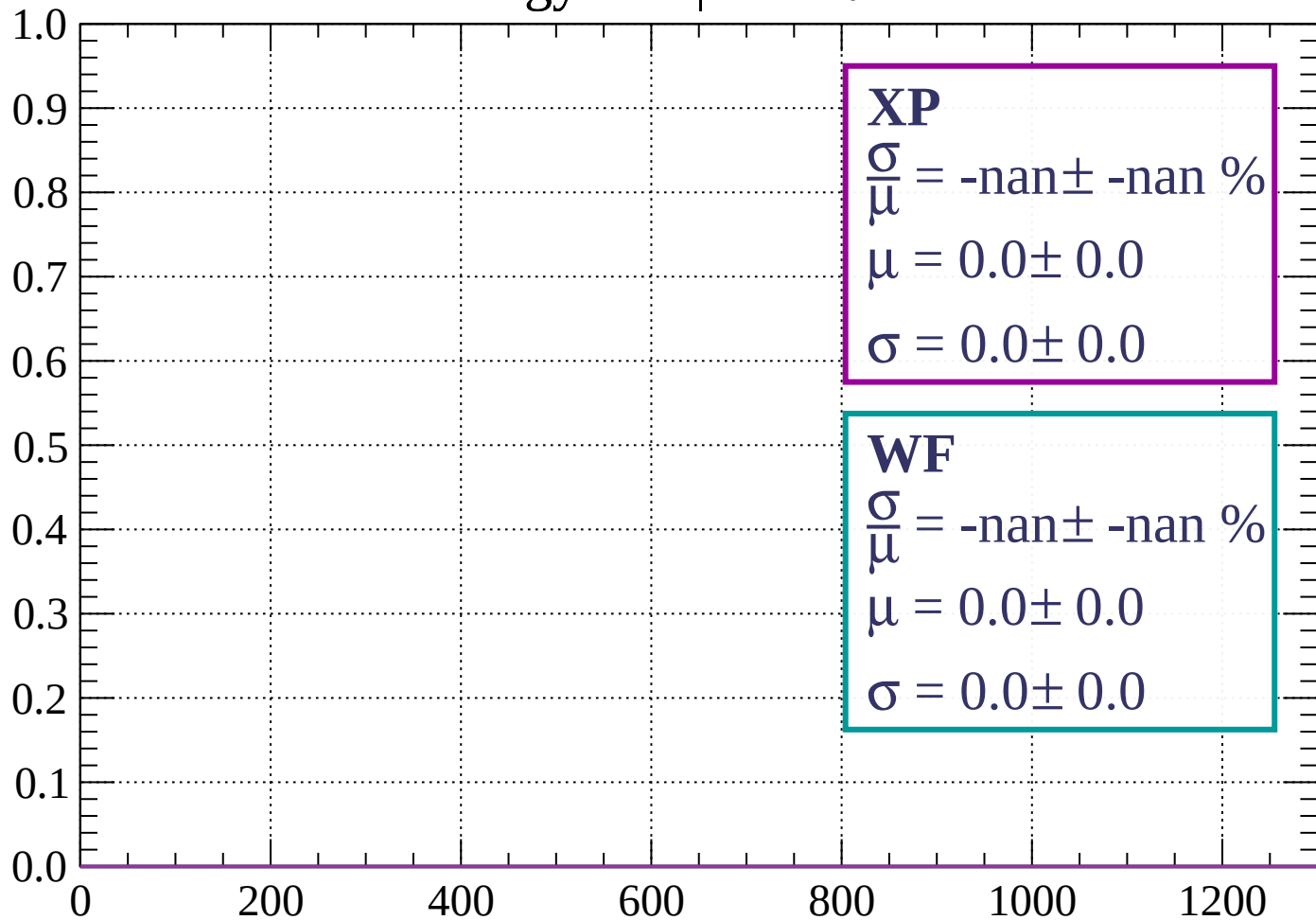
Energy loss | $72 < \theta < 75$

Count



Energy loss | $75 < \theta < 78$

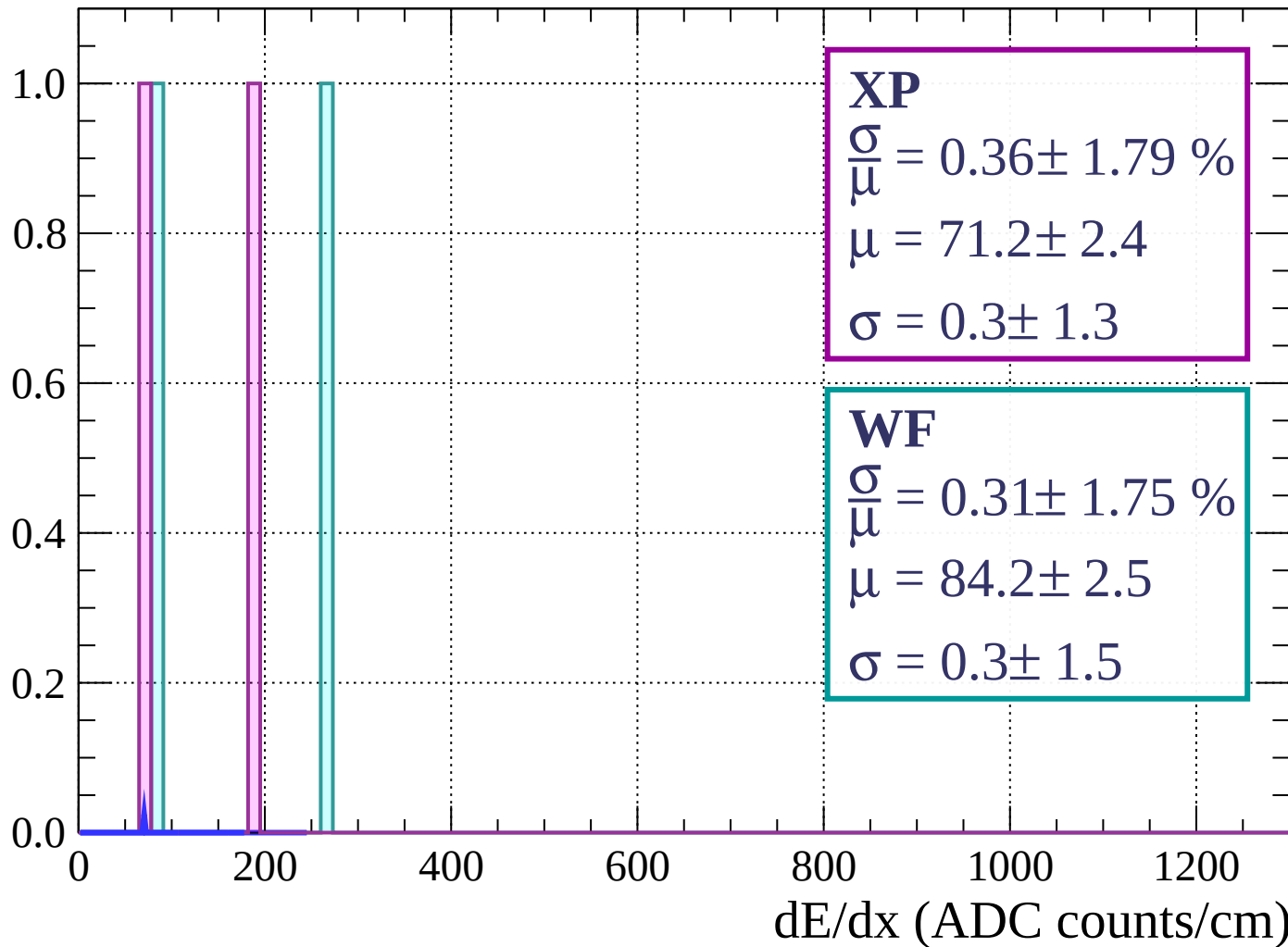
Count



dE/dx (ADC counts/cm)

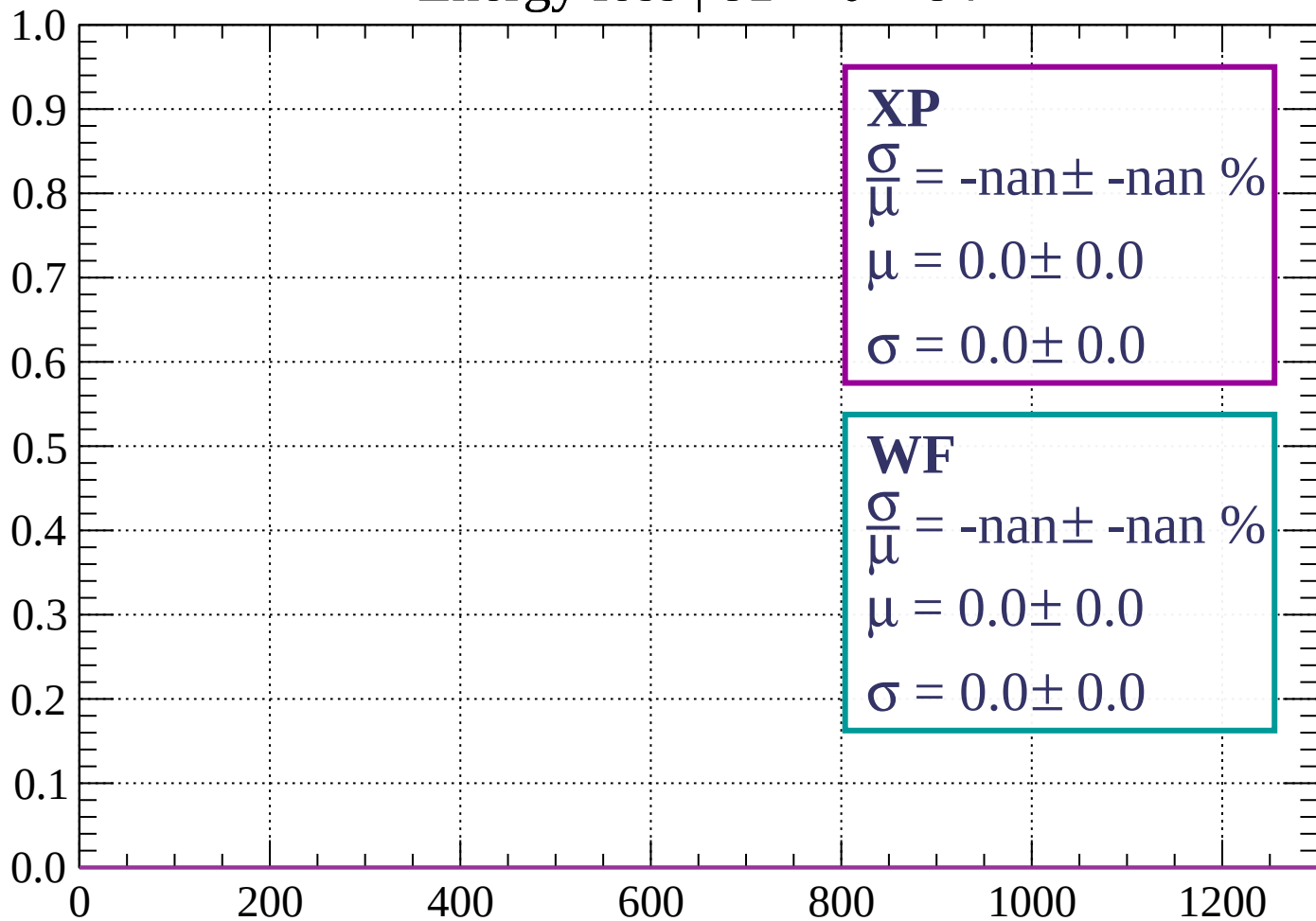
Energy loss | $78 < \theta < 81$

Count



Energy loss | $81 < \theta < 84$

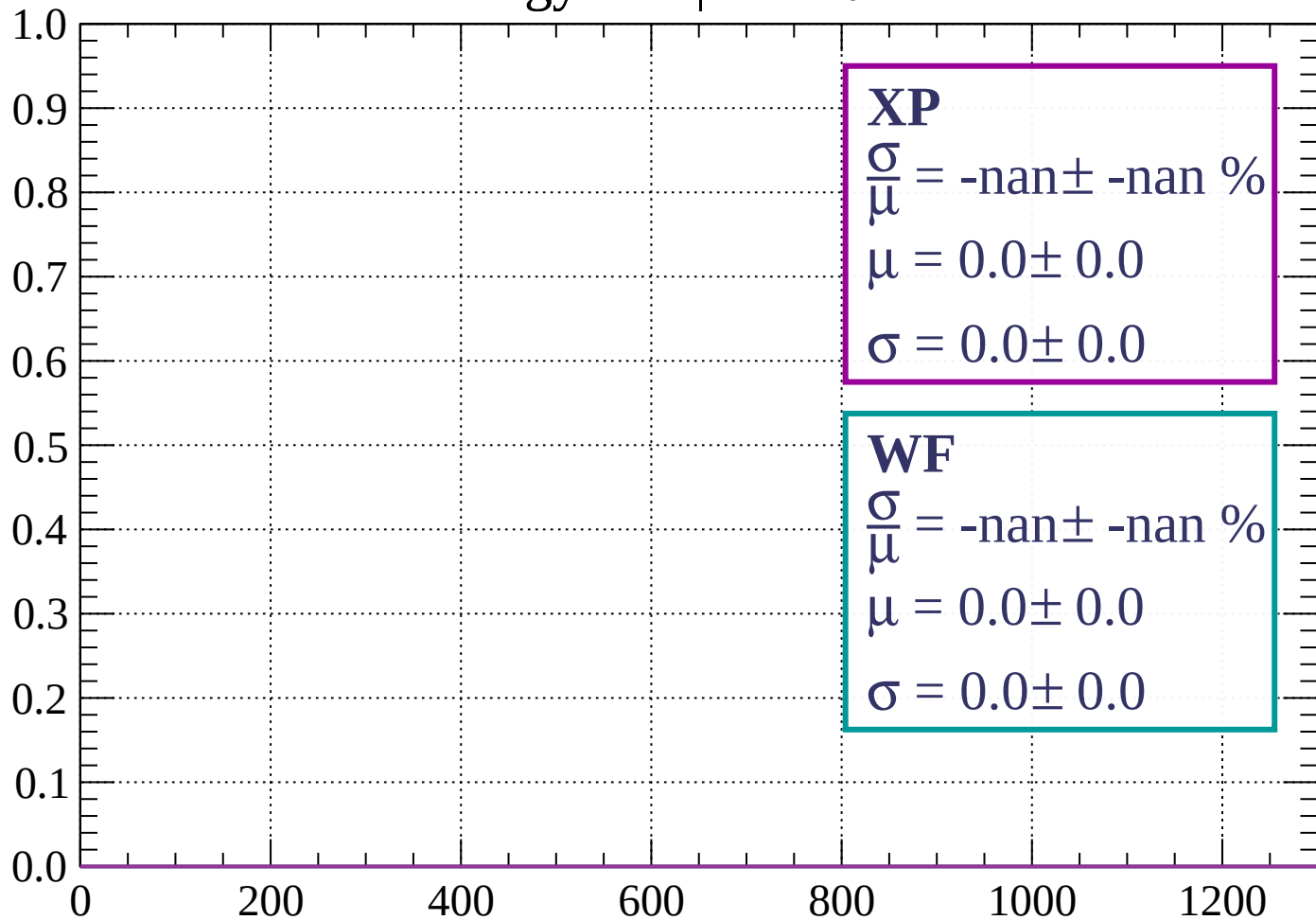
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 87$

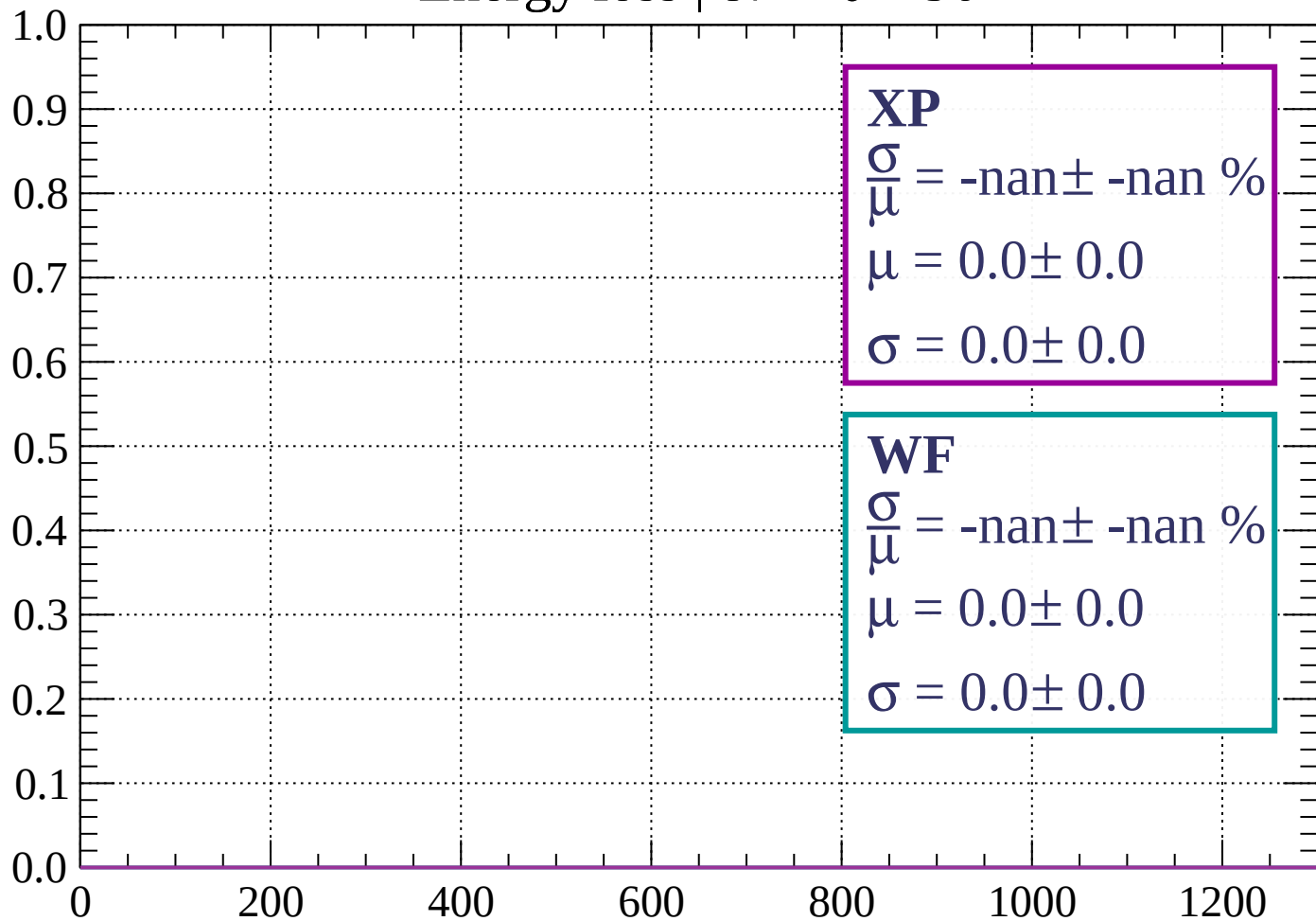
Count



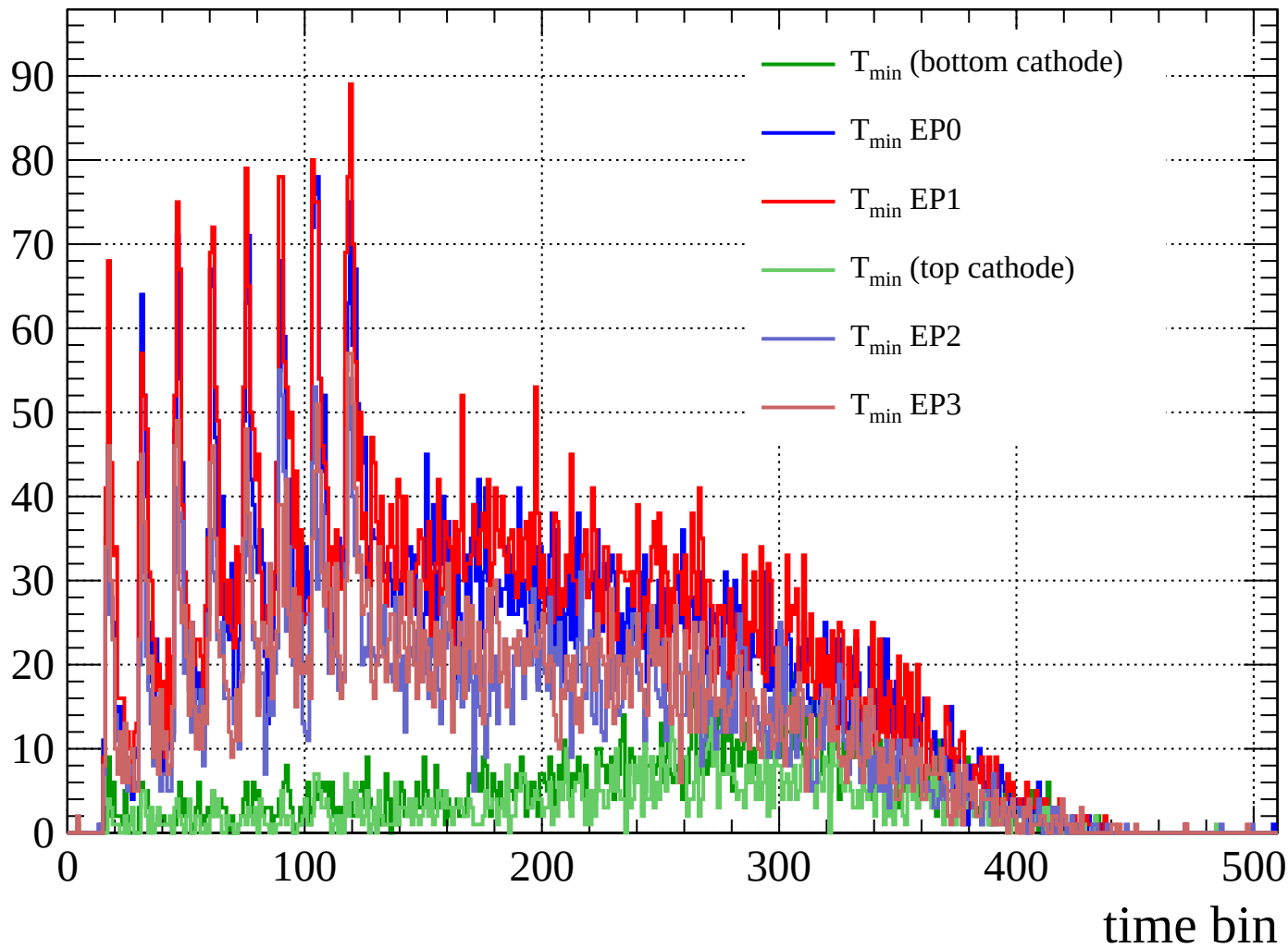
dE/dx (ADC counts/cm)

Energy loss | $87 < \theta < 90$

Count



Count



Count

